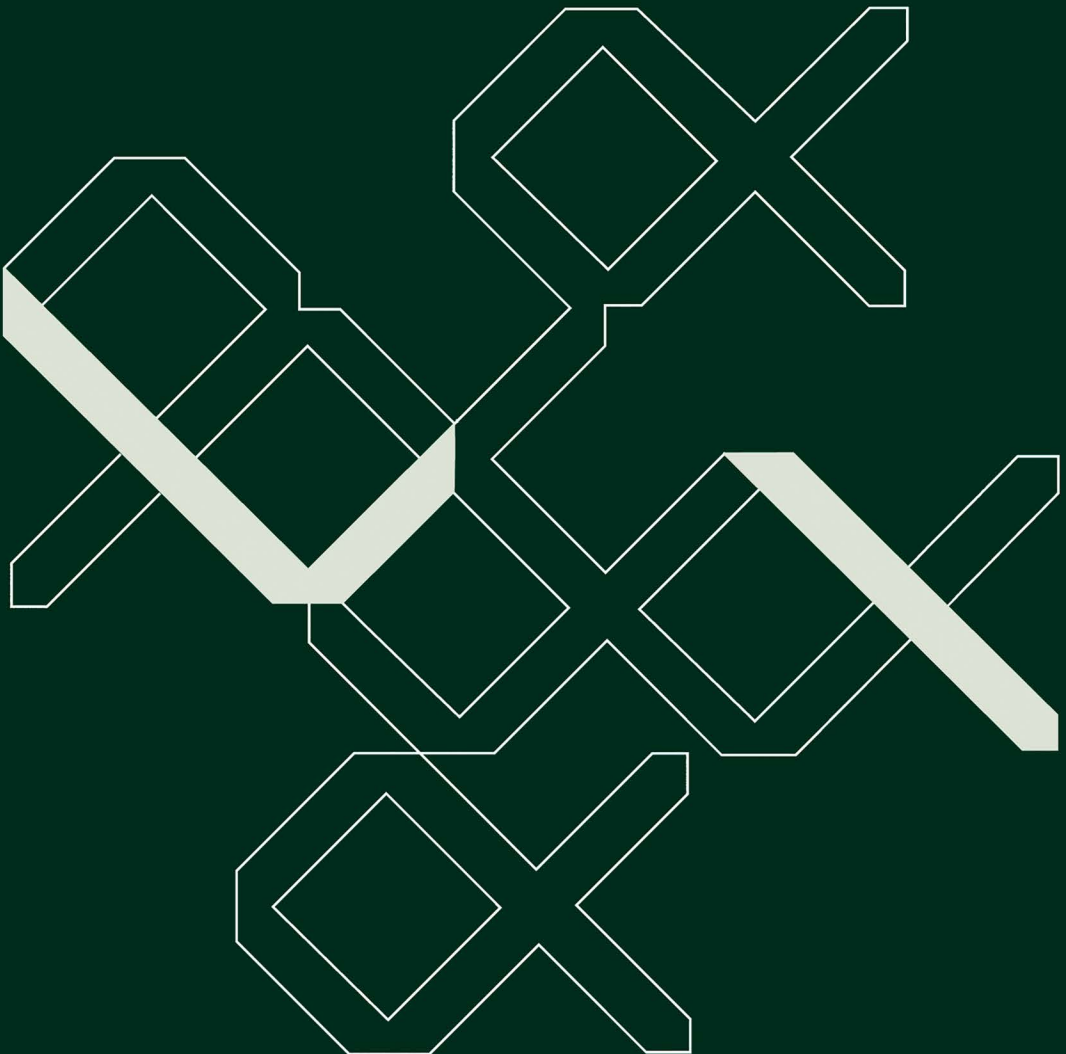




Linguistic Inquiry
Monograph Ninety

Mereological Syntax: Phrase Structure, Cyclicity, and Islands

David Adger



Mereological Syntax

Linguistic Inquiry Monographs

Samuel Jay Keyser, general editor

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Mereological Syntax

Phrase Structure, Cyclicity, and Islands

David Adger

The MIT Press
Cambridge, Massachusetts
London, England

The MIT Press
Massachusetts Institute of Technology
77 Massachusetts Avenue, Cambridge, MA 02139
mitpress.mit.edu

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The MIT Press would like to thank the anonymous peer reviewers who provided comments on drafts of this book. The generous work of academic experts is essential for establishing the authority and quality of our publications. We acknowledge with gratitude the contributions of these otherwise uncredited readers.

This book was set in Times New Roman by Westchester Publishing Services.

Library of Congress Cataloging-in-Publication Data

Names: Adger, David author

Title: Mereological syntax / David Adger.

Description: Cambridge, Massachusetts : The MIT Press, 2025. | Series:

Linguistic inquiry monographs ; 89 | Includes bibliographical references and index.

Identifiers: LCCN 2024052293 (print) | LCCN 2024052294 (ebook) | ISBN 9780262553278 paperback | ISBN 9780262384667 pdf | ISBN 9780262384674 epub

Subjects: LCSH: Grammar, Comparative and general--Syntax | Whole and parts (Philosophy) | Linguistics--Philosophy

Classification: LCC P291 .A34 2025 (print) | LCC P291 (ebook) | DDC 415.01--dc23/eng/20250422

LC record available at <https://lccn.loc.gov/2024052293>

LC ebook record available at <https://lccn.loc.gov/2024052294>

EU Authorised Representative: Easy Access System Europe, Mustamäe tee 50, 10621 Tallinn, Estonia | Email: gpsr.requests@easproject.com

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Series Foreword

We are pleased to present the ninetieth volume in the series *Linguistic Inquiry Monographs*. These monographs present new and original research beyond the scope of the article. We hope they will benefit our field by bringing to it perspectives that will stimulate further research and insight.

Originally published in limited edition, the *Linguistic Inquiry Monographs* are now more widely available. This change is due to the great interest engendered by the series and by the needs of a growing readership. The editors thank the readers for their support and welcome suggestions about future directions for the series.

Samuel Jay Keyser
for the Editorial Board

Preface

This book is not what I originally intended it to be. The Leverhulme Trust kindly granted me a two-year Fellowship to run from 2020 to 2022 to write a book on a project I'd proposed: my plan had been to try to add a kind of computational memory to the fundamental syntactic operation Merge, and to use that to solve some long-standing theoretical problems with Merge that I'd been worrying about for a while. I couldn't make the idea work properly, though. I don't really think my brain was in good shape: my partner had been diagnosed with cancer just before the fellowship started, and COVID was ravaging the world. Strangely, throwing myself into complicated syntax was almost therapeutic, and, while trying and failing to make Merge do what I wanted it to do, I got interested in something else: what it means for one thing to be *part* of another.

I started reading about mereology—the theory of parthood—in philosophy and logic. I remember the moment I had the basic idea of this book, that syntactic objects should be thought of mereologically and not set-theoretically. I was drinking a really good coffee in the East London café Ramble (actually, as it was COVID times, I was in the café's outside space, which is basically a car park under some old railway arches), and I scribbled a version of Merge in my notebook that made something part of something else, rather than constructing an entirely new set-theoretic object out of old ones. I saw, almost immediately, that this provided an answer to some of the theoretical issues that I'd been failing to solve. The resulting theory then more or less wrote itself after that, and it did so fairly quickly, which was good because the Fellowship was coming to an end!

In 2022, I presented an early version of the idea at the CRISSP Lectures, KU Leuven in Brussels, and, in that same year, at the Tromsø Workshop on the Foundations of Extended Projection, as well as at seminars in the Universities of Frankfurt, Bielefeld, Geneva, and, again, Tromsø. I was lucky to have amazingly engaged and helpful audiences in all of these places, and I thank the

organizers and participants for all their input. I presented the core of chapter 3 and some of the material from chapter 4 at GLOW 46 in Vienna in 2023, and at two seminars at Queen Mary University of London also that year; some material from chapters 4 and 5 was presented at NELS 54, MIT, and at seminars in Oxford University and Cambridge University in early 2024, as I began to work on the final revisions. Again, many thanks to the organizers and participants of all of these for questions, suggestions, and new puzzles. I also took my MA syntax class through a draft of the book, and I thank them, and the auditing PhD students and visitors, for their forbearance and thoughtful comments. Just before I completed the revisions, I presented an overview of the general idea at the GLOWing 2024 Syntax lecture, where Heather Newell was my discussant and made me think much harder about how this system interacts with phonology; it was a fascinating conversation, and I'm very grateful to her, and all the other people who attended that lecture and gave comments. The final outing of this work was at a series of lectures at the ZAS-Leibniz Institute in Berlin that Tonjes Veenstra invited me to give. These lectures covered the whole system and the comments I got from the attendees included some new, and correct, predictions! I'm extremely grateful for the opportunity to discuss the ideas with such an informed and thoughtful audience. Thanks too to the library in Sabhal Mòr Ostaig for giving me a desk for a week in June 2023 to look at the beautiful view across the Sound of Sleat to Knoydart, while I worked on chapter 4.

As well as two anonymous referees for the MIT Press, I'd also like to thank the following people for their comments on the first draft of the book: John Collins, Thomas Graf, Terje Lohndal, Tom Meadows, Mal Shah, Mark Steedman, Giuseppe Varaschin, and Qiuhaoyan, and also Kyle Johnson for a particularly helpful long conversation on a very hot day in Kew Gardens and Peter Svenonius for many helpful long conversations in London, Tromsø and, elsewhere! Thanks also to the following for questions and comments at talks, help with data or citations, and general discussion and inspiration: Enoch Aboh, Luke Adamson, Artemis Alexiadou, Daniel Altshuler, András Bárány, Theresa Biberauer, Jonathan Bobaljik, Josh Boivin, Tanya Bondarenko, Hagit Borer, Carlo Cecchetto, Chris Collins, Jeroen van Craenenbroek, David Diem, Marcel den Dikken, Caterina Donati, Michael Erlewine, Antonio Fabregas, Susana Fong, Grant Goodall, Lulu Guo, Daniel Harbour, Jutta Hartmann, Stephanie Harves, Matt Husband, Dany Jaspers, Idan Landau, Magdalena Lohninger, Andrew McInnerney, Luisa Martí, Tom Meadows, André Meinunger, Line Mikkelsen, Sergey Minor, Andrew Murphy, Hazel Pearson, Genoveva Puskás, Gillian Ramchand, Anna Roussou, Oddur Snorrason, Coleman Sokol, Dominique Sportiche, Linnaea Stockall, Gary Thoms, Satoshi Tomioka, Coppe van Urk, Tonjes Veenstra, Susi Wurmbrand, Guido van den Wyngaerde and Stan Zompì.

I owe a huge debt of thanks to the Leverhulme Trust who supported this research with a Leverhulme Major Research Fellowship (2020-2022), but who also, in cooperation with the Faculty of Humanities and Social Sciences at Queen Mary, granted me an extension until September 2023 to finish the first full draft of the book, which had been delayed by responsibilities for my partner's cancer care.

I always seem to need a new café to think and write in for each book, and though the pandemic curtailed café-going during some of the time I was working on this one, I want to thank especially Lorna at Ramble in Forest Gate, mentioned above, and Kemal and the whole team at Mattcup, in Leytonstone, for caffeinic assistance, especially in the final hectic period of trying to make the initial deadline.

The book is of course dedicated to Anson.

cha dhuinn an dùbailteachd:
tha 'n coimeasgadh 'sa' ghaol.

(from *A Chiall 's a Ghràidh*
by Somhairle MacGill-Eain)

David Adger
London
July 2024

1 Introduction

1.1 What This Book Is About

In generative syntactic theory, a grammar is a device, embedded in the mind of a human being by evolution and experience, that can generate an unbounded collection of structures. These structures can then be used by other parts of that human being's mind to provide shape to certain mental formations (meanings) and physical formations (speech, sign, etc.). The grammar, then, generates structures that enable and negotiate the connection between meanings and forms. One term for these generated structures is syntactic objects (SOs). SOs are what the grammar produces, and the theory of syntax is fundamentally a theory of what produces them and how they connect to the other systems of the mind.

Linguists often represent an SO produced by a grammar via a tree—that is, a particular kind of object that can be given a mathematical characterization in graph theory. However, it's not always the case that the grammars, as theorized, produce trees, as mathematically characterized. Trees are a useful representation of SOs, but they are not necessarily the generated objects. In fact, as I'll discuss in chapter 2, early generative grammar took SOs to be sets of (sets of) strings, and minimalist approaches since the mid-1990s have taken SOs to be sets whose members are either lexical items or, themselves, SOs.

This book explores an operation for generating SOs that is familiar in some ways: it takes two preexisting SOs and combines them. However, rather than making them into a set, as does the Merge operation of Chomsky (1994) et seq., it makes one of them *part* of the other. I call this operation Subjoin, and it plays the same role as Merge plays in the standard Bare Phrase Structure theory that has served as the basis for minimalist syntax for about three decades now. Subjoin creates more complex SOs from simpler ones, and, just like Merge, it has an “external” and “internal” version. SOs in this new

system are not set-theoretic: they are mereological. Mereology is the philosophical and logical study of the part whole relation, and I adopt various ideas from the results of that field and adapt them so that they can provide a workable model for SOs. The book's purpose is to set out how this works and to explore some of the consequences of developing this system. In particular, I define a very simple constraint on mereological objects that I call Angular Locality. This constraint, I argue, has wide-ranging ramifications for what constitutes a possible SO, derives successive cyclicity as a theorem, and opens up a new approach to explaining why certain island phenomena behave as they do.

1.2 Overview

The argument of the book starts from some history. Chomsky's 1955 magnum opus, *The Logical Structure of Linguistic Theory*, delineated a view of syntactic objects as sets of strings. Chapter 2 traces the arguments about the nature of syntactic objects through the 1970s and 1980s, focusing on the development of the $\bar{X}$ -theoretic approach, and the reasons why order was removed from the theory of phrase structure. It then moves to the minimalist critique that Chomsky developed of $\bar{X}$ -theory in the early 1990s, his introduction of the operation Merge, and various changes of perspective since then about what Merge does, and why it does it. I then focus on two shortcomings of that model. The first is that, when Merge builds a set-theoretic SO, that new SO needs to be categorized (have a label) and that requires a new subtheory to do the labeling. The second concerns the way that Merge models long-distance dependencies via the construction of "copies" of syntactic objects. These copies also require a special subtheory to distinguish them from items that are formally identical but not constructed via Merge operating long distance. I argue that it would be better to have a system that didn't require these extra subtheories.

To this end, chapter 3 develops a new system based on the idea that syntactic objects are not set-theoretic; rather they are mereological. Usually, in linguistics, classical mereology is appealed to in the analysis of mass terms, plurals, imperfective aspect, and so on (e.g., Link 1983), and its attractiveness in these domains is because it provides a way of modeling undifferentiated parthood. However, syntactic objects are clearly highly differentiated, so classical mereology is insufficient. Instead, I introduce a particular kind of mereological system from the work of philosophers like Johnston, Koslicki, and Fine, and I argue that this is a better model for phrase structure than sets. This approach to mereology takes objects to consist not just of their parts but also of the ways that their

parts are composed, and it further allows parts to be part of objects in different ways (what I call *dimensions* of parthood). This provides the differentiation in the structure of SOs required by syntax.

Chapter 3 defines Subjoin as an alternative to Merge. Subjoin takes two SOs and makes one of them part of the other. I show how the time course of the syntactic derivation can be used to differentiate different dimensions of parthood. Subjoin provides the way that parts are parts of their whole, and dimensions provide different “flavors” of that combinatory mechanism. The first subjunction to an SO is in dimension 1 and roughly corresponds to the Extended Projection complement relation of Grimshaw (1991); the second subjunction is in dimension 2 and corresponds roughly to the specifier relation.

This new system, it turns out, is no more complex than the set-theoretic system, but it needs no special subtheory for labels (since, when you make one SO part of another, nothing changes in terms of the labels). It also derives syntactic objects where a single SO can be part of multiple other SOs, rather than requiring “copies.” A consequence of this is that it is straightforward to distinguish copies from non-copies. The new system is, then, at least as simple as the standard set-theoretic system but an improvement over it in that it requires no special subtheories to accomplish labeling or to distinguish between copies and non-copies.

I then turn to the implications of the mereological approach for theories of locality. I show how a simple extension of the most local relations between SOs in the system provides a theory of locality that answers four questions that are not properly answered by current theories:

- (1) a. Why are there locality domains?
- b. Why are they what they are?
- c. Why are there exceptions to locality domains?
- d. Why are these what they are?

In the mereological theory, there are locality domains because the part relation is logically transitive, but transitivity does not hold across dimensions of parthood. This means that some SOs are parts of another SO, but they are not part of that SO in a consistent dimension. If syntax “sees” parts only via dimensions, then those “disconnected” parts are invisible to it. That there are locality domains is a side effect of fundamental properties of the part relation.

The locality domains are what they are because of the two dimensions of syntax: essentially the innards of a specifier are in a different dimension from SOs outside of the specifier. This means that they are “invisible” from outside the specifier, from which it follows that specifiers, in general, are locality domains. This is, of course, a different theory from the standard view that there

are stipulated categories (phases) that constitute the domains, and much of the book is an empirical defence of this new approach.

Turning to (1c–d), there are in fact no special exceptions: locality domains can be “escaped” because of the logical transitivity of parthood within a dimension. If something is a specifier of a specifier of an SO, then transitivity implies it is a specifier of that SO. But then it is local to that SO, which makes the “edges” of specifiers transparent, while the insides of specifiers are opaque. Again, this is quite different from the standard theory, which simply stipulates that locality domains (phases) have exceptions (their edges), a stipulation that follows from nothing else. The fact that the specifier of a specifier, as opposed to some other position, is an “escape hatch” also answers the question in (1d) of why the exceptions to locality are what they are.

With this framework in place, I define a locality principle, *Angular Locality*, that ensures the results just discussed, and I show how, as a side effect, it also rules out super-local dependencies (such as that between a complement and a specifier of a category), and super-non-local dependencies (the equivalent in this system of Parallel Merge and Sideways Movement). I illustrate how the system works through an analysis of Gaelic wh-extraction. The system also extends elegantly to certain partial-wh-movement constructions and both adpositional and clausal pied-piping as a means of expressing long-distance phrase-structural dependencies. I close the chapter with an examination of how to map from mereological structures, which have only hierarchical dimensionality, to the linear structures that are required for most kinds of externalization of syntax (pronunciation or signing).

The bulk of the rest of the argument of the book focuses on how this system plays out in the domain of islands, one of the most central topics in syntax since the 1960s. I first show in chapter 5 how the system predicts wh-islands. Because of the interaction between the Transitivity axiom of parthood and Dimensionality, if an SO is subjoined to another SO in the second dimension (i.e., it is a specifier) but it has, itself, something subjoined to it in the second dimension (i.e., it has a specifier), it is impossible to subjoin anything else to it. This means that wh-islands (almost) follow from the system: a wh-island has a wh-expression subjoined to it, and a CP in this system is always a specifier (since it is a separate Extended Projection). The standard Bare Phrase Structure system does not have this consequence, since it allows multiple specifiers, and so has multiple positions at phase edges. This is why in the standard system, wh-islands are usually explained via minimality effects rather than by Phase Theory. In the mereological system, wh-islands emerge through the theory of phrase structure itself.

Over the past few decades, a number of exceptions have been discovered to the wh-island effect (especially Scandinavian languages, but also Italian, Hebrew, some Slavic languages, some varieties of English, etc.), and one might ask how these exceptions to wh-islandhood can arise in the mereological system. I adopt an analysis based on the idea that, in these cases, the “offending” wh-expression is not actually at the edge of the clause, because these languages allow a further category at clause edges that provides an extra specifier. One might think that this robs the theory of empirical bite; however, I show that it predicts a new effect, which I call the *Wh-Island Re-Emergence* (WIRE) effect, not predicted by the standard system. The WIRE effect arises because, although it is possible to subjoin an element outside of a specifier, it is not possible to subjoin two elements from the same specifier, because of the way that Transitivity and Dimensionality interact. I show how the WIRE effect can be detected in Norwegian, Danish, Hebrew, Italian, and English. This consequence of the theory (that it is not possible to extract two elements from a single constituent) would seem to be falsified by languages, like Bulgarian, that allow multiple wh-movement, but I proffer a new analysis of this phenomenon that I call the *declustering* analysis, which explains not only why multiple wh-movement of the Bulgarian sort is possible but also why, in such constructions, only absorbed readings of the wh-quantifier are possible. I also argue that this system predicts that in these languages the WIRE effect should not appear, and I show that this prediction is correct.

One of the consequences of the mereological system is that there’s no real geometrical difference between subjects and objects. However, there is a long tradition in generative syntax that says that extraction from objects is more acceptable than extraction from subjects. I argue that this is wrong in chapters 6 and 7.

I do this by first, in chapter 6, examining nominal island effects where the nominal is in object position. I analyze both extractions from prepositional complements to nouns and from clausal complements to nouns, and I argue that these all boil down to specificity effects. I show that, in the mereological system, such effects emerge in the same way as wh-island effects do: an internal part of the nominal subjoins to the SO that constitutes the nominal as a whole and, by Dimensionality, blocks any further subjunction from below it. I argue that the proposal I made in my 2013 book for nominal structure should be adopted (Adger 2013). This is the idea that apparent adpositional complements to nouns are actually introduced by functional structure that is relatively high in the noun phrase. I extend to clausal complements my previous work on adpositional phrases and show that, once this perspective is adopted, definiteness can be understood as involving subjunction to the category D. Indefinites do not

involve this subjunction, and so it is predicted by the system that subjunction (extraction) is possible from indefinites; definites and specifics, in contrast, have something subjoined to D, blocking extraction and deriving some of the empirical consequences of Fiengo and Higginbotham's (1981) Specificity Condition. I then extend the analysis to *wh-in-situ* cases like Persian and Mandarin, and show how the same effect emerges. I adduce further evidence from Mandarin that subjunction of any constituent to D gives rise to specificity effects and the concomitant blocking of extraction. With this in hand, I then show how the proposals extend to Slavic languages, and I provide an alternative to the widespread NP analysis of these languages. I argue that the mereological approach is an improvement over the NP analysis of these languages, as it makes a more precise prediction about the collocation of overt articles and extraction, and I appeal to data from Mayan, Slavic, and Austronesian languages that backs up this direction of analysis. The system also extends to some long-standing problems for the Specificity Condition, such as the Clitic Gate phenomenon in Galician and the *Ezafe* construction in Persian. The core difference between the systems is that, in the mereological approach, definiteness is syntactic, while in the Specificity Condition approaches, it is semantic. This chapter also extends the theory to the extraction effects found in Scrambling constructions in Germanic, showing that classical analyses based on Huang's Condition on Extraction Domains (Huang 1982) constraint or using the Freezing Condition (Wexler and Culicover 1981; Stepanov 2007) cannot be correct, and that syntactic definiteness captures the empirical patterns more effectively.

I further argue that the differences between weak and strong definite articles in Germanic can be understood as deriving from whether a Determiner category subjoins to D or not. I show how this correlates with extraction and then extend the idea to a new approach to the classical Specificity Condition paradigm in English, where the definite article shows a weaker effect than Possessors or Demonstratives. I argue that one way of thinking about this is in terms of the accessibility of a particular category to the parser in the speech event/context. In English, the definite determiner *the* is actually ambiguous, one instance being equivalent to strong articles in German and the other equivalent to weak articles. The former obligatorily subjoins to D, while the latter does not, so only the latter is compatible with extraction. The idea is that the weak version is less accessible in English, so the judgement of acceptability is essentially a side effect of this accessibility property: extraction is grammatical but requires a weak definite parse of the definite article, and that is more marked in English. I also show how this idea of weak and strong definiteness can explain a number of apparent exceptions to the Specificity Condition, such as with process nouns and superlatives, and how it can handle long-standing

problems such as extraction from definite nouns that are complements of verbs of creation. Finally, I show how this system extends to the classical Complex Noun Phrase Constraint (CNPC) phenomena and the exceptions to the CNPC noted by Ross in 1967. Overall, the core point of the chapter is that extraction from noun phrases is fundamentally regulated by whether the noun phrase has part of it subjoined to itself, a configuration that can be detected by how it is both pronounced and interpreted.

Finally, in chapter 7, I turn to subjects. Since subjects and objects are not geometrically distinct in mereological syntax (both are specifiers, or, more technically, they are parts in the second dimension), the system cannot appeal to anything like the CED to explain why subjects are more difficult to extract from than objects. The mereological system is also essentially multidominant in how it treats long-distance dependencies (“internal” subjunction). This means that it cannot appeal to Freezing type explanations of subject/object asymmetries, since these rely on chain links with distinct properties. I argue rather that subject NP islands must, like object NP islands, essentially be connected to syntactic definiteness, and I show how, with a careful evaluation of the data, this turns out to be correct. In the chapter, I draw on a great deal of experimental work on subject islands, much of which emerged in an attempt to test the kinds of predictions that CED and Freezing accounts predict (and more complex versions, such as Chomsky’s [2008] *On Phases* system). I review experimental work on German, English, Russian, and Hungarian. In addition to the experimental work, I gather evidence from English, French, Spanish, and Greek showing that none of these accounts capture the right generalizations. I also look at more recent approaches, including those that appeal to the information-structure syntax interface, and extend some ideas that were first proposed by Bianchi and Chesi (2014), who argue that the traditional distinctions of the *On Phases* theory (basically the differences between transitive, unergative, and unaccusative predicates) are insufficient. I argue that this approach is on the right lines but that a mereological interpretation of it has a number of advantages, and, of course, since that system cannot appeal to CED or Freezing type accounts, it has the advantage of ruling these out as possible analyses available to the language acquirer. To capture the empirical effects, I propose, following, ultimately, Diesing (1992), that English (for example) has two categories that host surface subjects (two T categories): one of these is similar to the scrambling clausal category I proposed for German, and the other is similar to the classical analysis of English T (it bears an EPP feature). I suggest that the former is correlated with an individual topic interpretation (a categorial construal), while the latter is interpreted as an event/situation topic interpretation (a thetic construal), providing a different implementation of Bianchi and Chesi’s ideas within

the mereological system. The theory correctly predicts that it is impossible to extract from subjects that are construed as topics, but possible to extract from subjects that are not. Just like I argued for definites in English, I take there to be an accessibility effect with these two T categories: it is easier to construe the subject as a topic (i.e., to use the T associated with this interpretation) so parses of extractions from subjects must involve the less accessible T, and that is why the experimental data tends to show there is an island effect. I show how this accounts for the experimental data discussed earlier in the chapter and extend it to the case of Hungarian, where I argue that Hungarian topics, which seem to be transparent to extraction just in certain cases, are actually always transparent to extraction because they bear the feature that checks topicality on D rather than on a lower category that subjoins to D (making Hungarian parallel to Persian in some respects).

The final chapter concludes with some suggestions about how to understand other CED-related island effects, namely relative clauses and clausal adjuncts. I adopt Sichel's (2018) idea that raising relatives can be extracted from, while matching relatives cannot, and show how this can be stated quite naturally in the mereological system, and then I extend the analysis of matching relatives to clausal adjuncts. I then conclude with a number of open questions. The theory of locality offered here is fundamentally geometric (sharing, in that respect, the intuition that underlies Connectedness accounts of locality like Kayne 1983), but there are other aspects of locality beyond cyclicity and islands that it doesn't touch on. Most obviously, the system as developed doesn't appeal to Relativized Minimality as an explanation of islandhood, though it could be the case that there is a role for some kind of minimality effect in how features are checked. There are also questions of the A/A-bar distinction, which is not touched on here, and whether Improper Movement effects can be integrated with the system (perhaps generalized to Williams Cycle effects as in Williams 1994; Keine 2020a; Meadows 2024). However, even though there is much more to explore, I hope the book shows that it can be fruitful to think about phrase structure in a different way from the standard set-theoretic approach.

2 Sets

2.1 Introduction

When I say a sentence, or comprehend one, something particular happens in my brain. There is a momentary configuration of mental matter. We might think of it as a “gesture of the mind” (Adger 2022). Some aspects of that mental gesture are grammatical. We have scant knowledge of the fine-grained physiological or neurological properties of what happens, though there are a number of architectures that have been proposed for how aspects of syntax might be neurally present (e.g., Murphy 2024). However, we do have well-developed, and to a great extent consensual, theories of the linguistic properties of this mental gesture, specifically its phonological, syntactic, and semantic attributes. Zeroing in on the syntactic aspect, linguists generally agree that it encodes constituency, grammatical category, particular kinds of dependencies between constituents, more global organization of constituents (such that certain categories are hierarchically superior to others), and sufficient information to connect with systems of meaning and pronunciation. There is also consensus that what I have in my head is not just a list of such mental gestures that I select from when producing or comprehending a sentence; rather I generate one as needed, using my mental grammar of, say, English. We can abstract out the syntactic properties of such mental gestures and think of them as “syntactic objects” (SOs). A syntactician, then, is engaged in the task of determining what the possible (types of) SOs are in a language, and what the grammar that generates them is. The task of the theoretical syntactician is to build theories of the grammars and the objects they generate.

The conception of SOs within approaches guided by the Minimalist Program (Chomsky 1992 et seq.) is that they are appropriately understood as set theoretic: they consist of nested sets, and the grammar is a grammar that generates such nested sets, via an operation called Merge. This implies that the properties

of such nested sets are homomorphic to the relevant properties of the actual syntactic objects (qua abstractions over mental gestures). Chomsky (1994) set out a number of arguments that syntactic objects are better modeled by sets than by other means, some of which I'll discuss in this chapter.

The purpose of this book is to argue for an alternative to sets: the idea is that SOs, those momentary mental gestures, are organized via a part-whole relationship, not a set-membership relationship, and that they are best modeled mereologically.¹

The fundamental relation of set theory is membership, which is intransitive (if a is a member of the set b , and b itself is a member of the set c , a is not a member of c). The fundamental mereological relation is parthood, which is transitive (if a is a part of b , and b is a part of c , then a is a part of c).² I will argue that SOs are objects with a part-whole structure, and a new SO is simply a preexisting SO to which another SO has been added as a part. I call the operation that does this *Subjoin*. Subjoin does the same kind of work as Merge, in that it builds SOs. The intuition I will work through in the book is that parthood is a better fundamental relation for modeling syntax than membership is, and that Subjoin is a better fundamental operation for the construction of complex SOs than Merge is. I will develop a system that is still able to capture the structured nature of syntax but that has parthood as its core relation. I will adopt Chomsky's idea that there is a single operation that builds SOs (Chomsky 2004), but I will argue that Subjoin is to be preferred to Merge and will show that it has interesting theoretical and empirical consequences.

Before turning to defining the theory and examining its properties, I will, in this chapter, provide a brief sketch of what motivates the current minimalist view that SOs are sets. The discussion will be partly historical, partly conceptual. I will then outline a couple of major problems for this approach, before turning to my alternative in the following chapter.

2.2 A Brief History of Phrase Markers

The theory of Phrase Markers that developed during the 1960s in transformational generative grammar assumed that linear order was baked into the grammatical system by the Phrase Structure component (e.g., Chomsky 1965, chapter 2). Phrase Structure rules in that theory concatenate linguistic symbols (categories and lexical items) into strings of such symbols. A Phrase Structure grammar, which is a collection of such rules, specifies derivations that rewrite strings of symbols as new strings, according to the rules of the grammar. From these derivations, Phrase Markers can be determined, and these constitute the legitimate Deep Structures of the language. Because of this architecture, Deep

Structures are linearly ordered, although their ordering is not the order of the utterances they underlie.

In Chomsky's original proposal (Chomsky 1955) in *The Logical Structure of Linguistic Theory* (LSLT), a Phrase Marker is a set: each line of a derivation (a string of symbols) is a member of this set, and, from the set, it is possible to extract information that determines the part-whole structure and linear order of a sentence. This particular architecture was influenced by the mathematical work on recursive function theory that had developed during the previous few decades. Chomsky's Phrase Structure rules used concepts developed by the logician Emil Post (Post 1943; see Pullum 2019 for some discussion). Post formalized logic via a system of Production Rules. These operate on strings of symbols and are sensitive to both the form of the symbols and their order. The string-theoretic basis of Phrase Markers in early generative grammar was partly a side effect of this mathematical technique. It was also partly motivated by the prevailing view of structural linguistics at the time (especially Chomsky's PhD supervisor, Zellig Harris [Harris 1951]) that order was a central property of language.

In addition to order, Phrase Markers need to capture hierarchy, since hierarchy is a fundamental property of sentences, and one of Chomsky's insights was that nonterminal symbols in Production Rules could capture that hierarchical information. These set-theoretic Phrase Markers were the theory of SOs throughout the 1960s and 1970s.

As an example, the rules in (1) determine the core divisions of some simple transitive clauses in English, where the symbol $\frown$ signifies left-to-right concatenation:

- (1) a. $S \rightarrow NP \frown VP$
- b. $VP \rightarrow V \frown NP$
- c. $NP \rightarrow \begin{cases} Lilly \\ Anson \end{cases}$
- d. $V \rightarrow \begin{cases} bit \\ scratched \end{cases}$

One well-formed derivation from this set of rules, where the start symbol S is systematically rewritten according to the rules, is (2):

- (2) a. S
- b. $NP \frown VP$
- c. $Lilly \frown VP$
- d. $Lilly \frown V \frown NP$
- e. $Lilly \frown V \frown Anson$
- f. $Lilly \frown bit \frown Anson$

An alternative expands the VP symbol before the initial NP symbol:

- (3) a. S
 b. $NP \frown VP$
 c. $NP \frown V \frown NP$
 d. $Lilly \frown V \frown NP$
 e. $Lilly \frown V \frown Anson$
 f. $Lilly \frown bit \frown Anson$

There are of course other possibilities in terms of which rule is applied when in the derivation, but it can easily be seen that they are equivalent, in the sense that each derivation connects the start symbol S to the final string, and each does so in a way that groups, for example, the verb and the object together, exclusive of the subject. The Phrase Marker, as defined in these early approaches, is the set of lines of all such derivations, and, given that it is a set, identical lines appear just once.

When we gather all the lines of all possible derivations for this set of Phrase Structure rules, we have the following Phrase Marker:

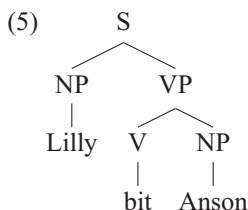
- (4) $\{S, NP \frown VP, Lilly \frown VP, Lilly \frown bit \frown NP, NP \frown V \frown NP, NP \frown V \frown NP, NP \frown bit \frown Anson, Lilly \frown bit \frown Anson, NP \frown bit \frown NP, Lilly \frown V \frown NP, Lilly \frown V \frown Anson\}$

From this Phrase Marker, it can be determined, by comparison of the members of the set, that, for example, the initial word *Lilly* is an NP, and that *bit Anson* is a VP, and that *Lilly bit* isn't any category, and so on. That is, the constituent structure of the sentence, including information about the type of constituent, is indirectly represented in the Phrase Marker as derived by the Phrase Structure rules. Phrase Markers like (4), then, contain information about both constituent structure and about order. These Phrase Markers can be mapped into other Phrase Markers by another component of the grammar, separate from Phrase Structure rules. This is the transformational component (also definable within the broad framework of Post-style Production Rules).

Further, given the way that context-free Phrase Structure rules work, only connected subsequences of any string of categories can be a constituent in a Phrase Marker. This means that (4) can be represented by a tree structure, where a tree is a particular kind of graph built out of connected nodes. Indeed, in the 1970s, formalizations of transformational grammar (putting aside Kupin 1978; Lasnik and Kupin 1977) typically ignored set-theoretic Phrase Markers and defined transformations directly on trees (e.g., Ginsburg and Partee 1969; Peters and Ritchie 1973). Trees are a natural way of representing Phrase Markers. They

have a unique root node, each of their nodes bears a label of some kind, and the nodes are related by predicates that specify direct domination and linear precedence between nodes. This allows trees to directly capture constituency, category, and linear order. Trees are further specified so that they obey certain constraints: the root dominates all other nodes, there are no loops allowed in the structure, and so on (Partee et al. 1990, pp. 437–448).

It is possible to construct a tree algorithmically from a derivation (cf. McCawley 1968) by taking the last line of a derivation, comparing it with the previous line to establish which nodes in the last line are directly dominated by the nodes in the previous line, and then applying this procedure up through the derivation. The resulting tree represents the constituency relations in its vertical dimension and the linear relations in its horizontal dimension.



In early transformational grammar, then, the tree structures, which later became ubiquitous, are really alternative ways of representing Phrase Markers, which are “officially” sets of strings.³

The transformational component also made crucial reference to linear order (whether these rules were defined on Phrase Markers or on tree structures directly). The transformational rules could reorder elements of the output of the Phrase Structure component, delete them, and insert further elements. Application of the transformational rules derived Phrase Markers from Phrase Markers, so both their input and output were crucially ordered.

There were then, in standard early transformational generative grammar, two different components that established and manipulated linear order: Phrase Structure and transformations. Indeed, Chomsky, in *Aspects* (Chomsky 1965, p. 124ff), explicitly argues that a system where Phrase Structure rules build unordered set-theoretic objects has no serious advantages over an ordered system, since order is empirically required for the functioning of transformations. Specifically, he argues that set-theoretical versions of (1), as in, for example, (6), should not be adopted:

- (6) a. $S \rightarrow \{NP, VP\}$
 b. $VP \rightarrow \{V, NP\}$

Such rules would generate Phrase Markers that are sets of sets, as opposed to sets of strings.⁴

Chomsky's argument that Phrase Structure rules should be understood as a mechanism for generating ordered strings held sway during the 1970s. In the early 1980s, however, Stowell (1981) presented a strong case that the Phrase Structure rules that generate D-structure should not include stipulations about the order of the categories they generate, and he concluded that D-structure Phrase Markers are therefore unordered. His argument was fundamentally based on redundancy and related considerations of learnability. He pointed out that there are two possibilities in an Aspects type theory for analyzing the prenominal order of (most) adjectives in English: the Phrase Structure rules generate the prenominal order, or the Phrase Structure rules generate a postnominal order and the transformational rules prepose a subset of adjectives:

(7) $NP \rightarrow AP \frown N'$ PS-rule

(8) a. $NP \rightarrow N' \frown AP$ PS-rule

b. $N' \frown AP \rightarrow AP \frown N'$ Transformational-rule

The first analysis in (7) certainly looks simpler if we just consider data involving adjectives and nouns. However, Stowell points out that prepositional phrases, relative clauses, participial modifiers, and so on are all postnominal in English, so a unified analysis of modifiers, presumably preferred by any reasonable evaluation metric, would take APs also to be postnominal and then preposed. The two theories are really more like this:

(9) a. $NP \rightarrow AP \frown N'$ PS-rule

b. $NP \rightarrow N' \frown \begin{cases} PP \\ RelClause \\ ParticipialPhrase \end{cases}$ PS-rule

(10) a. $NP \rightarrow N' \frown \begin{cases} AP \\ PP \\ RelClause \\ ParticipialPhrase \end{cases}$ PS-rule

b. $N' \frown AP \rightarrow AP \frown N'$ Transformational-rule

The latter theory is preferable on empirical grounds, given that APs can appear postnominally in English, and indeed must if they have their own complement:

(11) a. Any woman fond of cats is welcome.

b. *Any fond of cats woman is welcome.

Given that APs can appear on either side of the N, some transformational rule will be necessary in this *Aspects*-style framework, so a theory that places all modifiers postnominally and fronts a subset is both more parsimonious and more empirically adequate. Further, the transformational, but not the phrase-structural, analysis allows a uniform link between D-structure and the semantics.

However, this argument implies that the Phrase Structure rules define fixed orders for which there can be no surface evidence: it is the transformational component that determines order. One could, in fact, have an unordered D-structure that is then ordered by later rules, as long as the constituency structure defined by the Phrase Marker contains sufficient information for the functioning of transformations. Given that the transformations derive S-structure, which is ordered, a theory that places ordering in just one component of the grammar is preferable on general grounds of simplicity.

Stowell then went further and argued that aspects of surface orders can, in fact, be determined by various non phrase-structural components of the grammar and that developing a theory that takes D-structures to be unordered, combined with these other components, provides better generalizations about what is ultimately responsible for word order. This perspective allowed the elimination of the Phrase Structure component entirely, in favor of unordered $\bar{X}$ -theoretic structures projected from lexical items, with ordering information determined on a language-specific basis by properties of lexical items, such as the direction in which a verb assigns case, and so on.

$\bar{X}$ -theoretic principles in Stowell's sense are quite unlike the standard rewrite interpretation of Phrase Structure rules. Rather than running through a derivation (a computation) that generates a Phrase Marker (a set of strings) that can be represented as a tree, Stowell's principles directly license certain structures and not others as possible D-structures. This means that they act as constraints defining what the legitimate structures can be, in a way reminiscent of McCawley's node admissibility interpretation of Phrase Structure (McCawley 1968).

Furthermore, the principles of $\bar{X}$ -theory developed by Stowell and others do not need to specify SOs that have the formal properties of trees. Chomsky (1982) states the idea rather explicitly in *Concepts and Consequences*:

Still, we can ask whether D-structures, S-structures, etc., have the properties of tree structures. Insofar as they are determined by X-bar theory, this will be the case. But there are other factors that enter into determining their properties. Reanalysis and restructuring processes, for instance, may yield phrase markers that cannot be represented as tree structures. [...] Furthermore, X-bar theory can be constructed so that it does not require that phrase markers have tree properties. It has occasionally been suggested that coordination might be understood in terms of union of phrase markers (in effect, three-dimensional

projection), linear order being determined by a “spell-out” rule. The assumption would be, then, that if the very same language were to be used in a medium having a dimension in addition to linear time, this “spell-out” rule would be unnecessary. Such suggestions might be correct, and I think they merit examination. Much more radical departures from tree structures can be, and sometimes have been, proposed. I will not explore these questions here, but merely note that incompatibility of such proposals with the theory of phrase structure grammar stands as no barrier to them. (Chomsky 1982, pp. 14–15)

Notions such as endocentricity, binary branching, or the phrasal status of non-heads, are compatible with tree structures and restrict trees to particular forms, but, as Chomsky notes, there is no requirement, once one has jettisoned Phrase Structure rules, that the structure of phrase markers is straightforwardly representable in a tree-theoretic fashion. Rather, Phrase Markers are simply whatever the theory defines. For example, linear order is a crucial aspect of tree structures as they are axiomatized in, for example, McCawley (1968), but the D-structures that Stowell’s $\bar{X}$ -theory licenses are unordered, and hence, at least by McCawley’s axiomatization, not trees. They are just Phrase Markers, defined by the constraints of the theory.

The line of thinking initiated by Stowell eventually led to Speas’s (2000) Project α theory (see also Chametzky 1996), where $\bar{X}$ -theory is eliminated in favor of projecting information from lexical items, this information being combined with independent principles that license the relationships between structures (cf. Chomsky 1986b, p. 93ff, and constraint-based theories such as HPSG [Pollard and Sag 1994]). Project α simply states that a lexical item of category X is dominated by an uninterrupted sequence (hierarchy) of X nodes:

(12) Project Alpha:

A word of syntactic category X is dominated by an uninterrupted sequence of X nodes.

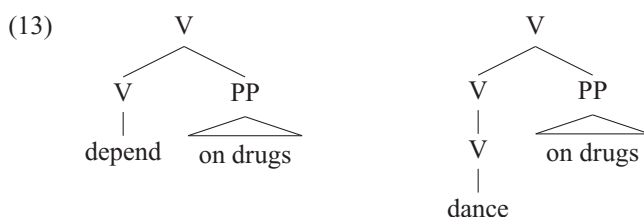
Unlike in the systems of Chomsky (1970), Jackendoff (1977), or Stowell (1981), there is no primitive notion of bar-level defined in Speas’s theory, rather maximal and minimal projections are defined as cutoff points for the sequence of X-nodes. A node of category X is maximal if all nodes dominating it are not equal to X, and minimal if X dominates a word.⁵

In Speas’s system, the nonhead daughters in a structure are licensed by relations they bear to nodes in the projection of the head. For example, the complement of a verb will be licensed by its thematic relation to the head, which is implemented by a requirement specified in the word, projected through the structure via Project Alpha.

Speas’s system has some interesting consequences that make it very different from a system based around Phrase Structure rules. It allows an indefinite

number of projections from a head, and it makes projections that are neither minimal nor maximal into elsewhere cases.

The licensing nature of the system comes out clearly in Speas's analysis of PP complements of V, versus PP adjuncts to V. There is a theta-theoretic principle that means that anything in complement position has a theta-role of the head, so, in order to have a PP adjunct, the head must project nonbranching structure so that the PP is not a sister to the immediate projection of the head, and is hence not assigned a theta-role. This leads to different structures being licensed by the interaction between Project Alpha and the relevant licensing principles that connect Phrase Structure with theta-theory, as in the following cases:



Speas's theory allows any number of projections of a head, so the number of nonhead daughters is unbounded, with potentially unlimited numbers of complements, specifiers, and adjuncts. Any actual restrictions on the number of nonhead daughters have to emerge from other constraints interacting with the features of the head.

D-structures in this theory, then, are not Phrase Structure trees. The SOs that Speas defines are hierarchical structures licensed by the projections of the properties of lexical items, together with the principles of grammar. They determine hierarchical relations and the connections of the syntax to both semantic interpretation and linear order. The linkage between the argument structure properties of lexical items and the syntax is reminiscent of early Categorical Grammar (e.g., Bar-Hillel 1953), though Categorical Grammars, like Phrase Structure rules, are typically taken to involve ordering as part of the specification of lexical items (though see Lewis 1972), unlike Speas's D-structures.

Though the D-structure Phrase Markers in Government and Binding Theory, developed through the work of Stowell, Speas, and others, are unordered, the S-structure Phrase Markers do contain ordering information. This is generally determined by properties of heads in the structure, which select, or assign Case, to their left or right within the constituent they head. These heads themselves may reorder with respect to those constituents via movement (Travis 1984; Koopman 1984). Such ordering information, however, is not constitutive of the nature of the Phrase Marker itself; it is rather additional. Chametzky (1996) argued that Phrase Markers should, in fact, be defined only in terms of

dominance, pointing out that while precedence relations are parameterized, that is never so for dominance. Precedence is not, in his view, part of syntax itself. He provides an axiomatization of Phrase Markers that takes them simply to be a structure involving connected nodes with a single root, ordered by a dominance relation. This is a very permissive view of Phrase Markers, very much in the spirit of Chomsky's remark in *Concepts and Consequences*. It also allows Phrase Markers where branches cross (cf. McCawley 1982; Huck 1985; and the papers in Huck and Ojeda 1987).

These two general ideas, that the objects of the theory can be directly defined by sets of constraints, and that different constraints hold on different levels of representation, underpinned much of the work in Government and Binding Theory. In that theory, an SO is an ordered collection of Phrase Markers for a sentence (a 4-tuple, consisting of a D-Structure, and S-Structure, a Phonetic Form, and a Logical Form); certain constraints hold on particular Phrase Markers within this tuple, and not on others. For example, the Case Filter constrains S-Structure and not D-Structure, while $\bar{X}$ -theory constrains all levels. The relationship between phrase-markers is given by Move- α , and certain constraints hold of this relationship too (for example, Subjacency). The SO that relates form and meaning in this conception is highly complex and multidimensional, and the question arises as to whether that complexity is warranted by the explanatory work the SOs are meant to do.

2.3 The Minimalist Critique

One way to understand the critique that Chomsky initiated of the Government and Binding theoretic approach to syntax is to see it as an argument for a return to the kind of architecture he had developed in LSLT (Chomsky 1955). In that architecture, which was maintained through the 1960s and most of the 1970s, Phrase Markers are sets of strings. If Stowell and others are correct in arguing that linear order (that is, strings as a data structure) should not be incorporated into Phrase Markers, one might ask what the LSLT architecture would look like if order were removed from it.

If order is removed from LSLT-style Phrase Markers, the remaining aspects of the data structure that are left are: (1) the distinction between a constituent and the category it belongs to (i.e., what label a constituent might have), and (2) sets, which model part whole structure by being embedded within each other, in an obvious fashion:

(14) { a, {{b, c}, d} }

Then the question becomes how to define a grammar that will generate labeled set-theoretic SOs that are unordered but have structure sufficient to explain syntactic phenomena?

One way to develop such a system, while keeping to the levels-based architecture of Government and Binding Theory would be to specify different levels of linguistic structure (e.g., D-structure, S-structure, Logical Form) as set-theoretic structures, related to each other via transformational operations. In such a theory, (14) might be generated via top-down rules that build labeled sets and then be transformed into (15), via a transformation targeting the set $\{b, c\}$ and making it a member of the outermost set, as follows:

(15) $\{a, \{d\}, \{b, c\}\}$

Such a theory could mimic the transformational theories of earlier years, with, for example, (14) serving as the D-structure and (15) as the S-structure. A D-structure might be derived via a general rule that defines what a well-formed set-theoretic structure can consist of, perhaps with the labeling information given by a designated element of the set. This could then feed into a transformational rule that derives (15) from it. Such a rule could, maintaining the basic LSLT architecture, consist of a number of elementary transformations that, say, insert some dummy element at the top level in (14), substitute $\{b, c\}$ for that dummy element, and delete $\{b, c\}$ in its “base” position:

- (16) a. $\{a, \{\{b, c\}, d\}\} \rightarrow \text{INSERT } \emptyset$
 b. $\{a, \{\{b, c\}, d\}, \emptyset\} \rightarrow \text{SUBSTITUTE } \{b, c\} \text{ for } \emptyset$
 c. $\{a, \{\{b, c\}, d\}, \{b, c\}\} \rightarrow \text{DELETE } \{b, c\} \text{ in the 'base'}$

Indeed, Chomsky (1993) suggests this kind of transformational operation. There, he defines a generalized transformation that builds $\bar{X}$ -compatible structures and a singulary transformation that modifies those structures along the lines of (16). However, he does not adopt the two-level architecture.

This is because, in the early 1990s, a number of empirical arguments were constructed that suggested that the functions of D-structure are better understood as properties of Logical Form. These arguments generally involve phenomena where expressions appear to be interpreted in their position at Logical Form, but not in their putative D-structure positions. That undermines motivation for having D-structures in the first place.

One interesting case, not discussed by Chomsky, but referred to by him, is that of exceptives, like (17):

- (17) Every good book about set theory was published last year except the one by Halmos.

Reinhart (1993) shows that the correct interpretation of examples like (17) requires *every good book about set theory* and *except the one by Halmos* to be a constituent. However, she further shows that the syntax of such sentences cannot involve extraposition of the phrase *except the one by Halmos* from the subject position. One powerful argument for this is that an exceptive is well formed with a strongly quantified subject, as (17) shows, while extraposition is blocked from DPs headed by strong quantifiers:

(18) *Every good book was published last year about set theory.

Reinhart argues that the correct analysis of exceptives requires generation of the subject and the exceptive as separate constituents, followed by a movement operation that brings them together into a single constituent at Logical Form (LF). This then gives the correct structure for interpretation, as follows:

(19) [t_i was published last year] [[every good book about set theory]_i [except [the one by Halmos]]]

If this is the correct analysis, then the relevant interpretable constituent is not built at D-structure. If, however, it is not necessary to build, at D-structure, constituents that are interpreted as single units at Logical Form (LF), the motivation for having a level of D-structure is profoundly undermined.

Chomsky (1993) mentions a range of other empirical arguments that also suggest that the notion of D-structure is problematic. He gives the *Tough*-construction example in (20), arguing that the standard analysis is inconsistent with D-structure:

(20) John is easy [_{CP} Op_i [_{IP} PRO to please t_i]]

If *easy* is a one-place predicate taking a clausal argument at D-structure, and the embedded clause is as is shown here, then *John* must be inserted in a non-theta position, and yet be assigned a theta-role from *please*. But D-structure is (partially) *defined* by a constraint that theta-marked arguments appear in theta-positions. So there is simply no D-structure available for (20), yet it is perfectly well formed (this argument, of course, depends on (20) being the right analysis of the construction; there are alternatives, e.g., Hicks 2009).

In addition to empirical arguments like these, Chomsky, in the material that was published as Chomsky (1995), some of which had been written over the early 1990s, advanced theoretical arguments that an architecture with fewer or no internal levels of structure is preferable on two distinct grounds. The first is that such a system is simpler. The second is a two-pronged hypothesis about the nature of the human language faculty: (1) the idea that the structure of the language faculty is almost entirely determined by its function in the overall

architecture of the human cognitive system, specifically in how it interfaces with motor systems involved in articulation, and with semantic systems involved in thought; (2) the hypothesis that the language faculty is an optimal solution to the requirements imposed by these interfaces. Together, the two elements of this conjecture are known as the Strong Minimalist Thesis.

To the extent that one accepts that the Strong Minimalist Thesis is worth pursuing, then D-structure, which has no role in either of the interfaces (its semantic properties being relegated to Logical Form), can be considered otiose, along with any other internal levels of structure that play a role in linguistic explanation but that do not feed into semantic or phonological interpretation (e.g., the NP-structure and Wh-Structure levels of van Riemsdijk and Williams 1981).

If we accept the conclusion that there is no D-structure and, more generally, that there are no internal levels, and if the language faculty must connect to both meaning and to external expression (whether by sound, sign, or in some other fashion), then the question is how to construct SOs that have the relevant properties. The Strong Minimalist Thesis motivates the needs for an operation that will (1) construct SOs that have the necessary properties to connect appropriately with the interfaces, and (2) be such that natural optimization constraints on this operation explain linguistic phenomena.

These empirical arguments and theoretical ideas led to the Minimalist Program, which adopted a particular execution for how SOs are constructed within these boundaries: SOs (1) are set-theoretic (since that is the data structure that is left when order is jettisoned from the system); (2) are hierarchical, to capture the part-whole nature of syntax; (3) are labeled (to capture empirical facts about the different distribution and interpretation of different categories); (4) have interface-relevant properties, and (5) are such that the operations that build them are optimal in some sense. In *Bare Phrase Structure*, Chomsky (1994, p. 7) proposes the operation Merge as a means of satisfying this set of requirements:

One such operation is necessary on conceptual grounds alone: an operation that forms larger units out of those already constructed; call it Merge. Applied to two objects α and β , Merge forms the new object γ . What is γ ? γ must be constituted somehow from the two items α and β ; the only alternatives are that γ is fixed for all α , β , or that it is randomly selected; neither is worth considering. The simplest object constructed from α and β is the set $\{\alpha, \beta\}$, so we take γ to be at least this set, where α and β are the constituents of γ .

Chomsky (2004) makes a further theoretically significant proposal, which is that Merge is not only responsible for local combination in the syntax but

also for the combination of SOs that are distant from each other in structure. This idea means that it is possible to model, with a single operation, both the building of structure and the changing of structure that had been the purview of the operation *Move* in previous versions of the theory. To see how this works, take the following inductive definition of syntactic object (see e.g., Chomsky 2008, p. 12; the version I give here is from Hornstein 2018):

- (21) a. if α is a lexical item then α is a syntactic object (the lexicon)
- b. If α is a syntactic object and β is a syntactic object then $\text{Merge}(\alpha, \beta)$ is a syntactic object.
- (22) For α, β , syntactic objects, $\text{Merge}(\alpha, \beta) = \{\alpha, \beta\}$

If the two SOs are distinct, then *Merge* licenses a new structure, a set constructed from the two SOs (this is commonly called “External” *Merge*). If one of the SOs is a constituent of the other, it can still serve as an argument of *Merge*. Such “Internal” *Merge* licenses a new SO:

- (23) $\text{Merge}(\beta, \{\alpha, \beta\}) = \{\beta, \{\alpha, \beta\}\}$

As (23) shows, when β , which is a member of $\{\alpha, \beta\}$, is Merged with $\{\alpha, \beta\}$ itself, the resulting SO has a “connection” between two of its subconstituents: they are formally identical (often called copies) and derived via *Merge*. Internal *Merge* replaces “movement” in the theory. The upshot is that there is a single operation that both builds and alters structure, unifying the previously distinct structure-building (phrase-structure) and structure-changing (transformational) components of the grammar.

As just discussed, *Merge* in the 2000s was given as part of the inductive definition of syntactic objects. As the minimalist approach developed, the *Merge* operation was further rethought in terms of what its computational properties might be. Rather than an operation on SOs directly, it became understood as an operation that maps from a kind of memory cache (a *Workspace*), within which SOs are stored, to a new cache, which is exactly the same in content, except that two of the SOs are now Merged into a new set-theoretic SO. Chomsky et al. (2023) give the following informal characterization:

- (24) *Merge*:
 - a. “looks inside” the workspace WS that it is applying to,
 - b. targets material within WS,
 - c. builds from that targeted-material a set-theoretic object (i.e., it builds a non-atomic structure), which is now

- d. a new object, which is a member of the WS, thereby modifying WS to WS'

(24c) specifies that Merge “builds . . . a set-theoretic object”: it does so by collecting the targeted material into a binary set. Chomsky et al. (2023, p. 15) state it as follows:

$$(25) \text{ Merge}(P, Q, \text{WS}) = \text{WS}' = [\{P, Q\}, \dots]$$

Since Merge, in this perspective, is an operation that applies to SOs “in” a workspace, it must find the relevant objects. Chomsky et al. define a Search operation, which is also binary: it finds one object, then seeks another. An application of Search to a workspace will find, say P, and then seek Q. Because Internal Merge involves one SO that is embedded inside another, and because set-membership is intransitive, Chomsky et al. define a notion of *term*, which is what an application of Search returns. Termhood is the ancestral of set membership, as follows:

- (26) X is a *term* of Y iff X is either (i) a member of Y or (ii) a member of a term of Y.

With this in hand, we can take some workspace, for example, $\{a, \{b, c\}\}$, and apply Merge to it as follows:

- (27) a. $\text{WS} = \{a, \{b, c\}\}$
 b. $\text{Search}(\text{WS}) = a \wedge \{b, c\}$
 c. $\text{Merge}(a, \{b, c\}, \text{WS}) = \text{WS}' = \{\{a, \{b, c\}\}, a, \{b, c\}\}$

However, we now have a workspace that has a new object in it, as well as the old ones. To restrict this, Chomsky et al. propose that the output of Merge is restricted by a principle they call Minimal Yield. Minimal Yield essentially prunes, out of the output workspace, anything except the newly built object, so (27) is actually (28):

- (28) a. $\text{WS} = \{a, \{b, c\}\}$
 b. $\text{Merge}(a, \{b, c\}, \text{WS}) = \text{WS}' = \{\{a, \{b, c\}\}\}$

Internal Merge works just the same. If we Merge the SO in WS' with its term c, we get, not (29c), but (29b):

- (29) a. $\text{WS}' = \{\{a, \{b, c\}\}\}$
 b. $\text{Search}(\text{WS}') = c \wedge \{a, \{b, c\}\}$
 c. $\text{Merge}(c, \{a, \{b, c\}\}, \text{WS}') = \text{WS}'' = \{\{c, \{a, \{b, c\}\}\}, \{a, \{b, c\}\}\}$
 d. $\text{MinimalYield}(\text{WS}'') = \{\{c, \{a, \{b, c\}\}\}\}$

Part of the motivation for this approach is that the inductive definitions usually given for Merge are too loose, allowing a wide range of Merge operations that then have to be filtered out (Chomsky 2019; see Adger 2021 for discussion). This computational perspective has the potential to be more constrained via a definition of Merge that is very general, but which is subject to constraints that are, at least potentially, not specialized to syntactic objects (that is they are “Third Factor”; Chomsky 2005). The restriction to binarity in the constructed SO might be thought, for example, to be a result of a kind of general principle of parsimony (i.e., the minimal structure is binary, and by parsimony, the maximal structure is therefore binary). Similarly, the notion of Minimal Yield might be thought to be derivative of a more general restriction in cognition, that computation “forgets” the results of previous computations. This newer version of Merge, then, is intended to satisfy the Strong Minimalist Thesis in a more thoroughgoing way than the earlier approach did.

2.4 The Labeling Question

As mentioned above, in Bare Phrase Structure, Chomsky takes Merge of α and β to be at least the set $\{\alpha, \beta\}$. However, given the idea that SOs are constructed so that they are suitable for the interfaces, the fact that verb phrases, for example, appear to be semantically and syntactically different from noun phrases, requires that complex SOs like $\{\alpha, \beta\}$ contain within them the wherewithal to signal this. This role is traditionally afforded to syntactic categories (such as V and N), which “project” to the complex SO, or are otherwise assigned to it. Keeping to this idea entails that category information must somehow be associated with the entire SO $\{\alpha, \beta\}$. How might this work?

$\bar{X}$ -theory, of course, did away with the idea that phrases are *assigned* a special category by particular rules of the syntax (e.g., that the phrase consisting of a noun phrase and a verb phrase is assigned the category S); rather it suggested that category information associated with the phrase is always *derived* from an immediate constituent of the phrase. The consequence of this is that all phrases are headed, in the sense that they share category information with one of their constituents. In this way, classical $\bar{X}$ -theory enforces universal endocentricity: the $\bar{X}$ schemata ensure that the information encoded by X is projected to XP. However, in a set-theoretic system, there is no analogue of the $\bar{X}$ schema. How then is the category information associated with $\{\alpha, \beta\}$ derived from the category information associated with its head, and how does the system ensure that that information is structurally present at the phrasal level?

The earliest approach to doing this was to say that Merge produces a more complex SO than just $\{\alpha, \beta\}$. In this early version of the theory (Chomsky

1994), the output of Merge is a set, but it is a more complex set than just $\{\alpha, \beta\}$: it consists of an SO, which will be interpreted as the category, plus $\{\alpha, \beta\}$:

$$(30) \text{ Merge}(\alpha, \beta) = \gamma = \{\delta, \{\alpha, \beta\}\}$$

Here δ encodes the category of γ , so δ is the label of γ . How the label is determined is a separate question from how it is represented in the SO. In the Bare Phrase Structure system, following the $\bar{X}$ -theoretic intuition that all phrases are endocentric, δ is one of either α or β . Take the arguments of Merge to be, for example, a verb, say *cut* and a nominal object, which I'll assume for purposes of illustration here is simplex, say *oranges*:

$$(31) \text{ Merge}(\text{cut}, \text{oranges}) = \gamma = \{\delta, \{\text{cut}, \text{oranges}\}\}$$

What is δ ? Clearly the phrase *cut oranges* could either be a verb with an object, or a participle modifying a noun. On the first interpretation, δ needs to signal that the category of the whole phrase is verbal, because the category of *cut* is verbal; for the second interpretation, where the category of the whole phrase is nominal, the head is the nominal *oranges*. The simplest way of specifying this is to make δ be whichever of the lexical items is the syntactic head of the phrase, so there are two outcomes (continuing to assume that these words are syntactically simplex):

$$(32) \begin{array}{ll} \text{a. } \text{Merge}(\text{cut}, \text{oranges}) = \gamma = \{\text{cut}, \{\text{cut}, \text{oranges}\}\} \\ \text{b. } \text{Merge}(\text{cut}, \text{oranges}) = \gamma = \{\text{oranges}, \{\text{cut}, \text{oranges}\}\} \end{array}$$

There were numerous attempts to improve on this approach over the next decade or so, because it is unsatisfactory in a number of ways. The fundamental argument against it is that it makes the operation Merge a more complex operation than one that would produce the “simplest” object (which would just be $\{\alpha, \beta\}$), a point noted by Chomsky (1994), but perhaps most forcefully made in Collins (2002), which will I return to directly.

Further, as noted by Langendoen (2003, p. 3), if the label is one of either α or β , the result of Merge is an ordered pair, as opposed to a simple set. This is because $\{\alpha, \{\alpha, \beta\}\}$ essentially creates an asymmetry in the structure (and, Langendoen argues, is hence set-theoretically equivalent to the ordered pair $\langle \alpha, \beta \rangle$, via Kuratowski's encoding of ordered pairs as sets). Perhaps that is what is required, that there is an asymmetry, but then binary set formation doesn't seem to be the right model for SOs, since binary set-formation is symmetric.

A further issue is that the label in the structure is a bone-fide syntactic object (it is a member of a set), but that would suggest that it should be able to undergo syntactic processes (say movement/Internal Merge). A derivation

like the following should then be possible, where the bolded **a** is derived by movement/Internal Merge of the label of the constituent {a,b}:

$$(33) \{c, \{c, \{a, \{a, b\}\}\}\} \rightarrow \{\mathbf{a}, \{c, \{c, \{a, \{a, b\}\}\}\}\}$$

As far as I'm aware, no such derivation has ever been suggested, and it is unclear how to interpret it. For all of these reasons, Chomsky's initial suggestion for labeling the outcome has not been followed up (though see Hornstein (2024) for a recent defence of a similar approach).

In an attempt to rescue the simpler system, Collins (2002) (see also Seely 2006) argued that no labels are needed beyond categories of lexical items; that is, the outputs of Merge don't need to be annotated by labeling information. Collins's suggestion is that projection, whether it is à la $\bar{X}$ -theory or via Chomsky's extra label, is unnecessary because all the information needed to distinguish phrasal SOs is available from the syntactic relations those SOs independently enter into, combined with certain principles that govern the derivation.

For example, Collins proposes what he calls the Locus Principle, following an idea in Chomsky (2000, p. 132), which ensures that in any derivation the selectional properties of the lexical item that has just entered the derivation must be satisfied before the derivation can allow a new lexical item to enter into it. To take a concrete case, Collins assumes that the inflectional head *T* has the selectional property in English that it requires a subject. This means that it is impossible to Merge a complementizer with a constituent headed by *T* until that property has been satisfied. This in turn means that it is impossible to build a constituent where *C* Merges with what one might think of in $\bar{X}$ -theory as $\bar{T}$, deriving, in a different way, part of what is stipulated in $\bar{X}$ -theory (that complements of heads are XP , and not X or $\bar{X}$).

Other principles that Collins appeals to regulate subcategorization, allowing the system to do away with labels at the phrasal level. Take, for example, the verb *tell*, in the incarnation where it selects a PP headed by *on* (in the idiom for "inform," *tell on you*). How is the information that *on you* is a PP made available to *tell*? Usually, it is assumed that that information is carried by the label of the whole phrase *on you*, which is *on*, and that that label matches the selectional requirement of *tell* that it combine with a prepositional phrase (see, e.g., Adger 2003 for a fully fledged system that works in this way).

Collins's idea in his label-free system is that there is an operation that allows the verb to look inside its sister, and to ascertain whether there is a lexical item of the relevant sort. The verb *tell* combines with {*on*, *you*}, and the operation identifies the selectional property of *tell* (that it selects a PP headed by *on*) and searches in the syntactic object that is *tell*'s sister for *on*.

Collins implements his system using the the operation Agree (Chomsky 2000; Chomsky 2001). Agree is implicated in a wide range of syntactic dependencies, such as agreement, case assignment, wh-movement, and so on. Collins defines a particular constraint on Agree when it is used in selection that ensures that the selected lexical item is “accessible.” An SO is defined as accessible if it was involved in the immediately previous operation in the derivation via its selectional or other features (this is the Locus Principle at work). Since *on* both selects and case licenses *you* in {*on*, *you*}, it is accessible and so can serve as the syntactic object that *tell* selects. In this way, there’s no need to have the label at the phrasal level; the operation Agree, incorporating accessibility, will do the relevant work.

Applying this system to our previous ambiguous example of the phrase *cut oranges*, if *cut* selects a nominal object as part of its lexical specification, then the whole phrase will behave for the purposes of the semantic systems as a verb phrase (as *cut* is accessible and its lexical features tell us that it is a verb). If *cut* is rather a participial modifier, it is an adjunct. Though Collins leaves adjunction aside, we might take adjuncts to be immune to the Locus Principle, so it is the modified noun that is “involved” in the immediately previous operation, and the syntactic and semantic systems treat the phrase as nominal.⁶ With all of this in place, Collins’s system opens the door to taking Merge to produce a simple set as its output; this simple set contains the two arguments of Merge and nothing else.

The basic thrust of Collins’s system has been maintained in developments of Bare Phrase Structure since, with improvements that attempt to simplify that system further. Whereas Collins’s system deconstructs the notion of label, farming it out to various independently needed syntactic relations, other approaches that have been developed since have attempted to simplify this so that there is fundamentally one mechanism in the system responsible for labeling. The outcome is a set of fairly similar theories that maintain the idea that the data structure that Merge produces does not include any more information than is present in the arguments of Merge, beyond the fact that they comprise a set (cf. Chomsky’s condition of Inclusiveness, Chomsky 1995, p. 225). Labels, then, are not part of the data structure but are derived from its properties.

For example, Chomsky (2013) and its companion paper Chomsky (2015) offer an approach where the label is assigned by an algorithm that operates at certain points in a syntactic derivation. This approach extends Collins’s idea that labeling can involve an operation that searches through structure, and proposes that labels are uniformly given by such a search mechanism. The main difference in the systems is that, rather than deconstructing the notion of label into various distinct syntactic dependencies, Chomsky proposes a

specialized Labeling Algorithm, whose function is to determine the label for any particular SO.

Chomsky's algorithm searches top-down through an SO and halts when it determines a unique label (note the similarity to the Search operation that selects the elements in a workspace for the operation of Merge). It determines a unique label when it finds a lexical item (where no further search is possible), or when it finds a pair of lexical items that are featurally the same. Take, for example, (34):

(34) {eat {the, orange}}

The label of this structure is *eat*: working from the top down, the algorithm scans both constituents of the whole structure in parallel. One of these is a lexical item (*eat*), while the other requires further search. Given that the label can be determined at this first step of search, search terminates, and the label of the whole structure is the lexical item *eat*. In a situation where both components of a structure are lexical items, as in (35), an ambiguity would appear to arise, as search can terminate with either item (or both).

(35) {the, orange}

Chomsky (2015) suggests that the system is set up so that, in fact, this never happens. He adopts the idea that nonfunctional lexical items, like *orange*, are labelless roots (following a proposal most comprehensively defended in Borer 2005a; Borer 2005b; Borer 2013; see also Marantz 1997). When a functional item like a determiner, Aspect, or the Agent introducing category *v*, Merges with a root, the algorithm can return a label with no further search, the label of the functional item, since roots, by hypothesis, lack any capacity to label.

(36) {the, $\sqrt{\text{orange}}$ }

Here, the whole structure has *the* as its label.⁷ It follows from this approach that many, apparently simplex, lexical items (like, for example, *oranges* in *eat oranges*) are actually complex structures.

As well as situations where there are two lexical items Merging, there are also situations where there is no lexical item uniquely identifiable by the algorithm. This is generally the case in structures with specifiers. For example, in a finite clause with a subject in the specifier of TP, neither of the immediate constituents of the structure is a lexical item (VP here is just an abbreviation for whatever complex set-theoretic structure has been built up in earlier stages of the derivation):

(37) { {the, $\sqrt{\text{cat}}$ }, {T, VP} }

The algorithm will, in parallel, search into the two subconstituents and find two lexical items (*the* and *T*). Chomsky's suggestion is that the algorithm terminates in such a case only if those two lexical items have some (relevant) feature in common. Say they do (perhaps the ϕ features associated with subject-verb agreement, which are both on the determiner and on *T*), then the label of the whole structure that is returned by the algorithm is a pair of identical features.

Obviously, further questions arise: How is the relevant feature determined? What happens when there is no such feature? What is the semantic interpretation of the pair of labels in a specifier structure? Chomsky proposes answers to some of these questions that lead to novel theories of locality, but I leave them aside here (see Adger 2013, chapter 1 for critical discussion). The important take-away from this is that labels do not have to be represented as part of the syntactic structure at the phrase level: the algorithm identifies the lexical item that is the head of the phrase, and that is sufficient. Simplest Merge can be maintained.

An alternative approach to determining the labels, again stemming from the idea that it is properties of lexical items that are relevant, is that of Cecchetto and Donati (2015). Unlike in Chomsky's system, where roots lack labels, Cecchetto and Donati argue that all words have categorial labels and some also have grammatical features with the special property that they must set up a syntactic dependency with some other constituent. Such features are called "probes," and an item with a probe undergoes the syntactic operation Agree, which connects it with an item bearing a feature that matches the probe. This operation is at play elsewhere in the grammatical system (it is used for selection, for case assignment, for agreement, for movement dependencies, etc.), so the notion of probe is plausibly independently needed. Cecchetto and Donati's idea is that probes determine labels, an idea connected to Collins's Locus Principle.

The way that this works is through a definition of the notion of label as follows (Cecchetto and Donati 2015, p. 30):

(38) *Label*

When two objects α and β are merged, a subset of the features of either α or β become the label of the syntactic object $\{\alpha, \beta\}$. A label

- a. can trigger further computation, and
- b. is visible from outside the syntactic object $\{\alpha, \beta\}$.

This definition works in tandem with their algorithm for determining labels, which is as follows:

(39) *Probing Algorithm*

The label of a syntactic object $\{\alpha, \beta\}$ is the feature(s) that act(s) as a probe for the merging operation creating $\{\alpha, \beta\}$.

Given this definition, Cecchetto and Donati say, Merge “can remain symmetric and minimal.” For example, a verb can Merge with a simplex direct object (e.g., {drink, water}); the verb *drink* contains a probe (a feature that sets up a selectional relationship with the direct object) that matches with the categorial feature of the direct object, and hence the resulting phrase has a subset of the features of *drink* (presumably at least its categorial feature) as its label. Those features are visible from outside the SO (cf. the very similar system in Adger 2003).

Cecchetto and Donati also argue that, in general, structures built by Merge will involve a probe, since otherwise such structures would be unlabeled, and so effectively inert for further syntactic computation, and hence (almost) unusable. They are not explicit about how the features of an item inside the phrase “become” the label of the phrase, but in later work adopt a system like Chomsky’s (Cecchetto and Donati 2022). This system too, then, eschews syntactic representation of labels and posits instead an extra theoretical mechanism to determine them.

What is shared by the various approaches just reviewed is that the output of Merge is simple: there is no structural representation of the label. For the verb phrase *cut oranges*, rather than (31), we have (40), with various syntactic operations plus the Labeling Algorithm conspiring to treat the structure as a verb phrase:

$$(40) \text{ Merge}(\text{cut}, \text{oranges}) = \gamma = \{v, \{\sqrt{\text{cut}}, \{n, \sqrt{\text{oranges}}\}\}\}$$

There have been a number of arguments against the label-free systems. I reprise one of these here, an extension of the argument made in Adger (2013).

Consider a possessor structure in English, like *the cat’s tail*. Putting aside various movement operations and other tangential issues of analysis, the structure for this looks as follows:

$$(41) \{\{\text{the}, \{n, \sqrt{\text{cat}}\}\}, \{', \{n, \sqrt{\text{tail}}\}\}\}$$

In Chomsky’s system, the two subconstituents, {the, {n, $\sqrt{\text{cat}}$ }} and {', {n, $\sqrt{\text{tail}}$ }}, are both headed (labeled) by elements with the category D. Chomsky briefly mentions such structures in a footnote, suggesting that they might be analyzed analogously to the subject of a clause, being labeled via agreement of ϕ -features as $\langle \phi, \phi \rangle$. Cecchetto and Donati (2022) suggest instead that the algorithm returns a label of D (alternatively the pair $\langle D, D \rangle$).

We know that the specifier of this structure may be extracted or raised in various languages (e.g., Gavrusseva 2000; Deal 2013). However, as far as I am aware, we don’t find cases where the sister of the specifier (what would be $\bar{D}$ in $\bar{X}$ -theory) is extracted. Aissen (1996, p. 456) gives, for example, the Tzotzil construction in (42):

- (42) a. I-k-il-be s-tot li Xun-e.
 CP-A 1-see-IO A 3-father the Xun-ENC
 ‘I saw Xun’s father.’
 b. Buch’u av-il-be [s-tot ⟨buch’u⟩]?
 who A2-see-IO A3-father
 ‘Whose father did you see?’

In (42a), the possessor is a postnominal subconstituent of the object, and in (42b) it is wh-extracted to the leftmost position in the clause, stranding its head noun.⁸ I know of no language where what is extracted is systematically the possessee, as opposed to the possessor.

In English, there may be independent reasons for this (perhaps it would lead to stranding of the possessive clitic), but the absence of movement of constituents that are sisters of a specifier is fairly general. Indeed, this generalization was elevated to a general claim about human language in earlier versions of the theory: the $\bar{X}$ -Invisibility Hypothesis of Chomsky (1995, p. 242) (see also the lengthy discussion in [Epstein and Seely (2006)]). However, under the Bare Phrase Structure system augmented with a Labeling Algorithm, both constituents would seem to be equivalent, and it is a mystery why we don’t find possessee extraction in the same way that we find possessor extraction, or $\bar{T}$ raising, instead of subject raising.

Beyond this, there is a more fundamental issue: the set-theoretic construction of Merge requires a theory of labeling if it is to be empirically adequate. However, adding a theory of labeling to the theory of what builds SOs introduces extra complexity into the system. This is the core of what one might call the Labeling Question:

(43) The Labeling Question

Is there a way to implement the construction of SOs so that there is no need for a supplemental theory of labeling?

2.5 The Copy Question

Recall that the most recent views of Merge take it to apply as a mapping between workspaces of the following sort (e.g., Chomsky et al. 2023, p. 18; see also Collins and Stabler 2016, p. 49, Definition 14; Chomsky 2022):

$$(44) \text{ Merge}(P, Q, WS) = WS' = (\{P, Q\}, X_1, \dots, X_n)$$

Merge takes WS as input, and gives WS' as output by targeting P and Q in WS and making them into a set in WS' . P and Q can themselves be complex SOs

built by earlier applications of Merge. $X_1, \dots, X_n$ in (44) are the elements of the workspace that are unaffected by the application of Merge.

Consider the following derivation in this system (I'll follow Chomsky et al. and use square brackets for the workspace, to help make the structures somewhat more readable).

- (45) a. $WS = [a, b] \rightarrow \text{Merge}(a, b, WS) \rightarrow WS' = [\{a, b\}] \Rightarrow$
 b. $WS' = [\{a, b\}] \rightarrow \text{Merge}(\{a, b\}, b, WS') \rightarrow WS'' = [\{b, \{a, b\}\}]$

Here the derived syntactic object $\{b, \{a, b\}\}$ contains two occurrences that are identical (both have the form b). Merge in (45b) has, as an argument, an SO selected from another one of its arguments, so this is Internal Merge and the two occurrences are the same SO, since only one SO b has been introduced into the derivation. The two occurrences of b in such a derivation are often referred to as *copies*, but, in fact, there is a single SO being manipulated in the derivation (see Gärtner 2022 for discussion).⁹

An alternative derivation of the same SO is possible using just External Merge. In this case there are two distinct SOs that are inscriptionally identical in the initial workspace (in which case the workspace is a multiset, as opposed to a set; alternatively, we can keep the workspace as a set if we assume that a new inscription of b is added to the workspace once the other b has been Merged):

- (46) a. $WS = [a, b, b] \rightarrow \text{Merge}(a, b, WS) \rightarrow WS' = [\{a, b\}, b] \Rightarrow$
 b. $WS' = [\{a, b\}, b] \rightarrow \text{Merge}(\{a, b\}, b, WS') \rightarrow WS'' = [\{b, \{a, b\}\}]$

Once again, the derived syntactic object $\{b, \{a, b\}\}$ has two terms that are identical in form. In this case, however, they are distinct SOs. The problem, of course, is that the final SO does not contain its derivational history, so there is no way of distinguishing the two.

However, it is in fact necessary to distinguish the two cases in examples like the following, where we have an unaccusative verb in (47a) and a (pragmatically unusual but grammatical) transitive version of the same verb in (47b):

- (47) a. The window broke $\langle$ the window $\rangle$
 b. The window broke the window

In (47a), the standard analysis is that the single argument of the unaccusative verb is Merged as the verb's object and then Internally Merges with TP to become the specifier of T, and hence the surface subject. In (47b), however, there are two identical constituents, one Externally Merged as the verb's object, the other as its subject. The latter Internally Merges with TP. At that point, the two structures are, for relevant purposes, set-theoretically identical. The question is how the interface systems can distinguish the SO $\{\text{the window}\}$ in the

complement of V position as a copy of some higher SO or as an independently Merged SO:

- (48) a. $\{\{\text{the, window}\}, \{\text{T}, \{\{\text{the, window}\}, \{\text{v}, \{\text{V}, \{\text{the, window}\}\}\}\}\}\}$

The system must have some mechanism to distinguish these for both phonological reasons ((47a) involves a trace and (47b) doesn't) and semantic ones (the two "copies" in (47a) are interpreted differently from the two "repetitions" in (47b)).

Collins and Groat (2018), in their careful discussion of the copies/repetitions issue, consider a number of possibilities. One simple idea would be to index the distinct SOs as they are taken from the lexicon (this was the earliest approach, suggested in Chomsky 1995), which would lead to two distinct representations, viz.:

- (49) a. $[\{\text{b}_1, \{\text{a}, \text{b}_1\}\}]$ —copies
 b. $[\{\text{b}_1, \{\text{a}, \text{b}_2\}\}]$ —repetitions

A closely related idea is to introduce a special relation between identical inscriptions that is added to the structure just in the case of movement, so that occurrences of an SO that are linked by movement are in this relation (usually called a "chain") while those that are not derived by movement are not in the relation. As Collins and Groat note, this simply stipulates the difference much as indexing does.

A more promising approach might be to allow the interpretive systems, which need to know whether two identical inscriptions are copies or repetitions, to inspect not just the generated SO, but the derivation (or at least part of the derivation). If these systems can look back into the derivation, they will be able to determine whether one SO or two was introduced in the derivation, and therefore whether the identical inscriptions are copies or repetitions. But this is, of course, a lot of power to give the interfaces and is not, as Collins and Groat again note, motivated by anything else than the need to distinguish copies and repetitions. Indeed, one of the fundamental ideas of minimalism, the Strong Minimalist Thesis, would seem to bar the interfaces from inspecting the innards of a derivation.

A final approach would be to appeal to semantics. In the examples I gave above, the copies are both assigned a single thematic role, while the repetitions are assigned distinct thematic roles. Chomsky et al. (2023) and Chomsky (2024) suggest an approach along these lines via an interpretive rule, *Form Copy*, that assigns the relation *Copy* to identical inscriptions, where an inscription is just a term for an SO that is being computed with. *Form Copy* interacts with a version of the Theta-Criterion that is framed as follows:

- (50) A θ assigner cannot assign more than one theta role to a single element.
Chomsky (2024, p. 26)

Form Copy selects a member of the syntactic object it operates on, and then searches through that SO and, when it finds an inscriptionally identical SO, it assigns the relation Copy to the pair of SOs. Chomsky treats this pair itself as a syntactic object.

This approach will distinguish between our two examples. Form Copy in (47a) will make the copies into a single syntactic object, but, given that a single SO can be assigned no more than one θ -role, (47b) must involve two SOs *the window*, each of which has a θ -role. The phonological interface will only pronounce one of the two copies, so in the unaccusative example, there is a single pronounced instance of *the window*, while in the transitive example, there are two.

However, as Collins and Groat note when discussing a less developed version of this idea, adjuncts may also give rise to the copies/repetitions issue. They provide the example:

- (51) On Mondays I think that on Mondays he drinks.

Here there are two identical inscriptions of the PP *on Mondays*. The interpretation of (51) is that each PP contributes a distinct modification in the semantics, one in the lower and one in the upper clause. However, the syntactic object corresponding to the sentence in (52), where the lower PP has moved, leaving a copy, has an interpretation where the PP modifies the embedded predicate and not the matrix one.

- (52) On Mondays I think that ⟨on Mondays⟩ he drinks.

The Theta-theoretic approach will not help in distinguishing the relevant syntactic objects.

Chomsky's proposal is an improvement over the diacritic or chain-based approaches, since, Chomsky argues, Form Copy has explanatory utility for other phenomena, such as control. However, the system requires two types of SO: one constructed by Merge and another constructed by Merge plus Form Copy; the extra type of SO is an enrichment of the system that it would be better to do without. Collins and Groat develop a similar critique of discussion of Muñoz Pérez's (2018) feature-based system, which also requires an extra type of syntactic object.

The fundamental issue is that two distinct derivations, which need distinct phonological and semantic interpretations, actually generate the same SO. That is why the solutions are to (1) provide the interpretive systems with access to the derivations, to (2) alter the SOs somehow so as to make them distinct, or to

(3) allow the interpretive systems to distinguish the SOs on some other grounds (like Theta-theory). None of these approaches, as Collins and Groat make clear, are satisfactory, but it is not obvious, within the set-theoretic approach to Merge, how else to make the necessary distinctions. This is the Copy Question:

(53) The Copy Question

Is there a way to implement the construction of SOs so that there is no need for a supplemental theory to distinguish copies and repetitions?

Both the Labeling Question and the Copy Question are signaling the same kind of dissatisfaction: set-theoretic SOs do not serve as a good models for syntax. Labels are crucial, but the theory of SOs needs to be supplemented to account for them; copies and repetitions need to be distinguished, but the theory of SOs needs to be supplemented to do this.

2.6 Summary

This chapter has provided a very brief history of the notion of syntactic object, and the reasons why the prevailing current view is that syntactic objects are set-theoretic. I have pointed out two quite serious problems with that view, which can certainly be solved, but only by adding extra theoretical apparatus to the system.

3 Parts

3.1 Sets vs Objects

The previous chapter sketched out the current set-theoretic approach to syntactic objects (SOs) in Minimalist syntax and noted two rather major problems. The first is that a supplemental theory of labeling is required for set-theoretic SOs; the second is that set-theoretic SOs do not have enough information in them to distinguish cases that are, in fact, empirically distinct, specifically to distinguish copies from repetitions. In order to do this, set-theoretic SOs require augmentation by some other mechanism. One could attempt to motivate these extra subtheories (labeling and, say, indexing or derivational memory) by showing that they are independently needed. One could also try to reduce these theories to something else. The third possibility is what I will pursue here: rethink the nature of SOs and the mechanism that builds them in such a way that the issues of Labeling and Copies simply do not arise. In this chapter I show how to do this, and the remainder of the book explores some of the ramifications of that theory.

Mereology is the theory of how parts and wholes of objects are related. Much like set theory attempts to provide a formal underpinning for thinking about the notion of a collection of objects, mereology attempts to provide a formal underpinning for thinking about the parts of an object. Unlike the version of set theory used in Bare Phrase Structure, which makes a distinction between an object and a set containing an object, mereology keeps the “type” of the whole and part *qua* objects the same.¹ In mereology, there is no notion analogous to that of a singleton set distinct from the object that it has as a member. More generally, in set-theoretic thinking, *a* can be in the set $\{a\}$, and that can be in the set $\{\{a\}\}$, and so on, generating a hierarchy of sets. In mereology, if there is the object *a*, then *a*’s existence does not license the possibility of some object *b* that *a* is part of (or, indeed, that is part of *a*).

The distinction between an object and a set is important in Bare Phrase Structure: it is the distinction between a lexical item, which is not set-theoretic but is a bone fide syntactic object, and a set containing that lexical item (and perhaps other items). For example, we might have the following kinds of situations:

- (1) a. D , n , and $\sqrt{\text{cat}}$ are primitive syntactic objects;
- b. $\{n, \{\sqrt{\text{cat}}\}\}$ is a complex syntactic object, labeled by n because $\sqrt{\text{cat}}$ has no capacity to label;
- c. $\{D, \{n, \{\sqrt{\text{cat}}\}\}\}$ is a complex syntactic object, labeled by the primitive object D because the complex object $\{n, \{\sqrt{\text{cat}}\}\}$ does not provide a label.

(1) shows that there are effectively two “types” of syntactic object in Bare Phrase Structure, and that Labeling Theories capitalize on this difference.² There is no such distinction available in the mereological theory.

In the same way that mereological theories eschew the kind of abstraction that differentiates an object from the set containing the object, they also tend to eschew an equivalent of the empty set: there is no “null object” that would be part of all objects, analogously to the empty set being a subset of all sets (though some theorists have explored how one might add such a notion to mereology, see the discussion in Cotnoir and Varzi 2021, section 4.5). The empty set has tended not to be used in minimalist syntactic theory (but see De Belder and van Craenenbroeck 2015 for one attempt), although it is, of course, a subset of every SO constructed by Merge, since it is a subset of every set.

More generally, mereological theory is interested in “stuff”: objects and their composition, as opposed to abstractions over objects. I think this makes it an interesting alternative basis for syntax, especially if we think of syntactic objects as configurations of mental “stuff,” however briefly present in our minds/brains.

The theory of classical (extensional) mereology emerges from the work of Leśniewski (see Simons 2018 for accessible discussion) and from the work of Leonard and Goodman (1940) in the first half of the twentieth century, with antecedents in the work of philosophers like Husserl and Whitehead. Mereology provided, for these logicians, an alternative basis for mathematics to that provided by set theory. For Leśniewski, set theory was plagued by paradox and contained the, to him unpalatable, notion of the empty set; for Leonard and Goodman, set theory was committed to abstract objects in a philosophically objectionable way. Over the past century, the tenets of mereology have been explored both formally and philosophically and have undergone various developments and clarifications (see Cotnoir and Varzi 2021), though all of these versions of mereology have the same fundamental differences from set theory. In what follows, I adopt and adapt some of these ideas.

Classical extensional mereology is typically analytical. It takes the objects that are in the world and attempts to ask how they are constituted by their parts. Wholes are seen as the fusion of their parts. For example, classical mereology takes there to be a cat-fusion, an object (the smallest one) that consists of all and only cats. All the cat parts in that fusion (whiskers, tails, etc.) are also a part of the cat-fusion object, and it is in that sense that the fusion of cats is different from the set of cats (while whiskers are parts of the cat fusion, they are not members of the set of cats). Classical mereology has proved useful in semantics, especially in the analysis of mass terms, plurals, imperfective aspect, and so on (e.g., the influential work of Link 1983, and research building on that). The notion of a whole being constituted by all of its parts in an undifferentiated way is, however, quite obviously not going to be helpful when thinking of syntactic objects, which are highly structured: a noun phrase containing a verb phrase (in a relative clause perhaps) has to be kept distinct from a verb phrase containing a noun phrase. However, over the last few decades the intuitions embedded in classical mereology have been reappraised in ways that make the resulting framework of ideas, I will argue, useful for syntax.

In what follows, I'll adopt a mereology that differs from classical extensional mereology in that it is both hylomorphic and pluralistic (Johnston 2006; Koslicki 2008; Fine 2010).

Hylomorphism is the idea, going back to Aristotle, that the whole is constituted by its parts plus some principle that makes that aggregation the kind of whole it is. One of the philosophical arguments for taking the hylomorphic view is that classical mereology admits, as primary objects, such odd combinations as my cat's tongue and Io, a moon of Jupiter; this object is partially located (at least as I write this) inside my cat who is on the chair in my office and out beyond the asteroid belt; this seems odd, to say the least (the same issue holds, of course, for set theory, though it seems intuitively less odd to have an arbitrary collection than an arbitrary object). Hylomorphism, at least of the sort whose aggregative principle is more robust than simple summation, suggests that there is no such object because there is no organizing principle that generates it. Examples of robust principles of organization are the chemical bond that organizes hydrogen and oxygen into water, or the functional and engineering principles that make a gun and a toy gun distinct kinds of objects. Hylomorphic theories of mereological objects amount to a specification of the organizing principle, or principles, that capture the properties of the objects (e.g., Koslicki's 2008 suggestion that organizing principles are tied to natural kinds).

Pluralism in mereology (Fine 2010) is the idea that there are many ways of being a part. It is closely tied to hylomorphism, since, typically, things can have parts that are organized in different ways from other parts (a person can

be an (organized) collection of molecules, or of body parts put together in a certain way, or indeed, in a more abstract sense, a person can be understood as a collection of social and legal responsibilities, and so on). The hylomorphic idea that an object is not just a collection of parts but that the parts are also organized via some principles, and the pluralistic idea that there is more than one way of being a part, fit together well and, I will argue, give us new purchase on what syntactic objects are.

Classical mereology, in contrast, is monistic and non-hylomorphic (beyond the very general notion that summation is an organizing principle). Fine (2010) argues that classical mereology does not provide a general enough basis for understanding parthood. We need to distinguish, for example, the way that the letter *n* is part of the written word “no” from the way that the letter *n* is part of the alphabet, so there are different ways of being a part. We also want to distinguish the way that the letter *n* is part of the written word “no” from the way it is part of the written word “on.” We have the same parts in both cases, but some principle of organization (in this case order) distinguishes them. The classical approach doesn’t provide an intuitive way of thinking about these differences, which seem to require pluralism for the former case and hylomorphism for the latter.

A further useful example is the relation of parthood that holds between a heart and a human being on the one hand, and between a human being and an orchestra on the other. Though a heart can be part of a human being, and a human being can be part of an orchestra, it seems to be missing something important to say that the heart is part of the orchestra. Trading off the same idea: this particular letter *n* is part of this book, but it’s part of the book in a different way from the idea that Phrase Structure is mereological is part of the book, and in a yet different way from the pages of the physical book, or the subsections of the argument, and so on.

If we adopt both hylomorphism and pluralism, then the notion of a “cat-fusion” will depend both on the principle that organizes the cat-fusion object and the way that its parts are parts of it. For the cat-fusion, we’d say that the fusion has as parts (fusions of) cats and their parts. For the collection (set) of cats, we’d say that being a member of the collection is being a part in a different way than the way in which the body parts of a cat are part of it. Classical mereology, in fact, can be thought of as a hylomorphic mereology where the organizing principle is simply summation, and set theory can be thought of as a hylomorphic mereology where the organizing principle for parthood is membership (again see Fine 2010).

Once we specify an organizing principle by virtue of which something is a part of a whole, and adopt the idea that parts might be parts of wholes in

different ways, mereological ideas can do some interesting work for us when we come to look at syntactic objects. One advantage, I will argue, that a mereological perspective has over a set-theoretic one is that the part relation is transitive while the member relation of set theory is not. This will open up a way of understanding certain locality effects that I'll explore in the next chapter. Further, parthood is at least plausibly asymmetric: if x is part of y , then, assuming x is distinct from y , y is not part of x . Intuitively, the embedding of syntactic constituents inside other syntactic constituents is transitive and asymmetric. Set membership, while asymmetric, is not transitive, which is why in Bare Phrase Structure theories one needs to define a new notion of *term*, as the ancestral of membership, which allows reference to constituents within constituents. Exploring the consequences of a mereological, as opposed to a set-theoretic, view of syntactic objects, I'll argue, gives us a different and improved perspective on what kind of objects best serve to model Phrase Structure.

In what follows, I will not, however, adhere to particular axiomatizations of mereologies or focus on formalizing the different philosophical ideas that have been explored in that domain (again, see Cotnoir and Varzi 2021 for an in-depth discussion). I'm not, at least for the purposes of this book on syntactic theory, particularly interested in mereology qua logical or philosophical theory. What I want to do, rather, is to think somewhat laterally about the notion of a syntactic object, drawing inspiration from the ways that philosophers and logicians have thought of how parts relate to wholes, and to see if exploring some of those ideas can provide a better basis for understanding the properties of Phrase Structure than set theory does. Fundamentally, I want to see if taking "part" as the primary syntactic relation, as opposed to "member," leads to a system that improves over the current one. I will argue that it does.

3.2 The Basic System

3.2.1 Subjoin

Since Phrase Structure is, at least on classical conceptions, hierarchical, if we are to conceive of it mereologically, we need a principle of organization that provides hierarchy. This is precisely what Merge does in classical minimalist approaches: it serves as a way of ensuring that the elements in Phrase Structure are organized hierarchically, and it does so by providing a recursive generative system that gives us the relevant SOs. My critique here of classical minimalist approaches is not that the idea of recursive construction of objects is wrong (in fact, I think it is very much right); it is merely that the kinds of objects built by set-theoretic Merge are not the best models for syntax.

Let's rethink Merge, then, replacing it with an analogous operation that builds mereological objects, as opposed to set-theoretic ones: the organizing principle for these mereological objects is one of a part-whole hierarchy. I'll call the relevant operation Subjoin.³ The core intuition is that when an object is subjoined to another object, it becomes part of it; that is, the target of the subjunction is modified.

We can think about this via the following analogy: my house has changed over the years I've lived in it, mainly in its internal constitution. We've added a new bathroom, and a cat-flap, and a new window. In all of these cases, the object that is my house is in some sense the same object, but in another sense it has changed, in that it has gained new parts. This is the intuition behind Subjoin: after subjunction, an object is the same object but has a new part.

We can symbolize this way of thinking about Subjoin(x, y) as follows:

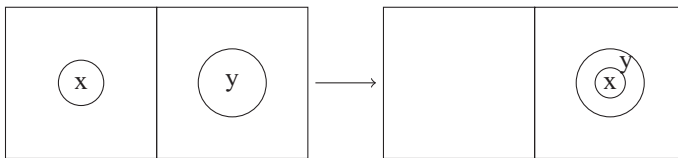
$$(2) \text{ Subjoin}(x, y) \Rightarrow y : x < y$$

Subjoin carries an object y from one state, where it does not have x as a part, to a new state, where it does.

The $<$ symbol is the "proper part" relation of mereology (Cotnoir and Varzi 2021). I'll use $<$ as the core relation, as opposed to using $\leq$ (which would allow x to be part of y if $x = y$). Proper parthood captures the (I think common) intuition that a syntactic constituent is not a constituent of itself, though, of course, issues of reflexivity of categories in structure have been a matter for debate over the decades in definitions of dominance and so on (e.g., Cushing 1978).

Visually, we can represent the derivation as follows:

(3)



The relevant properties of $<$ are usually specified via a set of axioms that ensure that the relation is:

- (4) a. irreflexive ($\forall x. \neg x < x$),
- b. transitive (if $x < y$ and $y < z$, then $x < z$)
- c. asymmetric (if $x < y$, then $\neg y < x$)

Irreflexivity states that nothing is part of itself. Asymmetry states that if some x is part of some y , then it cannot be the case that y is part of x . Transitivity

ensures that if x is part of y and y is part of z then x is part of z .⁴ These axioms act as constraints on what a possible SO formed by Subjoin can be.

Recall that I’m assuming a pluralistic version of mereology where there are different ways of being a part. I call these different ways of being a part *dimensions of parthood*. Irreflexivity restricts parthood irrespective of Dimensionality (so x is not a part of x in any dimension). However, as in other pluralistic versions of mereology (Fine 2010), the following holds:

(5) Dimensions restrict transitivity of parthood.

To see this, let’s extend our previous example about hearts and orchestras a little: a valve is part of a heart which is part of a person. By Transitivity, then, a valve is part of a person. A person may be part of a string section, and a string section is part of an orchestra, so, again by Transitivity, a person is part of an orchestra. But that set of part relations does not entail that a valve is part of a string section, or a heart part of an orchestra. The part-of relations here are in different dimensions. Transitivity holds within but not across dimensions.

Putting these ideas to use, take two syntactic objects, a and b . We can Subjoin a to b (or b to a), and, after that subjunction operation, we now have one object b , which is more complex, in that it has a as a part. We can represent the constructed object as a “tree,”⁵ where the asymmetry of parthood is represented by the vertical dimension of the “tree”:



In this structure b is not a “projection” of a in any sense; it’s simply the object b altered so that it has a as a part.

Parthood is irreflexive. If we attempt to Subjoin some object to itself, then the output will violate the irreflexivity of the proper part relation:

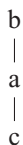
(7) Subjoin(a, a) $\nRightarrow a : a < a$ (violates Irreflexivity)

Parthood is also asymmetric: because of this, the arguments of Subjoin, unlike those of Merge, are ordered. Subjoin(a, b) is distinct from Subjoin(b, a). We can think of subjunction as targeting the second argument, altering it by making the first argument a part of the second.

We can now ask what happens if the righthand argument of Subjoin is not a simple object, but is itself the result of a previous subjunction; for example, if we wish to Subjoin c to the object b constructed in (6).

Since the target of Subjoin is the object b , c subjoins to b , not to a . I will assume that the following is not a legitimate output (even though in (8), c is part of b by Transitivity) and return below in section 3.2.3 to why:⁶

(8) Subjoin(c, b) $\nRightarrow$



Rather than (8), subjunction of c to b , makes c part of the object b :

(9) Subjoin(c, b):



However, c is part of b in a different way than a is part of b , specifically in that it was added at a different derivational point. I'd like to make the proposal that the Dimensionality of the part relation in syntax is read off of the derivational step, adding the following constraint on the output of Subjoin:

(10) Dimensionality:

- a. The first application of Subjoin to an object x is in dimension 1;
- b. If an object x already has a part in dimension n , then an application of Subjoin to x will be in dimension $n+1$;
- c. There are 2 dimensions in syntax.

Dimensionality entails that in (9), b has the part a in dimension 1, and the part c in dimension 2. Obvious questions arise about whether subjunction in dimensions 3 and higher are available. I will assume not, at least for the development of the system in this book, and take the “branching” of structure to be maximally binary: since there are two dimensions, it is impossible to subjoin three objects to a single object.⁷

In the diagrammatic representations, I will put elements in dimension 2 to the left of elements in dimension 1, though an alternative notation (e.g., labeling the parthood lines with 1 and 2, using full vs dotted lines, colors, etc.) would be equally effective. The directionality of the line marks Dimensionality, not linear order. For consistency, I'll slant dimension 1 parthood relations (1-parts) to the right, even in the absence of a part in dimension 2 (2-parts). I'll occasionally write the Dimensionality aspect of the parthood relation $<$ as $<_1$ or $<_2$, when relevant, and I'll refer to these two relations as 1-part and 2-part, respectively.

What are these two dimensions that are determined by the derivation? I propose to think of them as follows:

- (11) a. Dimension 1 is interpreted as the Extended Projection Complement relation (Grimshaw 1991 et seq.) (except transitive).
- b. Dimension 2 is interpreted as the specifier relation (except transitive).

If x is a 1-part of y , this means that x is in the same Extended Projection as y (and, usually, is “lower” in that Extended Projection, though that is not strictly required by (11)). If x is a 2-part of y , x is a specifier (or a specifier of a specifier, etc.) of y . These relations differ from their counterparts in more standard approaches in that they are transitive. Parthood in this system differs from the classical tree-theoretic notion of dominance, which is also transitive, but not dimensionally restricted.⁸

Let’s run through the derivation of a simple sentence using these ideas to make them more concrete. I’ll assume that subjects of (unergative) verbs are introduced via a functional category v , as is standard.

Recall that the first part of Dimensionality runs as follows:

(12) Dimensionality (a)

The first application of Subjoin to an object x is in dimension 1 (a 1-part understood as the Extended Projection Complement relation, but transitive).

Take a verb like *jump*, which is of category V . Given Extended Projection, it will be the complement of v :

$$(13) \text{ Subjoin}(V, v) \Rightarrow v$$

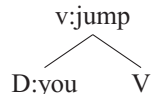

At this point it is already clear that this system avoids the Labeling Problem that standard Bare Phrase Structure had. V subjoins to v , but because the operation simply makes V a 1-part of v , the resulting object is still just v , with the label that v has always had. I’ll discuss the ramifications of this in more detail below.

The second clause of Dimensionality is:

(14) Dimensionality (b)

If an object x already has a part in dimension 1, then an application of Subjoin to x will be in dimension 2 (a 2-part: similar to specifier, but transitive).

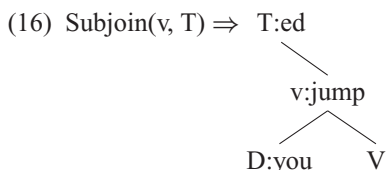
Let us now add the (agent) subject of *jump*, say a pronoun like *you*, which I will assume for simplicity is a simple D category, as the 2-part of v , in *you jump*:

$$(15)$$


I have placed (the orthographic representation of) the phonology of the actual words next to the relevant category after a colon, following the notation I

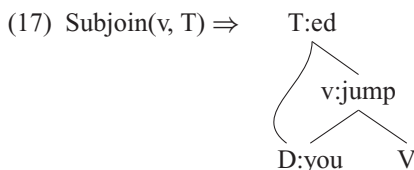
introduced in Adger (2013). I'll return to how the linear order of these structures is calculated below.

Now let us add tense information, assuming v is the complement of T . This gives the following structure for *you jumped*.



Recall that Dimensionality restricts to two the number of subjunctions to any category, so we cannot subjoin T to v (which would anyway violate the order of the Extended Projection for the clause, however that is implemented). We can however subjoin v to T as a 1-part (since it is the first subjunction to T).

The standard analysis is that the subject in English is a specifier of T . We derive this in the mereological system by simply subjoining the 2-part of v to T (adding a second part relation to T , which will be a new 2-part relation between T and $D:you$).



This “Internal” Subjunction is the analogous operation to “Internal” Merge. Subjunction maintains the core advantage of the Merge system: the unification of local and nonlocal dependencies. I'll return to some further consequences of this below.

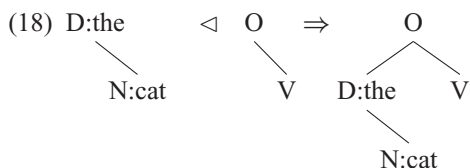
The idea that parthood in dimension 1 corresponds to the Extended Projection complement relation, as seen above, is somewhat intuitive. On the one hand, objects headed by categories lower in an Extended Projection are standardly interpreted as constituents, hence parts, of objects headed by categories higher in it (VP is a constituent of TP , and so on); on the other, there have been proposals that take the meanings of objects lower in an Extended Projection to be parts of the meanings of objects that are higher. For example, the influential proposal of Ramchand and Svenonius (2014) takes the events that VP s denote to be parts of the situations that TP denotes, which themselves are parts of the propositions that CP s denote. If the syntax is built via the part-relation, there is a pleasing transparency to the syntax-semantics relationship.⁹

In terms of dimension 2, it is again the case that specifiers are usually taken to be constituents of the projection of the category that heads them, and hence

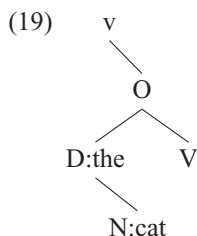
in some sense part of that category. The idea, then, that a 2-part of a category is akin to a specifier of that category seems plausible. Semantically, at least some specifiers are taken to be arguments of their heads, and the denotata of arguments can be thought of as part of whatever verbal projections denote (for example, as in Situation Semantics, where individuals are particular parts of larger situations, cf. Kratzer 1989). If specifiers are the structural realization of conceptual Figure-Ground relations (e.g., Ramchand and Svenonius 2002), then again that is a straightforward mapping.

Let us take another example, but this time look at the subjunction of two complex objects: the sentence *the cat grew*. This consists of a nominal phrase combining with a verbal phrase of some sort. I'll assume *grow*, in this sentence, is an unaccusative verb. Since 1-parthood is associated with the Extended Projection relation, the sole argument of *grow* cannot be its complement. Instead the theory forces us into an analysis where arguments like *grow*'s incremental theme are specifiers, or, more accurately in the mereological system, 2-parts. Such analyses are well developed in the literature (e.g., Borer 2005b; Ramchand 2008; among many others). I'll remain agnostic about the actual content of the category that has, as a 2-part, the argument and will just call it O, mnemonic for "object."

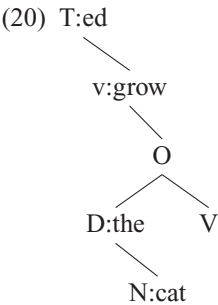
With this in place, we subjoin N to D so it becomes a 1-part of D as in (18) below, to give the structure for *the cat*, and we subjoin V to O to give the structure for the unaccusative verb. Subjunction of D to O will result in D becoming 2-part of O, since O already has the 1-part V. I will sometimes use $\triangleleft$ for typological ease to notate the Subjoin operator when the arguments of Subjoin are complex:



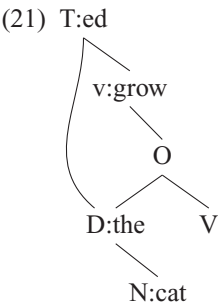
Adopting the analysis that the v category can lack a subject, and that this gives us an unaccusative vP (e.g., Adger 2003), we then subjoin O to v:



As before, we can subjoin the result to T, giving the structure underlying *the cat grew*:

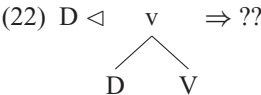


Finally, given that Subjoin, like Merge, can have, as one of its arguments, part of its other argument, we can internally subjoin D in (20) to T, giving (21) as the structure for *the cat grew*, with the subject being a 2-part (a specifier) of T as in the standard analysis for English:



The SOs generated by the mereological system are very similar to the telescoped representations introduced by Brody (2000) in his Mirror Theory, and discussed by Adger et al. (2009) and Adger (2013). They contain all of the information one might expect in a more standard Bare Phrase Structure representation (or indeed in an $\bar{X}$ structure), but there is no terminal node associated with the “head.” The phonologization of the structure is done via a separate mechanism, discussed below.

I’ve discussed the first two clauses of Dimensionality, so I will just briefly here mention the third clause. This would be a structure where we attempt to subjoin an object (whether simple or complex) to an object that already has a 2-part:



Dimensionality provides no notion of 3-part; 2 is the highest dimension. Further subjunction to a complex object with a 2-part is therefore impossible given Dimensionality. As in the Merge system, “branching” is maximally binary, though Dimensionality is functionally more similar to the restriction to one bar-level in classical $\bar{X}$ -theory: it allows one 1-part and one 2-part per category.

3.2.2 Subjunction vs Merge

The mereological system I’ve sketched out here is, in some ways, quite close to the Merge system, but, in others, very different. Like Merge, Subjoin applies to two preexisting objects and operates on these creating a relationship between them that is familiar in $\bar{X}$ -theoretic terms (sisterhood for Merge, head to nonhead for Subjoin). While Merge builds a set, and Subjoin builds a mereological object, both sets and mereological objects are linearly unordered, and in that respect are similar.

However, while Merge creates a new object from two existing ones, Subjoin modifies two objects introducing an asymmetrical part relation between them. On most interpretations at least, an object created by Merge has a symmetric relation between its two members (though there have been attempts to add asymmetry into Merge, e.g., Zwart 2011; De Belder and van Craenenbroeck 2015). There is, of course, an asymmetric relation of set membership between the newly created set and its members, but not between the arguments of Merge itself. Subjoin, in contrast, creates an asymmetric parthood relation between its arguments. I have noted that this has advantages when it comes to the theory of labeling, and will explore that further below.

There are other differences: in the Merge system, as briefly mentioned above, a new notion of *term* has to be defined (as the transitive closure of membership) in order to provide the right arguments for (especially Internal) Merge. This is because the relation that is given by Merge (membership) is not transitive, but, if we are to have terms, transitivity is required.¹⁰ In the mereological system, in contrast, because the part relation is transitive, no special new kind of object needs to be defined, and I will argue that the transitivity of parthood will provide empirical dividends for us in many different ways as I develop the theory.

A further difference is that while Merge makes a distinction between an object and a set containing that object (they can be fundamentally different things: e.g., one a lexical item, the other a set), that is not true for Subjoin, which maps from objects to objects. In the mereological system, then, there is only one kind of object, while in the Merge system, at least standardly, there are two (lexical items and binary sets).

When we think about issues of simplicity, hard though these are to draw any firm conclusions about, it is difficult to argue that Subjoin is any more complex

than Merge. Subjoin adds one new part-of relation to an existing object, while Merge creates a new object by adding two new (member-of) relations. Both operations are binary. Though set membership is more familiar than mereological parthood, neither can be definitively said to be more “fundamental” (e.g., Simons 1995).

The one area where there is a potential difference is that, because I am adopting a pluralistic mereology, there are two types of part relation ($<_1$ and $<_2$), but, in the Merge system, there is only one type of membership relation. However, the two types of part relation are derived from the timing of subjunction operations: first subjunction to an object is $<_1$, and second is $<_2$. In Bare Phrase Structure, the same structural distinction is keyed to the distinction between a head and a nonhead, which relies on the difference between an object and a set, a difference that the mereological system does not have. In the mereological system, that work is done by the two types of parthood, which are derivative of derivational steps. Again, the trade-off here seems roughly equivalent and I think there’s no obvious argument that one approach is fundamentally simpler than the other.

On a final historical note, in the LSLT system of Chomsky (1955), syntactic objects are built out of elements via the core relation of concatenation. Concatenation involves both linear ordering and a kind of “joining”: two (or more) existing elements are brought together in a linear dimension so that they constitute a new object:

(23) $\text{concatenate}(a, b) \rightarrow a \frown b$

As discussed briefly in the previous chapter, the hierarchical aspects of syntax in LSLT are modeled by sets of concatenated primes. Much of LSLT is devoted to emphasizing the importance of the *sui generis* nature of hierarchy in syntax: it is not derived by some kind of generalization of concatenated units, but is rather generated via rules. LSLT used the notion of concatenation algebras as a means of formalizing the relevant linguistic objects for good reason: concatenation was an operation that seemed natural given the presuppositions of Harris (1951), Wells (1947) and others, and their views on how to understand their findings.

The move to the $\bar{X}$ system, and then to the Merge-based approach, removed the concept of linear ordering as part of the fundamental nature of syntactic objects. However both theories kept the notion of “joining” units together to make a new unit: the arguments of $\text{Merge}(x, y) = K$ are joined together to make the constituent parts of K , modeled by sets in the Merge-based approach.

Subjoin replaces this last vestige of concatenation with a relation that directly expresses part-whole relationships. An analogous notion to “joining” does not exist in the mereological system because there is no new object created. The

empirical phenomena that “joining” was used to capture are instead modeled by 2-parts being parts of a whole in a different way from 1-parts. The mereological system, then, is even more fundamentally divorced from linear “externalization” than the classical Merge-based system: the last vestige of thinking of syntactic structure as derivative of abstractions over physical events coming together in time is gone. In this, the Mereological system is somewhat similar to ideas that have been explored in Dependency Grammar (Tesnière 1959; Hays 1964; and more relevantly here Manzini 1995), though these systems are in other ways very different. The dependencies in Mereological Syntax are between potentially complex objects, while dependencies in Dependency Grammar are between basic objects (typically words).

3.2.3 Cyclicity and Subjoin

Subjoin is further to be distinguished from External Merge in that the derivations it sanctions don’t obey certain formulations of the Extension Condition, although the issue is somewhat more complex than it might appear at first blush, and requires some prior excavation of the relationship between Extension and cyclicity.

The Extension Condition, as it was initially defined, constrains operations that apply to some object α to derive some object β in a way that ensures that α is a proper part of β (informally stated thus in Chomsky 1993, p. 22; in more consistent set-theoretic terms, α is a member of β , cf. Collins and Stabler 2016, p. 59, Theorem 9). Subjoin does almost the opposite of this: it adds information *inside* the object to which it applies, so that the targeted object is the derived object.

The 1993 version of the Extension Condition just mentioned, unlike later versions, does not actually enforce strict cyclicity of derivations, although it is clearly intended to. It allows an operation to target some object internal to a larger object, extending the targeted object. Take for example, (24):

(24) $\{x, \{w, \{u, v\}\}\}$

It is possible to derive (25) from (24):

(25) $\{x, \{v, \{w, \{u, v\}\}\}\}$

Here an operation has targeted $\{w, \{u, v\}\}$, which is a member of (25), internally Merging v with it, to derive $\{v, \{w, \{u, v\}\}\}$. It is clear that, after that operation, $\{w, \{u, v\}\}$ is a proper part (a member) of $\{v, \{w, \{u, v\}\}\}$, and the derivation is therefore consistent with the version of the Extension Condition given by Chomsky. Similarly, nothing would rule out (26), derived from (24):

(26) $\{x, \{\{z, y\}, \{w, \{u, v\}\}\}\}$

Here the object $\{z, v\}$ has been Externally Merged with $\{w, \{u, v\}\}$, extending it to an object $\{\{z, v\}, \{w, \{u, v\}\}\}$, which is a member of the larger object that also has x as a member. The derivation extends the target.

Typically, however, derivations within minimalist systems have been assumed not just to require extension of the target, but rather extension of a target that is not the result of a previous computation: that is, derivations are assumed to be cyclic, unable to map from (24) to (25) or to (26).

There are reasons to block such countercyclic derivations. For example, as is well known, there are many languages that allow A-raising from non-finite clauses, but where that movement is blocked by an intervening experiencer. McGinnis (1998) gives the following from Icelandic (see also Rizzi 1986 on Italian):

- (27) a. Ólafur virðist [_{TP} vera gáfaður]
 Olaf.NOM seems to.be intelligent
 ‘Olaf seems to be intelligent.’
 b. *Ólafur virðist mér [_{TP} vera gáfaður]
 Olaf.NOM seems me.DAT to.be intelligent
 ‘Olaf seems to me to be intelligent.’

If a countercyclic derivation were possible, then (27b) could simply be derived from (27a) by countercyclically externally merging the experiencer *mér* after raising of the lower subject to the matrix specifier of TP. A-movement would have already taken place in a legitimate fashion, and by hypothesis countercyclic Merger of the experiencer is allowed.

Many other similar derivations could be created that subvert intervention effects in this way. For example, consider the following Superiority paradigm:

- (28) a. What did Anson force Merkel [_{<what>} PRO to eat _{<what>}]?
 b. Who did Anson force _{<who>} [PRO to eat fish]?
 c. *What did Anson force who [_{<what>} PRO to eat _{<what>}]?

A classical analysis would rule out (28c) on the grounds that movement always raises the relevant element that is closest to the final landing site (Shortest Move, see, e.g., Chomsky 1995, p. 296). The phrase structural analysis assumed is that the item *what* in the specifier of the embedded control infinitive is more distant in binary Phrase Structure than *who*, which is Merged higher, for example, in a VP shell, (see, e.g., Adger 2003, p. 323, for a summary of the binary branching analysis of object control verbs). Since *who* is closer to the matrix C than *what*, it is *who* that raises. However, if countercyclic Merge is allowed, then a derivation that extracts *what* to the matrix and then inserts *who* as the internal argument of *force* would be legitimate:

- (29) a. What did Lucy force [_(what) PRO to eat _(what)] →
 b. What did Lucy force who [_(what) PRO to eat _(what)]

There are, of course, alternative analyses that will rule out both the experiencer intervention and Superiority cases. For example, if the relevant constraints are filters on final structures, as opposed to constraints on legitimate derivational steps, then one could argue that both structures are ruled out on the grounds of Relativized Minimality (Rizzi 1990) or some other relevant filter.

In fact, Freidin (1999), building on a form of argument laid out in Freidin (1978), argues more generally that cyclicity of the relevant kind is epiphenomenal, and its effects are rather to be derived from other principles of grammatical structure. Some approaches allow certain elements to be countercyclically added, while disallowing others. For example, Bošković and Lasnik (1999) suggest examples like (28c) are ruled out because of a ban on countercyclic Merge of arguments (which they derive from a particular view of theta-roles), whereas constituents that do not receive theta-roles can Merge countercyclically (cf. Lebeaux 1990; Takahashi and Hulsey 2009; and many others).

For those theories that take the Extension Condition to be required, there have been various approaches to ensuring that syntactic derivations apply in such a way that they always extend an unembedded SO. In fact, the general idea has long been a feature of transformational approaches to syntax. The Strict Cycle Condition of Chomsky (1973), for example, forbade any rule of grammar from applying to a subpart of the derivational output (a so-called cyclic node) that had already been passed in the derivation.¹¹ Later versions of the Extension Condition require operations to target and extend the root of the structure (that is, that SO that is not contained in any other SO). Like the Strict Cycle Condition, that version of the Extension Condition is sometimes simply stipulated (e.g., Chomsky 1995, p. 248; Bobaljik 1995).

Similarly, Collins and Stabler (2016, p. 49, Definition 14), in their definition of a derivation, require that, if Merge maps between stages of a derivation, one of the arguments of Merge must be a member of the workspace (hence itself not be contained in any other SO). This definition allows them to state the Extension Condition as a theorem, but its effects are essentially stated in the definition of a derivation. Obviously such a constraint on Merge, whether External or Internal, will rule out countercyclic derivations, since nodes internal to an SO can't be the target of the operation.¹²

A number of authors have attempted to derive Extension-type cyclicity from other properties of syntax or from syntax external factors, as opposed to stipulating it as a specific constraint on operations. In the system in Chomsky (1995, p. 234), for example, the suggestion is that the kind of feature that motivates

a syntactic operation (a “strong” feature) has to be immediately eliminated by that operation before the derivation proceeds any further. In (24), for example, if w has a strong feature, then v must raise to w to eliminate that strong feature before x is Merged, forcing a cyclic derivation (see also Bošković and Lasnik 1999). Such an architecture would make it impossible to derive Shortest Move violations via countercyclic Merge.

Watanabe (1995) suggests that Extension-type cyclicity can be derived by an economy condition that prefers derivations that make the least possible changes to the structure (cf. Chomsky’s [2005] No Tampering condition). Kitahara (1997), along similar lines, suggests deriving Extension-type cyclicity from a general economy condition. He argues that a countercyclic derivation (such as one that maps from (24) to (25)) requires a *replacement* of the category that was x ’s complement by a new category (one that is not identical with the object $\{w, \{u, v\}\}$). Given that such a derivation involves both movement and replacement, it is more complex than a derivation that just involves movement, and, since simpler derivations are preferred, Merge must always be to the root (the argument depends on further premises, but I leave these aside). Collins (1996, pp. 82–84) also provides a definition of Merge that enforces cyclicity by ruling out replacement of categories. In the same vein, Chomsky (2004, p. 109) states, as part of the definition of Merge, a Strong Extension Condition, such that when some object α Merges with β , β is completely unchanged. This obviously entails that α cannot be Merged inside β , as that would change β , rather it must be Merged at the “edge” of β , that is with β as a whole.¹³

With this rather lengthy discussion in hand, we can now ask the question of how Subjoin is to be understood in terms of Extension-type cyclicity. Clearly, as mentioned at the start of this section, a subjunction derivation does not extend a root SO: the target of subjunction does not become embedded after subjunction. In fact, the target of subjunction, if it is a root SO, remains a root SO. However, we might still want subjunction derivations to be cyclic, in the sense that they always target an unembedded SO. Subjoin on its own, just like Merge, will allow countercyclic derivations, and, just as in a Merge based system, if we want to force Extension-type cyclicity, we need to add something extra to the system, or, ideally, derive the cyclicity from some other, deeper, property.

Given that Subjoin has an inherent asymmetry between its arguments, it is straightforward to require that its second argument must not be a part of anything else. In fact, there is some naturalness to the idea that, in creating a part-whole relation, the whole itself must not be a part. Let’s simply state this as a kind of mereological equivalent of the Extension Condition. I will restrict

Mereological Extension to “external” subjunction for the moment, returning to the question of whether it is needed for “internal” subjunction in the next chapter.

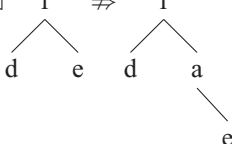
- (30) Mereological Extension: if x is not a part, then to apply $\text{Subjoin}(x,y)$, y is also not a part.

This will rule out a derivation where subjunction targets an SO that is part of another SO.

Take a situation where we have two distinct SOs. There are a number of cases to be distinguished. First, we saw previously that Dimensionality means that subjunction to an object that already has a 2-part is impossible since the maximal Dimensionality is 2:

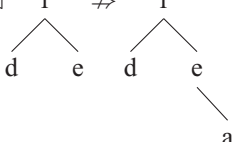
$$(31) a \triangleleft f \Rightarrow ??$$


Dimensionality also makes it impossible, as already noted, to Subjoin a to f as a 1-part. Each derivational step adds Dimensionality, so a kind of “Tucking In” in dimension 1 is not possible:

$$(32) a \triangleleft f \not\Rightarrow f$$


The object a cannot Subjoin to f in the first dimension, becoming a 1-part of f , since Dimensionality is derived from the sequence of operations, and a subjunction has already applied making e a 1-part of f .

These illegitimate outputs are already ruled out in the system. However, say the subjunction here is not $\text{Subjoin}(a,f)$, but is rather $\text{Subjoin}(a,e)$ as follows (or, indeed, it involves subjunction of a to d):

$$(33) a \triangleleft f \not\Rightarrow f$$


This derivation doesn't fall foul of the same problem. Subjoin can't add a new part relation to something that already has a part relation in the relevant dimension, but here, e doesn't have any parts, so that constraint doesn't apply. What rules out this derivation is Mereological Extension: the second argument

of Subjoin is e , and e is a 1-part of f . I'll adopt Mereological Extension in what follows, although it's important to bear in mind Freidin's arguments that it may not be required and that its empirical effects may be the result of other constraints. We will see that, when it comes to "internal" subjunction, Extension is subsumed by the theory of locality.

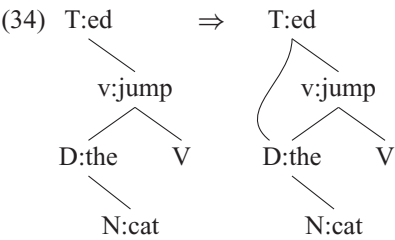
3.2.4 Multiparthood

I now return to another way in which the mereological system is similar to the Merge-based system: its treatment of "movement" dependencies. I already mentioned, when discussing how the mereological system analyses "movement" of a subject to become a 2-part (specifier) of T (see 21), that an internal version of subjunction is available, analogously to the internal version of Merge, but it will be useful to explore the ramifications in more detail.

One of the successes of the standard Minimalist approach to Phrase Structure, discussed in the previous chapter, is the unification of structure building and structure change (movement) via the definition of Merge. To briefly recapitulate, Chomsky (2004) proposed a single operation Merge that can either apply to two distinct objects, A and B, bringing them together to create a new object $\{A, B\}$, or that can apply to an object B and some object A that is internal to B (a term of B), again to create a new object $\{A, B\}$. Though there is a single operation, the cases are distinguished by what the operation applies to: A is external to B in the first case (External Merge), but is drawn from inside B in the second case (Internal Merge); the relevant notion of "inside" here involves the transitive closure of set-membership to access the "terms" of B. There is also the case of Self-Merge (Guimaraes 2000, Kayne 2010, Adger 2013), which is a variant of Internal Merge where the object internal to B is itself B, so that from B, the object $\{B, B\}$ is constructed, which is of course equivalent to just $\{B\}$ in set-theoretic terms.¹⁴

The conceptual advance of unifying the Merge and Move operations of Chomsky (1995) et. seq. (essentially the phrase structural and transformational components of classical transformational grammar) is enormous, and the same basic idea is clearly portable to the mereological system, as we have seen. I will however argue that applying this idea in the mereological system has some theoretical advantages when it comes to thinking about the vexed issue of copies versus repetitions (Chomsky 2013, p. 40; Collins and Groat 2018; and section 3.4 below).

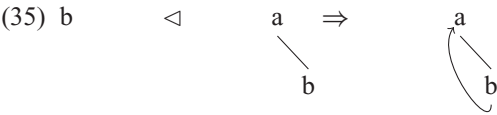
Let us return to the previous example where a subject of an unergative has subjoined to T:



This is generated via an application of Subjoin that has the subject *the cat* as its first argument and the SO *T* as its second. It adds a new 2-part relation (since this is the second subjunction to *T*), making *the cat* a 2-part of both *v* and *T*.

It is clearly straightforward within the mereological system to create non-local relations via the same operation that is used to create local relations, analogously to how Merge, in the set-theoretic system, can apply externally or internally. However, there is a difference, because of the kinds of objects the two systems build. A mereological object constructed using “internal” subjunction has an organization that is reminiscent of the multidominant structures of other syntactic frameworks, especially the Phrase Linking Grammars of Engdahl (1986), though the object is multipart as opposed to multidominant (also McCawley 1982; Johnson 2012; Gärtner 2002; Johnson 2020; Citko and Gračanin-Yuksek 2021; and others). In fact, the system developed here provides an extremely simple mechanism for generating multipart structures, with a single Subjoin operation. The multipart nature of the generated objects will turn out to be an advantage when it comes to distinguishing copies and repetitions in section 3.4.

Before we turn to other matters, it’s worth going through two other logical possibilities for “internal” subjunction. The first case is what happens when we apply subjunction to the same SOs “twice.” Take, for example, the operation Subjoin(*b*, *a*) applied to the SO *a* we have already constructed, which has *b* as a part:

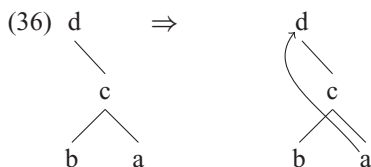


Dimensionality will allow this derivation since the second subjunction of *b* to *a* is to an SO that is not part of anything else: in the resulting structure, *b* is a 1-part of *a* and a 2-part of *a*. Such a derivation is the equivalent in the mereological system of the Anti-Local movement of a complement of

a functional category to that category's specifier discussed (and ruled out) by Abels (2003).

There are immediate questions about how *b* might be linearized and interpreted both as a 1-part and as a 2-part of *a*. However, as we will see in chapter 4, this super-local derivation will actually be ruled out by the theory of locality.

We can ask a similar question about a complex object where we Subjoin a lower 1-part of that object as a 2-part to it:



This is the equivalent of a “roll-up” derivation, where some constituent (e.g., VP) in the Extended Projection moves to become a specifier of some higher category (for example, something that licenses VP-topicalization) in that same Extended Projection. In this case, again, there is the question of whether, and if so how, *a* might be linearized and interpreted both as a 1-part of *c* and as a 2-part of *d*. I return to these issues in the next chapter, along with various other derivational possibilities, including lowering and sideward movements.

The upshot of this discussion is that Subjoin, like Merge, allows External and Internal variants. The major difference between the two systems, however, is not in the architecture of the theory itself: both are generative systems with a single operation, where derivations are subject to cyclicity and the operation has both external and internal subtypes. The difference is in the kind of SOs the operation creates. Subjunction, because it delivers multipart structures, has no equivalent of distinct “copies” in at least some of the senses of that term that have been explored in the literature (e.g., Nunes 1995). The SOs are resolutely singular in nature, though they may bear dimensionally restricted parthood relations to multiple other SOs. I return to this issue in section 3.4.

3.3 Atomic Objects and Labeling

The mereological system as I’ve defined it so far generates structured objects. I’ve not, however, said much about what the primitive objects are, out of which these more complex objects are constructed.

I take the basic objects of the system to be a set of syntactic categories of a familiar sort: V, v, T, N, D, C plus whatever other categories particular analyses require. I assume that both “lexical” categories and “functional” categories

are in the relevant set of objects and that they can be ordered via Grimshaw's theory of Extended Projection (Grimshaw 1991). These are the "basic" SOs but they are also potentially the "complex" SOs, since the Subjoin operation takes these categories as its arguments (I'll sometimes use the word "category" as a synonym for SO when focusing on the type of SO, as opposed to the parts).

This is different from how the classical Merge system works: systems like that in Chomsky (1995) et seq. take the basic objects to be lexical items, which are triples of phonological, semantic, and categorial information. Obviously, it is not straightforward to take the basic objects in a mereological system to be lexical items in this sense, since we would end up saying that, for example, *cat* is part of *the* in the phrase *the cat* (or perhaps vice versa, depending on analysis).¹⁵ We would need a notion of part that is quite different from the intuition we began with, that part is to be understood (roughly) as constituent.

There are, then, no "lexical items" *in the syntax* in the mereological system: just categories and the structures that categories build. There are however "lexical items"—that is, stored triples of syntactic, phonological, and semantic information—at the interfaces. Rather than entering into syntactic derivations, a lexical item is a skeletal morphophonological or semantic interpretation of an SO. The system, then, adopts a kind of Late Insertion, as is familiar from Distributed Morphology (e.g., Marantz 1997). An advantage of such a view is that it enforces quite a strict autonomy of the syntactic system (Adger 2017), explaining why syntax is phonology-free (Zwicky and Pullum 1986) and, in all likelihood, semantics-free.

I assume that the relevant syntactic information in the lexical item is stored as a (possibly single-category) sequence of categories. When the syntax constructs a structure where T, O, and V are in a 1-part relation, that sequence of categories can be accessed by a lexical item, or by a number of lexical items, and be pronounced (eventually) as, say, *grew*:

(37) $V \neg O \neg T[\text{past}] \Rightarrow /gru/$

How (37) comes to have the phonology that it does depends on some specifics about the mapping that, I think, are still open questions. One approach is that each of the categories in the sequence has stored abstract phonology. The phonological component then computes over the linear sequence of exponents to derive the outcome; this is essentially the idea of Mirror Theory (Brody 2000). For a fusional example like *grew*, either the phonology of each component is abstract enough that the apparently irregular form is regularly derived (e.g., Newell 2024), or a series of readjustment rules apply (Chomsky and Halle 1968), or a single phonological unit inserts over a "span" of categories (where

a span is a sequence of categories, as in Williams 1994; Bye and Svenonius 2012; Adger 2014; Merchant 2015). I won't choose between these approaches here.¹⁶

On the semantic side, I assume the same basic architecture holds. The various elements of the sequence will be interpreted by the relevant semantic mechanisms computing how the meanings of V, O, and T[*Past*] come together to give, very roughly:

$$(38) \text{ grow}(e,x) \wedge \text{Theme}(e,x) \wedge \text{Past}(e) \Leftarrow V \neg O \neg T[\text{past}] \Rightarrow /gru/$$

I'm assuming, then, that information relevant to the interface systems is delivered to them by the SO, and that they compute over this information to provide the interpretive systems with what they need to create or interpret the correct meanings and forms for SOs.

If the basic SOs are categories, then the traditional distinction between a label and an SO that is labeled, which comes from classical Phrase Structure grammar, is not available. The atomic object and the category (the label) are one and the same. The notion of category in this system is reified: V is not a type of various words, or lexical items, because the system does not work on lexical items. It works on syntactic objects. This means that there are numerous objects that are formally identical to each other, and those objects are all Vs. A good analogy is atoms of an element. There are numerous oxygen atoms in the world, which are (putting isotopes aside!) qualitatively identical to each other but numerically distinct. From this perspective, two instances of an object (e.g., D) in a structure (say in two separate clauses, or as a subject and object in a single clause) are to be understood as two separate syntactic objects that, though identical in their inherent formal properties, are still distinct objects, just as two atoms of oxygen, one in the air and the other in my cup of tea, are distinct objects, though they are identical in type. Keeping to the analogy, V is formally distinct from T in syntax in much the same way that H is formally distinct from O in chemistry. Categories are just atomic objects that are built into larger complex objects by Subjoin. Actual verbs, nouns, tense morphemes, and so on are interpretations, both morphophonological and semantic, of syntactic objects built by Subjoin.

A consequence of this way of thinking is that a single SO in a structure (say a complex object D with N as a part) is a single SO, even though it is, perhaps, a 2-part of both v and T in a clause where it is the grammatical subject of an unergative verb. There is a single object D, as opposed to two. This is analogous to a single atom of oxygen that bonds to distinct atoms of carbon and hydrogen in an alcohol molecule.

Unlike work in, for example, Nanosyntax (Starke 2009; Caha 2009), I assume a difference between features and categories, following Adger and Svenonius (2011). I'll take features to be secondary properties of categories, as opposed to being categories themselves. Features can vary across categories and are a primary locus for syntactic differences: they are the technology language acquirers use to make their analyses of what they hear consistent with the principles of the grammar. The core use I'll make of features in this book is to set up syntactic dependencies; I won't be too concerned with either the set of features, or their content, beyond a couple of cases, and even these I won't theorize too deeply.

I'll use the classical notation of uninterpretable [uF] vs interpretable features [F] for this (e.g., Chomsky 1995; Stabler 1997; Pesetsky and Torrego 2001; Adger 2003, and others). An [F] on a category may affect the interpretation of that category by the semantic or phonological systems, and the interpretation of features, combined with the fact that they set up syntactic dependencies, will do a fair amount of analytical work in the system. A [uF] on a category may affect the interpretation of the category phonologically but cannot affect it semantically (as in, for example, agreement). Or it may just affect the syntactic dependencies (i.e., parthood relations) the category enters into.

Take, for example, the syntax of English tense. In the system I adopt, this is analyzed via the category T. T has variable features on it that affect its phonological and semantic interpretations (e.g., the features [Past] and [Present], among others). The actual interpretations of T bearing these features are quite complex, including, on the morphophonological side, allomorphy and, on the conceptual semantics side, polysemy (see, e.g., Jespersen 1924, p. 56). In addition, T has complex logical interactions with other elements of clause structure (as in sequence of tense phenomena). T in English also bears, I'll assume, a feature [uD]. This feature does nothing more than set up a dependency with some element bearing D, as, following the standard notion of feature checking, [uF]s must be in a checking relation with a matching [F]. This means that the kinds of SOs Subjoin applies to look as follows:

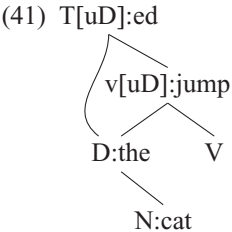
(39) T[Past, uD]; T[Present, uD]

The matching and checking of features is the primary way that the system implements syntactic dependencies. I assume a very simple checking theory where a [uF] needs to be in a 2-part relation with an [F]. This is why the subject DP subjoins to T, because, once it does so, the [uD] on T is in a 2-part relation with [D] on D.¹⁷

(40) Checking

A category bearing [uF] must be in a 2-part relation with a category bearing [F].

Applying this to our earlier example, we update the structure in (34) as follows:



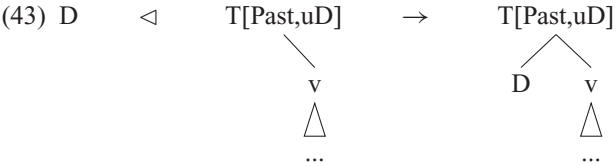
In what follows, I'll usually omit the [uD] features on v and O that effect selection of their 2-parts, so that the structures remain relatively unencumbered by predictable features.

With this discussion under our belt, lets return to the Labeling Question:

(42) The Labeling Question

Is there a way to implement the construction of SOs so that there is no need for a supplemental theory of labeling?

In the system I just sketched, this question has a positive answer. Consider a case where we Subjoin a D to a T that already has v as a 1-part:



There is never an issue about what the label of a new object is, as the SO's features never change. The SO T[Past,uD] remains T[Past,uD] when D is subjoined to it, because that is how subjunction works. There's no need for a theory of labeling. The headedness of a structure emerges directly from the asymmetry of subjunction and the fact that the arguments of Subjoin are SOs.

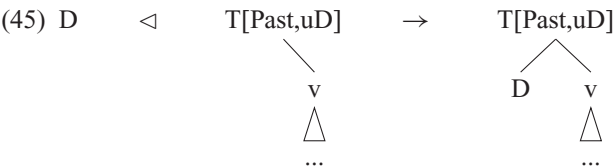
I mentioned above that the structures built in Mereological syntax are telescoped and similar to the structures of Brody's Mirror Theory. There are differences, however, between Mirror Theory and the system developed here. The mereological system builds the telescoped structures derivationally (an approach I originally suggested in Adger 2013), while Mirror Theory licenses the structures representationally. In Mirror Theory, the syntax is effectively filtered by the morphology, while the proposal here is that the morphology is a

spellout of the syntax. A more major difference is that the notions of 1-part and 2-part are transitive, while that is not true for the concepts of complement and specifier that Brody’s Mirror Theory appeals to.

Telescoped structures have another advantage, which is how $\overline{X}$ constituents are treated. $\overline{X}$ constituents are typically argued to be “invisible” to operations (Fukui 1986; Chomsky 1995, p. 242). This is odd in the classical Bare Phrase Structure system, since they are legitimate terms in that approach. If a specifier is Merged to a head-complement structure, as in (44), that head-complement structure is a $\overline{X}$ level constituent, but it is not, in any sense, different from any other constituent, and there’s no obvious reason why it should be invisible to further operations:

(44) $\text{Merge}(\{x,y\}, \{z,w\}) = \{\{x,y\}, \{z,w\}\}$

In the telescoped structure, however, there simply is no such object. Look again at the structure where D is subjoined to T:



In the output of subjunction, there is no substructure corresponding to the equivalent of $\overline{T}$, and therefore no way to target the equivalent of $\overline{T}$ for further subjunction in a way that will not include its 2-part (specifier). More generally, the equivalent of $\overline{X}$ categories in this system just don’t exist, so $\overline{X}$ -Invisibility follows. This is the reason that while we have Possessor Extraction (or Possessor Raising), we don’t have Possessee (that is $\overline{D}$) Extraction or Raising, solving the problem I raised at the end of section 2.4.

3.4 Copies vs Repetitions

We have seen that the set-theoretic system and the mereological system are similar in that they use a single operation (Merge or Subjoin) to build both local and nonlocal syntactic dependencies. There is, however, a major difference between the two systems. As discussed in the previous chapter, the set-theoretic system can create an SO with inscriptionally identical elements in distinct positions in two different ways. The constructed object, however, does not contain information that allows one to determine how it was constructed. Unfortunately, that information is necessary to explain the empirical phenomena. Briefly recapping, consider the following derivation:

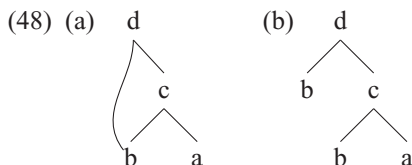
- (46) a. a
 b. b
 c. $c = \{a, b\}$
 d. $d = \{b, \{a, b\}\}$

Here the syntactic object d has two terms that are inscriptionally identical (both have the form b). There are two derivational routes to d : External Merge of b twice, or Internal Merge of b . There is no way to tell which took place by inspecting (46d), unless one adds in that information somehow. But the information is necessary to distinguish, for example, a structure that involves a single object in two positions, or two inscriptionally identical objects in different positions, as in the examples we saw previously.

- (47) a. The window broke ⟨the window⟩
 b. The window broke the window

As discussed in the previous chapter, Collins and Groat (2018) show that this is a real problem for the set-based approach to Merge.

The mereological approach has the immediate advantage that there are indeed two distinct structures that correlate with the two distinct derivations:



These two objects are distinguished because they have distinct part relations. In one, a single object b is a 2-part of both c and d . In the other, there are two b s: one is a 2-part of c but not of d , and the other is a 2-part of d but not of c . They are distinct objects though they are inscriptionally identical, since they bear incompatible part relations to other objects. No indices, occurrences, memory, or special rules are required.

Collins and Groat (2018) provide a number of arguments that multidominance systems aren't a good solution to the copies problem, and the subjunction structures in (48) are similar to multidominance structures. However, I don't think that Collins and Groat's arguments hold for the approach I've developed here.

One argument that Collins and Groat provide is that minimalist systems are set-theoretic, and it's incoherent to think of multidominance in a set-theoretic fashion (though see Gärtner 2022 for a different perspective). A related argument is that the set-based minimalist systems are superior to multidominance systems (or graph/tree theoretic systems in general) as the latter

complicate the definition of SO, requiring the specification of sets of ordered pairs that specify the dominance relations between nodes. Set-based systems eschew such complexity.

However, as I hope I've shown, the mereological system developed here is no more complex (and perhaps less so) than the Merge-based system, and the kind of SOs it creates are actually simpler than sets in some senses. These two arguments, then, while they might apply to classical multidominance systems, do not apply to the mereological approach.

Collins and Groat's third argument is that multidominance requires extra information anyway. They write (p. 4):

Such a move does not provide any new insights into the distinction we desire to make, but only forces us back to diacritics. Imagine the case of two occurrences of Z in a multidominance structure, such that Z is dominated by both A and B (i.e. the graph includes the ordered pairs <A, Z> and <B, Z>). What information could be added here that would tell us whether these two relations pointed to copies or repetitions of Z? Again, it would seem that the only way would be via some diacritic on Z, say Z1 vs. Z2. Of course this is no improvement. (p. 4)

The argument here is that if we look at a multidominance tree as a collection of dominance relations (a set of ordered pairs), we can't tell, just looking at those dominance relations one by one, whether the graph contains one Z or two Zs. To distinguish the Zs, we'd need indices.

To see this, let's take our window example, and assume a simple classical Phrase Structure analysis, without adopting the VP-internal subject hypothesis:

- (49) a. [_{TP} The window [_{VP} broke <the window>]]
 b. [_{TP} The window [_{VP} broke the window]]

In the unaccusative, we have two immediate dominance statements

- (50) <TP, 'the window'>, <VP, 'the window'>

But in the transitive we have exactly the same:

- (51) <TP, 'the window'>, <VP, 'the window'>

Collins and Groat's argument then is that there is nothing to distinguish these.

This argument does not apply to the mereological system, which is in the business of directly constructing objects from other objects in a system where objects can be of the same type but still distinct (like atoms of oxygen). Collins and Groat's argument is that the propositions in the tree logic for a multidominance system can't be distinguished in the two cases (that is that they have the same models). That may well be (although it will, I think, depend on whether the set of

propositions includes negation). However, Subjoin doesn't build propositions. It builds SOs directly.

Let's take the case at hand. The interface systems see, in the unaccusative, a single object D that is part of two other objects, while in the transitive they see a different set up of part relations which entails that the D that is not a 2-part of O is different from the D that is a 2-part of O. Since the system works on objects, not on linguistic symbols, there is no contradiction here. Collins and Groat are right that that is not the case for set-theoretic objects, where there is no notion of qualitative vs numerical identity.

The mereological system, then, doesn't require any new kind of syntactic object (such as a chain, or a "copy pair"), nor any appeal to semantics. The system delivers a formal difference in objects that is reflective of the formal difference in the derivations. We have, then, an positive answer to the Copy Question:

(52) The Copy Question

Is there a way to implement the construction of SOs so that there is no need for a supplemental theory to distinguish copies and repetitions?

Subjunction gives us exactly the kind of distinction in SOs needed to distinguish copies and repetitions.

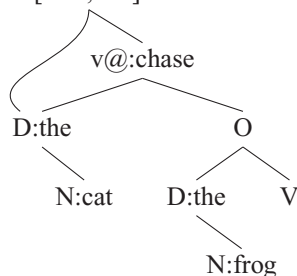
One empirical question that comes to mind in a multiparthood system (or a multidominance one) is how to deal with cases where we see phonological representation of both "copies." Plausible examples of this are phrasal spellouts of intermediate traces (such as the Africans examples discussed by du Plessis 1977), predicate clefts (Landau 2006; Kandybowicz 2008; Bleaman 2021; among many others), and various other constructions. I return to these briefly in the next chapter.

3.5 Linearization

The core data-structure that interfaces with pronunciation in the mereological system is the Extended Projection, that is, a sequence of syntactic categories organised by the 1-part relation. The morphophonological system will then have to compute over material given to it via the morphophonological exponents of these categories. It has to work in this way, as there are no lexical items qua terminal nodes.

Let us look at a transitive sentence like *the cat chased the frog* and see how this might work. I will explain the @ and * annotations directly:

(53) T*[Past, uD]:ed



Up to now, I have simply put orthographical representations of morphemes next to the relevant categories. However, as discussed above, phonological information is not “in” the syntax, which operates only over categories and features. The morphophonology is associated with the relevant categories via the spellout systems. The question is how.

The spellout systems have, as I see it, two jobs. One is to properly account for the linear order of elements in sentences; I’ll call this linearization. The second is to account for how those elements are pronounced; I’ll call this phonologization. My main concern here is with linearization, so I’ll merely gesture at how phonologization might work.

In terms of linearization, languages, as we know, differ in where, in an Extended Projection, the head of that projection, and various other elements, are pronounced. The standard view, since Pollock (1989) developed Emonds’s original proposal (Emonds 1978), is that English main verbs are pronounced somewhere inside the verb phrase (the consensus these days being at the *v* position), French verbs are pronounced higher and are associated with the *T* position, while verbs in Celtic languages are pronounced yet higher. There is somewhat less consensus about the position of verbs in head final languages, though for those adopting the approach argued for in Kayne (1994), these verbs are very low, coming after their objects. The different heights of verbs are usually dealt with via the mechanism of verb movement, with parametrically varying diacritics on verbs or functional heads, or both, determining the height to which the verb moves (e.g., the suggestion of Chomsky 1995 that French *T* has a “strong” feature on it that causes the verb to move to *T*, while English *T* lacks this).

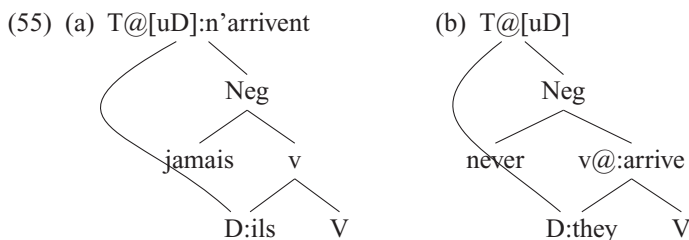
The order of the word-internal elements (e.g., where the verb root comes with respect to prefixes and suffixes) is often assumed to follow from the idea that head movement adjoins a moving head to an affix on one side or the other. In some theories (e.g., Kayne 1994), the head always left-adjoins, and suffixation arises; in others (e.g., Harley 2011), the affix can specify the direction of its linear order with respect to the moved head.

In telescoped systems of Phrase Structure, as noted by Brody (2000), there are no heads, and hence no head movement. Rather than a diacritic on T determining that the verb moves to T, there is a diacritic on T that determines that the Extended Projection (or some fragment of it in some approaches) rooted at V is linearized at T. The question to be answered is what “at T” means.

I will adopt Brody’s diacritic, @, which signifies that the Extended Projection (or some fragment of it) is to be pronounced at the category bearing @. In English, v bears @ while in French T bears @. This means that the verb, plus its tense inflection is pronounced in different places in the two languages accounting for classical contrasts such as (54), where the negative adverb, assumed to be in the specifier of a negative projection, precedes the verb in English, but follows it in French:

- (54) a. They never arrive on time.
 b. Ils n’arrivent jamais à l’heure.
 they NEG-arrive.3PL.PRES never at the-time
 ‘They never arrive on time’

If negation is in the same position in the clausal Extended Projection, somewhere between T and v, the outcome of pronouncing the Extended Projection at T in French but not in English, is that the verb and its tensed inflection will follow negation in English but precede it in French, just as in the more standard theory. The relevant structures are as in (55):



In an VSO language like Scottish Gaelic, some category higher than the linearization site of the subject bears @—I return to this below when I discuss how 2-parts are linearized. If we follow Kayne’s suggestion, that in verb final languages the verb remains low, then, in, for example, Malayalam or Japanese, which are OV, V bears @. The proposal that the verb is linearized low is certainly correct for some head final languages, as argued for extensively by Adger et al. (2009) for Kiowa.

The @ diacritic, then, ensures that the pronunciation of the Extended Projection (or fragment of it), as a coherent phonological unit, is sited at a particular category. In addition, we need a theory of the linearization of the categories that

make up the Extended Projection, that is the categories that will be realised as morphemes within a word.

The order of morphemes in a word in English is homomorphic to 1-parthood in a way that gives suffixation: if x is a 1-part of y , then x precedes y , so the V or N precedes its affixes. The morpheme sequence (word internal linear order) for an English past tense main verb like *chased* is as in (56):

(56) $v@: V \frown O \frown v \frown T[\text{Past}, uD]$

Linearization of morphemes in a word, then, generally follows the order of the Extended Projection (as expected in a Mirror Theoretic type approach). This is distinct from phonologization, the insertion of morphophonological exponents. For (56), the phonologization is as follows (where the phonological structures will of course be more abstract than what I have represented here):

(57) $v@: tʃes \frown 0 \frown 0 \frown d$

The categories O and v in English have no associated morphemes, and I have represented this fact with 0s; V and T do have associated phonology: /tʃes/ and /d/ respectively. The morphophonological system then computes the output of this in a fairly standard way, via whatever mechanism is necessary to give the final output form, for my Scottish English pronunciation, /tʃest/.

This gives us a workable approach to phenomena that are usually dealt with via verb movement in classical Bare Phrase Structure theory: a sequence of categories in a 1-part relation with each other is phonologically realized at the category that bears @ as a sequence of exponents of those categories. This is what Ramchand (2014) calls a Direct Linearization theory, where order is not the result of movement, but rather of instructions by the syntax to the systems of linearization. There is some debate over whether Head Movement feeds semantic interpretation, which would not be expected in Direct Linearization theories, but I think there are fewer and fewer arguments that it does (see Hall 2015 and Thoms et al. 2023). Adopting Direct Linearization for 1-parts, however, still leaves open the question of how to linearize 2-parts. I will develop a Direct Linearization account of this too.

The general principle for English is that the linearization of 2-parts precedes the linearization of the SO they are a 2-part of (equivalent to specifiers preceding heads), and that the linearization of a 1-part follows the linearization of the SO it is a 1-part of (intuitively, heads precede their Extended Projection complements). If this were true universally, this would be the same as Kayne's proposal that specifiers precede their heads, which in turn precede their complements. Let's state this for ease of reference:

(58) Linear Order (English)

- a. Linearization of a 2-part precedes the linearization of the SO it is a 2-part of;
- b. Linearization of a 1-part follows the linearization of the SO it is a 1-part of.

We are now in a position to see how example (53), *the cat chased the frog*, is linearized: in English, *v* bears @, so the linearization of the Extended Projection is “at *v*.” @ instructs the spellout systems to access the lexicon, and to associate a morpheme sequence with *v*.¹⁸ The morpheme sequence is that in (57).

What of the specifiers, *the cat* and *the frog*? The latter is linearized as the 2-part of O, and the former as the 2-part of T. In both, the phonologization of the D category is a single morphological exponent /ðə/. The @ diacritic here is trivial, and simply instructs the lexical access operation to associate this morphophonological exponent with D. Since the linearization of categories precedes the linearization of their 1-parts (58b), /ðə/ will precede both /kat/ and /frɒg/. I will omit such trivial @s in future.

Turning to the relationship between the DPs and the Extended Projection they are 2-parts of, the subject *the cat* is a single SO which is a 2-part of two other SOs (*v* and T). Just as we needed a diacritic to tell us where the (fragments of the) Extended Projection can be linearized, we need a diacritic to tell us where the subject can be linearized: is it as a 2-part of *v*, a 2-part of T, or both. I'll use the * diacritic (often used for “strength”) for this, and assume it has the effect in (59):

(59) *-Linearization:

- a. If an SO *x* is an immediate 2-part of each member of a set of SOs $\Sigma = \{y_1, \dots, y_n\}$, *x* is linearized immediately preceding the linearization of the *y_j* bearing * such that *y_j* is not a part of any other $SO \in \Sigma$ bearing *;
- b. If an SO *x* is an immediate 2-part of each of the members of a set of SOs $\Sigma = \{y_1, \dots, y_n\}$, and no *y_j* bears *, *x* is linearized immediately preceding the linearization of the lowest *y_j*.

This has the effect of linearizing an SO which is a 2-part of multiple SOs in such a way that it immediately precedes the spellout of the highest SO it is a 2-part of that bears *; otherwise *-Linearization linearizes the SO preceding the lowest SO it is a 2-part of.

In our example, English T bears *. The DP *the cat* is a 2-part of the members of the set $\{T^*, v@\}$. This means that DP will be pronounced preceding the linearization of T, given that T bears * and is higher than *v*. The * diacritic here then is roughly analogous to the EPP property of T. Since *the frog* is only a 2-part of one SO (O), it is pronounced preceding that SO's linearization (that being the lowest SO it is a 2-part of).

We can now put all this together: in English the linearization of the 2-part of T precedes the linearization of the category T itself. However, T has no phonology in its own linearization (it doesn't bear @). We derive:

(60) *the cat* $\prec$ T

T's linearization precedes that of v, so we then move to v@ (following (58b)). The 2-part of v@ does not have a linearization, since *-Linearization has already linearized it as the 2-part of T*. This means that, in English at least, there is nothing to linearize. The linearization of v@ follows that of its 2-part (because of (58a)), and in this case it is pronounced (after phonological computation) as /tʃest/. Given that a cycle of phonological computation happens on /tʃest/, I enclose it in square brackets to set it off from the other precedence statements:

(61) /ðə kat/ $\prec$ T $\prec$ [tʃes/∧0∧0∧d]

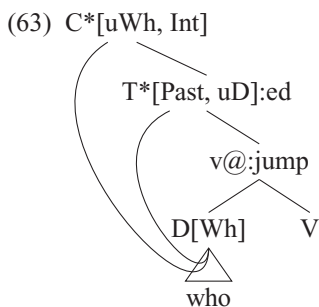
We then move to the linearization of O. The linearization of O's 2-part precedes O's linearization, so *the frog* precedes the linearization of O, which is empty, as is the linearization of V.

(62) /ðə kat/ $\prec$ 0 $\prec$ 0 $\prec$ [tʃes/∧0∧0∧d/] $\prec$ /ðə frɒg/ $\prec$ 0 $\prec$ 0

Putting all this together, the result is that /ðə kat/ precedes /tʃest/ which precedes /ðə frɒg/.

As mentioned above, for a suppletive verb (e.g., *ate* in *the cat ate the frog*), we might say either that the phonology is abstract enough so that the suppletion can be handled via phonological rules (Newell 2008) or that the lexical access operation inserts a suppletive form that “spans” the sequence of categories from T to V (Svenonius 2016). In any event, it is the @ diacritic of English that ensures that the verb is linearized where it is.

Turning to cases where there is overt movement to C of, for example, a wh-expression like *who*, we have (63) as the representation for *who jumped*:

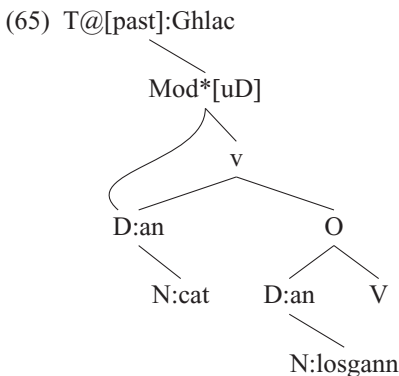


In this structure the wh-expression *who*, which bears a Wh feature ensuring it is interpreted as a wh-quantifier by the semantic systems, subjoins to T to check T's [uD] feature, and also subjoins to C to check C's [uWh] feature. C also bears [Int] ensuring that the clause is interpreted as an interrogative. D[Wh] is then an immediate 2-part of each of the members of the set $\{C*[uWh,Int], T*[uD], v@\}$. *-Linearization ensures that it is linearized as the 2-part of the highest of these, enforcing an analysis where the wh-subject has moved, as seems generally correct (e.g., Agbanyani 2000). We will return in the next chapter to what happens when a wh-expression is the immediate 2-part of SOs in different Extended Projections.

Turning now to variation in basic word order, we can ask how a VSO language like Gaelic works in this system? Consider the translation of *the cat caught the frog* into Gaelic:

- (64) Ghlac an cat an losgann
 catch.PAST the cat the frog
 'The cat caught the frog.'

Adger (forthcoming) argues that the position that the subject moves to in Gaelic is the specifier of a functional category Mod(ality), which hosts future/conditional features, and that the higher category is T, hosting past/non-past features. I'll adopt that perspective here for concreteness. Given this, T in Gaelic bears @, while it is Mod that bears *. The structure then is as follows:



The algorithm works in just the same way as for English. In Gaelic, since T bears @, the linearization of the relevant sequence of categories is at T. In Gaelic, the T[past] morpheme precedes the verb, causing lenition of the verb root *glac*, /gla^hk/, “catch” to *ghlac*, /ɣla^hk/. I’ll assume that this is a morpheme particular realization for the past tense exponent as a prefix (see Bye and Svenonius [2012])

for discussion or prefixes and infixes from this perspective). Aside from this morpheme, the order of affixation is the same as English:

(66) $T@: T[\text{past}] \neg V \neg O \neg v \neg \text{Mod}^*[\text{uD}]$

Each of these categories has a morphophonological exponent as follows:

(67) $T@: /y/ \neg /gla^hk/ \neg 0 \neg 0 \neg 0$

The exponent y causes lenition of the following consonant and deletes (unless that lenition creates a vowel initial structure, see Adger 2010). In the case at hand, the morphophonology computes the phonology of *ghlac* as $/y la^hk/$.¹⁹

Since the linearization of $T@$ precedes that of the constituent Mod^* , and the 2-part of Mod^* is part of Mod^* and precedes the spellout of the category Mod^* , it follows that the verb will precede the subject. The rest of the derivation is trivial: the zero spellout of Mod^* precedes that of v , which precedes that of O , but the linearization of the object *an losgann*, “the frog” precedes the spellout of O , which all entails that the subject precedes the object, deriving the VSO order:

- (68) a. *Ghlac* *an cat an losgann*
 catch.PAST the cat the frog
 “The cat caught the frog.”
 b. $y la^hk \neg \text{an } ka^ht \neg \text{an } l\text{oskan}$

The linearization of an OV structure, assuming a Kaynian analysis, will work similarly, although the $@$ will be placed on V (or perhaps O), so that the verbal Extended Projection will be linearized after the object (see Adger et al. 2009 for examples from Kiowa).

Of course, much more needs to be said here about how to model the various restrictions on the relationship between hierarchy and order (including the restrictions on that relationship discussed by Cinque 2005, Cinque 2023, among many others). I have assumed in the discussion above that *-Linearization linearizes 2-parts immediately before the linearization of the SO they are a 2-part of, but one could, or course, place directionality on the $*$, so that some languages linearize 2-parts leftwards, and others linearize them rightwards (Abels and Neeleman 2012). I think this is probably the right analysis of the mobile placement of adverbials to the left or right of their modifiee in English, and is perhaps the correct analysis of adverbials that occur in scope order to the right, which are sometimes handled by roll-up movement (Cinque 1999). One could also do the same to the $@$ diacritic, giving the effect of rightwards head movement, as has been argued for some head-final languages (Han et al. 2007). See Svenonius (2024b) for relevant commentary and difficult cases.

3.6 Summary

This chapter has laid out the bones of a system of grammar that builds mereological rather than set-theoretic SOs. I have adopted a hylomorphic, pluralistic version of mereology, and proposed a single operation, Subjoin, that generates more complex objects by making one object part of another, where parthood is irreflexive, transitive, and asymmetric. Parthood comes in two dimensions that are roughly analogous to the Extended Projection complement relation, and to the specifier relation. As is familiar from the standard theory, I've adopted a constraint on subjoin (Mereological Extension) that enforces a certain kind of cyclicity. Also similar to the Bare Phrase Structure system, there is an "internal" version of Subjoin that parallels Internal Merge. The structures built by Subjoin are telescoped, like the structures in Brody's Mirror Theory, although the way the system works overall is quite different to Brody's system. One area, however, where the two are similar is in linearization via appeal to diacritics that do the work of determining how high in an Extended Projection an element is pronounced. I extended this Direct Linearization idea to the issue of which position a multiply subjoined 2-part is pronounced in.

The mereological system is, I've argued, no more complex than the standard theory but has some advantages over it. Subjoin is an asymmetric operation unlike Merge, and that seems closer to what is required in syntax. Mereological SOs are also of one "type," while there are two types of SO in the set-theoretic system (atomic units and sets) and possibly three if the notion of "term" also has to be defined (possibly more if we need chains, or Form Copy outputs). More importantly, the mereological system does not require a separate theory of labeling, and nor is it subject to the copies/repetitions problem that besets the standard Merge-based theory. Mereological ideas are, I'd argue, more suited to modelling syntactic objects than set-theoretic ones are.

4 Angles

4.1 Locality

4.1.1 Two Types of Locality

Some core syntactic relations are highly local, a fact recognized in all generative grammatical theories: in classical Phrase Structure theory and its descendants (Gazdar et al. 1985; Müller et al. 2021), rules build larger constituents out of contiguous smaller constituents, licensing local trees. $\bar{X}$ -theory inherited this property of locality from Phrase Structure grammar, and its two primary relations are local: the sisterhood relationship between a head and its complement, and the mother daughter relationship between a phrase and its specifier. In Categorical Grammars, categories combine with other categories under a condition of string contiguity (Steedman 2023) while, in Bare Phrase Structure-based theories (Chomsky 1994 et seq.), the output of Merge is always a set with two members in a maximally local relationship.

From a slightly expanded perspective on locality, a whole range of syntactic diatheses are obligatorily internal to finite clauses or noun-phrases—double object constructions, passives, raising, and so on. For example, while a passive may apply within a finite clause, it cannot apply across one (putting aside constructions such as hyperraising, Zyman 2023):

- (1) a. Anson believes that Merkel was fed (Merkel).
- b. *Merkel is believed (by Anson) that was fed.

This notion of locality is not contiguity, but rather what we might call domain locality: an operation can apply freely within a domain (where a domain is just a labeled constituent) but not across it. Domain locality is captured by the classical notion of a cyclic category in transformational syntactic theories: transformations, like passive, apply within the constituent labeled by a cyclic category. Once the transformation has operated in that domain, it may reapply within a higher cyclic node, but not in such a way that the dependency crosses

the domain (with exceptions I will return to). If the cyclic categories are the finite sentence and the noun phrase, this models empirical data like (1). The locality here is not strict contiguity, but rather rule application within a domain.

As is well known, there are also apparently nonlocal relationships that extend beyond the finite clause or the noun phrase: constituent questions, relativization, topicalization, comparative formation and various other syntactic relationships can all involve dependencies that cross finite clauses, or noun phrases, and that are, in principle anyway, unbounded (Chomsky 1977).

However, transformational theory has, over the decades, argued that these apparently unbounded syntactic relationships are actually composed of connected domain-local relationships. The transformational rules are still bounded by cyclic nodes, but the conditions that enforce domain locality have certain exceptions. In early versions of the theory (e.g., Chomsky 1973), a condition on the application of transformations (Subjacency) blocked rules applying across two cyclic nodes. However, if there was a “stop off point” (sometimes called an “escape hatch”) for the dependency after it had crossed one domain, the rule could apply successively, accessing the contents of the “stop-off point.” This gives a mechanism to build a long-distance dependency out of shorter ones, where the shorter dependencies only ever cross one domain. Schematically, Subjacency rules out a derivation like (2a), but if the transformational rule applies in such a way that its output crosses one domain, it may reapply, accessing that output to cross the next, as in (2b):

- (2) a. *X ... [Domain ... [Domain ... ⟨X⟩ ...
 b. X ... [Domain ... X ... [Domain ... ⟨X⟩ ...

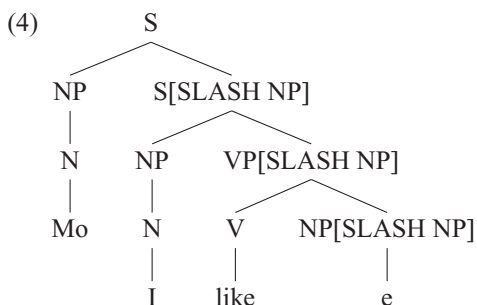
In later approaches (e.g., Chomsky 2008), the domain (renamed a “phase”) is opaque to operations except for its “edge”:

- (3) a. *X ... [Phase ... ⟨X⟩ ...
 b. X ... [PhaseEdge X [Phase ... ⟨X⟩ ...

The accessibility of the “edge” of the phase is ensured by the Phase Impenetrability Condition (see below for more discussion), which plays the same role as Subjacency did in earlier theories in terms of enforcing cyclicity. In both earlier and later versions of the theory, long-distance dependencies are built up out of shorter, domain-local, dependencies.

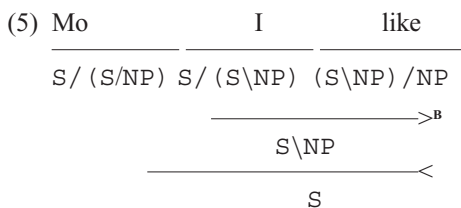
Assuming this idea, that long-distance dependencies are built out of shorter ones, there are two quite distinct approaches to just how short the dependencies can be; that is, how big the locality domains can be. One view is that dependencies are actually always within phrasal projections (or even that they always relate mothers to daughters in a tree). This means that every constituent

projected by a head is a locality domain (every nonlexical node is cyclic, or phasal). This is how Generalized Phrase Structure Grammar and Head-Driven Phrase Structure Grammar work. In these approaches, long-distance dependencies involve a special category-valued feature (SLASH), which is passed up through each category dominating the foot of the dependency until it reaches the top of the dependency, as in (4):¹



In this (rather simplified) representation for the topicalization *Mo, I like*, the verb *like* selects a noun phrase, and if that noun phrase is a special (phonologically null) category, it causes the feature [SLASH NP] to be instantiated on the VP (alternatively, the selecting verb can be directly specified as [SLASH NP], in which case it no longer selects its object as a syntactic complement). The feature then percolates up the tree, instantiating on every node, and a special rule uses it to introduce the topic *Mo*. Each category intervening between the foot and the top of the dependency participates in the dependency by virtue of bearing SLASH.

Similarly, in Categorical Grammars, long-distance dependencies are built up out of local selectional dependencies (see, e.g., the papers in Oehrle et al. 1988 and also Neeleman and Van De Koot 2010): each category in a selectional relationship between the foot and the top of the dependency is connected to the next by a special rule (Functional Composition).



Here the transitive verb *like* has a category that seeks an NP to its right (the object) and, if it gets one, the result will be a predicate category (i.e., a category that needs a subject to complete it). There is no NP to the right of *like* in (5);

Abels (2012b) calls theories like those just discussed *Uniform Path* theories, as every category (or projection) between the foot and top of the dependency is obligatorily and uniformly affected in some way by the dependency (they each bear a SLASH feature, are involved in Functional Composition, or are Internally Merged to).

Most approaches within transformational generative grammar are not Uniform Path theories. They are what Abels calls *Punctuated Path* theories. In these, as mentioned above, not all categories are cyclic.

The classical analysis of a relative clause like that in (8), for example, is that long-distance relativization involves the head of the relative clause (or a null element associated with that head) being first extracted to a position that is local *within* the embedded finite clause boundary CP₂. The intermediate categories (or strings of words) are not affected by this operation—they effectively act like a variable in the structure, as noted by Ross (1967). In Bare Phrase Structure, this means that there are copies not only in the theta-position and in the final surface position but also at the “edge” of certain intermediate phrases ((8) assumes that CP is the relevant cyclic domain):

- (8) The frog [CP₁ that [TP₁ I [_{VP1} thought [CP₂ ⟨the frog⟩ that [TP₂ the cat [_{VP2} caught ⟨the frog⟩]]] croaked.

The extraction that connects the final position of the head of the relative clause to its trace in the specifier of CP₂ of course crosses the finite clause boundary CP₂, but it does so from a position that is intuitively very local to that boundary. The path that the dependency creates is “punctuated”: there are “stop-off points” but not at every category (for example, TP is not assumed to be a stop-off point, although in certain approaches vP is).

Abels points out that it is actually quite difficult to distinguish empirically between Uniform Path and Punctuated Path theories. To show that a movement path is punctuated requires one to show that not all categories between the foot and top of the dependency participate in that dependency. But determining whether a category participates in a dependency or not is nontrivial. It is perfectly possible for a category to participate in a dependency but for it to be difficult or impossible to determine empirically that it does so (that is, that there is a copy at its edge, or it bears a feature, or has the right complex category to compose).

We often see overt morphological evidence of a category participating in a dependency that starts below and terminates above it. For example, C categories often change form when they intervene in *wh*- or relative dependencies, as in the Gaelic examples in (9):

- (9) a. Thuir̥t Daibhidh gum buail Calum an cat.
 say.PAST David THAT hit.FUT Calum the cat
 ‘David said Calum will hit the cat.’
- b. An cat a thuir̥t Daibhidh a bhuaileas Calum.
 the cat REL say.PAST David REL hit.FUT.REL Calum.
 ‘The cat that David said Calum will hit.’
- c. *An cat a thuir̥t Daibhidh gum buail Calum.
 the cat REL say.PAST David THAT hit.FUT Calum
 ‘The cat that David said Calum will hit.’

(9b–c) show that the complementizer of the clause selected by the matrix verb must change its form if a relative clause dependency crosses it. The usual declarative complementizer *gu(n)*, “that” in (9a) is obligatorily replaced by the particle *a* in (9b). However, we also see that the form of the finite verb changes (it takes on what is traditionally known as its relative form, partially signaled by the *-eas* suffix in (9b)). Does this mean that T also participates in the relative dependency, or is the form of the verb an allomorph that appears after the relative particle?

If we look at a more complex case, we see that it is, in fact, only the verb adjacent to *a* that changes form. Lower elements do not:

- (10) An cat a thuir̥t Daibhidh a bhios Calum a’ bualadh
 the cat REL say.PAST David REL be.FUT.REL Calum PROG hit.VN
 ‘The cat that David said Calum will be hitting.’

Here the auxiliary adjacent to the relative particle *a* appears in the relative form *bhios* and we might think that this suggests that T, and perhaps therefore all the intervening categories, participate in the dependency (i.e., that they bear a relevant feature, or must be Merged to). However, if we look lower in the clause, the aspectual element *a’* is in the same form as it would be if there were no relative dependency (this is, in fact, true of all other categories except T). We can’t know whether ASP does actually participate in the dependency, but simply doesn’t show any morphological marking, or whether the absence of morphological marking is because it doesn’t participate in the dependency. Given that the relative form of the verb might just be allomorphy, the only real conclusion we can make is that C, at least, participates in the dependency, though we have no evidence that the other categories don’t. Morphology is a restricted and potentially unreliable guide to whether dependency paths are punctuated or uniform.

Abels and Bentzen (2009) provide an interesting argument from Norwegian, not on the basis of morphology, but rather on the basis of reconstruction, that paths are in fact punctuated. They show that moved quantifiers in Norwegian can

reconstruct below an adverb like *sannsynligvis*, “probably” but above a modal verb like *må*, “must”:

- (11) ... at noen gutter sannsynligvis må ha dratt til Roma.
 ... that some boys probably t_2 must have t_1 gone to Rome
 ‘... that some boys probably must have gone to Rome.’

This sentence has a reading that can be paraphrased *it is probable that some boys are such that they must have gone to Rome*. Even though the quantified DP *noen gutter*, “some boys” appears to the surface left of, and hence hierarchically above, *sannsynligvis*, “probably,” it reconstructs for interpretation to a point hierarchically below it. Abels and Bentzen argue that this means that the quantified DP, whose Theta-position is inside the verb phrase, has “stopped off” at an intermediate position below *sannsynligvis* and above *må*, “must,” and it is this stopping off that allows the reconstruction needed for the correct meaning to be derived.

This doesn’t distinguish between punctuated and uniform paths yet. However, Abels and Bentzen note that if the phrase *mot sin vilje* “against his own will” is added, the reflexive anaphor *sin* requires a higher binder, blocking reconstruction below it.

- (12) .. at noen gutter sannsynligvis mot sin vilje må ha
 ... that some boys probably t_3 against REFL will t_2 must have t_1
 dratt til Roma.
 gone to Rome
 ‘... that some boys probably must have gone to Rome against their will.’

In (12), the quantified subject cannot scope below *mot sin vilje*, into positions t_2 or t_1 , as it does in (11). This is expected, since it needs to bind the anaphor. However, in contrast to (11), it also cannot scope directly below the adverb *sannsynligvis*, “probably.” This suggests that while positions t_2 or t_1 are potential reconstruction positions, t_3 is not. But if t_3 is not a potential reconstruction position, that means that whatever constituent it is the specifier of does not obligatorily participate in the dependency. In the Merge system, we would say that that specifier was skipped. The conclusion is that the path is not uniform.

Of course this kind of argument is not available for every category in a clause structure, but it does suggest that the mechanism that creates long-distance dependencies does not do so via uniform paths. In other words, if we model dependencies via an operation like Internal Merge, Abels and Bentzen’s argument suggests that Internal Merge of some object α does not target every category between the base position of α and its final position.

In what follows I'll take this argument to be correct: punctuated path type theories are on the right lines. Assuming this, long-distance dependencies are built up out of shorter dependencies, and certain categories require a "stop off." Other categories may allow, but not require, such a stop off (so the movement is successive, but not cyclic), and yet others may disallow a stop off (so movement across them is not successive). The task will be to construct a theory that has punctuated paths as a consequence, and moreover, tells us where the punctuated categories are.

4.1.2 Four Questions for Locality Theory

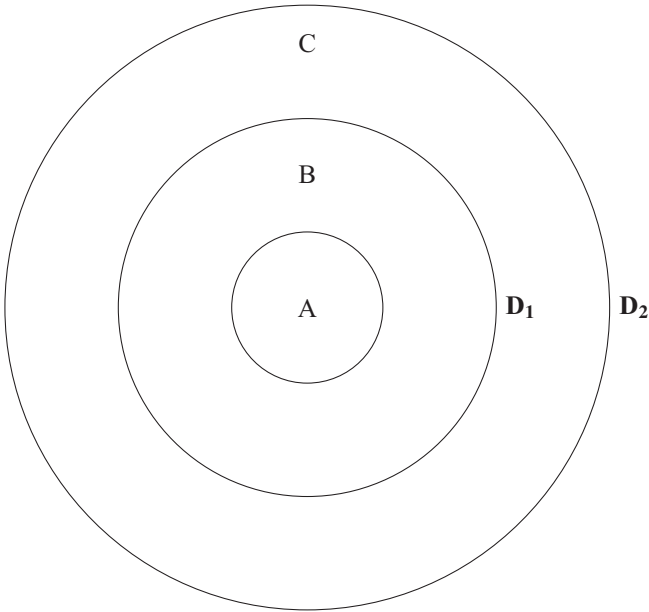
Given this discussion, the following question arises: What kind of theory ensures that successive cyclicity is both (1) available to the syntactic system and (2) forced for the relevant cases?

To answer this question, it's worth thinking about what we might like a theory of locality to do. We would like it first to answer the question of why there are locality domains (i.e., constituents labeled by cyclic categories) at all. Secondly, we would like it to answer the question of why, if there are locality domains, it is not the case that all operations are internal to them—that is, why are syntactic operations not always confined to (for example) finite clauses? Why are there "escape hatches"? Let's call these two questions the Etiology and Exceptionality questions for easy reference below.

- (13) a. Etiology of Locality: Why are there locality domains at all?
- b. Exceptionality of Locality: Why do locality domains have escape hatches?

The way that human language actually works looks like an odd mishmash. One might be tempted to propose that perhaps the reason that locality domains can sometimes be obviated is for communicative purposes, so that the system is able to express wide scope questions or long-distance relativizations. That, however, seems highly implausible: there are, of course, other ways to express long-distance relativization without movement (using, for example, bound pronouns, or, in many cases, prolepsis), and further, these mechanisms are available for use independently in other circumstances. Why not just co-opt them? Such a system would be simpler.

To see the issue more clearly, let's think about domains of rule application in a more general way, abstracting away from particular syntactic assumptions. Consider a structure where there are two domains of rule application, D_1 and D_2 where D_1 is embedded inside D_2 . D_1 and D_2 are the equivalent of cyclic domains.



Inside D_1 itself, there is further structure, though that structure does not itself constitute a domain. Outside that structure we have an object B, and inside it we have an object A. Outside D_1 , but inside D_2 , we have an object C. The question is how we might relate the various objects by possible syntactic rules.

One type of relation (let's call it type R1) relates A and B. That relation holds inside domain D_1 , and, given it takes place inside a domain, it is legitimate. R1s are the local (cyclic) relations/rules/operations. Already the question arises as to why all operations are not like this: Why are there domains like D_1 and D_2 at all? A simpler system would omit them.

However, assuming that there are domains, consider a type of relation (R2) that relates A to C. That takes place across the domain D_1 . If there is no constraint on the locality of relations, or on the domain boundedness of rules, then R2 relations should be fine. However, they are not, by hypothesis, so we need some constraint on rule-application or on the span of relationships to ensure this. Here's a very simple version of this:

- (14) Simplest Locality
 - a. Domains are defined as $P(D)$
 - b. Relations cannot hold across a domain

$P(D)$ here is simply a set of propositions which define domains, perhaps something as simple as a list of cyclic categories, perhaps something more complex.

Simplest Locality, which is a very strict version of locality, will make R2s impossible. In such a system, there can be no domain spanning relations.

If we think of the relations as being rules, then we would have a system where each set of rules applies within a domain, and none of them apply to elements of a larger domain. Once the rules have applied in the smaller domain, a new set of rules (perhaps the same ones) applies to the larger domain that includes the smaller one. This is the classical conception of a cycle of rules: the rules exhaustively apply within one domain before they can apply at the next. When augmented with locality, the further restriction is that (most of) the output of the first set of rules is unavailable to the second. Using our example, in a system constrained by Simplest Locality, D_1 is opaque after the cycle of rules has applied: no rule can affect what is part of D_1 once all the rules that apply in D_1 have been exhausted. Clearly that kind of locality is too strict, as it rules out all cross-domain relations.

Now consider a rule that relates a special prominent subpart of D_1 , that part where B is, to C. We might think of a situation where a type of relation (call it R3) is possible from that prominent subpart of D_1 , but not from other parts of D_1 . This, of course, is the kind of locality usually assumed in syntax, where the subpart of D_1 that contains B is the “escape hatch,” “phase edge,” and so on. Systems that allow R3 have a different kind of locality imposed on them from that in (14), something more like (15):

- (15) a. Domains are defined as $P(D)$
- b. Domains have an exception defined as $E(D)$
- c. Relations cannot hold across domains, unless ...
- d. $E(D)$ is an argument of the relation

$E(D)$ is a function that takes in a domain and returns its exceptions. Usually we think of $E(D)$ as some subdomain, but the way I’ve phrased it here highlights the stipulative nature of the system. There needs to be a principled reason why $E(D)$ is what it is, or else the theory really isn’t much of a theory.

(15) is essentially a version of the Phase Impenetrability Condition (Chomsky 2000 and following) and is often combined with the following assumptions:

- (16) a. the domains defined by $P(D)$ are vP and CP and;
- b. their exceptions are given by some function that applies to a domain and returns its head and any specifiers (the exceptions in this approach are the “edge” of the domain).

The Phase Impenetrability Condition itself states that the complements of the heads v and C (VP and TP) are opaque for a dependency that spans vP or CP ; only v and C and their specifiers can be reached by dependencies from outside

vP and CP. Rephrasing Chomsky's somewhat opaque formulation, we can state it as:

- (17) Phase Impenetrability Condition (PIC): The complement of a phase head (v or C) is not accessible to operations which span the maximal projection of the phase head (vP or CP); only the phase head itself and its specifiers are accessible to such operations. (cf. Chomsky 2000, p. 108).

The approach to locality embedded in (17) is fairly complex, and it's not at all obvious why language would have (17) as opposed to the far simpler condition in (14) (or indeed neither!).²

This system doesn't respond in a satisfactory way to the Etiology or Exceptionality questions I raised above. The answer, in classical Phase Theory, to Etiology of Locality is that third-factor effects on memory reduction in computations are responsible (e.g., Chomsky 2005). That, however, is not an answer given by the theory of locality or of phrase structure. Some independent theory provides it. Perhaps that's correct, but given that the independent theory of memory reduction appealed to here is not particularly explicit, it's somewhat unsatisfying.

This system also does not answer the Exceptionality of Locality question. It states that locality domains can be subverted and tells us when that is possible, but it doesn't say why the theory allows exceptions in the first place. It also doesn't say why they are what they are.

The classical system then also raises a third and a fourth question beyond Etiology and Exceptionality: Why are the locality domains defined as what they are and why are the exceptions defined as what they are? Let's also name these for reference:

- (18) a. Domain Content: Why are the locality domains what they are?
b. Exceptions Content: Why are the exceptions to locality what they are?

The answer to the Domain Content question is sometimes assumed to be connected to propositionality, though other suggestions have been made. Chomsky (2008) suggests that phases are propositional, and that limitations on memory in the computational procedure ensure that a phase is transferred out of the current memory buffer once the phase is complete. Abels (2003) and Boeckx and Grohmann (2007) both note an issue with the notion of propositionality as an explanation of what domains are: what is transferred in Chomsky's theory (the complement of the phase head) is not actually what is propositional. There is a mismatch between phases and propositions, unexplained by the theory.

An alternative motivation for the answers given to the Domain Content question is that certain categories delimit cycles for the rules of the phonological and

semantic component, and these categories are the phase heads (on phonology, see especially Marvin 2002; Newell 2008; Newell and Piggott 2014). However, these approaches simply accept the syntactic identification of phase heads, and so do not provide a motivation for the identity of phases.

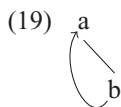
The answer to the Exception Content question is also not given by the theory. Why, for example, does the theory not leave accessible the highest positions where elements check their features? Or why isn't the highest A-position relevant? Or just the head? Or just the specifiers? Or the closest specifier to the head? Or the most distant one? The answer is that such proposals would be (arguably) empirically incorrect, but only stipulation makes this so. Ideally we would like the theory to entail the identity of the domains and their exceptions.

What I propose to do in the next section is show how the mereological system developed in the last chapter provides motivated answers to the four questions I have raised about locality: Etiology, Exceptionality, Domain Content, and Exception Content. I will develop a theory of locality that answers these questions and will show that it also has empirical advantages.

4.2 Angular Locality

I will tackle the development of a theory of locality in Mereological syntax by first looking at the nature of the relationships that hold between minimal numbers of SOs in a very local structure. With this in place, I will then suggest a constraint that ensures that these minimal relationships are maintained when the structures are nonlocal. The theory of locality, then, is a strengthening of the basic relations in structure.

First consider two objects already in a structure, where one is a 1-part of the other. Can we internally subjoin *b* to *a* a second time as shown in (19)? (I mentioned this question in the previous chapter.)

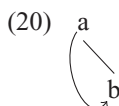


So far in the development of the theory, the restrictions on the nature of derivations are twofold: (i) Dimensionality means that if something already has an *n*-part, further subjunction to it gives an *n*+1-part (where the highest *n* is 2). In the case of (19), Dimensionality does not stop *b* subjoining to *a* as a 2-part (though if *a* already had a 2-part, subjunction would then be impossible); (ii) I defined Mereological Extension so that it only applies to external subjunction. If we were to extend it also to internal subjunction, then in (19), subjunction of *b* to *a* would clearly satisfy it: *a* is not part of *a* or of anything else. (19),

then, would be well formed, under Dimensionality and this internal version of Mereological Extension.

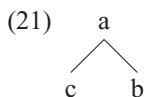
Subjunction of b to a in (19) is, interestingly, equivalent to a derivation in Bare Phrase Structure that violates the kind of Antilocality proposed by Abels (2003): in this analogy, a is a functional head, b is its complement, and b is raising to a 's specifier. Abels argues that empirically there is good reason to block such derivations. I'll show how this result emerges from the theory of locality below.

Subjunction of a to b ("downward") is legitimate as far as Dimensionality goes, since b is a simple object (has no parts); it would, however, violate a version of Mereological Extension extended to internal subjunction, as b is part of something (a):



However, this subjunction is also, in fact, independently ruled out by Irrelexivity and Transitivity, since it will make a a 1-part of b , and since b is a 1-part of a , and the 1-part relation is Transitive, a will then be 1-part of a , which is impossible by Irreflexivity. There is a redundancy here between the basic axioms and a version of Mereological Extension that applies to internal subjunction, suggesting that the scope of Mereological Extension might best be confined to external subjunction.

Now consider a case where we have three objects in structure to see whether further subjunction within this structure is possible:



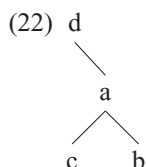
There are various logical possibilities, but they are all ruled out already by Dimensionality, Mereological Extension, or by the fundamental axioms of the mereology.

Subjunction of b to a is ruled out because a already has c as a 2-part, so cannot, by Dimensionality, have any further subjunction, while subjunction of a to b is ruled out by Irreflexivity and Transitivity as above.

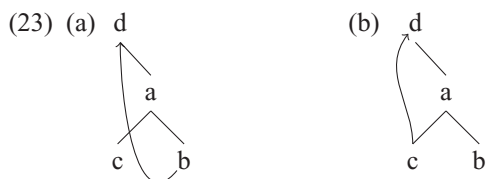
Similarly, subjunction of c to a is ruled out by Dimensionality, as a already has a 2-part (which is, in fact, the object c), and subjunction of a to c is ruled out by Irreflexivity and Transitivity, this time of the 2-part relation. What about subjunction of b to c or c to b ? These would both be ruled out by Mereological Extension, if we extended that constraint to internal subjunction. I will return

to this case of (putative) “sideward” subjunction below and show that it will be blocked by the theory of Locality I will develop. Given that second redundancy, the case for Mereological Extension as a condition on internal subjunction is weakened.

Up to now, I’ve discussed some minimal kinds of internal subjunction, the most local: subjunction of an object to the next object “up” within an Extended Projection. If we extend (21) by making it a 1-part of an object *d*, then more possibilities open up:



The issue of subjunction of *a* to *d* has already been discussed (it violates neither Dimensionality nor Extension, but I will return to it below and show how the theory of Locality rules it out). Neither the subjunction of *b* to *d*, or of *c* to *d* is problematic. Both would add a 2-part to *d*, so the following structures are legitimate:

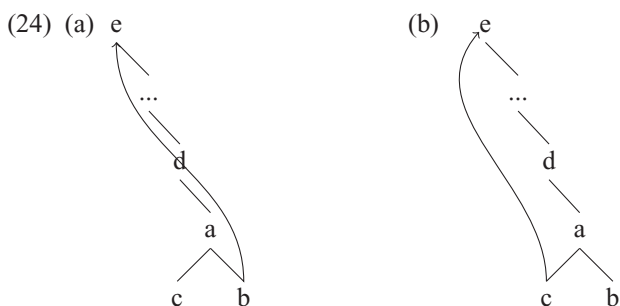


The leftmost structure corresponds to “movement” of a subpart of an Extended Projection (say a VP) to a higher specifier in that same Extended Projection (say the specifier of a topic functional projection)—a roll-up movement of the sort familiar from the work of Kayne (1994) and others. The rightmost structure corresponds to a specifier of some element in an Extended Projection (say the specifier of vP) “moving” to the specifier of a higher element in that same projection (say the specifier of TP). We discussed such examples of internal subjunction in the previous chapter.

Downward versions of the internal subjunctions in (23) would be ruled out by Mereological Extension extended to internal subjunction. A downward version of (23a) will also be ruled out by Irreflexivity and Transitivity, since *d* would become part of itself, but that is not the case for a downward version of (23b), since *c* is a 2-part of *a* and *a* is a 1-part of *d*, and Transitivity does not cross dimensions. Downward (23b) is a situation where the extended version of Mereological Extension would do some work in the system in constraining

internal subjunction. However, I will argue below that this is another situation that will be ruled out by the theory of Locality. I conclude that extending Mereological Extension to internal subjunction is not warranted, and that it can be restricted to external subjunction, if, indeed, it is needed there.

In this minimal extension of subjunction from “within” a category to the next category up, either a 1-part of an object in a very small Extended Projection (sequence of 1-parts) subjoins to that object, or a 2-part of an object within the very small Extended Projection subjoins to that object. Given that 1-parthood is transitive (within the first dimension), exactly the same relations are preserved if we allow subjunction to a higher category in that Extended Projection:



In both cases a maximum of one dimension is crossed: an object is a 1-part of some other object, and that 1-part dimension is maintained until the target of subjunction is reached, or an object is a 2-part of some other object, and that object is itself a 1-part of the target of subjunction.

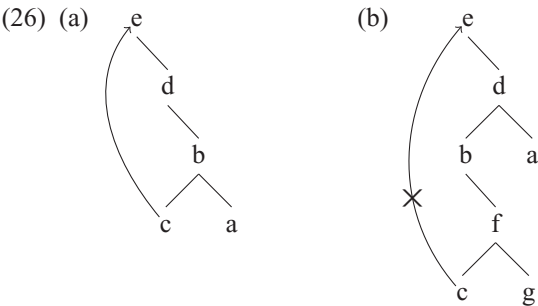
Let's make this the cornerstone of our theory of locality, and say that the most minimal kind of subjunction is also the most maximal: subjunction never crosses more than one dimension. We can do this by imposing a principle of locality on “internal” subjunction as follows:

(25) Angular Locality

if γ is a part, then γ can subjoin to β only if there is an α s.t. γ is a n -part of α and α is a 1-part of β .

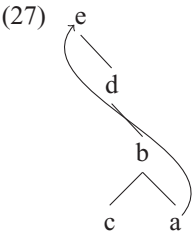
Here n -ranges over the two possible dimensions, so this principle allows the two kinds of subjunction in (24). In the (a) case of (24), b is a 1-part of a and since a is a 1-part of e , b can subjoin to e . In the (b) case, c is a 2-part of a and since a is a 1-part of e , c can subjoin to e . The geometry of locality is maintained, but the transitivity of the part relation allows an extension of its range.

I've called this principle “Angular Locality” precisely because its effect is to impose a kind of geometry on nonlocal subjunction: such subjunction is possible only if there is a single “angle” in the structure. Consider (26):



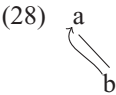
In the leftmost structure, because 1-parthood is transitive, Angular Locality allows subjunction of *c* to *e*, since *c* is a 2-part of *b*, and *b* is a 1-part of *e* (by Transitivity of its being a 1-part of *d*, which is a 1-part of *e*). In the rightmost structure, in contrast, *c* is a 2-part of *f*, but *f* is not a 1-part of *e*, since *f* is a 1-part of *b*, but *b* is not a 1-part of anything that is a 1-part of *e*. Recall that Transitivity does not cross dimensions. That means that *c* is neither a 1-part of *b*, nor a 2 part of *b* (*c* is *disconnected* from *e* in such circumstances). The same is true for both *f* and *g*: since *b* is not a 1-part of *e*, subjunction from inside *b* to *e* is impossible. The object *b* then acts as a locality domain: in general the internal parts of a 2-part of an object cannot be subjoined outside of that object (with one crucial exception that I return to in the next section).

As noted above, the “roll-up” type of movement is allowed by Angular Locality.



Since *a* is a 1-part of *b*, and *b* by Transitivity is a 1-part of *e*, subjunction of *a* to *e* as a 2-part is allowed: visually the angle is “flat.”³

Let’s now return to the case where we have a second subjunction of an object that is a 1-part of another object. Consider again a two-object structure where *b* is a 1-part of *a*, and we attempt to subjoin *b* to *a*:



I repeat here the definition of Angular Locality for ease of reference:

- (29) if γ is a part, then γ can subjoin to β only if there is an α s.t. γ is a n-part of α and α is a 1-part of β .

If we were to subjoin b to a in (28), this time as a 2-part, then Angular Locality would require there to be some object (α in the definition) that b is a part of and that object must be a 1-part of a . But that object could only be a , and a is not a 1-part of a , since parthood is Irreflexive. So the operation is illegitimate and superlocal subjunction of this sort is ruled out. As mentioned above, such subjunctions are the counterpart in this system of Antilocality movements that move a complement of a head to its specifier.⁴ Such Antilocal operations are usually assumed to be ruled out either because of Last Resort constraints on feature relations (Abels 2012a) or because of a constraint on interpreting copies in too local a relationship (Grohmann 2000). In the mereological system, they are ruled out because of the Irreflexivity of the part relation interacting with the theory of locality.

This result is, I think, a theoretical advantage of this system, since the principle that imposes locality in general (and, we will see, derives which categories will be locality domains) interacts with the properties of the core syntactic relation to derive both the possibility of nonlocal dependencies within an Extended Projection, and the impossibility of superlocal dependencies.

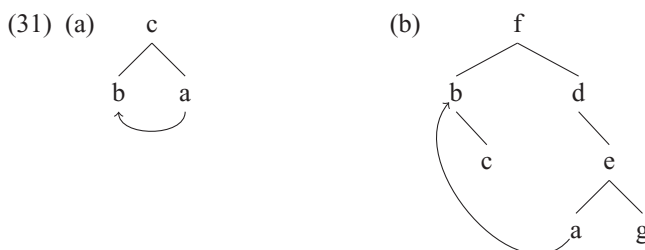
Angular Locality will also rule out the subjunction equivalents of parallel merge derivations (Citko 2005 et seq.):



In (30), for a to Subjoin to d , b , which a is a 2-part of, must be a 1-part of d , and it's not. The system then rules out parallel merge derivations, even though, as discussed in the previous chapter, it is essentially a multidominance system. The multidominance of the system is restricted to single rooted structures (exactly the opposite, in fact, of the kind of multidominance system developed in Johnson 2012 et seq.).

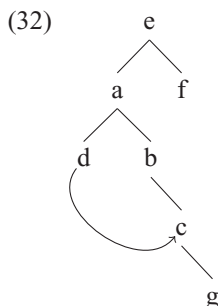
Subjunction to an object that itself is a 2-part (i.e., subjunction to a specifier) is also ruled out (this would be the equivalent of the sideward movement operation of Nunes 1995). In the following examples, if a is to subjoin to b (or vice versa), a must be part of some object that is a 1-part of b , but in the leftmost example the only other object is c , which is not a 1-part of b . In the righthand example a is a 1-part of e , but e is not a 1-part of b , so Angular Locality is

violated. In general, subjunction to a 2-part (specifier) is ruled out by Angular Locality:



This is another situation where Angular Locality makes Mereological Extension redundant for internal subjunction.

Carrying on this theme, Angular Locality will also rule out lowering subjunctions in general. Take a case where *d* is a 2-part of some object *a*, and we subjoin *d* as a 2-part to some lower object *c*:



Here *d* is an *n*-part of *a*, but *a* is not a 1-part of *c*, so *d* cannot subjoin to *c*. This is the same logic as we saw in the case of sideward subjunction extended to the lowering case.

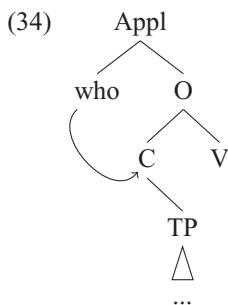
To make this more concrete, take the analysis of an example like the following:

(33) *We asked ⟨who⟩ [who [Anson wrote the book]]

Here, the *wh*-expression has moved (subjoined) to the lower spec CP. Since that CP now has a *wh*-element in its specifier, it can be interpreted as a question. All the other various thematic and selectional requirements are met: *who* is assigned a Goal theta-role and case by *ask* (as in *We asked Anson if they were there*), and the lower clause likewise has all of its thematic and case requirements satisfied. Further, there are no problematic “chains” constructed in (33) that might be ruled out by a condition like Relativized Minimality. What has gone wrong is

just that the derivation involves a “downward” operation (or that the outcome has an unbound trace).

In the mereological system, the structure underlying (33) would involve subjoining, to C, the 2-part of the category that selects the indirect object of *ask* (which I’ll assume is an applicative category Appl). This will not violate Irreflexivity and Transitivity, or Dimensionality. It does, however, violate Angular Locality, since C is not a 1-part of Appl.



In fact, Angular Locality will rule out all lowering subjunctions as well as sideward subjunctions. This means that there are no cases where an appeal to a version of Mereological Extension extended to internal subjunction is required. We can then restrict Mereological Extension to External subjunction, or, perhaps, as suggested by Freidin (1978), jettison this kind of cyclicity from the system entirely, if alternative accounts of its empirical effects can be found.

The various consequences of the theory that we have derived so far are as follows:

- (35)
- a. “Downward” subjunction is ruled out by Angular Locality.
 - b. “Superlocal” subjunction (encompassing some versions of Antilocality) is ruled out by the interaction of Irreflexivity and Angular Locality.
 - c. “Sidewards” subjunction (to a 2-part/“specifier”) is ruled out by Angular Locality
 - d. “Parallel” subjunction (to an unattached object) is ruled out by Angular Locality
 - e. “Long-distance” subjunction (from inside a 2-part/“specifier” of some object to a higher object (i.e., subjunction across Extended Projections) is ruled out by Angular Locality.

Angular Locality is a simple constraint on subjunction that has wide-ranging effects in restricting the basic structures that the system can generate. It is

motivated by a simple strengthening of the geometry of the most local configurations to all extensions of those configurations.

Turning to our four questions for the theory of Locality, this system provides an answer to the Etiology of Locality question: Why are there are locality domains at all?

As we have seen, the innards of a 2-part (with one exception I will address in section 4.3) are not related by transitivity to anything outside that 2-part. The 1-part and 2-part relationships don't interact in terms of Transitivity, so any dependency built on these relationships will be restricted to being in one dimension. It follows that there will be locality domains, since 2-parts are disconnected from the object they are a 2-part of. Angular Locality in effect loosens this restriction, allowing dependencies to cross at most one dimension.

Further, the system provides an answer to the question of why the locality domains are what they are (the Domain Content question). Locality domains are 2-parts, because (almost) all of their parts are disconnected from anything outside them. We now have theoretical answers to the Etiology and Domain Content questions raised earlier in the chapter.

A major conceptual advantage of this system is that the theory of locality is brought within the theory of phrase structure. There is no separate Phase Theory regulating what information is available to higher structures, and no appeal is made to particular category labels. The theory of locality simply takes the most basic local relations available and requires that the properties of those relations are maintained in any more complex relations. The fundamental axioms of the part relation, especially Transitivity, but also Irreflexivity, do the rest of the work.

We have not yet answered our other two Locality Questions: why are there exceptions to locality domains, and why are they what they are? I turn to these in the next section.

4.3 Successive Cyclicity

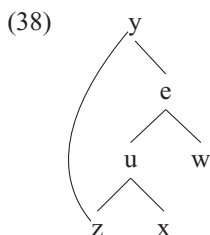
So far, the system appears to constrain syntactic dependencies to Extended Projections. However, we know that syntactic dependencies can cross 2-parts. In the phrase structure system I've been developing, verbal arguments such as nominal phrases, prepositional phrases, and finite clausal complements are 2-parts (specifiers) of objects in higher Extended Projections. This follows if DPs, PPs, and CPs "top off" their Extended Projections, and so cannot be 1-parts of anything else.

Certain kinds of extraction are possible from these SOs, such as subextraction from prepositional phrases (36), or extraction from clausal complements (37) (which, in the system here, is essentially a kind of subextraction too):

- (36) a. Whose nostril did the doctor look [[very far] into ⟨whose nostril⟩] ?
 b. How far did the doctor look [(how far) into [your nostril]]?
- (37) a. Which cat did you say [⟨which cat⟩ fell into the pond]?
 b. Which poem did you say [that Anson wrote ⟨which poem⟩]?

These appear to have the structure that is ruled out by Angular Locality: extraction from the innards of a 2-part (specifier). It seems, then, that, much as in the classical system, we might have to specify an “exception” to the locality domains.

However, although it is correct that a subjunction from inside a 2-part is ruled out by Angular Locality, there is in fact a derivation that will allow a dependency from inside a 2-part to outside that 2-part. Consider the following situation, where we subjoin a 2-part of a 2-part to some higher object⁵:

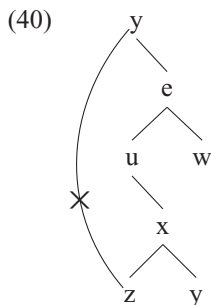


Recall that Angular Locality is defined as follows:

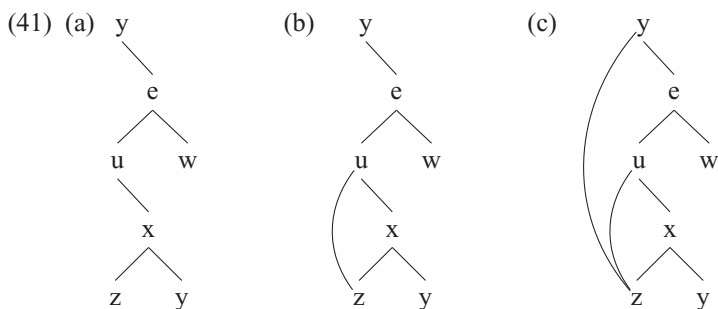
- (39) if γ is a part, then γ can subjoin to β only if there is an α s.t. γ is a n-part of α and α is a 1-part of β .

Angular Locality requires that, to subjoin z to y , there must be something that z is an n-part of (in this case a 2-part of) which is a 1-part of y . There is such an object: e . Because z is a 2-part of u , and u is a 2-part of e , by Transitivity z is a 2-part of e . So e is α in the definition of Angular Locality, and of course e is a 1-part of y . It follows that subjunction of z to y is legitimate. Only one dimension has been crossed; visually, there's a single angle in the structure between the two objects that are undergoing subjunction. Notice that that's not the case for, say, x . x is an n-part of u , but in this case it's a 1-part of u ; u is a 2-part of e , but since transitivity does not cross dimensions, x is neither a 2-part nor a 1-part of e . There is nothing in the structure that corresponds to α in the definition of Angular Locality, and x cannot therefore subjoin to y .

If it's possible to subjoin from a specifier of a specifier, it follows that prior subjunction to a specifier of some object will allow that object to then subjoin outside of the specifier. Take an example that is disallowed:



Although z cannot subjoin to y for the reasons we just looked at, if that step is preceded by subjunction of z to u , the derivation is legitimate. Consider the following:



In (41b) z has been subjoined to u , so is a 2-part of u , and, by Transitivity, it is therefore also a 2-part of e . The SO e is a 1-part of y , so Angular Locality allows subjunction of z to y , deriving (41c).

This system, then, because of the interaction between a very simple locality principle and the Transitivity of parthood, derives the consequence that a long-distance dependency out of a 2-part (a specifier) is possible just when the lower object is a 2-part of that 2-part. In other words, we derive an outcome which is very similar to that stipulated by the classical theory of phases: only the edge of a locality domain is accessible to operations outside of that domain. However, the “exception” to the locality domain is not actually an exception. It’s just Transitivity of parthood working exactly as it does everywhere else in the system. Unlike in the classical system, there’s no need for a special definition of phase edges, or for those edges to be made exceptions to the locality domains. This radically simplifies the theory of locality and provides an answer to the question

of why locality domains have exceptions (the Exceptions to Locality question), and why the exceptions are what they are (the Exception Content question).

We can now add this to the collection of consequences derived from the theory I listed above.

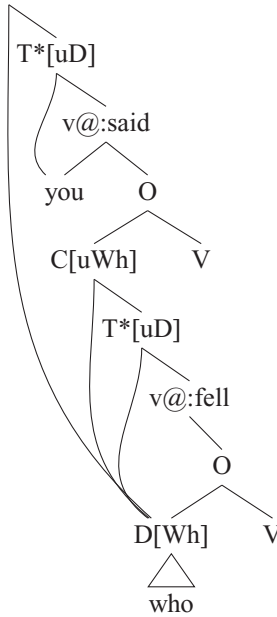
- (42) a. “Downward” subjunction is ruled out by Angular Locality;
 b. “Superlocal” subjunction (encompassing some versions of Antilocality) is ruled out by the interaction of Irreflexivity and Angular Locality;
 c. “Sideward” subjunction (to a 2-part/“specifier”) is ruled out by Angular Locality;
 d. “Parallel” subjunction (to an unattached object) is ruled out by Angular Locality;
 e. “Short-distance” subjunction within an Extended Projection is permitted by the way Transitivity interacts with Angular Locality;
 f. “Long-distance” subjunction (from inside a 2-part/“specifier” of some object to a higher object (i.e., subjunction across Extended Projections) is ruled out by Angular Locality;
 g. “Escape hatches” for subjunction out of a 2-part are permitted by the way Transitivity interacts with Angular Locality.

To see how this idea of intermediate subjunction might work in an actual example, take the following case:

- (43) (guess) who you said fell?

Assume that *fall* is unaccusative and that *who* is introduced as its object via O. Assume too that the clausal complement of *say* (*C who fell*) is introduced in the same way, again by O (this is a simplification; see Moulton 2015 and chapter 6). In (43), the sentence embedded under *guess* is interrogative, encoded by an interpretable feature [Int].

In (44), since the clausal complement of *say* is structurally a 2-part of the higher O, it is not possible to subjoin the wh DP *who* from its base position to the topmost C directly. This is because, in its base position, *who* is a 2-part of the lower O, but O is not a 1-part of that topmost C. However, if *who* is subjoined to the lower C, then it becomes a 2-part of that C, and hence, by Transitivity, a 2-part of the higher O. The higher O is a 1-part of the topmost C and so Angular Locality allows subjunction of *who* to C*[uWh, Int]:

(44) $C*[uWh, Int]$ 

The *wh*-expression *who* is a 2-part of a number of distinct SOs: O, $T*[uD]$, $C[uWh]$, and $C*[uWh, Int]$. This raises the issue of how to linearize it. The formulation I gave of *-Linearization in the last chapter was as follows:

(45) *-Linearization:

- If an SO x is an immediate 2-part of each member of a set of SOs $\Sigma = \{y_1, \dots, y_n\}$, x is linearized immediately preceding the linearization of the y_j bearing $*$ such that y_j is not a part of any other $SO \in \Sigma$ bearing $*$;
- If an SO x is an immediate 2-part of each of the members of a set of SOs $\Sigma = \{y_1, \dots, y_n\}$, and no y_j bears $*$, x is linearized immediately preceding the linearization of the lowest y_j .

*-Linearization as it stands will apply to cases like (44) as follows:

- $D[Wh]$ is an immediate 2-part of each member of the set of SOs $\{O, T*[uD], C[uWh], C*[uWh, Int]\}$
- the SO bearing $*$ which is not part of any other SO bearing $*$ in this set is $C*[uWh, Int]$
- so $D[Wh]$ is linearized immediately preceding the linearization of $C*[uWh, Int]$
- the outcome is that *who* is linearized before the linearization of the remainder of the sentence.

I have specified the various categories that require dependencies with other categories using the [uF] notation, where, in order to check a [uF] feature on a category, an [F] feature must be subjoined to that category as laid out in the last chapter, repeated here:

(47) Checking

A category bearing [uF] must be in a 2-part relation with a category bearing [F].

The [uWh] feature on the intermediate C[uWh] allows the intermediate subjunction, without which the movement of the wh-expression out of its Extended Projection would be impossible, because of Angular Locality. If a language lacks [uWh] on embedded C, all wh-movement will be local to the clause.

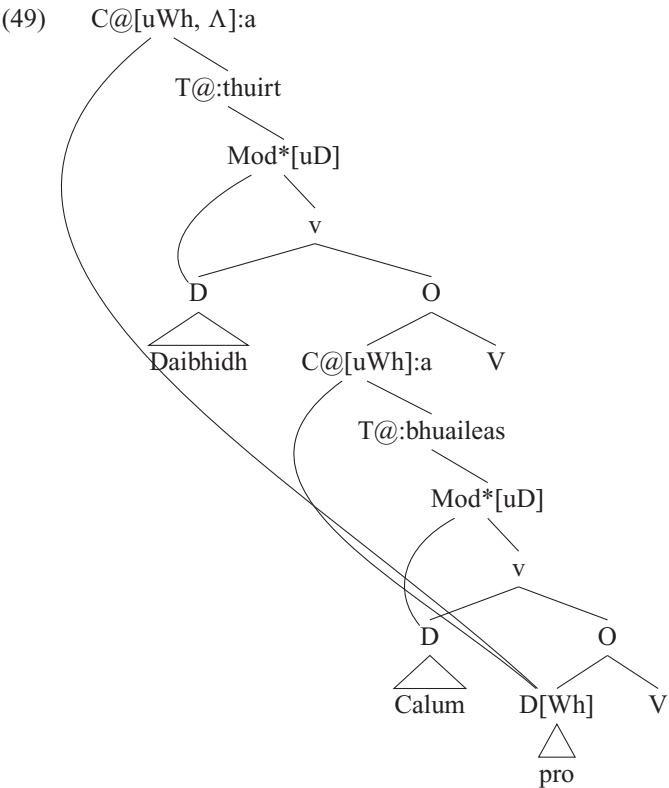
We usually think of movement out of clausal complements as involving a series of small “chains” of movements, each one going from the specifier of a phase to the next specifier of a phase. What is cyclic in the Mereological system is the ordering of the derivational steps, as opposed to the positions themselves, though the outcome is much the same. One difference between this system and classical successive cyclicity is that there is no relationship between the “intermediate” position and the final position: everything is mediated through the base of the dependency, so the structure has multiple relationships in parallel between the base object and the higher objects to which it subjoins (in this, the theory bears a resemblance to the parallel movement chains discussed by Chomsky 2008 and Aboh and Dyakonova 2009).

The system neatly captures those languages, like Scottish Gaelic, where at each clausal position in a long-distance extraction, we find an obligatory marker that the extraction has taken place (cf. McCloskey 2002; Adger and Ramchand 2005). Let us look again at the examples of an embedded clause and a corresponding relative from Gaelic:

- (48) a. Thuirt Daibhidh gum buail Calum an cat.
say.PAST David THAT hit.FUT Calum the cat
‘David said Calum will hit the cat.’
- b. An cat a thuirt Daibhidh a bhuaileas Calum.
the cat REL say.PAST David REL hit.FUT.REL Calum
‘The cat that David said Calum will hit.’
- c. *An cat a thuirt Daibhidh gum buail Calum.
the cat REL say.PAST David THAT hit.FUT Calum
‘The cat that David said Calum will hit.’

As we have already seen, the embedding complementizer is *gu(n)* (or rather its allomorph *gum* preceding a labial) in (48a), but, when there is an $\bar{A}$ -dependency crossing it, as in the relative in (48b), it obligatorily takes the form *a*, and the

following verb appears in a special relative form. The Phase Theory system takes the relativized DP (or a null operator) to extract first to the intermediate specifier of CP, and then on to the topmost C in the relative (which is also obligatorily *a*). In the mereological system, the relativized DP (or a null pronoun) subjoins to the intermediate C, and, since that then makes it a 2-part of the topmost clause, it can subjoin to the topmost C. Both Cs are phonologically spelled out as *a* rather than *gu(n)*:



I’ve followed Adger and Ramchand (2005) who argue that, in Gaelic, relatives involve a null pronominal (*pro*) specified with an uninterpretable feature. I’ve just repurposed the feature [Wh] for this, though it may be that we want to distinguish *wh*-features from relativization features (for Adger and Ramchand it is a feature that marks the pronoun as a variable that needs to be bound by a λ -operator). Unless this *pro* subjoins to the intermediate C, Angular Locality will block it subjoining to the topmost C. That intermediate subjunction is, then, purely formally motivated, and has no effect on interpretation. I’ve implemented the dependency via an uninterpretable version of the [Wh] feature, and

the morphological exponent of $C[uWh]$ is *a*, while the morphological exponent of *C* is *gum*, capturing the pattern. The core variable binding relationship is that between the topmost *C*, and the *pro*, so that topmost *C* bears a feature which I've notated $[\Delta]$, following Adger and Ramchand. This feature is interpreted as binding the variable introduced by *pro*, creating a predicate which modifies the head of the relative *an cat*, "the cat" (not shown).

This approach is a little different from Adger and Ramchand's (2005) analysis. There, we argued that Gaelic relatives were built via Agree, and not movement. The question in the current system becomes moot, since there is no Agree, or movement, just Subjoin. The lack of connectivity effects we reported on in Adger and Ramchand (2005) are dealt with by the assumption that the relativized element is a pronoun in Gaelic, and not a full nominal phrase. Our previous theory dealt with the intermediate movement by assuming that intermediate *C* bore a feature interpreted as a λ -binder, but, via a principle we called Interpret Once Under Agree, the interface rules treated both instances of that feature as just one binder, interpreted in the position where it was interpretable. In the current system, subjunction to the intermediate *C* is triggered just so Angular Locality can be met, and the feature that implements this is just the uninterpretable counterpart of the $[Wh]$ feature on the pronoun that distinguishes it from other pronouns.

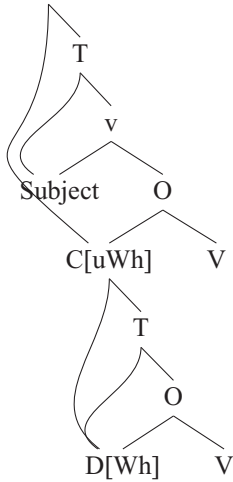
In this analysis the topmost *C* bears a different set of features from the intermediate *C*s, though both are spelled out as *a* in Gaelic. This predicts we might find languages where the topmost *C* differs in form from intermediate *C*s. This is the right prediction. There are in fact languages which have the same property as Gaelic (that $\bar{A}$ -extraction across intermediate *C*s results in a different *C* from that seen in embedding contexts), but which also make a difference between the topmost *C* and the intermediate ones. For example, in Kĩĩtharaka, there is a preverbal marker *n-* that appears in clauses where $\bar{A}$ -extraction has taken place, including intermediate clauses (Muriungi 2005, p. 45ff). However, that same *n-* cannot appear on the verb where the $\bar{A}$ -dependency terminates:

- (50) N-uu (*n)-u-ku-thugania ati John n-a-ug-ir-e Lucy
 FOC-who (N)-2SG-PRES-think that John N-SM-say-PERF-FV Lucy
 n-a-ring-it-e
 N-SM-beat-PERF-FV
 'Who do you think that John said that Lucy beat?'

A further kind of derivation is possible for successive cyclic dependencies in the system I've developed here. Let's look at the structure for a cross-clausal extraction of the object of an unaccusative, as in (43). In the case of English, I assumed that the *wh*-expression subjoins to the intermediate *C*, and then, since

Angular Locality is satisfied, it subjoins to the matrix C in order to check the [uWh] feature there (see (44)). However, if the intermediate C itself subjoins to the matrix C, then, by Transitivity the interpretable [Wh] feature on D is in a 2-part relation with both uninterpretable [uWh] features on the Cs:

(51) C*[uWh, Int]



This derivation will satisfy the Checking condition, since D[Wh] is, by Transitivity, in a 2-part relation with both C[uWh] and C*[uWh, Int]. In this structure it is the clausal complement that subjoins to the matrix C, but because the wh-expression has subjoined to the “edge” of that clausal complement, there is a sequence of 2-part relations connecting the interpretable [Wh] feature of the wh-expression to the uninterpretable [uWh] features on each C. It follows that this should be another route to deriving wh-constituent questions. If the highest C bears *, what we expect to see linearized as its 2-part is the embedded clause, rather than the wh-expression itself.

Such clausal pied-piping has been argued to be one of the strategies for long-distance extraction in languages like Basque (Ortiz de Urbina 1989). In that language, a wh-expression can be extracted to a position to the left of the finite verb (either a specifier of CP, or of a Focus category), as in (52) (examples from Arregi 2003):

- (52) Se pentzate su [CP esan dabela Mirenek [CP idatzi rabela
 what you-think said has Miren.ERG written has
 Jonek]]?
 Jon.ERG
 ‘What do you think Miren said Jon wrote?’

However, rather than successive cyclic movement of the *wh*-expression, it is possible to pied-pipe the embedded CP to the edge of which the *wh*-expression has extracted:

- (53) [_{CP} Se idatzi rabela Jonek] pentzate su [_{CP} esan dabela
 what written has Jon.ERG you-think said has
 Mirenek]?
 Miren.ERG
 ‘What do you think Miren said Jon wrote?’

This raises the question as to why such clausal pied-piping is not available in, for example, English (54). This must boil down to an issue of parametric variation in terms of the features of the matrix C, perhaps a requirement that [*uWh*] co-occurs with a [*uC*] vs a [*uD*] feature in the different languages (an idea reminiscent of the Feature Cooccurrence Restrictions of early Generalized Phrase Structure Grammars, Gazdar et al. 1985). I leave the question of exactly how to featurally implement the variation to one side here.

- (54) *[who left] did you say?

Given this approach to pied-piping, I’d like to turn back to Scottish Gaelic. I have already discussed cases where an object or subject is extracted, showing that intermediate Cs in the extraction path take on a special form. In cases where the extractee is the object of a preposition, there are three possibilities. I will focus initially on the analysis of the stranding versus pied-piping cases in (56a–b), and then turn to the kinds of examples in (56c):

- (55) Thuirt thu gun do bhruidhinn thu rithe.
 say.PAST you THAT PRT speak.PAST you to.3FS
 ‘You said that you spoke to her’
- (56) a. Cò a thuirt thu a/*gu bhruidhinn thu ris?
 who REL say.PAST you REL/*THAT speak.PAST you to.3MS
 ‘Who did you say that you spoke to?’
 b. Cò ris a thuirt thu a/*gu bhruidhinn thu?
 who to.3MS that.REL say you REL/*THAT speak.PAST you
 ‘Who did you say that you spoke to?’
 c. Cò a thuirt thu ris an do bhruidhinn
 who REL say.PAST you to.DEF DEP PRT speak.PAST
 thu?
 you
 ‘Who did you say that you spoke to?’

In (55), the embedding complementizer is *gu(n)*, which, as we have already seen, takes a particular verb form after it, traditionally called the *dependent* form. In the past regular paradigm, that form requires a prefixal particle *do*. In the extraction construction in (56a), the intermediate and matrix complementizers obligatorily appear as *a*, followed by a different verb form, traditionally called the *relative* form, with no *do*. The preposition is stranded and appears in its default 3MS form, irrespective of the number and gender of the extractee (see Adger and Ramchand 2005 for details).

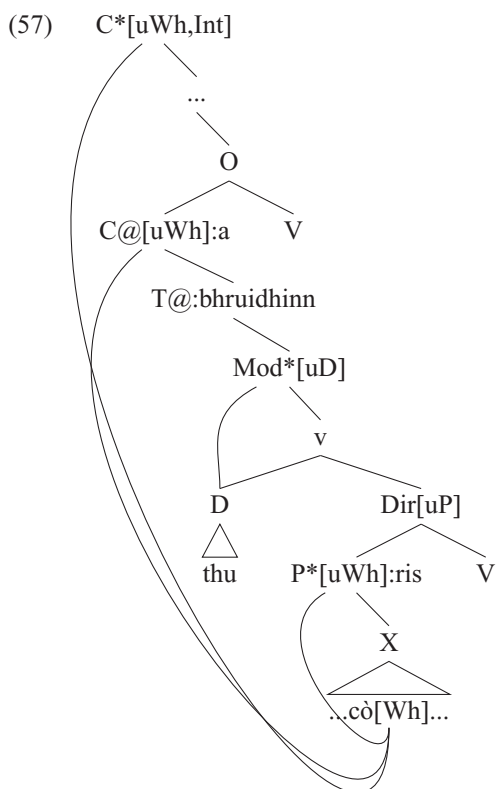
In (56b), the *wh*-element forms a constituent with the preposition, again in its default form, although the *wh*-expression unexpectedly precedes rather than follows the preposition, giving [*cò ris*]. In this construction, just as in (56a), the Cs take the form *a* not *gu(n)* and the verb lacks the prefix. This is a kind of pied-piping construction, but the order of the preposition and *wh*-expression is inverted.

In (56c), the preposition appears at the edge of its own clause. I will show below that it actually appears in a special definite form (syncretic with the default form for the preposition *ri* “to/with,” but not in general). This preposition is followed by a third form of C, *an*, and the verb is in the dependent form, prefixed by *do*. In this construction, the *wh*-element extracts to the matrix apparently ‘stranding’ the preposition in an intermediate position. Just as in the other two constructions, any intermediate Cs appear as *a*, followed by the relative form of the tensed verb.

The subjunction system with Angular Locality provides, I think, an interesting way to think about these three options.

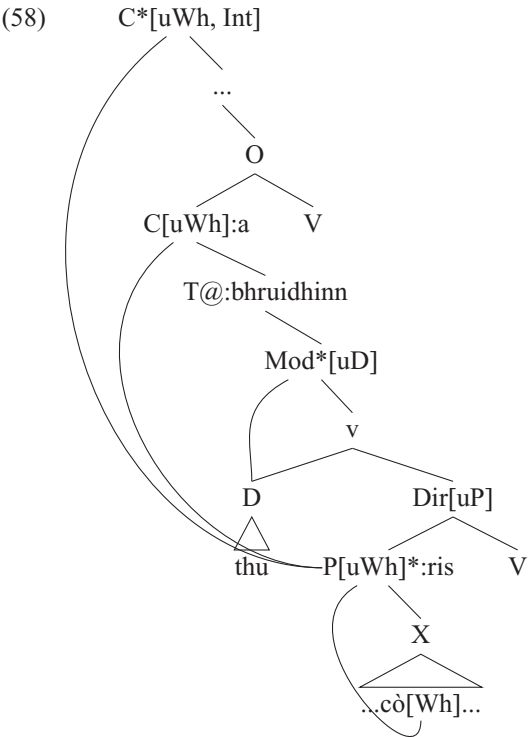
In order to capture the selectional requirements of the verb *buidhinn*, “speak” for a prepositional argument, I will assume a category, call it *Dir[uP]*, which is part of the Extended Projection that the verb spells out. This is a way of expressing that this verb selects an obligatory prepositional argument: the PP argument is a 2-part (specifier) of *Dir* and checks *Dir*’s [*uP*] feature.

The *wh*-expression *cò*, “who” is, I assume, inside the complement of the preposition *ri*, “to,” which I’ll just represent as *X*, remaining agnostic about its category for now, though see Adger (2013) and chapter 6 for discussion. Example (56a) is then analyzed as (57):



In (57), the wh-expression *cò* bears a [Wh] feature, and, by virtue of it subjoining to P, it becomes a 2-part of Dir. Because it is a 2-part of Dir, it can, by Angular Locality subjoin to the intermediate C. This, in turn, means it is a 2-part of matrix O, by Transitivity, which means that it can subjoin to matrix C and linearize there. The obligatory morphological forms of the intermediate Cs signify the subjunctions required by Angular Locality. Even though P bears a * feature, *-Linearization will ensure that the wh-expression is linearized at the highest *-bearing category that it is a 2-part of.

The system also makes an alternative derivation available as in (58):



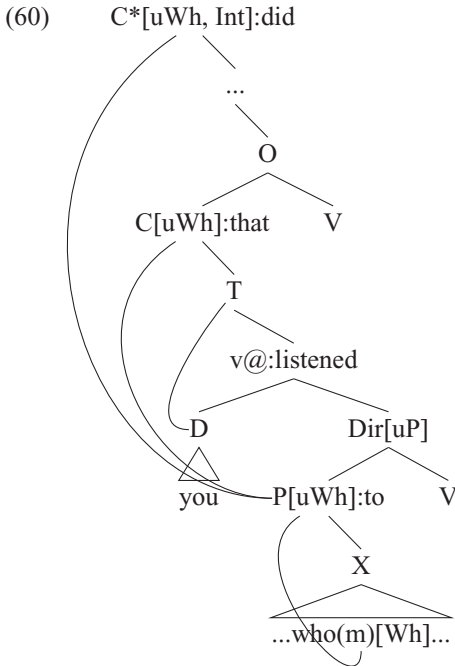
In this structure, *cò* subjoins to P as before, but rather than *cò* then subjoining to the intermediate C, the PP itself subjoins to the intermediate C. This is allowed by Angular Locality, since only one dimension is crossed, and the Transitivity of the 2-part relation allows the [uWh] features on the intermediate and matrix Cs to be checked by the feature [Wh] on *cò*, deriving (56b). The difference between the two structures is whether the wh-expression or the PP subjoins to matrix C. *-Linearization will ensure that C*[uWh,Int]’s immediate 2-part is linearized preceding its spellout, so the PP is linearized to the left of the matrix C. Within that PP, the immediate 2-part of P*[uWh] is also linearized to P*[uWh]’s left, giving the inverted order.

What of a language like English, where, when pied-piping is well formed, the wh-expression does not appear in the specifier of P?

- (59) a. To whom did you say (that) you listened?
b. *Who to did you say you (that) you listened?

In the pied-piped case, P in English bears [uWh] just like in Gaelic, but it does not bear *. Although *who(m)* subjoins to P and then P subjoins, just as in Gaelic,

to the higher Cs, it is never linearized to the left of P. *-Linearization linearizes immediate 2-parts, so P is linearized to the left of matrix C, deriving (59a) with the structure in (60).



We now have an analysis of the first two constructions in Gaelic that involve extraction of or from a PP. In the third Gaelic construction, the P appears after the wh-word, giving an inversion order, but it cannot appear higher than its own clause.⁶

As previously mentioned, the complementizer that appears in this construction is neither *gu(n)* nor *a*, but rather *an*, and the verb form is the same verb form as appears after *gu(n)*. The construction within a single clause looks as follows:

- (61) Cò ris an do bhruidhinn Calum?
 who to.DEF C PRT speak.PAST Calum
 ‘To whom did Calum speak?’

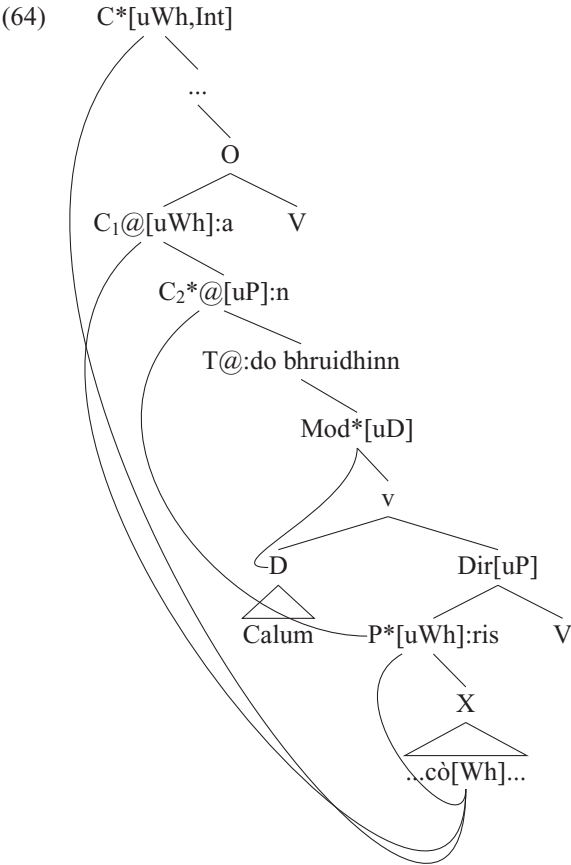
As mentioned, this strategy is not possible cross-clausally:

- (62) *Cò ris an do chrèid thu a bhruidhinn/ an do
 who to.DEF C PRT believe.PAST you REL speak.PAST/ C PRT
 bhruidhinn Calum?
 speak.PAST Calum?
 ‘To whom did you believe that Calum spoke?’

Rather, the preposition appears at the left periphery of the local clause, with the *wh*-expression extracting to its scope position,⁷ and the usual relativizing complementizer *a* appears at any intermediate Cs and at the matrix C:

- (63) Cò a chrèid thu ris an do bhruidhinn Calum?
who REL believe.PAST you to.DEF C PRT speak.PAST Calum
‘Who did you believe Calum spoke to?’

This construction has always been puzzling (to me at least), but I think it is fairly straightforwardly analyzed in the subjunction system. The proposal I’d like to make is that the PP fronts to a position that is lower than the highest C in the embedded clause, and from inside that position the *wh*-expression subjoins to that C, from where it can, by Angular Locality, subjoin to the matrix, or to intermediate complementizers, all of which will then be realized as the relative particle *a*. The structure, then, is as in (64):



In this structure, the PP has subjoined to the lower of two categories that I have labeled C_1 and C_2 giving a kind of CP-recursion. In the next chapter I will argue that there are two C-type categories in the left periphery of the clause associated with wh-expressions, following Vikner (2017) and others. In Gaelic, the lower of these is specialized to Ps, and triggers subjunction of a PP as a 2-part. Since the wh-expression *cò* is subjoined to P, as we saw with the pied-piping cases earlier, and since PP is subjoined to C_2 , *cò* can, by Angular Locality, subjoin to C_1 in the embedded clause. Once it has done so, it may then subjoin to the matrix C, deriving the correct structure. In this structure, then, unlike in the other inversion case, the wh-expression and the PP are not in constituency in the final structure.

This approach captures the fact that the PP in these constructions can only appear fronted in the lowest clause. Because C_2 is not the topmost projection in its Extended Projection, the PP will never escape from that projection: Angular Locality will, as usual, block it from subjoining outside its Extended Projection. This is the reason why (62) is impossible.

However, the wh-expression it contains can escape by first subjoining to C_1 and then to higher Cs. This predicts that all of the intermediate Cs in such examples should be *a*, since the structure is exactly the same as that for long-DP extraction. This is correct, although some speakers report that these are awkward or literary:

- (65) *Cò a thuirt thu a bha Daibhidh a' creidsinn ris*
 who REL say.PAST you REL was.PAST David ASP believe.VN to.DEF
an do bhruidhinn Calum?
 C PRT speak.PAST Calum
 'Who did you say that David believed that Calum spoke to?'

One final note on the morphological makeup of the complementizer *an*. In the proposal here it involves two syntactic "layers": one is identical with the relative complementizer *a*, both phonologically and syntactically (it is $C[uWh]$); the other is phonologically a nasal, which undergoes exactly the same assimilation processes as the nasal in the declarative complementizer *gu(n)*. Plausibly the two phonological pieces correlate with the two syntactic pieces, so the higher C is realized as *a* and the lower as just a nasal (orthographically *n*, but realized phonologically in various ways).⁸

Linearization of the structure follows the pattern we have already seen: the wh-expression *cò* is an immediate 2-part of the elements of $\{P*[uWh], C_1[uWh], C*[uWh,Int]\}$. The highest of these is $C*[uWh,Int]$, so that is where *cò* is linearized. The PP *ris* is an immediate 2-part of the elements of $\{Dir[uP], C_2*@[uP]\}$. The highest of these bearing * is $C_2*@[uP]$, so that is where that PP is linearized.

One issue before we leave successive cyclicity is the question of clearly phrasal spellouts of apparent intermediate traces (as opposed to the spellouts of the categories to which subjunction takes place that we have seen in, e.g., Gaelic).

Consider the Afrikaans example in (66), reported by du Plessis (1977), where the *wh*-expression *met wie*, “with whom” appears not just in its scope position but also at each intermediate clausal boundary between the scope position and the position where it is selected by the verb *trou*, “marry”:

- (66) [met wie] het jy nou weer gesê [met wie] het Sarie gedog [met
with who have you now again said with who has Sarie thought with
wie] gaan Jan trou?
who go Jan marry
‘Whom did you say (again) did Sarie think Jan is going to marry?’

Such examples suggest that it is possible to spell out a single object in multiple sites simultaneously. Of course, this is not the usual case. Usually movement (and subjunction) have the property that Johnson (2020) calls Terseness:

- (67) Terseness:

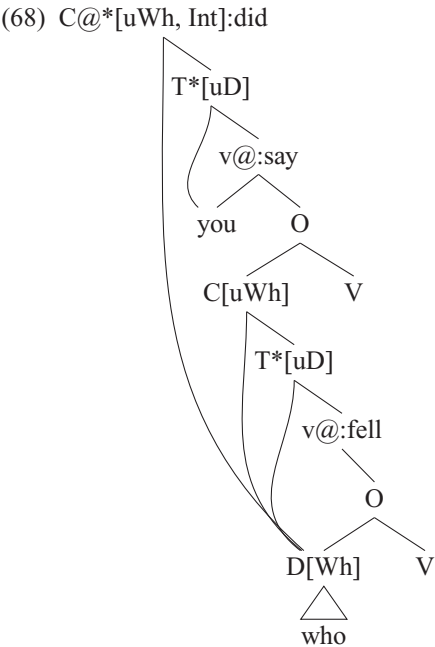
When a term is moved from one position to another, it gets spoken in only one of those positions.

Nunes (1999) provides a system that derives Terseness (and some exceptions he discusses). His proposal is that movement involves copies, and that copies give rise to a linearization problem, since linearization requires a proper linear order, following Kayne’s Linear Correspondence Axiom (Kayne 1994). If an item has moved, then, perforce, one of its copies will precede something and the other one will follow it. But that would violate what a linear order is, since an object cannot both precede and follow another object in a linear order. Nunes suggests that human languages solve the problem by deleting all but one of the copies.

This approach is not possible in the mereological system, since there are no copies. There is just a single object. The solution instead is to allow that object to be pronounced in multiple places, as in the Afrikaans system, or to be pronounced in just one place, as Terseness would have it, leaving the option as a parametric choice, as opposed to deriving Terseness as an invariant constraint.

The *-Linearization proposal as it stands encodes Terseness, since it places the linearization of a multiply-subjoined object at the highest SO bearing * of which it is a 2-part. However, we can allow examples like the Afrikaans cases if, rather than assuming that there is a unique linearization of the object, there are simply multiple linearizations of it. *-Linearization will then place the linearization of the multiply subjoined object at the highest *-bearing category in each Extended Projection (rather than the highest *-bearing category tout court).

In the kind of example I discussed earlier, such as the English long-distance question *who did you say fell*, the wh-expression *who* is a 2-part of the matrix C*, the embedded C, the embedded T* and the embedded O, so it has the structure in (68):



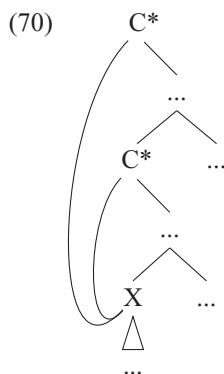
*-Linearization allows only one realization of *who* at the highest C. However, we might instead allow *who* to be realized not at the highest SO bearing *, but at the highest SO bearing * in each Extended Projection. We would in essence allow the *-Linearization operation to be parameterized in the classical sense of that term (Chomsky 1986b):

- (69) *-Linearization:
- a. If an SO *x* is an immediate 2-part of each member of a set of SOs $\Sigma=\{y_1,...,y_n\}$, *x* is linearized immediately preceding the linearization of **every** *y_j* bearing * such that *y_j* is not a 1-part of any other SO∈ Σ bearing *;
 - b. If an SO *x* is an immediate 2-part of each of the members of a set of SOs $\Sigma=\{y_1,...,y_n\}$, and no *y_j* bears *, *x* is linearized immediately preceding the linearization of the lowest *y_j*.

I have altered the definition in (69a) where it is marked in bold. Let's apply this to the English example first. The SO *who* is an immediate 2-part of each

member of the set $\{O, T^*[uD], C[uWh], C@*[uWh, Int]\}$. It will be linearized immediately preceding the spellout of any SO bearing $*$ that is not a 1-part of any other SO in the set that bears $*$. There are two SOs bearing $*$: $T^*[uD]$ and $C@*[uWh, Int]$. Neither is a 1-part of the other, so, if we apply this version of $*$ -Linearization to English, we'd predict that *who* is linearized to the left of the spellout of both of these, and the outcome would be *Who did you say who left?*. This is obviously not grammatical in English, where we have a single linearization of the extracted wh-expression.

However, if we assume that in Afrikaans, each intermediate $C[uWh]$ may bear $*$, we predict the right result for that language. In such a case we have the schematic structure in (70) where, the single object X , which is the wh-expression, could appear multiple times, linearizing at each intermediate C , deriving (66):



X is an immediate 2-part of both C s, both C s bear $*$, and neither is a 1-part of the other, so we linearize X preceding both the spellout of the higher and the lower C .

This approach to the spellout of multiply subjoined SOs, will allow both the Afrikaans example and the English example, as different ways that the two languages “solve” the linearization problem. English linearizes at the highest $*$, while Afrikaans linearizes at each highest $*$ in each Extended Projection.

An interesting twist to this is what happens in the lowest “trace” position. It is impossible, even in Afrikaans, to pronounce the lowest trace:

- (71) $*[met\ wie]$ het jy nou weer gesê $[met\ wie]$ het Sarie gedog $[met\ wie]$ gaan Jan trou $[met\ wie]$?
 with who have you now again said with who has Sarie thought with
 who go Jan marry with who
 ‘Whom did you say (again) did Sarie think Jan is going to marry?’

The version of *-Linearization parameterized for Afrikaans predicts this distribution, since, in the lowest clause, both SOs that the *wh*-expression is a 2-part of (i.e., the verb phrase internal category that licenses its introduction as an argument, and the C*[uWh] of the lowest clause) are in a 1-part relation with each other. *-Linearization predicts a Terseness effect, so only one can be pronounced. Terseness, then, varies across languages depending on whether it is restricted to an Extended Projection or not.

This section has of course just provided a bare sketch of a theory of multiple spellouts for the mereological system, and there are more complexities in the Afrikaans data that remain to be explored. There are also other open questions, such as cases where a subpart of an Extended Projection has been argued to raise and attach to that projection, which Terseness might also be thought to rule out (e.g., the V-doubling constructions discussed by Landau 2006 and others). My main aim here has been to provide a way of understanding why one occasionally sees multiple “copies” spelled out, given that, in the mereological system, there are no copies, just a single SO. The solution is that a single SO may be linearized in a number of distinct positions.

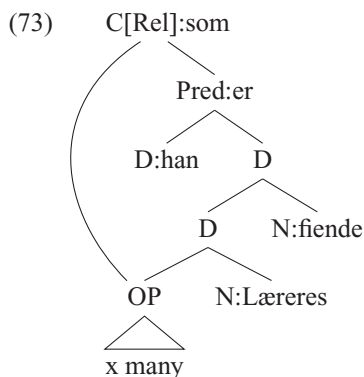
4.4 Some Further Consequences

One interesting consequence of the way that Angular Locality works is that it predicts that it should be possible for dependencies to reach into not just a 2-part of a 2-part, but also into a 2-part of a 2-part of a 2-part and so on. One might imagine that such cases are anyway challenging to parse, but I think that there are actually two constructions that show exactly this behaviour. This first is Comparative Subdeletion in Norwegian discussed by Taraldsen (1978, ex. (64)). Taraldsen gives the following kind of example, which he argued was derived via a free deletion operation, as deriving it by movement would violate Subjacency:

- (72) Han er [likke mange] studenters ven som han er [DP₁ [DP₂ læreres]
 he is as many students' friend REL he is teachers'
 fiende]
 enemy
 'He is as many students' friend as he is that number of teachers' enemy'

The comparative operator needs to create a dependency into the possessor of a possessor to generate (72). Such a dependency violates classical formulations of Subjacency (and indeed current Phase-based theories of locality, if DP is a Phase). In terms of Angular Locality, however, the Transitivity of the $<_2$ relation between the quantifier [*x many*] and DP₂, between DP₂ and DP₁ and between DP₁ and the predicate allows it to be a 2-part of the comparative clause headed

by *som*. The analysis of (72) in the mereological system is roughly as in (73), leaving out various clausal projections for tense and so on, and details of the possessive marking in the DP:



I think the same kind of example is just about possible, though awkward, in English, though I need a heavy pause just before *teachers'* to help with the parse:

(74) He's as many students' friend as he is teachers' enemy

The second phenomenon that is susceptible to the same kind of treatment is the possibility, in some varieties of English, of subextracting possessors (Davis 2021). Davis notes the acceptability, for some speakers, of examples like (75):

(75) Mary is the author [_{CP} who they said [_{CP} [_{DP} 's new book] is good]].

Davis shows that this type of example does involve extraction (for example it is subject to a higher constituent being an island) and obeys certain other restrictions that will not concern us here.

The mereological theory makes a prediction not discussed by Davis: it ought to be possible, given the way that Angular Locality works, to subextract from the possessor of a possessor, since Transitivity will make that possessor of a possessor a 2-part of *v*, and hence able to subjoin to a higher *C*.⁹ This turns out to be correct:

(76) who do you think [_{CP} [_{DP} 's neighbor's cat] ate the most frogs.

Here the *wh*-expression *who* is a 2-part of the complex subject *whose neighbor's cat*. That complex subject subjoins to *T*, but *who* further subjoins to *C*. The fact that some varieties of English allow this has to do with variety-specific conditions on linearization: my English, which does not allow these structures, requires the immediate 2-part of *D* to be linearized at *D*; other varieties do not impose this restriction on *wh*-expressions, allowing them to be linearized higher

at C. I leave exactly how to capture the complexities Davis discusses for another occasion, but the core intuition is that the linearization requirements of C can override those of D for these varieties.

Before concluding this chapter, I want to make a comment on a difference between the system developed here and the phase-theoretic system of Chomsky (2001), Chomsky (2008), et seq. All “specifiers” in the subjunction system become locality domains with “escape hatches,” but categories within the same Extended Projection (i.e., in a 1-part relation) are all in the same locality domain. This means that, in general, vP will not be a locality domain unless the language makes vPs into specifiers (perhaps of an auxiliary verb, creating a kind of biclausal structure). One might ask, then, about the evidence that has been adduced to argue that vP is a phase. That evidence has tended to be of two types: morphological evidence that marks the edges of phases (see Van Urk 2020 for review), and interpretive evidence for there being positions for reconstruction to the “edge” of vP (e.g., Fox 2000, p. 164; Legate 2003).

It may, of course, be the case that a language forces or allows movement to a position even though that position is not a structural locality domain. For example, one analysis of floating quantifiers (e.g., Koopman and Sportiche 1991) assumes that they are stranded when a moving nominal “stops off” in various clausal positions, giving well-known patterns such as (77):

(77) You (all) might (all) have (all) been (all) dancing at the wedding.

This does not, in most approaches anyway, entail that these positions are locality domains, just that they are possible positions for intermediate movement (or trace-binding, or quantifier scope, and so on).

For those languages where vP has been shown to act as a “stop-off” point for successive cyclic $\bar{A}$ -dependencies (see, for example, the discussion of Dinka-Bor in Van Urk and Richards 2015; van Urk 2015), it may be the case that the vP provides such a stop-off point without being a locality domain, much as the EPP feature on T forces movement into the specifier of TP. That these languages have a V2 type syntax might be thought to support such a view. Alternatively, for some of these languages, it may be that the vP is a specifier, and Angular Locality forces extraction to proceed through the relevant 2-part. In any event, in the theory developed here, it is full Extended Projections that must be 2-parts (since there is no way to extend them into further 1-parts), and vP is not a full Extended Projection. This means that we expect DPs and CPs to behave differently from vPs in terms of their behavior as locality domains.

While I think this issue is somewhat open, there is some recent processing evidence that has suggested that while CP is treated as a locality domain for processing, vP is not (Keine 2020a). Overall, I think that the consensus is that

the evidence for phasehood of vP, compared to that of CP, is generally somewhat weaker (though see Van Urk 2020 for a different assessment). I will continue to work with the theory as it is, with no extra stipulations to make vP a locality domain, though, of course, this may ultimately be proved wrong.

4.5 Summary

In this chapter I developed a system that provides answers to the four questions we raised about locality theory. These questions were: Why are there locality domains at all? Why do the locality domains have exceptions? Why are the domains what they are? And why are the exceptions what they are? In Phase Theory, the answers to these questions involve appeal to other factors which are undermotivated (memory limits on computation for the first, propositionality for the third) or are simply not given except as stipulations (the second and fourth questions).

The theory I have developed here answers the four questions via the core concepts of the system: Dimensionality and Transitivity. There are locality domains because the inner parts of 2-parts of some SO are disconnected from that SO, and that is because Transitivity does not cross dimensions. This same property of the system is what makes the locality domains what they are: SOs that are 2-parts of some other object. SOs in an Extended Projection are always accessible to other SOs in that Extended Projection, because all the SOs are related by the transitivity of 1-parthood. The moment that the Extended Projection becomes a 2-part of some SO, dimensionality switches, and the transitivity of 1-parthood is stopped in its tracks.

Locality has exceptions because of the transitivity of the 2-part relation: this entails that a 2-part α of another 2-part β of some object γ enters into the same relations that β does with γ , which means α is not disconnected from γ , and can subjoin to objects that γ is a 1-part of. This is just a theorem of the system, so the exceptions to locality, and the nature of those exceptions, are, in fact, not exceptional.

The theory of locality itself simply capitalizes on these properties of the system. Angular Locality constrains nonlocal relations to be transitive extensions of the possible local relations. A bonus of the way that Angular Locality is specified is that it rules out superlocal relations, and it rules out super-nonlocal relations (subjunction to 2-parts, or to unattached objects, as in parallel subjunctions) because these end up going beyond the minimal extension of the local relations.

Overall, the system provides an elegant theory of locality that both allows and forces successive cyclicity without appeal to phases or other stipulated locality domains.

5 Wh-Islands

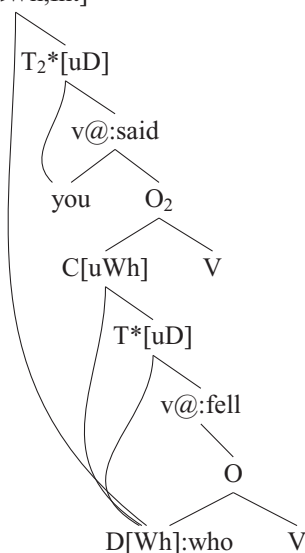
5.1 Core Cases

Angular Locality predicts wh-island effects. Consider the contrast in (1):

- (1) a. (guess) who you said fell?
 b. *(guess) what you asked who bought?

Extraction of *who* to the edge of the embedded clause as in (1a) is grammatical, and is derived in the way I outlined in the last chapter, repeated, for this example, in (2).

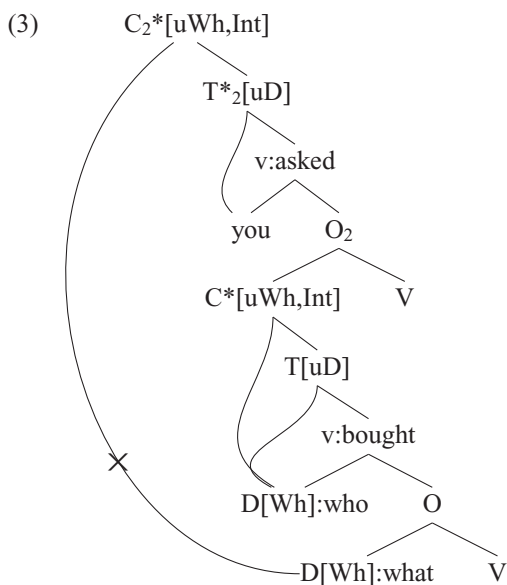
- (2) $C_2*[uWh,Int]$



In (2), *who* is the object of the unaccusative verb, and is introduced as a 2-part of a functional category O. It subjoins to T, and then to C, checking [uD] and [uWh]

features, respectively. The latter operation makes *who* a 2-part of the higher category O_2 (I use a subscript here to make reference to the different objects easier), which introduces the clausal object of *say*. Because *who* is a 2-part of O_2 , Angular Locality allows it to subjoin to C_2 , deriving the embedded question.

However, if the embedded clause already has a 2-part, extraction is predicted to be impossible. Consider the analysis of (1b):



In (3), the embedded verb is transitive, and *who* is its subject. *Who* is introduced as a 2-part of v , then subjoins to T and C . By Dimensionality, an SO can only have one 2-part, and the embedded C has *who* as its 2-part. By Angular Locality (and Transitivity), *what* in this structure, which is introduced as the object of *buy*, cannot subjoin to *who* (this would be sideward subjunction). This means it can never be a 2-part of O_2 and hence, again by Angular Locality, can never subjoin to the matrix C . This correctly predicts the wh-island effect.

Compare this to the analysis of wh-island effects within the classical Minimalist system. Given that Bare Phrase Structure allows there to be multiple specifiers of any syntactic object, and that the Phase Impenetrability Condition allows any element at the edge of a phase to escape that phase, the theory does not immediately capture wh-islands. Typically, within Minimalism, these are therefore understood to be a result of a different kind of locality condition, the Minimal Link Condition, which ensures that if a head attracts some XP to its

specifier then it will attract the closest XP of the relevant type (e.g., the system in Chomsky 1995, p. 311).

Consider again the structure of the *wh*-island violation from above, assuming a classical Bare Phrase Structure type structure:

(4) (guess) what C you asked [_{CP} who C [_{TP} ⟨who⟩ bought ⟨what⟩]]?

In (4), *who* has raised to the specifier of the embedded C as it is the closest *wh*-expression to that C. The matrix C has a *wh*-feature, but the closest *wh*-expression to that C is *who*, not *what*, so (4), which requires *what* to move, is ungenerable.

However, even the Minimal Link Condition is insufficient to rule out *wh*-islands. Consider a derivation where the embedded C has a feature that first attracts *who* to its specifier, and then *what*, so that both are raised to become specifiers of the embedded C, and then either *what* is extracted to the higher C deriving (5a), or, alternatively *who* is extracted, deriving (5b):

(5) a. (guess) [_{CP} what C you asked [_{CP} ⟨what⟩ [_{CP}who C [_{TP} ⟨who⟩ bought ⟨what⟩]]]]?

b. (guess) [_{CP} who C you asked [_{CP} what [_{CP} ⟨who⟩ C [_{TP} ⟨who⟩ bought ⟨what⟩]]]?

There is no Minimal Link Violation here (at least one of the *wh*-expressions in the lower CP will be “closest” to the higher C, or they are equidistant). Rather what has gone wrong is that English does not permit multiple specifiers of C, though phrase structure theory does. Chomsky (1995, p. 268) notes that parameterizing the number of specifiers a functional head may have allows one to account for languages that violate *wh*-islands (following Reinhart 1981). Indeed, Sabel (2002) extends Chomsky’s system to account for adjunct/argument asymmetries in *wh*-extraction across *wh*-islands, allowing multiple specifiers in specific cases.¹

The classical Bare Phrase Structure system, then, does not predict the existence of *wh*-island effects, though it can capture these effects with some supplementation: they are a result of the interaction between a requirement for a single specifier of C (in some languages), and the Minimal Link Condition’s effect on movement operations that take place for reasons of feature checking. This is quite different from the mereological system, which, as it stands, blocks any subjunction out of a 2-part, except for 2-parts of that 2-part. If a *wh*-expression is the “outermost” 2-part of a clausal argument, movement from inside that clause is blocked.

One might think that the classical approach is empirically better than the mereological system, in that it allows space for violations of *wh*-islands, which we know are possible in some languages, as I discuss in some detail below. I will argue that, within the mereological system, it is still possible to capture these violations by adopting the idea that some languages, and indeed some constructions, have an “extra” category in embedded questions, which functions as an escape hatch (e.g., Preminger 2010, Vikner 2017). I will show that this idea, when combined with Angular Locality, makes new, correct predictions about *wh*-island violating languages that the classical system does not have a straightforward way of capturing.

5.2 *Wh*-Island Violations

The mereological theory I’ve set up predicts *wh*-islands to be strong islands: if a clause (or DP) has a *wh*-expression as a 2-part, subjunction from inside that clause is blocked. However, one might question whether this prediction of the system is a good one. It is certainly true that *wh*-islands block many kinds of extraction, but they are typically understood to be selective: some kinds of syntactic object appear to be able to violate *wh*-islands. How are these to be understood in the current system?²

As is well known, in many languages, including, for many people, in English, extraction from a *wh*-island is acceptable if the extractee is a complex *wh*-expression or if the extraction type involves relativization or topicalization. This was first noted by Maling (1978), Rizzi (1982), and Rudin (1981) for Scandinavian, Italian, and Bulgarian, respectively, and the observations were extended to certain cases in English in Cinque (1990b). More recently formal experiments have supported these conclusions (e.g., Hofmeister and Sag 2010; Sprouse and Almeida 2013; Villata et al. 2016; Kush et al. 2018, among many others), though there is still no consensus on exactly what makes a “complex *wh*-expression” (see, e.g., for proposals, Comorovski 1989; Cresti 1995).

To begin with English, extractions of a *which*-DP, relativization, or topicalization from the following cases of *wh*-islands seem relatively acceptable (at least to me):

- (6) a. Which cake did Anson forget [that we had put ⟨which cake⟩ in the fridge]
- b. ?/% Which cake did Anson forget [who had put ⟨which cake⟩ in the fridge]
- c. ?/% This is the cake that Anson forgot [who had put ⟨which cake⟩ in the fridge]
- d. ?/% THAT CAKE, Anson forgot [who had put ⟨that cake⟩ in the fridge]

Simple *wh*-expressions, adjunct extractees, or bare quantifiers, are typically judged to be unacceptable in these contexts:

- (7) a. What did Anson forget [that we had put ⟨which cake⟩ in the fridge]
- b. *What did Anson forget [who had put ⟨which cake⟩ in the fridge]
- c. *When did Anson forget [who had put the cake in the fridge ⟨w⟩hen]
- d. *NOTHING, Anson forgot [who had put ⟨nothing⟩ in the fridge]

Other languages are yet more permissive. Scandinavian languages are a well-studied case. Maling (1978) gives a range of examples from various Scandinavian languages that show acceptable cases of island violations, and further work since then has tended to confirm and extend these observations. For example, Engdahl (1986, p. 15) gives (8) from Swedish, which is comparable to the English case with a complex *wh*-expression, and reports that it is perfectly acceptable:

- (8) Vilken film var det du ville veta vem som ⟨vem⟩ regisserat
 which film was it you wanted know who REL directed
 ⟨vilken film⟩?
 ‘Which movie was it you wanted to know who directed?’

Christensen, Kizach, and Nyvad (2013) provide examples from Danish where there seem to be no *wh*-island effects, with extraction of the adjunct *hvor*, ‘where,’ across a *wh*-island being as acceptable as extraction of a bare argument *hvad*, ‘what’:

- (9) a. Hvad ved hun hvor man kan leje ⟨hvad⟩ ⟨hvor⟩
 What knows she where you can buy
 ‘What does she know where you can buy?’
- b. Hvor ved hun hvad man kan leje ⟨hvad⟩ ⟨hvor⟩
 Where knows she what you can buy
 ‘Where does she know what you can buy?’

Kush, Sant, and Strætkvern (2021) give the following naturalistic example from Norwegian of a topicalization from a *wh*-island:

- (10) Var det den (som) vi ikke visste hvor vi kunne finne ⟨den⟩ ⟨hvor⟩?
 was it that REL we neg know where we could find.INF
 ‘Was it *that* one that we didn’t know where we could find?’

In contrast to Christiansen et al.’s observation about Danish, Kush et al. found that there is an adjunct/argument asymmetry in Norwegian:

- (11) *Hvordan spurte han [hvem som hadde oppført seg ⟨hvordan⟩]?
 how asked he who REL had behaved self
 ‘How did he ask who had behaved themselves?’

Many other languages have been argued to violate wh-islands. I give some representative examples from Hebrew (Reinhart 1981), Romanian (Comorovski 1986), Bulgarian (Rudin 1988), Hungarian (Horvath 1998), and Slovenian (Stepanov, Mušič, and Stateva 2018):

- (12) *eyze sefer Yoni taha le-mi ha-mora natna?*
 which book Yoni wondered to-whom the-teacher gave
 ‘Which book did John wonder to whom the teacher gave?’ Hebrew
- (13) *Maria, cu care știu ce crezi că am discutat?*
 Maria with whom know-1SG what think-2SG that have-1SG discussed
 Maria, with whom I know what you think I discussed. Romanian
- (14) *Koja ot tezi knigi se čudiš koj znae koj prodava?*
 which of these books wonder-2SG who knows who sells
 Which of these books do you wonder who knows who sells? Bulgarian
- (15) *Kivel kérdezted hogy hol találkozott János?*
 who-with asked-2SG that where met John.NOM
 ‘With whom did you ask where John had met?’ Hungarian
- (16) a. *Kaj se je Peter spraševal, kako je Špela popravila?*
 what REFL AUX Peter wonder-PST how AUX Špela fix-PST
 ‘What did Peter wonder how Špela fixed?’
 b. *Kako se je Peter spraševal, kaj je Špela popravila?*
 how REFL AUX Peter wonder-PST what AUX Špela fix-PST
 ‘How did Peter wonder what Špela fixed?’ Slovenian

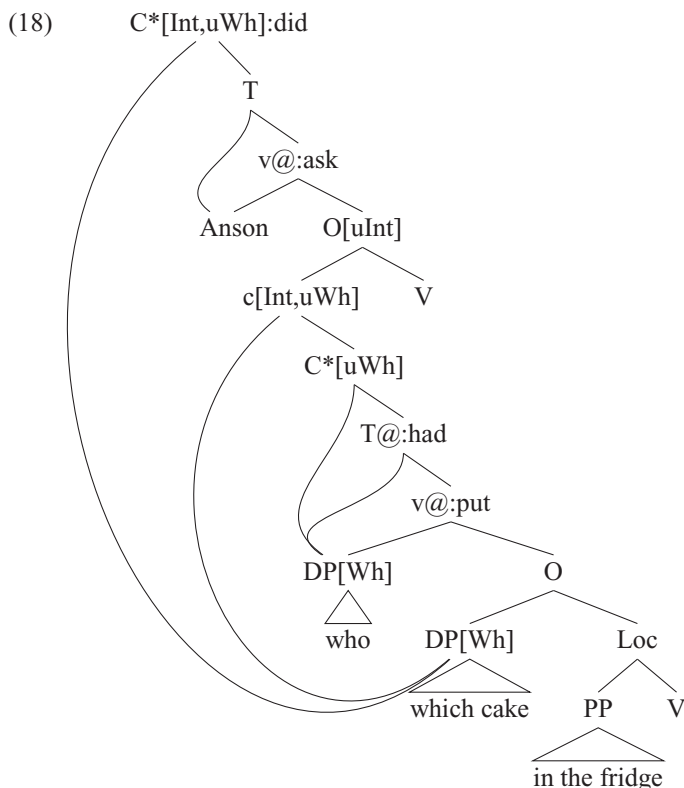
Why are these cases well formed? The consensus is that an approach like that of Cinque (1990b) involving a non-movement dependency (an A-bar bound pro in the gap position) is untenable, since these extractions typically show different patterns of behavior from resumptive pronouns (see, e.g., Engdahl 1997 for general discussion).

The obvious alternative to a non-movement account is that these languages have an “extra” escape hatch that is projected above the island. Reinhart (1981) was the first to suggest that there might be an extra specifier position for CP, an idea developed in more depth by Sabel (2002). Closely related, but conceptually distinct, are a series of analyses that suggest that certain languages allow the projection of an extra head above the landing site of the embedded wh-expression (Preminger 2010; Vikner 2017).

In the mereological system, Dimensionality blocks the possibility of extra specifiers, along the lines suggested by Reinhart and Sabel, but not the Extended Projection of V containing an extra category. Take again an example where a complex wh-expression has extracted across a wh-island, such as (17):

(17) ?Which cake did Anson ask who had put in the fridge?

If we assume the existence of an extra category that can bear the [Int] feature that the higher verb selects, the structure underlying this sentence can be represented as follows:



Much as we saw before, *who* in the embedded clause has subjoined from its “base” position to T and to C, and it is C that bears a [uWh] feature and the * that ensures that its 2-part is linearized to the left of its own linearization (enforcing “overt movement”). However, C is not a 2-part of the higher O, rather another category, which, following Vikner, I have called *c* (“little *c*”) is the topmost category in the Extended Projection. Vikner (2017) and Nyvad et al. (2017) suggest that *c* is a recursion of the C layer, but we might also take it just to be a distinct category in the left periphery (Rizzi 1997 et seq.). Preminger (2010) suggests for Hebrew that the overt position of *wh*-expressions is the specifier of Focus (cf. Brody 1990), and that the higher category is the category we usually recognize as the complementizer. Either approach, or indeed many others, will work for my purposes here. I’ll adopt Vikner’s nomenclature in what follows.

The *c* category bears the [Int] feature that satisfies the selectional requirements of *ask*, and we can also, at least in some languages, endow it with a feature encoding discourse familiarity, ensuring that a definite expression must subjoin to it. This will allow us to capture the selectivity of these islands.

Given this analysis, the overtly moved element *who* is actually not at the true “edge” of the embedded clause (i.e., it is not the highest 2-part of the 2-part of O). The *wh*-expression that is the object of the embedded verb (*which cake*) can therefore subjoin to *c*, in which case it will be a 2-part of the higher O. By Angular Locality, *which cake* is then able to subjoin to the matrix *C*, apparently violating a *wh*-island, but, in actuality, obeying Angular Locality as required.

5.3 Long Wh-Island Re-emergence

One might think that the proposal I’ve just made vitiates the theory of empirical interest, making it effectively a notational variant of the standard theory, at least as far as empirical predictions go. Just like Classical Bare Phrase Structure allows multiple specifiers, but parameterizes this, the mereological system allows an extra category *c* that *C* subjoins to, and that provides the possibility of a “higher” 2-part, and parameterizes the presence of *c* (or the [uWh] feature on *c*).

However, the mereological system makes a different prediction from classical Bare Phrase Structure theory. In the latter, if we set the parameter so as to allow multiple specifiers, nothing will rule out multiple extraction from a single clause: that is, a case where two elements are extracted from an embedded clause, one to a higher specifier of CP, and the other to a yet higher one. In languages (or constructions) where *wh*-island violations are possible, the prediction is that such derivations should be legitimate.

In the mereological system, in contrast, once something has subjoined as a 2-part to a 2-part, Dimensionality rules out further subjunction. This makes a difference when multiple *wh*-expressions are extracted from a single domain.

Consider, for example, the following kinds of sentence. I’ve marked the *wh*-island inducing *wh*-expression in bold, and have used ⟨...⟩ to mark the “trace” position, as is standard. The crucial point is that, in (19a), the expression that induces the *wh*-island (*where*) has not extracted out of its own clause, while in (19b), it has:

- (19) a. Which cake [_{CP} were we told [_{CP} we should enquire [_{CP} **where** to put
⟨which cake⟩ ⟨where⟩]]]

- b. *Which cake [_{CP} did we enquire [_{CP} **where** she was told [_{CP} to put (which cake) (where)]]]

Although extraction from a wh-island is (for many speakers) fairly acceptable in English, if that wh-island is induced by an expression that has moved from a lower clause, the wh-island effect reappears. This appears to be universal for all the native English speakers that allow wh-island violations with definite wh-expressions that I've tested.³ Let's call this the *Wh-Island Re-Emergence (WIRE) Effect*:

(20) Wh-Island Re-Emergence (WIRE):

If x and y are wh-expressions that are clausemates, and x raises to induce a wh-island, then that wh-island can only be violated by extraction of y if x remains in its own clause.

I'm using the term clausemate in the restricted sense that two expressions are clausemates if they originate in the same clause.

WIRE states the impossibility of extracting both of two wh-expressions that originate in a single clause to positions outside that clause.⁴

The WIRE effect also appears in finite versions of wh-islands, though, as is usual, these are all slightly less acceptable:

- (21) a. ?These are the books [_{CP} that you told me [_{CP} that they were wondering [_{CP} **who** sells most cheaply]]]
 b. *These are the books [_{CP} that you were wondering [_{CP} **who** they told me [_{CP} sells most cheaply]]]
- (22) a. ?Here's the cat [_{CP} that they said [_{CP} you'd asked me [_{CP} who had taken to the vets]]]
 b. *Here's the cat [_{CP} that you'd asked me [_{CP} **who** they'd said [_{CP} had taken to the vets]]]

Again, in the (a) examples, the element responsible for the wh-island (in bold) is in the same clause as the initial trace of the long-distance extractee, while in the (b) examples, both wh-expressions have been extracted from the most embedded clause. This pattern of acceptability requires that both wh-expressions are extracted from the same clause. The following examples, where the wh-island inducing expression is local to its own trace but is not a clausemate with the extractee, have the status of the (a) not (b) patterns.

- (23) a. ?These are the books [_{CP} that you were wondering [_{CP} **who** told me [_{CP} that Gay's The Word sells]]]
 b. ?Here's the cat [_{CP} that you had asked me [_{CP} **who** had said [_{CP} they had taken to the vets]]]

Turning to languages like Norwegian (24) and Danish (25), where *wh*-island violations are generally acceptable, we find the same pattern, unnoticed, as far as I'm aware, in the literature on these languages:

- (24) a. Dette er kaka som hun spurte hvor hun skulle plassere.
 this is cake.DEF REL she asked where she should put
 'This is the cake that she asked where she should put.'
- b. Dette er kaka som vi ble bedt om å spørre hvor vi
 this is cake.DEF REL we were asked about to ask where we
 skulle plassere.
 should put
 'This is the cake that we were asked to ask where we should put.'
- c. *Dette er kaka som vi spurte hvor hun ble bedt om å
 this is cake.DEF REL we asked where she was asked about to
 plassere.
 put.
 'This is the cake that we asked where she was asked about to put.'
- (25) a. Hvilken båd foreslog naboen hvor billigt vi skulle
 which boat suggested neighbor-DEF how cheaply we should
 sælge?
 sell
 'Which boat did the neighbor suggest how cheaply we should sell?'
 (adapted from Vikner 2017)
- b. Hvilken båd sagde du at naboen foreslog hvor billigt
 which boat said you that neighbor-DEF suggested how cheaply
 vi skulle sælge?
 we should sell
 'Which boat did you say that the neighbor suggested how cheaply we
 should sell?'
- c. *Hvilken båd foreslog naboen hvor billigt du sagde at
 which boat suggested neighbor-DEF how cheaply you said that
 vi skulle sælge?
 we should sell
 'Which boat did the neighbor suggest how cheaply you said that we
 should sell?'

This pattern has actually been noted before in the literature on Hebrew, another language that violates *wh*-islands. Preminger (2010), extending the observations of Reinhart (1981), shows that it is possible to extract across a *wh*-island in Hebrew:

- (26) *ma Dina shaxexa le-mi Dan natan ⟨le-mi⟩ ⟨ma⟩?*
 what Dina forgot DAT-who Dan gave
 ‘What did Dina forget to whom Dan gave?’

However, when the wh-island inducing expression has been moved from its own clause, the wh-island effect reemerges, similar to the patterns we have noted for English and Scandinavian:

- (27) **eyze sefer shaxaxta [CP le-mi Rina xashva [CP she-Dan shalax*
 which book forgot.2SG DAT-who Rina thought that-Dan sent
 ⟨le-mi⟩ ⟨eyze sefer⟩]]
 ‘Which book did you forget to whom Rina thinks that Dan sent?’

Finally, Rizzi (1982, p. 56ff) noted in an early discussion of wh-island effects the following contrast in Italian:⁵

- (28) a. *Il mio primo libro, che credo che tu sappia a chi*
 the my first book that believe.1SG that you know.2SG to whom
ho dedicato, mi è sempre stato molto caro.
 have.1SG dedicated 1SG.DAT be.3SG always been very dear
 ‘My first book, which I believe that you know to whom I dedicated,
 has always been very dear to me.’
- b. **Il mio primo libro, che so a chi credi che*
 the my first book that know.1SG to whom believe.1SG that
abbia dedicato, mi è sempre stato molto caro.
 have.SUBJ.1SG dedicated 1SG.DAT be.3SG always been very dear
 ‘My first book, which I know to whom you believe that I dedicated,
 has always been very dear to me.’

In (28a), the wh-expression *a chi*, ‘to whom’ has extracted to the edge of its local clause, and its clausemate relative pronoun *che*, (or a null operator associated with it) extracts across it (via, in the system developed here, a subjunction to little c). In (28b), in contrast, *a chi* has extracted out of its local clause to a higher clause (the complement of *so*, ‘I know’). This is exactly the WIRE configuration.

Interestingly, Rizzi ties the ungrammaticality of these examples to that of other cases. These involve examples where a wh-expression extracts across two wh-islands:

- (29) a. *Non so proprio chi possa avere indovinato a chi*
 NEG know.1SG really who could have.INF guessed to whom
affiderò questo incarico.
 entrust.1SG this task

“I really don’t know who might have guessed to whom I will entrust this task.”

- b. *Questo incarico, che non so proprio chi possa avere
 this task that NEG know.1SG really who could have.INF
 indovinato a chi affiderò, mi sta creando un sacco di
 guessed to whom entrust.1SG, 1SG.DAT is creating a bag of
 grattacapi.
 headaches
 “This task, that I really don’t know who might have guessed to whom
 I will entrust, is getting me into trouble.”

There would not be a WIRE effect in the current system for these examples, since the *wh*-expressions never move out of their own clause. I think that this is correct in general. In English the equivalent questions are, for me at least, fairly well formed:

- (30) a. ?This is the cake that no one knew who to ask where to put.
 b. ?This is the cake that no one knew who we should ask where we should put.

There is a marked contrast if two *wh*-expressions are extracted out of the lowest clause creating the WIRE configuration. (31) is very bad:

- (31) *This is the cake that no one knew where we wondered who should put.

Engdahl (1986, p. 15) also reports as grammatical similar examples of extraction across two *wh*-islands in Swedish:

- (32) Vilken film var det du redan glömt vem det var som visste vem
 which film was it you already forgotten who it was REL knew who
 som regisserat?
 REL directed
 “Which movie was it that you had already forgotten who it was that knew
 who had directed.”

I’m not sure why the Italian examples that Rizzi adduces with extraction across multiple *wh*-islands are ungrammatical,⁶ but I take it that it is not connected to the WIRE effect itself, though there is much more to be uncovered here, perhaps related to finiteness of the embedded clauses (see also Browning 1987, p. 173 and pp. 304–305).

In Bare Phrase Structure theory, to allow for the violation of *wh*-islands in these constructions, the embedded clauses could, as suggested by Chomsky, allow multiple specifiers. However, in that approach, there’s no obvious difference between the relevant representations of the *wh*-island versus the WIRE

example to which we can appeal. Schematically the two possible derivations look as follows:

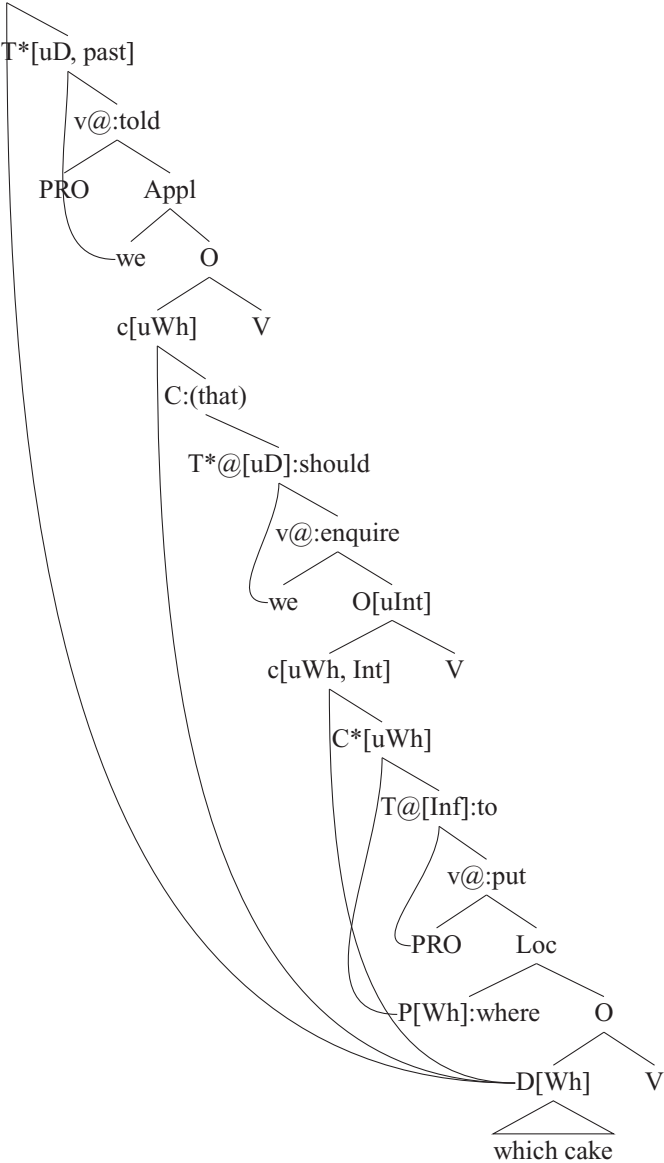
- (33) a. ?Which cake were we told [_{CP} ⟨which cake⟩ we should enquire [_{CP} ⟨which cake⟩ [_{CP} where to put ⟨the cake⟩ ⟨where⟩]]]
 b. *Which cake did we enquire [_{CP} ⟨which cake⟩ where she was told [_{CP} ⟨the cake⟩ [_{CP} ⟨where⟩ to put ⟨which cake⟩ ⟨where⟩]]]

The only difference is the fact that the adjunct *where* undergoes an extra extraction in the (b) example, but nothing in the system should disallow this, since multiple specifiers are available at each clausal boundary.

While the wh-island violation pattern in (33a) pattern is derivable in the mereological system, the WIRE-effect pattern in (33b) is not.

Let's first look at the (a) pattern:

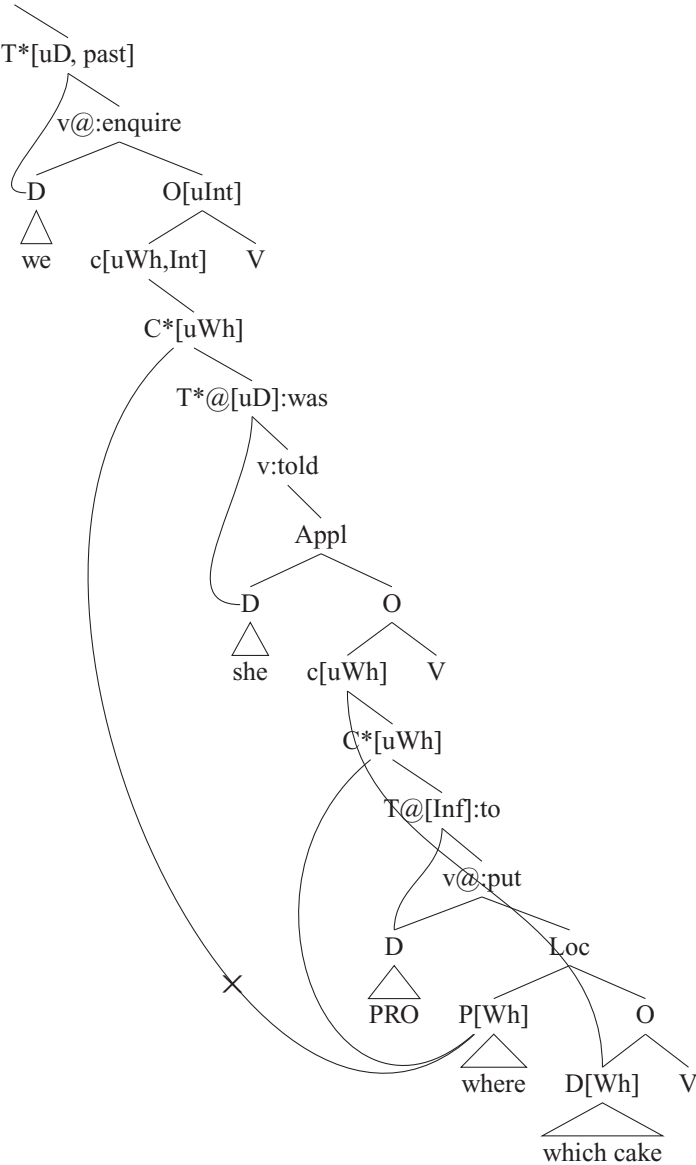
(34) C*@[uWh,Int]:were



As usual I'm assuming that clauses are introduced by O, as are thematic objects, though obviously extra distinctions need to be made to capture the particular selectional restrictions of verbs. I've assumed that a ditransitive like *told* involves the projection of two argument-introducing categories, and that the lower one licenses the clausal complement, and the higher is some kind of applicative category Appl licensing the indirect object. I've introduced the locative argument of *put* via a category Loc above O, though it may be below O, depending on analysis. The locative argument *where* subjoins to the local C* and is pronounced there. Since c is projected, and the wh-expression *which cake* is compatible with the checking requirements of c, *which cake* can subjoin successive cyclically to the higher c categories, surfacing in the highest one. This matrix C bears [Int] as it is a question, and both * and @, with * enforcing linearization of the extractee to the left of C, and @ linearizing the auxiliary segment of the Extended Projection of the matrix clause in C, giving the characteristic V2 effect of matrix questions in English.

Now consider what happens if we try to subjoin both wh-object and the locative argument out of the lowest clause. In such a situation, Angular Locality comes into play, and there is no derivation that works out. Consider the following structure, for example:

(35) C*[uWh,Int]:did



I have omitted the subjunctions from the *wh*-expression *which cake* to the higher *c* and *C** just so the representation of the structure is not too dense. The issue is that after initial subjunction of *where* to the lowest *C*[uWh]*, it cannot subjoin any higher, given that the *c[uWh]* in that same clause already has *which cake* as a 2-part. Further subjunction of *where* is therefore ruled out by Angular Locality, and any further subjunction of *which cake* is moot.

Recall our previous examples where the *wh*-expression that induces the island is local to its own trace, but the extractee is more deeply embedded. I repeat them here for ease of reference:

- (36) a. ?These are the books [_{CP} that you were wondering [_{CP} **who** told me [_{CP} that Gays The Word sells]]]
 b. ?Here's the cat [_{CP} that you had asked me [_{CP} **who** had said [_{CP} they had taken to the vets]]]

The same judgments hold when the island is induced by a *wh*-object as opposed to a subject:

- (37) a. ?These are the books [_{CP} that you were wondering [_{CP} **who** Anson told ⟨who⟩ [_{CP} that Gays The Word sells]]]
 b. ?Here's the cat [_{CP} that you had asked me [_{CP} **who** Anson informed ⟨who⟩ [_{CP} they had taken to the vets]]]

Angular Locality predicts these cases to be well formed (for those speakers that allow *wh*-island violations) since there is only one element extracted from each clause. If we do one more embedding, we find a WIRE effect again (though given the complexity of these examples, it does become challenging to separate processing issues from grammar, and perhaps the double whammy of both is why these are all so crashingly bad):

- (38) a. *These are the books [_{CP} that you were wondering [_{CP} **who** [Anson said [⟨who⟩ told me [_{CP} that Gays The Word sells]]]]
 b. *Here's the cat [_{CP} that you had asked me [_{CP} **who** no one thought [⟨who⟩ had said [_{CP} they had taken to the vets]]]]
 (39) a. *These are the books [_{CP} that you were wondering [_{CP} **who** [Anson said [David told ⟨who⟩ [_{CP} that Gays The Word sells]]]]
 b. *Here's the cat [_{CP} that you had asked me [_{CP} **who** no one thought [Anson had informed ⟨who⟩ [_{CP} they had taken to the vets]]]]

This shows that the issue isn't only a problem with double extraction; it's rather extraction over something that has already extracted from its own clause, just as predicted by Angular Locality.

This approach to wh-island violations is quite close in its overall structure to the analyses of Preminger for Hebrew or Vikner for Danish. Indeed Preminger blocks the Hebrew WIRE effect example via a stipulation that C may have only one specifier. However, as I hope to have shown here, WIRE follows from the interaction between Angular Locality, Dimensionality, and Transitivity and is a quite general effect across those languages that lack wh-islands because they allow projection of an extra category in the C-domain.⁷ It's important that the logic of the system entails that adding in more extra heads is not going to provide a means to circumvent the WIRE effect. Angular Locality imposes a hard constraint on the structure of the derived objects, irrespective of the richness of their functional structure.

5.4 Multiple Wh-Questions

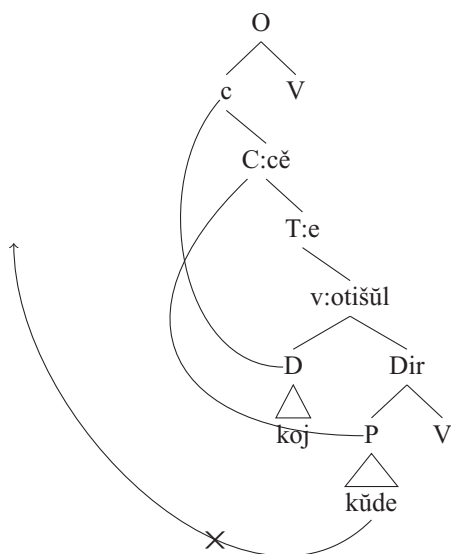
If multiple elements cannot be extracted from a locality domain, which is a consequence of Angular Locality and Dimensionality, then cross-clausal multiple wh-movement would seem to be an issue, since, in such cases, multiple elements appear to have been extracted from a single clause.

Take a Bulgarian example like the following:

- (40) *Koj kŭde misliš čē e otišŭl?*
 who where think.2SG that has gone
 ‘Who do you think went where?’

In this example the wh-expressions *koj*, “who” and *kŭde*, “where” are both interpreted as arguments of the participle *otišŭl*, “go,” in the embedded clause, but both surface in the matrix. Given that the embedded clause is a 2-part of some O that contains the matrix verb, Angular Locality should rule out subjunction of both elements outside of that embedded clause: one wh-expression will be a 2-part of C, the other a 2-part of c. The former will be blocked from subjoining higher, so it should be impossible for both to appear in the matrix, contrary to fact:⁸

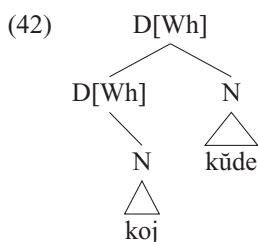
(41)



This structure will allow further extraction of *koj*, “who,” but *kŭde*, “where,” is blocked, by Angular Locality, from subjoining to any higher category. It seems, then, that the system rules out languages like Bulgarian, not a particularly good result.

One of the proposals for dealing with languages like Bulgarian has been that the wh-expressions undergo a syntactic operation that forms a cluster. Rudin (1988), in her original proposal, derives this cluster by first moving the highest wh-expression to the specifier of CP, and then adjoining further wh-expressions to this, creating a single constituent in the specifier. Grewendorf (2001) suggests as an alternative the idea that the wh-expressions form a cluster before the whole unit moves to the specifier of CP. In the example above, we might reformulate these ideas by saying that *koj*, “who” subjoins as a 2-part of *v*, and *kŭde* raises and adjoins to it, followed by raising of the cluster. However, both Rudin’s and Grewendorf’s proposals involve a kind of sideward movement unavailable in my system.⁹

However, there is a closely related alternative that the mereological system does allow. Consider a derivation like the following, where we subjoin one wh-phrase to the other as a 2-part, creating a complex wh-quantifier (possibly with a richer internal structure than is given here):

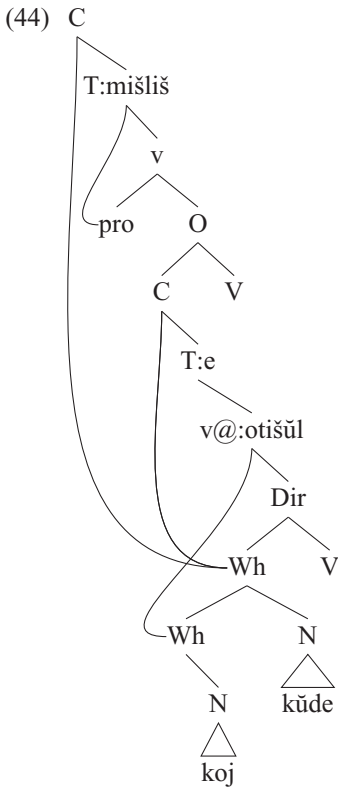


The indefinite expressions subjoin as 1-parts to D[Wh]. Then one subjoins to the other syntactically creating an “absorbed” quantifier (Higginbotham and May 1981, May 1990). Barss (2000) argues that “absorption” is both necessary and sufficient to provide pair-list readings for multiple wh-questions, and Rudin (1988) points out that this is exactly the reading that these kinds of questions get in Bulgarian. The meaning of (42) will then be paraphrasable as: *for which pair x, y , where x is a location and y is an individual, is it the case that ...*

The mereological system will not allow a wh-expression to subjoin from a position within a structure “sideward” into another wh-expression, as pointed out in chapter 4, since this would violate Angular Locality. However, the reverse of this, externally subjoining the wh-expressions to each other to create a cluster, and then declustering via internal subjunction is legitimate. Each part of an SO in the mereological system receives its interpretation via the object it is a part of. Typically, that will involve “external” subjunction to that object as a 1-part or 2-part. However, nothing rules out internal subjunction of some object as a 2-part to a category that provides it with the relevant interpretation. There is no restriction on “movement to a theta-position” in the system: objects simply need to be interpretable appropriately in all the positions they appear in (more accurately, as the relevant parts of all the objects they are parts of).

With this in hand, I suggest that the correct analysis of the Bulgarian example, repeated here in (43) is derived as follows: (42) is subjoined in the Dir position of the embedded verb. Its immediate part, the D whose 1-part is *kŭde*, receives an interpretation as the Goal argument of the verb. Then the wh-expression *koj* subjoins to become a 2-part of *v*, licensing its interpretation as the agent of *otišŭl*, “go.” At this point, the arguments of the verb are in the correct configuration for their thematic interpretation. However, the cluster is still a legitimate object, and that cluster then subjoins to the embedded C[uWh], making it a 2-part of that C. By Transitivity it is then a 2-part of O in the matrix clause. At that point in the derivation, Angular Locality will allow it to subjoin to the matrix C, deriving (43) with the structure in (44).

- (43) Koj kŭde misliš čē e otišŭl?
 who where think.2SG that has gone
 ‘Who do you think went where?’



Linearizing the cluster in the matrix C position gives the correct linear order, with the earlier subjunctions providing the relevant semantics, resulting in (40).

This approach also gives us some handle on why the ordering in the cluster in Bulgarian is isomorphic with the syntactic hierarchy, at least for 2-element clusters, in quite a different way from the “Tucking In” approach of Richards (1997). Subjunction is only upward (by Angular Locality), which means that the cluster must be constructed so that it can be subjoined to the lowest position, and then its subpart subjoined upward from there. Assuming that the linear order of 2-parts in Bulgarian is that they are linearized before the SO they are a 2-part of, this will mean that the cluster will have to be built so that the subject will be a 2-part of the object inside the cluster. The object can then externally subjoin to O, and then the subject can subjoin upward to v. The opposite order would lead to the subject not being able to be interpreted. So the system correctly predicts the classical ordering effects noted for Bulgarian by Rudin:

- (45) a. Koj kogo vižda?
 who whom sees
 ‘Who sees whom?’
 b. *Kogo koj vižda?

The order of three arguments would also follow on two further assumptions. The first is that the category that licenses indirect objects is sandwiched between *v* and *O* (that is, it is an applicative head, Pylkkänen 2000). The second is that the subjunction to *Appl* is of the subcluster, so that first the indirect object subjoins to the applicative, and then the subject to *v*. This second assumption might be an “Attract Closest” effect, though it has not been necessary to appeal to such a constraint elsewhere in the system. As is well known, the data become more complex when there are more than two *wh*-expressions, and I leave the issues open here.

As discussed by Rudin, there is a distinct class of multiple *wh*-movement languages that behaves differently in a number of respects. For my purposes, the crucial fact is that these languages do not allow multiple *wh*-expressions to move outside of a single clause. While examples like (46) are possible in Serbo-Croatian (I use that language name as it was what Rudin used in her original paper), multiple extractions from a single clause are not (47):

- (46) a. Koi želite da vam šta kupi?
 Who want.2PL to you what buy.3SG
 ‘Who do you want to buy you what?’
 b. Štai želite da vam ko kupi?
 What want.2PL to you who buy.3SG
 ‘Who do you want to buy you what?’
 (47) a. *Ko šta želite da vam kupi?
 who what want.2PL to you buy
 for ‘Who do you want to buy you what?’
 b. *Šta ko želite da vam kupi?
 what who want.2PL to you buy
 for ‘Who do you want to buy you what?’

Rudin distinguishes the two classes of languages via a parameter that essentially determines whether the specifier of CP can contain multiple *wh*-expressions (her [+MFS], Multiply Filled SpecCP, languages) or not ([−MFS]) languages.

There have been various approaches to the distinction between [+MFS] and [−MFS] languages. For Rudin, as mentioned, Bulgarian (and Romanian) allow a cluster in the specifier of CP, while [−MFS] languages allow one element in

the specifier of CP, with the others adjoined to TP (IP, as it was then). Various refinements of this picture have been suggested; Bošković (2002) suggests that the noninitial wh-expressions in [–MFS] languages involve movement to a Focus position (cf. Surányi 2002 on Hungarian).

In terms of the mereological system, the parameter would be whether the clustering is allowed or not: a simple property of D[Wh] in the language. Since Dimensionality entails that an object can have only one 2-part, in a non clustering language like Serbian, the noninitial wh-expressions cannot be a cluster that is a 2-part of C. Rather they must, as suggested by Bošković, be lower, perhaps in some focus-related positions.

The idea that [+MFS] languages form clusters predicts that these languages should not have the Wh-Island Re-Emergence (WIRE) effect. Recall that this is an effect where, in a language that usually allows violations of wh-islands, if the wh-island is created by a wh-expression that has moved out of its own clause, extraction of another wh-expression from that clause is blocked. However, clusters allow a derivation that subverts this effect. This is because the subjunction of the cluster to a higher C will make any wh-expression which is a 2-part of that cluster also a 2-part of whatever the higher C is a 2-part of, and hence, by Angular Locality, able to subjoin outside of that higher C. This is the correct prediction. Bulgarian allows examples like the following, which, as we've seen, are impossible in English, Scandinavian languages, Italian, and Hebrew:

- (48) Čoveka, kojto ne znaeš [CP kakvo kazvat [CP če e kupil]]
 man who NEG know.2SG what say.3PL that has bought
 'The man who you don't know what they say that he bought ...'

In this example the wh-expression *kakvo*, "what," originates as the object of *kupil*, "bought," while the relative pronoun *kojto* is the subject of that verb. If *kakvo* and *kojto* subjoin separately to the lowest clause's C and c categories, *kakvo* will be stuck there by Angular Locality. However, if they form a cluster, with the cluster subjoining as the object of the verb, and *kojto* then internally subjoins to little v, both can have the appropriate interpretations. Crucially, the whole cluster can then subjoin to C, and then to the intermediate C position, where *kakvo* spells out. Since the cluster has subjoined to that intermediate C, *kojto* is a 2-part of the cluster, which is a 2-part of C, which is a 2 part of the O that introduces the clausal argument of the verb *znaeš*, "know." Angular Locality is satisfied, and *kojto* can subjoin to the (relative) C in the topmost clause, generating the example:

cluster formation to take place by bringing together wh-expressions already “merged” into their base structures, as Rudin and Grewendorf suggest. It does, however, allow the cluster to be constructed, and then syntactically “deconstructed,” and this has the (correct) consequence that the WIRE effect will be absent in such languages. This is because the cluster itself creates the relevant parthood relations to allow Transitivity to sidestep Angular Locality.¹⁰

As well as no WIRE effect, [+MSF] languages are predicted to allow violations of wh-islands, since the cluster can simply subjoin to an intermediate C, and then its 2 part can subjoin from there to a higher position. This is correct:

- (50) Vidjah edna kniga, kojato se čudja koj znae koj prodava
 saw.1SG a book which wonder.1SG who know.3SG who sells
 ‘I saw a book which I wonder who knows who sells.’

In this example, the relative expression *kojato* subjoins to *koj* in the lowest clause and the cluster subjoins in object position, with *koj* then subjoining to v where it gets interpreted as the subject. The cluster then subjoins to c in the intermediate clause, and *kojato* subjoins from there to the higher c, where it spells out.

The way in which [+MSF] languages violate wh-islands is somewhat different from the way in which languages like English, Hebrew, and the Scandinavian languages do. [+MSF] languages create a cluster, and subextraction from the cluster is possible; English, Hebrew, and Scandinavian languages have an “extra” escape hatch and no cluster formation. This is the reason they differ in whether they show the WIRE effect.

A final issue about multiple questions concerns languages where the wh-expressions are in situ. As is well known, wh-in-situ elements in English may scope outside of wh islands. Consider (51), where the expression *which book* may be interpreted either in situ, as in (51b), or, in conjunction with the matrix wh-expression, as a complex wh-quantifier with matrix scope (Baker 1970):

- (51) a. Which student knows where Mary bought which book?
 b. Bill does/Bill knows where Mary bought which book.
 c. Bill knows where Mary bought Aspects and Sue knows where Mary bought War and Peace.

I think the consensus on such cases is that they involve variable binding without movement (whether of an indefinite or of a choice function, see Reinhart 1997 for arguments that the latter is empirically preferable and Dayal 2016 for more general discussion). In the system I’ve developed here, nothing would rule out a subjunction analysis, which could straightforwardly be given a choice function semantics. The judgments are a bit delicate, but I think there is a WIRE

effect, suggesting that the relevant semantics is constructed through subjunction rather than directly on the surface position of the *wh*-expressions, and that a declustering analysis is not available¹¹:

- (52) a. Which student knows where Lucy said that Mary put which book?
 b. ?Bill knows where Lucy said that Mary put which book.
 c. *?Bill knows where Lucy said that Mary put Aspects and Sue knows
 where Lucy said that Mary put War and Peace.

There is much more to be said about the analysis of multiple-*wh* movement languages, and indeed *wh*-in-situ languages, but the system developed here together with the declustering analysis captures the overall distribution of a large array of facts in a way that emerges directly from the underlying theory.

5.5 Summary

When a 2-part of some category itself has a 2-part, an island effect emerges. The examples we have looked at in this chapter involve cases where a CP argument of a verb has a *wh*-expression as a 2-part of C, so that anything inside T is unable to subjoin outside of the CP. However, it may be the case that a *wh*-expression is in the specifier of C, but C itself is a 1-part of a higher category which then provides a free 2-part “slot” for subjunction. In such cases, the *wh*-island effect vanishes, and I’ve suggested that this is what is responsible for violations of *wh*-islands (following Preminger and Vikner).

There are, though, still cases where this “extra escape hatch” strategy won’t work, and that happens when two *wh*-expressions are extracted from a domain, and then one of these must subjoin yet higher. In this situation there is no way to satisfy Angular Locality. I’ve also shown how to extend this system to languages with multiple *wh*-movement.

6 Nominal Islands

6.1 Introduction

The last chapter argued that some fairly complex properties of *wh*-islands were amenable to an analysis in terms of Angular Locality, Transitivity, and Dimensionality. The core logic of the proposal is an old idea in generative syntax: to escape a locality domain, an SO must access a particular position, but if that position is blocked, it cannot, and hence it cannot escape the locality domain. In this chapter, I'll argue that the same logic can be successfully applied to some long-standing puzzles in extraction from nominal phrases in object position.

Consider (1):

- (1) a. When asked, booksellers usually recommend a book on a display table.
- b. *Which display table do booksellers usually recommend a book on?

The relevant reading of (1a) is that booksellers are influenced by their recent activities, and usually will recommend one of the books they have stacked on a display table. Under this reading, however, extraction of the object of the preposition is resolutely ill formed: (1b) cannot be a question about the location of the recommended book (see, e.g., Davies and Dubinsky 2003, who characterize these cases as involving extraction from nominal phrases headed by a class of what they call “concrete” nouns, where “concrete” is to be contrasted with “abstract”).

Compare (1) with (2):

- (2) a. When asked, booksellers usually recommend a book on politics.
- b. Which subject do booksellers usually recommend a book on?

When the noun is abstract and is interpreted as relational (that is, the PP expresses what the content or subject matter of the book is), extraction improves dramatically (Erteschik-Shir 1981). However, it is not always possible to extract from such relational phrases. If the nominal is definite, for example, extraction

is again degraded (cf. the Specificity Condition of Fiengo and Higginbotham 1981—I return below to the relatively weak effect with the definite article compared to demonstratives and possessors):

- (3) a. *Which subject do booksellers usually recommend that book on?
- b. *Which subject do booksellers usually recommend Chomsky's book on?
- c. ?Which theorem did the logician have to read the proof of very slowly?

Finally, as noted by Erteschik-Shir (1981) and discussed more recently by Davies and Dubinsky (2003), when a definite nominal phrase is the object of a verb of creation, extraction is unexpectedly improved:

- (4) a. Which subject did you write that book on?
- b. Who did you paint those pictures of?
- c. Which theorem did the logician develop the proof of very slowly?

Davies and Dubinsky argue that extraction from nominal complements of creation verbs when the noun is made definite by a possessor (as in (3b)) behaves differently from when it is made definite by a demonstrative or determiner. They suggest that such extraction is only possible when the possessor is coreferential with an argument of the selecting verb, giving the following contrast (see also Huang 2018):

- (5) Who did Tom_i write his_{i/*j} story about?

I'm not convinced this is the correct generalization. Consider a context where we are discussing a politician called Thomson who's written a new biography of a Roman emperor. I know that it was actually a ghostwriter who produced it but I don't know which emperor was its subject matter. I think in that situation it's perfectly grammatical for me to ask:

- (6) Which emperor did the ghostwriter write Thomson's new book about?

Here there's no coindexing, but the possessor doesn't block extraction. I think the effects that Davies and Dubinsky discuss are a side effect of the semantic connection between the subject of a verb of creation and the creator of the object of such a verb. If you dissociate these, as in my example, it's possible to have extraction. Possessors, then, behave just like other definites when they are in the complement of a verb of creation: they don't block extraction.

My proposal for analyzing these rather complex effects is essentially the same as the proposal for *wh*-islands: an object DP is a 2-part of O, and extraction from inside that DP must proceed via its 2-part, because of Angular Locality. In the cases where extraction is impossible, that is because either the 2-part of D is not reachable by the extractee (which will be the analysis of extraction from

“concrete” nominal phrases), or it is otherwise filled (which will be the analysis of the “Specificity Condition” cases). Verbs of creation select an extra layer in the DPs they take as objects, opening up another 2-part in a way reminiscent of the little *c* analysis of *wh*-island violations.

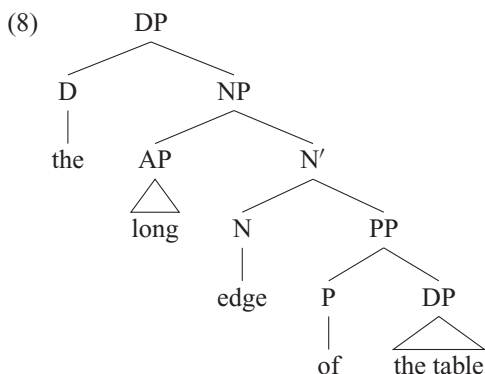
Much of this analysis will depend on a proposal for the structure of relational nominals that I made in Adger (2013), and I will briefly motivate this in the next section, before turning to the consequences of this idea for extraction.

6.2 Nominal Structure

Consider the examples in (7):

- (7) a. I sawed the long edge of the table in half.
 b. We met an uncle of the children in the park.
 c. She painted the picture of the castle in watercolor.

These examples involve relational nouns: *edge* is a relation that connects an object to a particular part of an object, *uncle* is a relation that connects one individual to another via a kinship relation, and *picture* relates an object to a representation of that object. Classically, these kinds of nouns are analyzed in a similar way to argument taking verbs: they have a semantics that involves two arguments (e.g., $\lambda x \lambda y. \text{edge}(x, y)$), and the prepositional phrase headed by *of* is taken to be a complement of the N, much like the object of a verb is taken to be a complement of the verb (see, for example, Chomsky 1970; Jackendoff 1977; or Abney 1987):



However, there are good reasons to reject this view. The first is what I called, in Adger (2013), the Optionality Generalization for Relational Expressions (OGRE): verbs can take their arguments obligatorily, while nouns never do. That is, there are vanishingly few cases of nouns that obligatorily take a PP argument

(the very few potential counterexamples are idioms). In the three examples in (7), the putative arguments of the relational nouns can all be dropped:

- (9) a. I sawed the long edge in half.
 b. We met an uncle in the park.
 c. She painted the picture in watercolor.

This is a mystery under the standard account, and is usually handled by a stipulation (e.g., type-shifting in the semantics, as in Partee 1997, Barker 2010, and for a somewhat more nuanced view, Vikner and Jensen 2002). To account for OGRE, I argued instead that all nominals are actually sortal (that is, they have no argument structure) and that their apparent semantic requirement for an argument is down to pragmatic constraints on their interpretation. This explains why it is so easy to coerce relational nouns into sortal nouns and vice versa (Löbner 1985). In the same way that the argument of relational nouns can be dropped, it is possible to give nonrelational nouns arguments:

- (10) a. A single cloud of the weather-front drifted into view.
 b. The glamour of the occasion surprised me.
 c. That teacup of the queen is too tacky to buy.
 d. The grand high wizard of Oz will now see you.
 e. Sitting on the roofs of Oxford she sees a bird, the daemon of a witch.

Here we see sortal nouns being coerced into part-whole, property, representation, role, and kinship syntax. The same sortal noun can also appear with different semantic relations, as in the following examples, where we see teacup in both representation and part relations:

- (11) a. (At a shop in the Royal Palace) How much is that teacup of the Queen?
 b. You've now broken two teacups of that new set we got from your mother.

In fact, it seems to be the preposition that drives the interpretation of the semantic relation, rather than the noun. A striking example of this comes from Norwegian, where different semantic relations are expressed by different prepositions (Adger 2013, pp. 68–69):

- (12) a. Jeg liker den ny-e høvding-en i by-en.
 I like DEF.M.SG new-DEF chief-DEF.PL in town-M.DEF
 'I like the new chief of the town.'
 b. Jeg liker den lang-e kant-en på bord-et.
 I like DEF.M.SG long-DEF edge-DEF on table-N.DEF
 'I like the long edge of the table.'
 c. Jeg liker den ny-e farge-n på romm-et.
 I like DEF.M.SG new-DEF color-M.DEF on room-N.DEF
 'I like the new color of the room.'

- d. Jeg liker hod-et til Ola.
 I like head-N.DEF to Ola
 'I like Ola's head.'

It doesn't seem to be the case that the nouns select the prepositions since, if we put the preposition (*av*), which usually denotes the substance from which an object is made, with a noun like *tekanna*, "teapot," we derive the pragmatically implausible meaning in (13), suggesting that the relation is inherent in the preposition, not in the noun:

- (13) *tekanna* *av vann*
 teapot.DEF of water
 'The teapot made of water.'

To account for this kind of data (and more), I suggested that what appear to be arguments of nominals are actually arguments of a functional category, itself with relational semantics, that is projected rather high up in the nominal Extended Projection, higher than modifiers like adjectives, numerals, and even demonstratives. I called this category $\bar{P}$ (pronounced as the Hebrew letter kof) and argued that (13) should be replaced by a constituent structure that looks more like (14):

- (14)
-
- ```

graph TD
 DP --> Pbar1[P̄]
 Pbar1 --> DetP
 Pbar1 --> Pbar2[P̄]
 DetP --> Det[Det]
 Det --> the[the]
 DetP --> NP
 NP --> AP
 AP --> long[long]
 NP --> N1[N]
 N1 --> edge[edge]
 Pbar2 --> PP
 PP --> Pof[P:of]
 PP --> DP2[DP]
 DP2 --> thetable[the table]

```

I will revise this structure below to make it more consistent with the mereological approach developed in this book, but the crucial difference from the more standard approach in (8) is that the PP is outside of the constituent formed by the head noun and its modifiers: there is no constituent containing N and PP to the exclusion of AP, determiners and demonstratives (and, as we'll see directly, other nominal modifiers). In addition,  $\bar{P}$  is not a part of the Extended Projection of the head noun *edge*, so it cannot be selected by that noun. This makes it different from little *v*, which can be selected by the verb which it is directly above in the verbal Extended Projection. I cashed out this difference in Adger

(2013) via a difference in the semantic mode of composition: following Kratzer (1996), the composition of the external argument of *v* involves Event Identification, while I argued that the two arguments of  $\bar{p}$  are introduced by recursing the category, introducing one argument via functional application and the other via Predicate Modification. In this sense  $\bar{p}$  is semantically composed like an adpositional phrase, taking its own argument and then combining with the projection of the noun as a modifier. However, since unselected functional structure can be projected or not (a noun can be singular or plural, a verb tensed or untensed), OGRE follows.

This structure also makes sense of the second generalization that I drew attention to in Adger (2013), the PP Peripherality Generalization (see also Giurgea 2009 for a related though somewhat different claim):

(15) PP Peripherality (Adger 2013, p. 93)

When AP modifiers and PP complements both occur to one side of N inside a noun phrase, the PP is separated from the N by the AP.

This generalization captures various typological facts, such as the finding that, in N initial languages, the order is typically  $N \neg AP \neg PP$ , as opposed to the expected  $N \neg PP \neg AP$  (which never occurs). In Gaelic, for example, we find (16a), where the adjective precedes the PP, but (16b) is totally ungrammatical:

- (16) a. An dealbh mòr de Mhàiri  
           the picture big of Màiri  
           ‘The big picture of Màiri’  
       b. \*An dealbh de Mhàiri mòr  
           the picture of Màiri big  
           for ‘The big picture of Màiri’

PP Peripherality also captures the fact that in N-final languages, the order  $AP \neg PP \neg N$  is typically unavailable.<sup>1</sup> PP Peripherality follows from the structure in (14), since the adjectival modifier is closer to the noun than the PP, but is mysterious under the more classical analysis.

This approach also explains certain constituency effects that suggest that the noun plus its modifiers act as a unit to the exclusion of the PP. For example, in Gaelic, demonstratives are postnominal, but must occur with a prenominal definite article.

- (17) \*(An) dealbh mòr siud  
           the picture big that  
           ‘That big picture.’

As with adjectives, the PP must occur after the demonstrative

- (18) a. An dealbh mòr siud de Mhàiri  
           the picture big that of Mhàiri  
           ‘That big picture of Mhàiri’  
       b. \*An dealbh mòr de Mhàiri siud  
           the picture big of Mhàiri that  
           for ‘That big picture of Mhàiri’

We have seen that the “argument” PP cannot occur between the N and the AP, or between the AP and a demonstrative. There is one position for the PP, and it is the rightmost position, as predicted by the  $\bar{p}$  analysis. It seems unlikely that this is extraposition of the PP, as it is obligatory and the PP is never found in its putative base position.<sup>2</sup>

Moreover, the unit consisting of the article, noun, and adjectives can be coordinated to the exclusion of the demonstrative as follows:

- (19) Na dealbhan mòra agus \*(na) h-ìomhaighean beaga siud  
       the picture.PL big.PL and the.PL image.PL small.PL that  
       ‘Those big pictures and small images’

Here the article must be repeated, suggesting that there is a constituent, [Det N AP], that is coordinated in a way that satisfies the demonstrative’s need to attach to a definite. In such cases, the demonstrative takes scope over both coordinates, so the structure is schematically as in (20):

- (20) [[the pictures big] and [the images small]] those

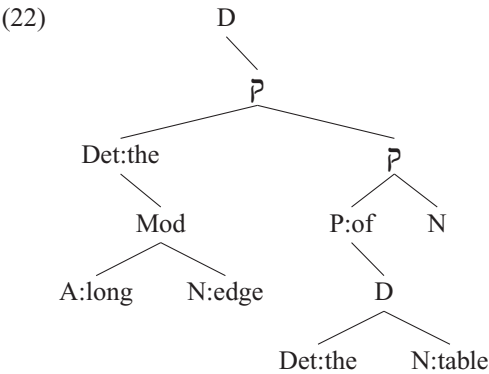
A PP “argument” obligatorily occurs outside of this coordinated structure:

- (21) Na dealbhan mòra agus na h-ìomhaighean beaga siud de Mhàiri  
       the picture.PL big.PL and the.PL image.PL small.PL that of Mhàiri  
       ‘Those big pictures and small images of Mhàiri’

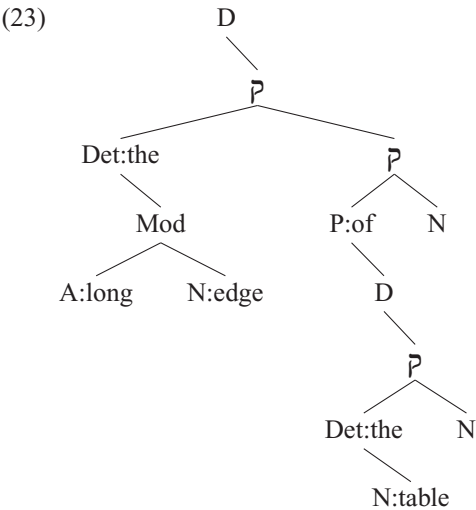
One might treat this as an Across-The-Board rightward extraposition of the PP, but, again, it would have to be an obligatory extraposition. A more straightforward analysis is available if the PP is introduced outside of the constituent containing the demonstrative.

Given these three arguments (OGRE, PP-Peripherality, and constituency), I’ll adopt the  $\bar{p}$  approach to relational nominal phrases: relationality is built up via fairly high functional structure.

If we rethink the structure in (14) in terms of the mereological system developed in this book, and assuming that adjectives are 2-parts of a category Mod in the nominal Extended Projection, it looks as follows:



However, now we have a structure where the topmost D has  $\bar{P}$  as a 1-part while the lower D has N as a 1-part. In Adger (2013) I suggested that actually all nominal Extended Projections that project to D involve  $\bar{P}$ , but for sortal nouns  $\bar{P}$  is intransitive, and the semantic relation is the one-place concept *THING* (or perhaps other semantic classes of intransitive concepts). If this is the right way to go, our final structure is that in (23):



The sequence of categories  $N-\bar{P}-\bar{P}$  in the main DP is semantically interpreted as a predicate which we can write as  $\lambda x \lambda y. PART(x, y)$ . One of the arguments of that predicate is the 2-part of the lower  $\bar{P}$ , *of the table*. The other argument is the 2-part of the higher  $\bar{P}$ , a constituent which contains the determiner, adjective and the noun *edge*. This is combined with  $\bar{P}$  via Predicate Modification. Semantically, the  $\bar{P}$  structure then delivers a meaning which one could represent as something like (24) (see Adger 2013 for details):

(24)  $\lambda x. \text{PART}(x, \text{the-table}) \wedge \text{edge}(x) \wedge \text{long}(x)$

(24) can then be bound by higher quantificational operators to deliver the meaning of the DP.

The sequence of categories N- $\bar{p}$  in the subordinate DP is semantically interpreted as a predicate which we can write as  $\lambda x. \text{THING}(x)$ .  $\bar{p}$  ensures that that is modified by  $\lambda x. \text{table}(x)$ , giving the lower DP the following rough meaning:

(25)  $\lambda x. \text{THING}(x) \wedge \text{table}(x)$

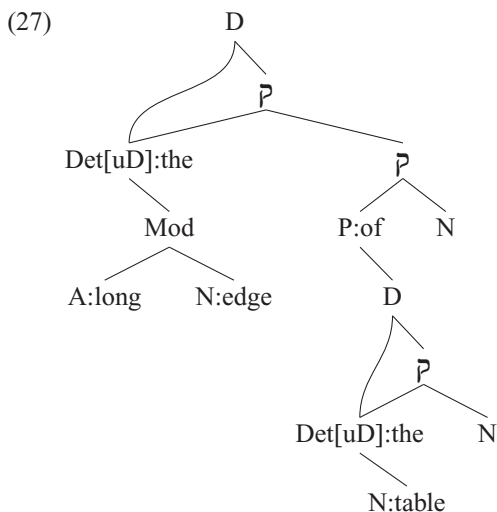
I have given the category of the 2-part of  $\bar{p}$  in (23) the name Det, again following Adger (2013). Det encodes morphosyntactic definiteness and is generally the position for spellout of definite articles. Semantically, Det does not encode Discourse Familiarity. That is the job of D. The whole constituent D is interpreted as definite in certain circumstances. I will assume for the moment that the relevant notion of definiteness is Discourse Familiarity (as in Heim 1982 or Kamp 1981, Enç 1991 et seq.), though we will return to the issue when we discuss weak versus strong determiners below.

I argued in Adger (2013) that one of the circumstances that makes D definite is when it has a 2-part. This was motivated by a number of morphosyntactic and semantic phenomena. I'll state the idea explicitly here, as the one-way implication in (26):

(26) D-Interpretation:

If D has a 2-part, then D is interpreted as discourse familiar

This implies that the analysis of *the long edge of the table* may be as in (27):



Det does not always subjoin to D (it does not, for example, in the case of nonspecific indefinites), so we need a way to distinguish the two situations. I suggest that Det may bear a [uD] feature, which is checked when Det subjoins to D. If Det does not bear [uD], it does not subjoin to D, and the DP is not required to be interpreted as discourse familiar. I will spell out in more detail the consequences of this set of ideas as I develop the account of extraction in the remainder of this chapter, as well as motivating it cross-linguistically.

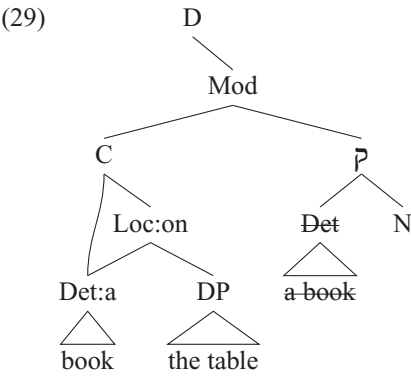
### 6.3 Extraction from Objects

#### 6.3.1 Concrete Nominal Phrases

Before turning to extraction from relational expressions, I briefly discuss sortally interpreted nominal phrases like the case we began with, *a book on the table*. These do not have a bipartite  $\bar{P}$  structure, since there is no argument structure involved: the locative PP is interpreted as a one-place predicate of the noun phrase. I've assumed up to this point that attributive APs are 2-parts of a category Mod (or a number of distinct categories, depending on how Cartographic one wishes to be). Locations are, however, predication, as opposed to attributive. Take, for example, (28), which shows that the predication relation between PPs and their NPs is susceptible to temporal modification (28a), while attributive adjectival predication is not (28b–c). (28d) shows that if an AP appears postnominally as a reduced relative, temporal modification of the predication relation is again possible:

- (28) a. The book on the display table yesterday has now been sold.  
 b. \*The new book yesterday has now been sold.  
 c. \*The new yesterday book has now been sold.  
 d. The books new to the store yesterday will sell quickly.

I'll continue to assume that this PP is a 2-part of a Mod category (perhaps specialized to this locative relation) in the Extended Projection of N, but I propose that we treat locational PPs in noun phrases as a kind of reduced relative and that the relativization is of the matching rather than the raising sort (Hulsey and Sauerland 2006) as follows:



(29) is a matching relative clause structure where the subject of the locative predication *a book* subjoins to C, creating a predicate that modifies the head noun of the main Extended Projection. The latter is not spelled out in head-initial relative structures (though it is in head-final structures, as in, e.g., Cinque 2020).

If this is the correct analysis for locative PPs, it will already block extraction of *the table* and rule out examples like (1b), repeated here:

(30) \*Which display table do booksellers usually recommend a book on?

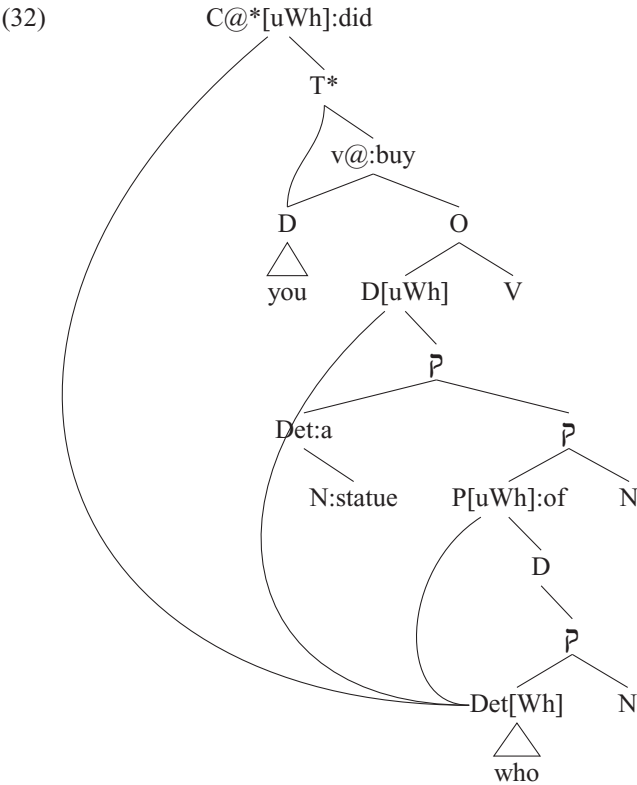
Since the constituent *a book* has subjoined to C, and C is a 2-part of Mod, *the table* (or a wh-version of that DP) cannot also subjoin out of that same C, as Angular Locality would be violated.<sup>3</sup>

With this in hand, let’s now look at extraction from nominal phrases that involve a relational interpretation.

6.3.2 Relational Nominal Phrases

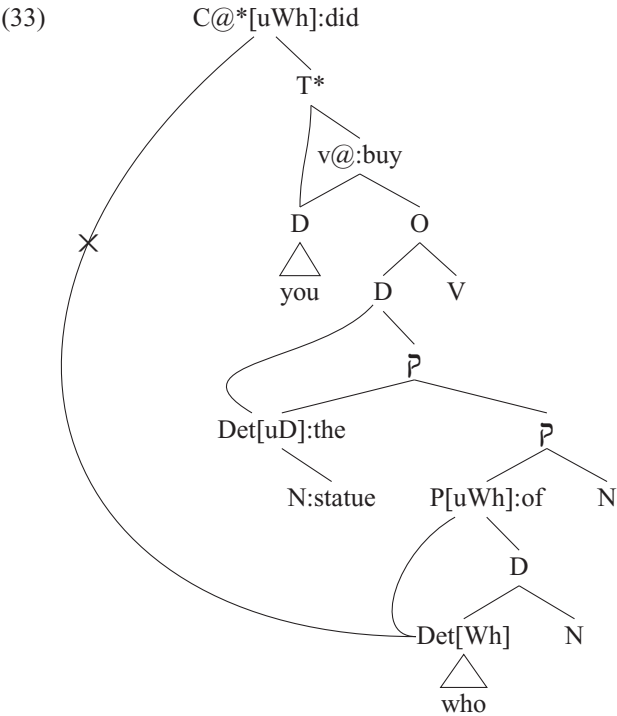
For a simple nominal object involving a relational reading, such as in (31), the prediction of the system is that extraction should be well formed, as nothing will bar subjunction to D of the wh-expression as in (32):

(31) Who did you buy a statue of?



In this structure, the nominal phrase from which extraction takes place (*a statue of who*) is indefinite, so its D has no 2-part. This allows the extractee *who* to subjoin to that D. To allow this, I’ve specified D with [uWh], so this D is similar in type to intermediate Cs in the path of long-distance extraction.<sup>4</sup> This structure also strands *of*, which, as we saw in chapter 4 involves subjunction to the lowest constituent containing *who*. This subjunction makes *who*, by Transitivity, a 2-part of *P*, allowing subjunction to D[uWh], and then to matrix C. In the pied-piping version of the sentence (*Of whom did you buy a statue*), it is the lower *P* that subjoins to the higher D, checking its [uWh] by Transitivity, as we saw when we discussed pied-piping in chapter 4.

Now let’s examine what happens when the object is definite (that is, discourse familiar). Things look quite different in this case as D-Interpretation will require subjunction of Det[uD] to D, as in (33):



As shown in the diagram, the subjunction of  $Det[uD]$  to  $D$  also has a syntactic effect. It “uses up” the 2-part of  $D$ , blocking subjunction of the lower  $wh$ -expression, and predicting that subjunction of the  $wh$ -expression from the nominal phrase should be blocked.

(34) ?Who did you buy the statue of?

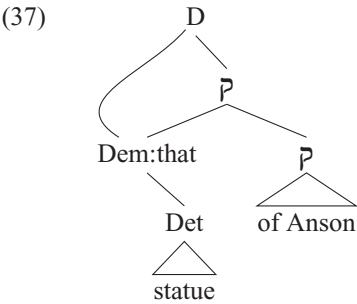
As mentioned in the last section, prenominal possessors (though not postnominal possessives) and demonstratives have the same effect<sup>5</sup>:

- (35) a. \*Who did you buy that statue of?  
b. \*Who did you buy Anson’s statue of?  
c. Who did you buy a statue of of Anson’s?

We find the same effect when the  $P$  is pied-piped, although for me, at least, the register of these makes them generally more awkward/literary:

- (36) a. Of who(m) did you buy a statue?  
b. ??Of who(m) did you buy the statue?  
c. \*Of who(m) did you buy that statue?  
d. \*Of who(m) did you buy Anson’s statue?

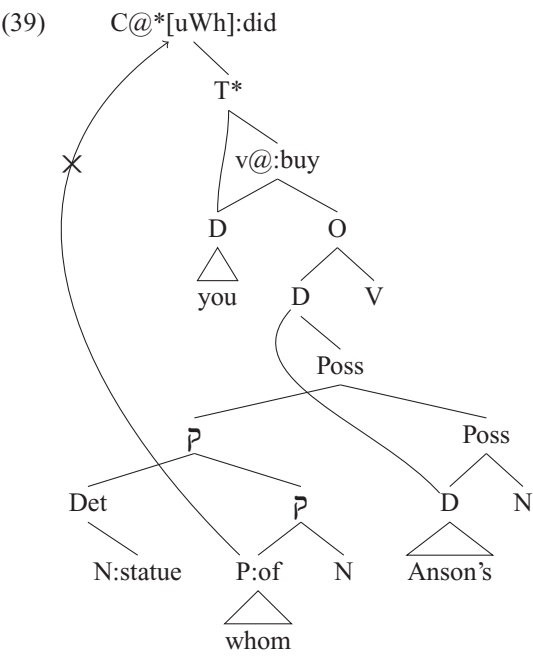
I'll follow Adger (2013) in taking demonstratives to be spell-outs of a category Dem that takes Det as a 1-part. Dem raises to D, in the same way as Det does in the structure above, with the same blocking effect:



In languages like English, the span of Dem and Det is spelled out as the demonstrative. In a language like Gaelic, where demonstratives obligatorily occur with definite articles, they are spelled out separately:

- (38) an cat sin  
the cat that  
'that cat'

I will take possessors to be introduced by a Poss category, which is relational just like  $\bar{P}$ , but which introduces a generalized possessor relation along the lines of Barker (2010). Poss takes  $\bar{P}$  as its higher 2-part and has the possessor as its lower argument. In English, the possessor subjoins to D, again blocking any further subjunction from inside D. This captures the absence of a definite article, since Det cannot also subjoin to D. See Adger (2013) for extensive argument for this kind of analysis of possessors. The analysis of (36d) is then as in (39).



Many other languages, although not all, have this same set of effects in extraction from nominal phrases. German disallows extraction from the same kinds of definite nominals as does English (examples from Müller et al. 2022):

- (40) a. Worüber hat der Fritz [NP ein Buch ] gelesen?  
about.what has the Fritz.NOM a book.ACC read  
‘What has Fritz read a book about?’  
b. ?\*Worüber hat der Fritz [NP das Buch ] gelesen?  
about.what has the Fritz.NOM the book.ACC read  
‘What has Fritz read the book about?’  
c. \*Worüber hat die Maria [NP Fritzens Buch ] gelesen?  
about.what has the Maria.NOM Fritz.GEN book.ACC read  
‘What has Maria read Fritz’s book about?’

This phenomenon is also found in wh-in-situ languages like Persian (Karimi 1999) where a wh-expression can take matrix scope from an indefinite, but not from a nominal phrase with a demonstrative:

- (41) a. Kimea diruz [NP ye she’r az ki ] xund?  
Kimea yesterday a poem by who read  
‘Who did Kimea read a poem by?’

- b. \*Kimea [<sub>NP</sub> in she'r az ki ] ro xund?  
 Kimea this poem by who SPC read  
 'Who did Kimea read this poem by?'

Similarly, in Mandarin, my consultants only get an echo question interpretation for in-situ-wh expressions inside nominal phrases headed by demonstratives.

- (42) a. Ni faxian-le [<sub>NP</sub> shei-de diaoxiang]?  
 you find-LE who-DE statue  
 'Who did you find a statue of?'  
 b. Ni faxian-le [<sub>NP</sub> na-ge shei-de diaoxiang]?  
 you find-LE that-CL who-DE statue  
 'You found that statue of who?'

Huang (1982) notes the same effect in cases of wh-in-situ inside relative clauses in Mandarin:

- (43) a. [<sub>NP</sub> shei mai-de shu] zui hao?  
 who bought-DE book most good  
 'The book that who bought is best?'  
 b. \*[<sub>NP</sub> shei mai-de nei ben shu] zui hao?  
 who bought-DE that CL book most good  
 That book that who bought is best?'

The wh-expression inside the relative can take matrix scope, and the sentence can be read as an information-seeking question in (43a), where the nominal expression is indefinite, but not in (43b) in the presence of the demonstrative.<sup>6</sup>

The following Japanese examples show the same effect in that language for relative clauses: matrix scope of the wh-in-situ expression is possible from an indefinite nominal phrase but not from a definite one with a demonstrative:

- (44) a. John-wa [<sub>NP</sub> dare-ga kaita hor]-o yonda no.  
 John-TOP who-NOM wrote book-ACC read Q  
 'John read a book that who wrote?'  
 b. \*John-wa [<sub>NP</sub> dare-ga kaita sono hon]-o yonda no.  
 John-TOP who-NOM wrote that book-ACC read Q  
 'John read that book that who wrote?'

The contrast between the indefinite and demonstrative examples in wh-in-situ languages can be accounted for by the standard assumption that the wh-expression takes clausal scope via a syntactic mechanism. In the theory developed here, that mechanism is subjunction. The wh-expression is blocked from subjoining to its clausal scope position (I assume C) in the examples with demonstratives, because it cannot subjoin out of D, as the 2-part of D is already taken up by the SO that makes D definite. I return in chapter 8 to why sometimes it is possible to subjoin from relative clauses.

Karimi notes that, for Persian, a language that allows scrambling, overt wh-scrambling is blocked in the same contexts, suggesting that overt and covert subjunction is equally affected:

- (45) a. az ki Kimea [<sub>NP</sub> ye she're] xund?  
           by whom Kimea a poem read  
           'By whom did Kimea read a poem?'  
       b. \*az ki Kimea [<sub>NP</sub> in she're ] ro xund?  
           by whom Kimea this poem SPC read  
           'By whom did Kimea read this poem.'

This general pattern of behavior, where extraction from or of a PP inside a definite object is degraded, has been tackled in previous work via the Specificity Condition of Fiengo and Higginbotham (1981), which bars free variables inside referential nominals (the variable must be bound inside the nominal). Bowers (1987) implements the proposal syntactically in the Barriers framework. Bowers argues that specific and definite nominal phrases (and also those containing strong quantifiers, demonstratives, and prenominal possessives) have a DP layer that is otherwise missing. That DP layer essentially protects the NP category from being governed by the verb, which turns the NP into a category that blocks movement (technically, it becomes a blocking category and hence a barrier, which entails that the DP layer becomes a barrier by inheritance; subadjacency then rules out extraction across two barriers). The resulting configuration accounts for the island effect, since it will stop extraction that crosses definite determiners, strong quantifiers, prenominal possessors, and demonstratives, all of which entail the extra D layer but allow extraction from NP with no D layer above it.<sup>7</sup>

The mereological system developed here captures the same effect via Angular Locality plus the idea that, sometimes, the D category has its edge used up.<sup>8</sup> We will see below that this approach is an improvement over the Specificity Condition, as there are cases where semantically definite or specific nominal phrases are transparent to extraction. This suggests that syntax is in play. We'll also see that the Angular Locality analysis is an improvement over a Bowers-style approach, which relies on the presence or absence of a D-layer, as there are phenomena where there is clearly a D-layer, but extraction is possible.

The mereological system relies on D's 2-part being filled, but the cases we have seen so far show that only rather abstractly, since the Det to D subjunction is very local and doesn't have an effect on the order of spelled-out material. There is, however, striking evidence from Mandarin for the correlation between D's 2-part being filled, the interpretation of D, and extraction from inside it. This evidence both backs up the formulation of D-Interpretation in (26), that

Ds with 2-parts are interpreted in a special way, and provides further evidence that when D has its 2-part “used up,” it is impossible to extract from.

Zhang (2006) shows that there are two surface positions for certain modifiers in Mandarin nominal phrases and argues that they are related by movement (see Zhang 2015 and Hall 2015 for more detailed discussion):

- (46) a. [dai yanjing de] san ge xuesheng  
           wear glasses DE three CL student  
           ‘three students who wear glasses’  
       b. san ge [dai yanjing de] xuesheng  
           three CL wear glasses DE student  
           ‘three students who wear glasses’

In (46a), the relative clause *dai yanjing de*, “who wear glasses” appears to the left of the numeral *san*, “three,” and its classifier *ge*, while in (46b) it appears to the right of these. Zhang calls the former case outer modification and the latter inner modification. She argues that both these examples can be indefinite, but that the nominal phrase with outer modification is obligatorily specific, that is, discourse familiar. As evidence for this, she notes that specific nominal phrases cannot appear in constructions of appearance (or disappearance), which she takes to be existential (47a). As the contrast between (47b) and (47c) shows, nominal phrases with outer modification are also barred from these contexts:

- (47) a. Lai-le liang ge ren / \*Daiyu / \*na ge ren.  
           come-PRF two CL person / Daiyu / that CL person  
           ‘Two people/\*Daiyu/that person have/has come.’  
       b. Lai-le san ge [dai yanjing de] xuesheng.  
           come-PRF three CL wear glasses DE student  
           ‘Three students who wear glasses have come.’  
       c. \*Lai-le [dai yanjing de] san ge xuesheng.  
           come-PRF wear glasses DE three CL student  
           for ‘Three students who wear glasses have come.’

A second argument for the semantic interpretation of NPs with outer modification as specific comes from verbs of creation. Compare the (a) and (b) subcases in (48) and (49):

- (48) a. Baoyu meitian yao xie san fen [guanyu shichang-jingji de]  
           Baoyu everyday must write three CL about market-economy DE  
           baogao.  
           report  
           ‘Baoyu must write three reports on market economy everyday.’

- b. \*Baoyu meitian yao xie [guanyu shichang-jingji de] san fen  
 Baoyu everyday must write about market-economy DE three CL  
 baogao.  
 report  
 ‘Baoyu must write three reports on market economy everyday.’
- (49) a. Baoyu mei nian wancheng wu pian [guanyu dongci de] lunwen.  
 Baoyu every year complete five CL about verb DE paper  
 ‘Baoyu completes five papers on verbs every year.’
- b. \*Baoyu mei nian wancheng [guanyu dongci de] wu pian lunwen.  
 Baoyu every year complete about verb DE five CL paper  
 ‘Baoyu completes five papers on verbs every year.’

These patterns follow rather neatly from the system developed here, on the assumption that verbs of creation in Mandarin obligatorily select a non-discourse-familiar nominal object (that is, their semantics is such that their object must be interpreted as newly introduced to the discourse, cf. Diesing 1992, and section 6.7 of this chapter). We simply say that D in Mandarin, when it has a 2-part, is interpreted as discourse familiar, following D-Interpretation (26), and that the subjunction of a lower modifier to D provides it with that 2-part (in effect signaling syntactically rather than morphologically the relevant semantico-pragmatic effect).

Adopting this analysis, we then also predict that outer modification of the nominal should correlate with the impossibility of extraction. If the modifier is a 2-part of D in the outer modification constructions, then extraction from lower in the nominal phrase is predicted to be impossible via Angular Locality. Zhang provides evidence that confirms this prediction. In (50a), we see internal modification correlating with the possibility of a focalization movement that fronts the head noun *lishi-shu*, ‘history book’ to a preverbal position. I’ve bolded the nominal in its base and fronted positions.

- (50) a. Akiu du-guo-le liang ben [guanyu Xizang de] **lishi-shu**.  
 Akiu read-EXP-PRF two CL about Tibet DE history-book  
 ‘Akiu has read two history books on Tibet.’
- b. Akiu (xingkui) **lishi-shu** du-guo-le liang ben [guanyu  
 Akiu fortunately history-book read-EXP-PRF two CL about  
 Xizang de].  
 Tibet DE  
 ‘Akiu has (fortunately) read two history books on Tibet.’

Compare this with (51), where we have outer modification. It is then impossible to front the head nominal:

- (51) a. Akiu du-guo-le [guanyu Xizang de] liang ben **lishi-shu**.  
 Akiu read-EXP-PRF about Tibet DE two CL history-book  
 ‘Akiu has read two history books on Tibet.’  
 b. \*Akiu (xingkui) **lishi-shu** du-guo-le [guanyu Xizang de]  
 Akiu fortunately history-book read-EXP-PRF about Tibet DE  
 liang ben.  
 two CL  
 ‘Akiu has (fortunately) read two history books on Tibet.’

Although Mandarin lacks a definite article, it uses syntactic means to signal the specificity/discourse familiarity of D via subjunction of a modifier as a 2-part of D. This then has syntactic consequences, as predicted by the theory, impacting selection and extraction.

#### 6.4 Articleless Languages

The mereological analysis of definiteness I’ve developed relies on there being a D category. However, there is a prominent approach to the Slavic languages that lack definite articles that takes these languages to lack a D category entirely. If this is correct, it might be thought problematic that D, in my system, is central to the calculation of definiteness, even in languages without overt articles like Mandarin. However, I’ll argue that the current system does a good, perhaps better, job of capturing the relevant generalizations in these languages too.

Bošković (2005) proposes that languages that lack definite articles lack a D layer, with concomitant syntactic effects. He assumes that NP is not a locality barrier (a phase) while DP is.<sup>9</sup> If the relevant Slavic languages lack DP, then we can explain why they allow APs or demonstratives, for example, to extract out of the nominal, as in (52) from Russian:<sup>10</sup>

- (52) Doroguju/tu on videl mašinu.  
 expensive/that he saw car  
 ‘He saw the expensive/that car.’

In the mereological system, an alternative perspective is available, and I think, to be preferred. Perhaps connected to the absence of articles, we might assume that in these languages, Det simply does not subjoin to D, leaving the 2-part “slot” of D free for extraction of the adjective.

Similarly, the mereological system predicts the absence of island effects for nominal phrases in object positions in these languages: D has a free 2-part “slot,” so extraction from nominal objects in such languages should be possible, as in the following example from Russian (Stjepanovic 1998):

- (53) [Iz kakogo goroda]<sub>i</sub> ty videl [<sub>NP</sub> devušek t<sub>i</sub> ]?  
 from which city you saw girls  
 ‘Girls from which city did you see?’

This same set of languages does not appear to show Fiengo and Higginbotham style specificity effects for strong quantifiers, demonstratives, or possessives, suggesting, again, that the D in these languages does not require a 2-part:

- (54) O kojem piscu je pročitao [svaku knjigu/ sve knjige/ (tu) tvoju  
 about which writer is read every book/ all books/ (that) your  
 knjigu]?  
 book  
 ‘\*About which writer did he read every book/all books/this book of yours?’

However, semantically, these nominal phrases are still definite. This suggests that D-Interpretation, repeated here, is a one-way generalization:

- (55) D-Interpretation:

If D has a 2-part, then it is interpreted as discourse familiar.

D can have the semantic properties of discourse familiarity, specificity, and so on, independently of having a 2-part, either simply optionally (and hence determinable by pragmatic context) or via some feature specification.

As I mentioned above, Bošković develops a theory that attempts to tie NP’s transparency to extraction of nominal phrases in these languages to the idea that there is no D layer, and therefore no DP phase; it follows that extraction is unproblematic. The theory also extends to an explanation of the impossibility of Left Branch extraction in a language like English, compared with its well-formedness in the relevant Slavic languages:

- (56) a. \*Expensive he saw car.  
 b. \*That he saw car.  
 c. \*David’s he saw car.

Bošković argues that extraction of an adjectival phrase to the edge of DP in (56a) is barred by a particular formulation of Antilocality. Antilocality requires movement to cross one full phrasal category. If AP is adjoined to NP it cannot move to the specifier of DP because, in doing so, it would not cross a full NP

category (it would only cross a segment). However, AP cannot exit DP without going through its edge, given that DP is a phase, so it is unable to extract. Bošković's proposal derives a putative generalization that languages with definite articles (i.e., with a D layer) will disallow left-branch extractions of APs. If demonstratives are also adjoined to NP, the same explanation would extend to them: there is no derivation that allows these elements to stop in the edge of DP, given Antilocality of the sort assumed. He states the generalization as follows (Bošković 2008, see also Bošković 2013):

(57) Only languages without articles may allow LBE.<sup>11</sup>

The mereological system doesn't derive this generalization. Rather it captures the following generalization:

(58) Extraction (subjunction) from inside a nominal phrase is impossible just when the nominal phrase is syntactically definite (that is, where D has a 2-part).

There are two points here that are worth highlighting. The syntactic operation that is blocked is subjunction, which has the effect that overt extraction is blocked. However, assuming subjunction is also the operation that places elements in their syntactic scope position (for example, clausal scope for *wh*-expressions), the generalization also extends to these. The second point is that it is syntactic definiteness that is at issue, and a D can be semantically definite without being syntactically definite (though not vice versa, if the generalization about D-Interpretation is correct). Putting these together, the generalization will allow the circumstance where there is extraction from some semantically definite phrases as in (54). This property of the system distinguishes it from Fiengo and Higginbotham's Specificity Condition.

Moreover, (58) is not a parametric generalization at the level of a language like (57), but is rather a cross-linguistic constraint on how extraction proceeds from nominal phrases irrespective of whether the language has definite articles or not. It will correctly block extraction of, for example, an AP from a definite DP in English, since Det will fill the specifier of D if the definite article is overt. It will not, however, block either extraction of an AP from an indefinite, or left-branch extraction of a demonstrative (or possessor) that is a 2-part of D (since the demonstrative or possessor will, by Transitivity, be a 2-part of the higher clausal structure, and therefore, by Angular Locality, be able to subjoin to that higher clausal structure).

I think that this is the right result: the generalization that Bošković draws about LBE and languages having, or not having, articles, is not correct. There are a number of languages that have articles but that allow Left Branch extraction. Little (2020), for example, discusses Ch'ol examples where a numeral may

extract leftward from its noun (possessors may also extract with slightly different restrictions on the syntactic context of the nominal phrase that is extracted from):

- (59) Cha'-kojty ta' i-k'el-e wakax aj-Rosa.  
 two-CLF PFV A3-see-TV cow NC-Rosa  
 'Rosa saw two cows.'

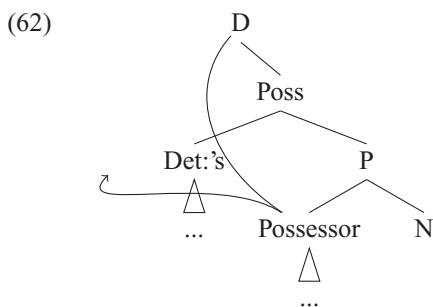
The language, however, has definite articles signaling Discourse Familiarity (Little 2020, p. 78ff)

- (60) jiñi cha'-kojty i-säsäk wakax aj-Rosa  
 DET two-CLF A3-white cow NC-Rosa  
 'the two white cows of Rosa's'

Closer to home, recall that I discussed Davis (2021) in chapter 4. Davis shows that, in certain varieties of English, Left Branch extractions of possessors are well-formed, giving examples like the following:

- (61) Mary is the author [<sub>CP</sub> who<sub>k</sub> they said [<sub>CP</sub> [<sub>TP</sub> t<sub>k</sub>'s new book] is good]].

This suggests that the 2-part of D position can, in fact, potentially be available to possessors as a means of extracting from DP, as the theory predicts. If possessors can subjoin to become 2-parts of D, and Det spells out in such contexts as 's, a continuation of a derivation that then subjoins the possessor higher is expected to be possible:



The argument however, is more indirect. Bosković's account is fairly narrow: it only applies to adjectival extraction from nominal expressions, while what we have in the Ch'ol and English cases are numeral and possessor extraction respectively. If numerals are in the specifiers of a numeral projection and possessors are in the specifier of a possessor projection, they would, in fact, cross full categories if they moved to the specifier of DP. This highlights the dependence of the Antilocality approach on the analysis of the extractee as adjoined to NP. The other side of the coin, of course, is that if APs are in dedicated projections, as proposed in, for example, Cinque (2010), or if demonstratives are, as argued

in Adger (2013), then the Antilocality approach won't get off the ground as an explanation of why these cannot (apparently) extract in languages with articles.

In fact, it turns out that there are languages with articles where left branch extractions of adjectives (and numerals) is licit. Bulgarian and Macedonian are both languages with articles, which are suffixed to the initial constituent of the nominal phrase. Stojković (2019) gives the following cases from Bulgarian

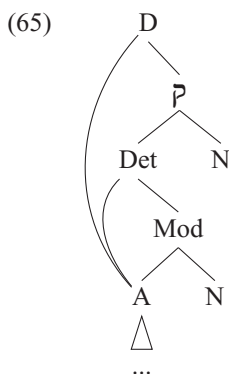
- (63) a. Červeni-te kupi [DP obuvki ]?  
           red-DEF bought shoes  
           'She bought the red shoes?'  
       b. Tri-te zavravi [DP čanti ]?  
           three-DEF forgot bags  
           'Did you forget the three bags?'

And Stanković (2013) reports the following parallel examples from Macedonian:

- (64) a. Crveni-te gi kupi [DP čevli ]  
           red-DEF them bought shoes  
           'You bought the red shoes?'  
       b. Tri-te gi zaboravi [DP torbi ]?  
           three-DEF them forgot bags  
           'Did you forget the three bags?'

It is not obvious how to circumvent the constraints imposed by Bošković's system so as to allow these. For him, if there is an article, that is evidence there is a DP layer. If there is a DP layer, and APs are adjoined to NP, AP should be blocked from moving by the conspiracy of Antilocality and the phasal nature of DP.

In the mereological system, there are various analytical approaches that could be taken. The most obvious is that Det takes a modified nominal as its 1-part, A subjoins to Det, then to D. Det is spelled out as an affix on the constituent A. Given A is a 2-part of D, which is a 2-part of some higher clausal object, A can then subjoin outside of D.



The only tricky part of this analysis is ensuring that Det is spelled out on A, although A is itself spelled out in a higher clausal position.<sup>12</sup> The absence of AP and Dem raising in, for example English, follows if English does not allow subjunction of AP or Dem to D.

There is further evidence from Chamorro that the correct generalization is one about the incompatibility of a definite article and extraction at the construction level, as opposed to the language level. Chung (1991) shows that, in Chamorro, a language with definite articles, it is possible to extract a possessor or complement of the nominal phrase just in case that nominal phrase lacks an article. In (66) we see that extraction from a nominal phrase with a definite article is impossible.

- (66) a. Ti man-mäguf [i famagon-ña si Jose].  
           not INFL.PL-happy the children-3SG ART Jose  
           ‘Jose’s children are unhappy.’  
       b. \*Hayi ti man-mäguf [i famagon-ña]?  
           who not INFL.PL-happy the children-3SG  
           for ‘Whose children are unhappy?’

Chung provides a full paradigm of all cases of overt articles: definite, quantificational, specific, and demonstrative, and shows that the same generalization holds. However, when the nominal lacks an article (and hence has an indefinite interpretation), extraction is well formed:

- (67) Hayi ti man-mäguf [famagon-ña]?  
       who not INFL.PL-happy children-3SG  
       ‘Whose children are unhappy?’  
       (68) Hayi un-yulang [munika-ña]?  
           who INFL.2SG-break doll-3SG  
           ‘Whose doll did you break?’

Again, Chung shows that this is the case for all situations where it is possible to have an indefinite lacking an article: object position, complement of existential verbs, (internal) argument of unaccusatives and passives. Subjects of transitives and unergatives are obligatorily definite or specific, and fall under the unextractability from definites generalization.

Chamorro shows that the correct generalization about Left Branch extraction is not about whether a language has articles or not. Rather the generalization is that left branch extraction is impossible from below D when D has a 2-part, but may be allowed if D has a free 2-part “slot”: this is why indefinites in general are not islands for extraction while definites often are. English, Ch’ol, Bulgarian, and Macedonian further show that Left Branch extraction may also be allowed if the extractee can subjoin to D, though whether it is then able to subjoin higher or not is a matter of parametric variation.

Since possessed nominals are generally semantically definite, the Chamorro pattern in (68) is further evidence that it is not the semantic definiteness of the nominal that blocks extraction, but rather the syntactic status of D, specifically its 2-part. If D has a 2-part, that 2-part may further extract, but nothing may extract from below it. The logic is, then, very much like the logic of the approach I took to *wh*-islands in the previous chapter.

## 6.5 Some Apparent Exceptions

This discussion takes us to some cases that appear to be exceptions to the generalization that it is impossible to extract from nominal constituents with definite articles. A striking example of this is what Uriagereka (1996b, pp. 270–271) refers to as *Cliticization Gates*. These involve situations where the D cliticizes to the verbal projection and extraction is unexpectedly licit.

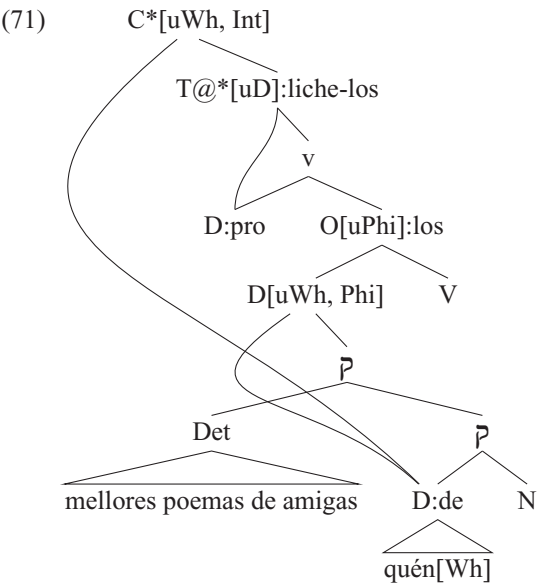
Uriagereka provides the following paradigm from Galician, which he calls a *Cliticization Gate*,<sup>13</sup> where extraction from a definite nominal phrase is ungrammatical unless the definite article is realized as a clitic on the verb:

- (69) a. (?)De quén liche-los mellores poemas de amigo?  
           of whom read.2SG-the best poems of friend  
           ‘Who did you read the best poems of friendship by?’  
       b. \*De quén liches os mellores poemas de amigo?  
           of whom read.2SG the best poems of friend  
           ‘Who did you read the best poems of friendship by?’
- (70) a. De que zonas liche-los mellores poemas de amigo?  
           of what areas read.2SG-the best poems of friend  
           ‘What areas did you read the best poems of friendship from?’

- b. \*De que zonas liches os mellores poemas de amigo?  
of what areas read.2SG the best poems of friend  
‘What areas did you read the best poems of friendship from?’

Stepanov (2012), in his treatment of this phenomenon, suggests that it should be analyzed as follows. If the determiner head-moves into the verb, it leaves no trace, as traces of functional heads are assumed to be interpreted only in their higher position. But then the constituent it has vacated has no head, which means it does not exist qua constituent: the previous structure is eradicated, and the pieces of that structure are no longer organized in the way they were. Given that, there is no DP layer to block movement (assuming that what was responsible for the island effect was the DP category). This proposal is theoretically problematic if we assume the syntactic system is constrained by some version of No Tampering, since the structure which has already been computed is radically changed.

In the approach developed here, if the definite article is spelled out as a non-clitic, then, this entails that Det has subjoined to D, so subjunction of the PP from lower is correctly predicted to be ungrammatical. However, if the incorporated article is on the verb, that suggests that it is part of the verbal spine: specifically it is a realization of O. It is then the D<O 2-part relation through which the definiteness of D is negotiated, rather than via the Det<D 2-part relation.<sup>14</sup> Consider the following structure:



The morphological pieces of the verbal Extended Projection are combined: tense, subject agreement inflection, the clitic in O, to provide the relevant morphological form *liche-los*. The availability of the clitic structure means that there is no syntactic reason for Det to subjoin to D, so there is no definite article, and the 2-part “slot” of D is then available for subjunction, allowing yet higher subjunction to take place without violating Angular Locality. This accounts for the Cliticization Gate effect. Of course, some languages may independently require subjunction of Det to D even when there is a clitic in O, and so will not have the Cliticization Gate effect.

More has to be said here about how the verbal form is morphologically constructed from the components of the verbal spine (see Harbour 2023 for a compatible approach), however, the system rather elegantly captures the effect without either head movement or mechanisms that destroy structure. Again we see that what matters is not semantic specificity, but rather syntactic definiteness.

Another extremely interesting exception to Fiengo and Higginbotham’s Specificity Condition comes from Persian. Karimi (1999) notes that, contrary to the Specificity Condition’s predictions, extraction is possible from specific nominal phrases in Persian, marked by the specificity marker *râ*, as long as there is no determiner in the structure. Nominal phrases marked with *râ* are interpreted as discourse familiar (identifiable to the hearer). Karimi gives the following cases of Ezafe constructions with possessors. In these constructions, a possessor occurs immediately after the head nominal, which is marked by a special morpheme, traditionally called the Ezafe morpheme (glossed -EZ here)<sup>15</sup>:

- (72) a. Kimea [<sub>NP</sub> ketâb-e kodum nevisanda] ro dust-dâr-e?  
           Kimea book-EZ which writer SPC friend-have-3SG  
           ‘Which writer x is such that Kimea likes the book by x?’  
       b. \*Kimea [<sub>NP</sub> in ketâb-e kodum nevisanda] ro dust-dâr-e?  
           Kimea this book-EZ which writer SPC friend-have-3SG  
           for ‘Which writer x is such that Kimea likes this book by x?’

As Karimi notes, both examples contain object nominal phrases marked by the specificity marker *râ*, but when the Ezafe possessor construction lacks a determiner, as in (72a), the wh-expression can take matrix scope. Karimi assumes this correlates with extraction at Logical Form (LF); when, however, a determiner is present, as in (72b), the wh-expression cannot take its required scope.<sup>16</sup>

Karimi also claims that matrix scope for the wh-expression is ungrammatical when there is an indefinite. I have not been able to replicate this, however, and

it appears that, for at least some speakers, such cases are grammatical, with the reading indicated:

- (73) Kimea [<sub>NP</sub> ye ketâb-e kodum nevisanda] ro dust-dâr-e?  
 Kimea a book-EZ which writer SPC friend-have-3SG  
 ‘Which writer *x* is such that Kimea likes some book by *x*?’

I’ll return to this variation in judgments, which is, I think, predicted by the account I’ll sketch below.

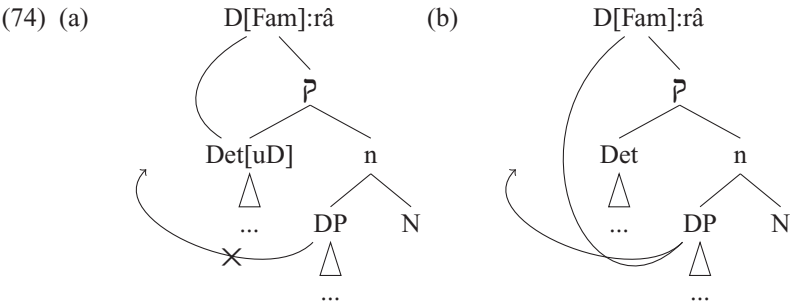
The contrast in acceptability between Ezafe constructions with and without demonstratives can be captured by the system I have proposed rather straightforwardly. Recall that D-Interpretation (26) is a one-way generalization. D can be discourse familiar and still have a free 2-part “slot.” This leaves open the possibility that some languages directly specify that certain Ds are discourse familiar; that is, certain categories have a property that the semantic systems interpret as discourse familiarity. This is just another way of saying that those categories bear a syntactic feature which causes the semantic systems to behave in this way. I will call that feature [Fam], for “discourse familiar” and assume that it involves connecting the Discourse Referent introduced by the D to some previously established, situationally present, or accommodated Discourse Referent (cf. Heim 1983, Kamp 1981, Enç 1991, among many others).

Let’s assume then that D in Persian can bear [Fam]. If D bears [Fam], D is spelled out as *râ* and interpreted as discourse familiar. If [Fam] is absent, D is not spelled out, and is interpreted as non-discourse familiar.

In the case of the category Dem, which I take to be inherently discourse familiar, Dem subjoins to D becoming its 2-part, D spells out as *râ*, and subjunction (i.e., extraction) from inside D is blocked. To ensure that the subjunction of Dem to D takes place in Persian, we can say that Dem bears [uD]. Checking, as always, takes place in a 2-part relation, so Dem[uD] subjoins to D[Fam].<sup>17</sup>

Indefinite numerals such as *ye* “a/some” optionally bear [uD]. I’ll assume *ye* is a realization of Det, but it may be a realization of a lower category Num and Det may be null. If Det is [uD], it subjoins to D[Fam] in the same way that demonstratives do, spelling out as *ye*. D[Fam] spells out as *râ* as usual. In such cases, extraction from inside D will be blocked by Angular Locality. However, if Det does not bear [uD], subjunction of Det to D does not take place, leaving the 2-part of D free for subjunction of the wh-expression. D, however, in Persian may still bear [Fam] and spell out as the specificity marker.

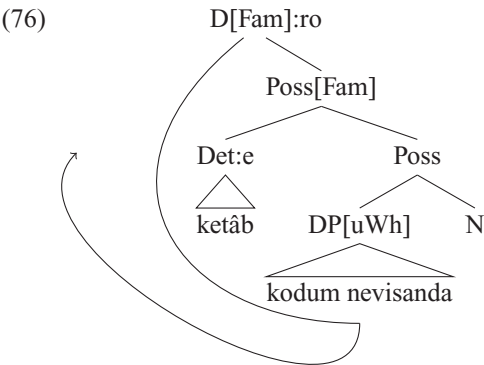
The relevant structures are as in (74):



This, I think, may be the source of the unclear judgments noted above. There is, in fact, a derivation that allows extraction, but if speakers allow [uD] on Det, then they also have a derivation that will not allow extraction. The variation in acceptability may simply be a result of individual speakers' initial or preferred choice of derivation. If this is true, however, it makes the prediction that even those speakers who initially find wide scope of the wh-expression unacceptable will be able to find a parse for it. I leave this for a more rigorous experimental investigation.

For the plain articleless Ezafe possessor constructions, these always co-occur with *râ*. This is presumably connected to the nature of the possessor relation, and we might say that in this case it is Poss that bears [Fam]. We can capture the example in (72a), repeated here as (75), as in (76):

- (75) Kimea [<sub>NP</sub> ketâb-e kodum nevisanda] ro dust-dâr-e?  
Kimea book-EZ which writer SPC friend-have-3SG  
'Which writer x is such that Kimea likes the book by x?'



Exactly how to ensure that when Poss bears interpretable [Fam], D does too is unclear to me. It may simply be a fact about the semantics of discourse familiarity, or it may require a theory of how features might percolate through the 1-part relation. Ezafe constructions are replete with many more mysteries than I can address, but the core point made by Karimi, that it is the syntax and not the semantics that gives rise to the Specificity Condition Effects, is I think fairly well captured in the system developed here.

## 6.6 Scrambling

As is well known, it is possible in German to scramble indefinite nominal phrases, in which case the indefinite is typically interpreted as discourse familiar, though the exact nature of the semantic effect associated with scrambled objects is in dispute. In (77a), we see the object appearing before the adverb *immer*, in a scrambled position, while in (77b) the object appears before the indefinite negative subject (examples from Müller et al. 2022). Both positions are scrambling positions in German.

- (77) a. dass Otto lange Romane immer gelesen hat  
           that Otto long novels always read has  
           ‘that Otto has always read long novels?’  
       b. dass ein Buch über dieses Thema keiner gelesen hat  
           that a book.ACC about this topic no-one.NOM read has  
           ‘...that no one has read a book about this topic.’

As noted by Diesing (1992, p. 129) and Müller (1998, p. 143), from whom the following examples are taken, such scrambled objects seem to be islands for extraction:

- (78) a. \*Was hat Otto [für Romane] immer gelesen ?  
           what has Otto for novels always read  
           ‘What kind of novels has Otto always read?’  
       b. \*Über wen hat der Fritz [ein Buch] letztes Jahr geschrieben?  
           about whom has the Fritz a book last year written  
           ‘Who did Fritz write a book about last year?’  
       c. \*Worüber hat [ein Buch] keiner gelesen?  
           about what has a book.ACC no-one.NOM read  
           ‘What has no one read a book about?’

One mode of explanation for such facts is via appeal to the structural difference between specifiers and complements. The idea is that once an object nominal phrase is moved, it is no longer in a complement position, but is rather in a configuration that is more like a subject, and hence, via the Condition on

Extraction Domains (CED) of Huang (1982), it becomes an island. Exactly what is responsible for the island status of non-complements is dependent on the theory. Huang's original proposal, developed into the *Barriers* theory in Chomsky (1986a), is that if a constituent is governed it is transparent for extraction. This was extended to rule out extraction in cases of scrambling by Diesing (1992): scrambled constituents have shifted away from their governed position and so are islands. Minimalist theories have also offered various ways to derive the effects of the CED via conditions on chains (Takahashi 1994), spellout-domains (Uriagereka 1999), the mechanics of phases and feature checking (Chomsky 2008; Boeckx 2003; Müller 2010); and so on. The mereological approach developed here does not predict that there are CED effects. In fact, since CPs and DPs are specifiers, extraction from them is licit as long as the extractee can subjoin to the category's edge. How, then, are we to account for the various scrambling islands just discussed?

In fact, I think there are good reasons to reject the idea that it is the CED that is at issue in these cases (and in Subject Islands, to be discussed in the next chapter). Fanselow (2001) (see also Abels 2007) points out that there are in fact cases where scrambling can feed wh-movement. The following examples are due to Fanselow

- (79) Worüber kann [einen Südkurier-Artikel selbst Peter nicht am  
 what-about can [a Südkurier-article] even Peter not at.the  
 Strand verfassen?  
 beach write  
 'For which topic is it the case that even Peter cannot write an article about it for the Südkurier when he is at the beach.' (from De Kuthy, Kordula and Meurers 1999, p. 27, adapted from Fanselow 1991, p. 188)
- (80) Über welchen deutschen Kaiser hatte [ein fertiges Manuskript]  
 about which German emperor had a done manuscript  
 leider keiner der anwesenden Historiker anzubieten  
 unfortunately none the present historians to.offer  
 'About which German emperor could none of the historians present offer a completed manuscript?'

Fanselow (2001) suggests that extraction from a scrambled object is well formed when it would be blocked if the object were in its base position. For the cases above, if the object were in situ, extraction across a quantificational element would lead to a so called Beck-effect (Beck 1996). If the object is scrambled, this overcomes the Beck-effect, and extraction is licit, even though it would violate the CED. One way of thinking about this is that the apparent restriction on extraction from scrambled objects is to be understood not via the CED, but

rather as falling under a Specificity Condition effect. In these particular examples, the object is, in fact, not necessarily specific. Fanselow himself concludes that it is the “referentiality” of the scrambled object that leads to the extraction problems.

In a sense, this conclusion is very close to what Diesing originally claimed, but from a different direction. Diesing attributed the island effect to the presuppositional or specific interpretation of the scrambled constituent, though for her this was because presupposed nominal needed to escape existential closure at the verb phrase level, and hence ended up in a high specifier. Given the derived syntactic position is a specifier, the scrambled constituent becomes an island for extraction via CED/*Barriers* as above.

Rather than tying the effect to the external syntactic configuration, I’d like to suggest that, just as we’ve seen throughout this chapter and the last, extraction is blocked because D ends up having its 2-part “used up.”

The core of the proposal, which is not at all novel (Adger 1994; Meinunger 1995; Miyagawa 1997; among many others) is that scrambling, when it is associated with an interpretative effect, is triggered by a clause-level category, which I’ll just call it Scr here. I’ll not take a position on whether there are multiple Scr categories or whether they are to be distinguished featurally. I think what I will propose can be implemented in either cartographic or noncartographic approaches, and one might even take Scr to be akin to old style adjunction, appearing in the Extended Projection of the verb above each category.

Building on the analysis I sketched for Persian, let’s say that Scr bears a feature associated with discourse familiarity of its 2-part (this is essentially the proposal of Adger 1994). Since Scr is a clause-level category, the feature is uninterpretable and needs to be checked, so the category is Scr[uFam]. Crucially, unlike, in Persian, I propose that D in German does not bear [Fam] (that is, there is no equivalent in German to *râ*). This means that checking [uFam] on Scr requires something bearing [Fam] to subjoin to D, so it can check Scr’s feature by Transitivity, since a [uF] cannot check another [uF] feature.

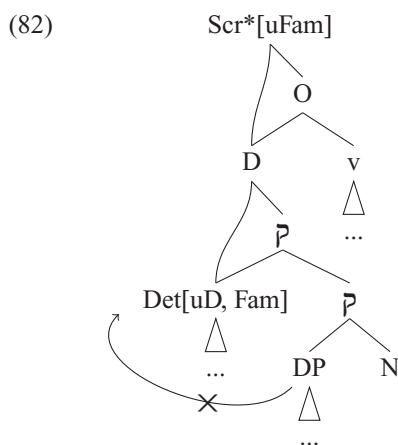
In German, the category Dem bears [uD, Fam] and so subjoining Dem to D will check [uD] on D (and block extraction from inside D). Scrambling a definite phrase headed by a demonstrative will then check [uFam] on Scr. Equally, such a phrase may remain in situ, and Scr in that case is simply not projected. In either case, extraction will be blocked.

Let’s now turn to a scrambled indefinite example like (81):

- (81) \*Über wen hat der Fritz [ein Buch] letztes Jahr geschrieben?  
 about whom has the Fritz a book last year written  
 ‘Who did Fritz write a book about last year?’

Indefinite determiners (or numerals or cardinal quantifiers) may have discourse-familiar interpretations. In English, I suggested that in such cases those determiners could bear a [uD] feature, so that they subjoin to D, and hence the whole DP is interpreted as discourse familiar given D-Interpretation. However, that is not sufficient for a scrambling language, if Scr bears [uFam]. Let us say instead that an indefinite category may bear [uD, Fam] optionally. If it does not bear [uD, Fam], no subjunction takes place, and no scrambling, ruling out scrambling of true indefinites (that is, non-discourse-familiar indefinites).

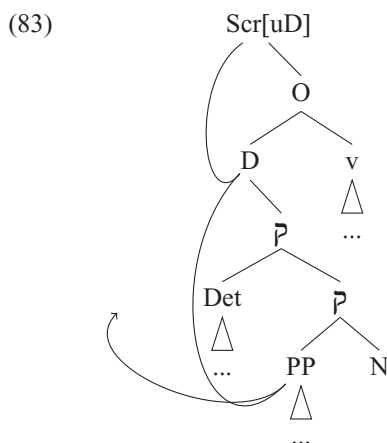
If, on the other hand, an indefinite Det (or, e.g., Num) does bear [uD, Fam], it will need to subjoin to D before D can subjoin to Scr[uFam] to allow checking of Scr's [uFam] feature via Transitivity with [Fam] on Det. Once D subjoins to Scr[uFam], Det[uD, Fam] becomes a 2-part of Scr[uFam], by Transitivity, and hence is able to satisfy Scr's checking requirement. Of course, since Det[uD, Fam] is then a 2-part of D, nothing else can extract from D, capturing the island effect and the association of the scrambled position with specificity:



If Det does not bear [uD, Fam], no subjunction takes place. In such a case, the [uFam] feature on Scr would remain unchecked and the derivation would not converge. A derivation that would converge with this feature combination, however, would be one that lacked Scr, in which case we have an unscrambled indefinite, which, as just mentioned, is correctly predicted to allow extraction.

Comparing this system to the Persian one, they are very similar, except that, in Persian, D can bear [Fam], and Dem and (optionally) Det can bear [uD], while in German, D does not bear [Fam] and Dem and (optionally) Det can bear [uD, Fam]. In terms of the placement of the [Fam] features on D, they are distinct, a fact that is presumably signaled by the phonologically overt specificity marker *râ* in D in Persian, and the absence of anything similar in the DP in German.<sup>18</sup>

For Fanselow's examples in (79–80), where extraction is possible from a scrambled indefinite, I think it is enough to say that Scr in these examples bears a [uD] feature, not a [uFam] feature (cf. Miyagawa's 1997 analysis of Japanese scrambling as being motivated by an EPP feature). This will cause D to subjoin to Scr, but nothing will force the Det constituent inside D to subjoin to its 2-part edge. This means that that edge is free for another constituent to subjoin to it, circumvent Angular Locality, and therefore further extract:



Of course, Det here could still bear [uD, Fam], forcing subjunction of Det to D, and extraction would be blocked. However, since there is an alternative derivation with no [uD, Fam] feature on Det, that derivation would always provide an alternative parse, predicting extraction is available.

This analysis reduces the problem of explaining the islandhood of scrambled constituents (while allowing Fanselow's exceptions) to the issue of when the [uFam] vs [uD] feature may be on Scr. One might say that in the unmarked case Scr will bear [uFam], but that [uD] is a (marked) option. Since the Fanselow examples are ungenerable with [uFam] on Scr, that marked option is taken when they are parsed. The question of whether a Scr with [uFam] or [uD] is chosen for the generation or analysis of a particular sentence will be a performance issue. As far as the grammar is concerned, both are possible, but one derivation is marked because the relevant category is less accessible to the parser. Judgments of unacceptability are really, at this point, judgments of markedness connected to the ease of access that the parser has to the resources it needs.<sup>19</sup> I'll extend this kind of explanation to Subject Islands in the next chapter, and discuss its implications in more depth.

I have illustrated the system above using an indefinite Det. However, I have not yet addressed the question of definite determiners.

There are two possibilities: either definite Det is like Dem and is inherently specified as [uD, Fam], or it is like an indefinite or cardinal quantifier (like *four*, or *many*), and has both [uD, Fam] and non-[uD, Fam] versions. There is good evidence that the latter option is correct. Schwarz (2009) draws attention to a number of languages that have distinct definite articles, one of which, called the “strong definite article” by Schwarz, has a discourse-familiar interpretation, while the other is merely required to be situationally unique (see also Jenks 2018 for the distinction in languages that lack articles). He calls the latter the “weak definite article.” In German, the two articles have distinct forms that can be clearly seen when they combine with prepositions, as in (84) (Schwarz 2009, p. 231):

- (84) In jeder Stadt, in der unser Zug hielt, wurde mir der Brief  
 In every city in which our train stopped was me the letter  
 vo-[m / #von [dem Bürgermeister] überreicht.  
 from-the<sub>weak</sub> / from the<sub>strong</sub> mayor handed  
 ‘In every city we stopped in, a letter was handed to me from the mayor.’

The weak/strong article nomenclature stems from work on Germanic where the strong definite articles are typically more morphologically substantial than the weak ones, recalling the stress difference between specific and nonspecific indefinites in these languages.<sup>20</sup>

Let us say, then, that Det may bear the feature [uD, Fam] along with a feature signaling situational uniqueness, and in such cases Det subjoins to D, just as a demonstrative does, and this results in a discourse-familiar reading, following D-Interpretation. Of course, the prediction is that this also makes the D an island for extraction, as it has no free 2-part.

Evidence that this consequence is correct comes from subextraction facts in Austro-Bavarian German (from Brugger and Prinzhorn 1996 as reported by Wiltschko 2013, p. 174). This language morphologically distinguishes weak vs strong definite articles, and, Brugger and Prinzhorn report, allows subextraction only when the nominal phrase has a weak article:

- (85) a. Von wem host du [s Possbild] nit gsegn?  
 Of whom have you D<sub>weak</sub> passport.foto not seen?  
 ‘Whose passport photo did you not see?’  
 b. \*Von wem host du [des Possbild] nit gsegn?  
 Of whom have you D<sub>strong</sub> passport.foto not seen?  
 for ‘Whose passport photo did you not see?’

In the current system when Det bears [uD, Fam] it will subjoin to D, resulting both in the D being an island for subextraction and in the morphology of the strong definite article. If the Det does not bear [uD, Fam] then this is

incompatible with the morphology of the strong definite article. Det will remain in situ, be spelled out as the weak definite article, and the whole DP will not be an island for extraction.

What of English, which does not make the strong/weak distinction morphologically? In that language, demonstratives and possessors are usually judged as creating quite strong specificity islands, while the definite article is judged as creating a weaker one. Recall the initial paradigm of judgments I gave:

- (86) a. ?Who did you buy the statue of  
       b. \*Who did you buy that statue of?  
       c. \*Who did you buy Anson's statue of?  
       d. Who did you buy a statue of?

It may be that this is because there are two variants of the definite article Det category in English too, one which bears [uD, Fam] and one which does not. The former subjoins to D blocking extraction from inside D, while the latter does not. Speculating further, the judgment in (86) for the definite article may be weaker because there is an alternative, though more marked parse.

Briefly summarizing, I've sketched out a proposal for scrambling in the system, which captures the general islandhood of scrambled objects, not through a CED-type effect (or a Freezing-type effect, as discussed by Corver 2017), but via an operation internal to the scrambled DP that blocks extraction out of that DP. This operation takes place for feature checking reasons, fundamentally driven by the properties of a clausal head Scr. The marked exceptions to the islandhood, and to the specificity, of scrambled DPs are taken to be due to a feature [uD] on Scr that, though present in the speakers' grammars, is less accessible to them when parsing. I've suggested more generally that the variable character of acceptability judgments in this empirical domain can perhaps be attributed to whether certain feature specifications of a category are more or less accessible to a speaker. This draws on some ideas from the theory of Combinatorial Variability (Adger 2006; Adger and Smith 2010; et seq.), developed for tackling the integration of sociolinguistic variation and minimalist syntactic theory. Accessibility here concerns how easy it is for the parser to access the resources required to provide a structure for the sentence (so it is fundamentally a concept of performance): the grammar makes options available, but the performance systems may systematically disprefer certain grammatical outcomes. This is why definite articles systematically show a weaker Specificity Condition type effect than demonstratives and possessors. Finally, the feature system sketched out here derives the generalization embodied in D-Interpretation, which becomes an outcome of the system, as opposed to a constraint embedded within it.

## 6.7 Further Apparent Exceptions

With this much in hand, I want to finally turn to cases where certain verbs in English appear to allow extraction from some definites. There are three cases, and it is the third that will concern us most here.

First, Davies and Dubinsky (2003) argue that complex event nominals always allow extraction, even when they are definite:

(87) What did they observe/hear about/remember/decry the production of?

If these nominalizations contain verbal phrases, as suggested by Borer (2013), then their objects are introduced by O, and there's no reason those should be opaque to extraction; however, such extraction should be blocked when there is an overt definite article, making examples like (87) potentially problematic.

It turns out, though, that the definite article in English generally doesn't make complex event nominals discourse familiar. Davies and Dubinsky show that apparently definite process nominals are perfectly well formed in existentials (their footnote 7):

(88) There was the revolution of a satellite around Mars on the video transmission, which the scientists had not expected.

I suggest, then, that these are cases of weak definite articles in English: Det remains in situ. Since the 2-part of D is empty, it can be used by the extractee to create the relevant structure that allows Angular Locality to be met, and extraction can take place. For English, we can't tell the difference between the weak and strong cases, but the prediction is correct for German (Schwarz 2009, p. 19), where process nominals strongly prefer weak definite articles:

- (89) a. Hans ist am Arbeiten.  
       Hans is at-the<sub>weak</sub> working  
       'Hans is working.'  
       b. Hans hat viel Freude am Tanzen.  
       Hans has much joy at-the<sub>weak</sub> dancing  
       'Hans really enjoys dancing.'

The conclusion is also, of course, consistent with the non-discourse-familiar nature of the whole nominal phrase as seen in (88).

Further confirmation that this might be correct is that demonstratives in these examples are not well formed, as expected if demonstratives always bear [uD, Fam] and subjoin to D:

(90) \*What did they observe/hear about/remember/decry that production of?

The second case of definite articles that are not truly definite was noted originally by Fiengo (1987) (see also Matushansky 2006). In superlatives, a definite article is obligatory, but does not block extraction:

- (91) (Of those present) Who did you take the \*(best) picture of?

Again, the definite article here is not discourse familiar, and I'll assume it's weak. Det remains in situ, so extraction is possible. Evidence that this is correct is that superlatives in German occur with the weak article (Schwarz 2009, p. 30):

- (92) Auf unserer Reise nach Tibet sind wir natürlich auch zum        / #zu  
       on our trip to Tibet are we of.course also to-the<sub>weak</sub> / to  
       dem höchsten Berg der Welt gefahren.  
       the<sub>strong</sub> highest mountain the.GEN world driven  
       'On our trip to Tibet, we of course went to visit the highest mountain of  
       the world.'

Perhaps the most interesting case, because it is not obviously attributable to a weak definite article, is when extraction takes place of an argument of a relational nominal phrase which is the object of a verb of creation. Davies and Dubinsky (2003, p. 2) point out contrasts like (93) (see also Bach and Horn 1976; Cattell 1979; Erteschik-Shir 1981) and characterize the generalization as follows: "Extraction is possible when representational nouns are complements of some predicates, but not others [...]. Extraction out of representational nouns appears not to be possible when they co-occur with definite determiners":

- (93) a. Who did you write/??read those essays about?  
       b. Who did you write/read essays about?  
       (94) a. Who did you tell/\*hear the jokes about?  
       b. Who did you tell/hear jokes about?  
       (95) a. Who did you paint/??see that portrait of?  
       b. Who did you paint/see a portrait of?

In these examples, extraction from definite representational nominal phrases is blocked unless that nominal is the object of a verb of creation.

Bach and Horn (1976), for similar examples, suggest that, in the acceptable cases, there is a parse where the PP is at the VP rather than NP level, so extraction is never from nominal phrases. That is, the well-formed examples above have the following kind of structure:

- (96) Who did you [<sub>VP</sub> write [<sub>NP</sub> essays] [<sub>PP</sub> about ⟨who⟩ ] ]

Erteschik-Shir (1981), however, argues that there must be extraction from nominal phrases, since certain verbs are incompatible with the PP at VP level, but allow extraction:

- (97) a. Who did they read/finish a book about?  
 b. \*They read/finished it about Nixon.

In (97a) extraction is possible, but (97b) shows that the PP is incompatible with the verb, since it cannot be adjoined to the pronoun. However, Erteschik-Shir's examples are indefinite, so don't constitute exceptions to Davies and Dubinsky's generalization.

Nevertheless, I'm skeptical that, even in cases of verbs of creation, there's always a parse where the PP is at VP level, although I think there may be in some cases. Consider the case of *tell jokes*: Heavy NP Shift seems very awkward with this collocation, though it is fine with a verb that clearly has a PP at VP level like *replace X with Y*:

- (98) a. ??I told about Suki those terrible jokes that we all thought were best forgotten.  
 b. I replaced with Neem that idiot of a waiter that we knew wouldn't last.

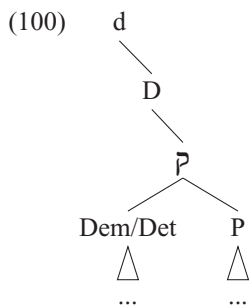
Further, for those nominals that co-occur with *of*-PPs, it's generally (although not always) the case that the *of*-PP is incompatible with the verb phrase, yet we still see the same pattern:

- (99) a. Who did you print/??admire that linocut of?  
 b. \*Who did you print it of?

I think that Davies and Dubinsky are correct. In each of the relevant cases, extraction from the definite nominal is degraded except when the nominal is an object of a verb of creation. It may be that some of the relevant examples (*write a book*, for example) do have the PP at VP level, as Bach and Horn suggest, but it's clear that that can't always be the case.

Davies and Dubinsky construct an analysis whereby the well-formed cases involve a controlled PRO inside the nominal that is the agent of the event associated with the nominal. When this PRO is present, they argue, a process of abstract noun incorporation takes place, which links the noun and the verb, just if the noun is a result of the action denoted by the verb. When this happens, it essentially erases the effect that the DP layer has in creating an opaque domain so extraction is possible (technically this is done by appealing to the Government Transparency Corollary of Baker 1988).

I think a simpler story is available in the system I've developed here. Let's say that there is a kind of D in English which is compatible with the introduction of a new Discourse Referent.<sup>21</sup> Verbs of creation will select (be semantically compatible with) that D, which, I'll call, little d, on analogy with the little c introduced when we discussed wh-islands. The interpretation of d is that the Discourse Referent it introduces did not exist prior to the event denoted by the verb taking place.



Dem and (optionally) Det may bear [uD, Fam], and subjoin to D, but d has a “free” 2-part so extraction from lower in the structure is possible. This will capture Davies and Dubinsky’s generalization: if a DP is the object of a verb of creation, then it may host a definite determiner or demonstrative and be interpreted as specific, but extraction from inside it is well formed. In the analysis proposed here, that is because verbs of creation select d, which provides further structure that can be subjoined to, allowing extraction while satisfying Angular Locality.

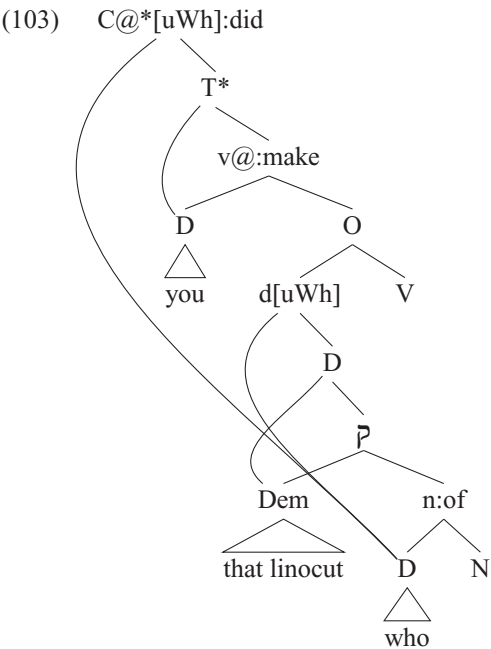
We don’t want to rule out a specific interpretation for the whole phrase, as it is clearly possible to refer back to an individual using a pronominal object even though the pronominal object is selected by a verb of creation, as follows:

- (101) I heard you got assigned three essays about syntax this semester. I’m so pleased you finally got around to writing them.

Here the pronoun *them* refers back to the three essays, but the second sentence in this short discourse has a verb of creation that causes the essays to come into existence. Further, we can see that extraction and specificity can go together in examples like the following:

- (102) Did you enjoy the exhibition of statues and portraits of famous people?  
What kinds of people did Vermeer paint those portraits of?

The relevant structure for an example like *Who did you make that linocut of?* is then as in (103):



A final comment on these verbs of creation: Diesing (1992) notes that objects of verbs of creation cannot be scrambled, but I’ve suggested that they can be discourse familiar. That is a surprise if, as Diesing proposed, it is the presuppositional or discourse-familiar nature of a DP that allows it to be scrambled. In the alternative I’ve proposed, this effect is to be understood as syntactic: d cannot bear [uFam] (or [Fam]). This means that it cannot scramble, since to check [uFam] on Scr, a [Fam] feature must become a 2-part of d, but that is impossible. The category d may be interpreted as discourse familiar, though, if it has, internal to it, a category D that does bear [Fam] (or is interpreted as discourse familiar via D-Interpretation). The discourse referent introduced by d is new with respect to the event, though it may be related to a discourse referent that is discourse familiar.

6.8 Clauses in Nominals

Consider the following patterning of data:

- (104) a. He resented [CP that he had not formulated the hypothesis himself].  
b. This is a hypothesis that he resented [CP that he had not formulated himself].

- c. [<sub>TP</sub> He resented the [<sub>NP</sub> claim [<sub>CP</sub> that he had not formulated the hypothesis himself] ] ].
  - d. \*This is a hypothesis which [<sub>TP</sub> he resented the [<sub>NP</sub> claim [<sub>CP</sub> that he had not formulated himself] ] ].
- (105) a. Kerry heard [<sub>CP</sub> that Kelsey was fond of Kim].
- b. Who did Kerry hear [<sub>CP</sub> that Kelsey was fond of]?
  - c. Kerry [<sub>TP</sub> heard the [<sub>NP</sub> rumor [<sub>CP</sub> that Kelsey is fond of Kim] ] ].
  - d. \*Who did [<sub>TP</sub> Kerry hear the [<sub>NP</sub> rumor [<sub>CP</sub> that Kelsey is fond of] ] ]?

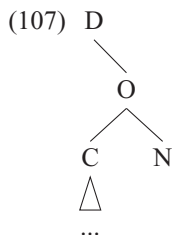
Since Ross (1967), such paradigms have been used to show that extraction from the clausal complement of a verb is well formed, while extraction from the clausal complement of a noun is unacceptable. Ross proposed the Complex NP Constraint (CNPC) to ensure that the grammar did not generate such examples.

One challenge for the CNPC is a set of violations noted by Ross himself. Contrast the unacceptable examples in the paradigms above with the following:

- (106) a. This is a hypothesis which he made the claim that he had formulated himself. (cf. Ross 1967, p. 139)
- b. Who did Kerry start the rumor that Kelsey is fond of?  
(Davies and Dubinsky 2003, p. 34)

Such examples have been characterized as involving “light” verbal predicates that, somehow, incorporate their nominal argument, thereby voiding the CNPC island effect. Ross suggested something like this in his dissertation and the idea was developed in more depth by Davies and Dubinsky (2003) who extended it, as we saw in the previous section, to extraction from NP more generally. I’ll suggest in what follows that the analysis for these too involves d.

The first question to address is what the structure for complement clauses to nouns would be in the mereological system. As discussed in chapter 3, I take semantic arguments to be 2-parts of the relevant category. So one approach would be to think of a CP complement to N as being the 2-part of a category (say O, as for verbal Extended Projections) that N is a 1-part of, as follows:



This would make clausal arguments of N structurally similar to clausal arguments of V: 2-parts of a functional category projected above the lexical category.

However, there is a consensus that the semantics of finite clausal complements to nouns is quite distinct from that of finite clausal complements to verbs and that the corresponding syntax in the two cases should also be differentiated. There have been various proposals since Stowell (1981) first pointed out the differences and suggested that finite CPs are not complements to nouns. Stowell proposed they were in apposition, and others have suggested that they are relative clauses (for example, Arsenijević 2009, Kayne 2010, Haegeman and Ürögdi 2010). Moulton (2015) collects and develops a number of arguments that there is indeed a syntactic and, he argues, semantic distinction between how verbs and nouns relate to clauses. For example, *so* appears to be a pro-form for finite clauses, at least with certain classes of verbs, as can be seen in (108):

- (108) a. I believe/claim/am afraid that pigs fly.  
b. I believe/claim/am afraid *so*.

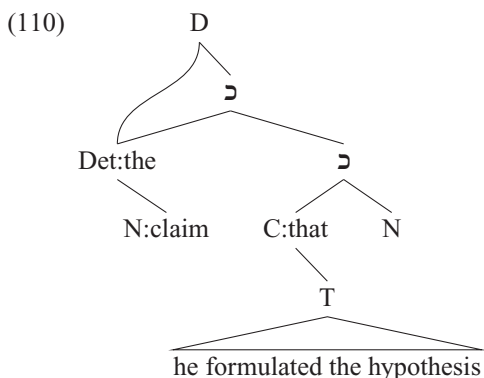
However, Moulton shows that no noun takes *so* as a pro-form for finite clauses, even those nouns that are associated with the classes of verbs that do take *so*. This gives rise to the following kind of pattern (see also Hallman 2006), suggesting a syntactic difference between CP complements to verbs, and those CPs that occur with nouns:

- (109) a. my belief/claim/fear that pigs fly  
b. \*my belief/claim/fear *so*

To account for the behavior of CPs that occur with Ns, Moulton argues that finite CPs are generally “predicates of propositional content” and that they combine with a subset of nouns that are also predicates of propositional content. Semantically, the combination is effected via Predicate Modification (as opposed to the CPs being semantic arguments of the noun). Moulton isn’t totally explicit about the syntax he assumes for nouns combining with their clausal complements, but it is, I think, to be understood as some kind of adjunction as opposed to classical complementation.

I’ll take this evidence, and the general consensus that CPs in nominal phrases are not true arguments, as motivation to treat CPs in a way that is analogous to the  $\bar{P}$  analysis of PPs. Just as in the case of PP “complements,” the nouns that take CP “complements” occur in various semantic classes (Moulton 2015 and especially Hankamer and Mikkelsen 2021), and I’ll assume that these nouns are combined via Predicate Modification with an abstract noun whose conceptual structure is consistent with the semantic class of the nominal. The analysis

of PPs and CPs with nominals will then be parallel, though I'll use a distinct category for CPs as there is a distinct set of relevant concepts, and (usually) a distinct morphosyntactic realization. I'll call the category that introduces CPs into the nominal structure Kaf, and represent it with the Hebrew letter כ. This means that the structure for, say, *the claim that he formulated the hypothesis*, will look as follows:



If this idea, that CPs and PPs are parallel, is correct, then the core cases of the complement clause CNPC must be reduced to specificity as we did in the PP cases. This will take care of the definite cases:

- (111) a. \*This is a hypothesis which he resented the claim [<sub>CP</sub> that he had not formulated himself]  
 b. \*Who did Kerry hear the rumor [<sub>CP</sub> that Kelsey is fond of]?

To derive (111a), the constituent *the hypothesis* in (110) needs to be subjoined outside of the entire nominal phrase. It can subjoin to the C, which has a free 2-part “slot,” but the topmost category of the nominal phrase is D, and D has Det as a 2-part. Angular Locality will therefore bar subjunction of *the hypothesis* outside of the nominal phrase.

Note that if we make the nominal indefinite, extraction improves dramatically, as expected:

- (112) a. There was a claim that he'd falsified the results.  
 b. I looked over the results that there was a claim that he'd falsified.
- (113) a. ?Kerry heard a rumor that Kelsey was fond of Kim.  
 b. ?Who did Kelsey hear a rumor that Kim was fond of.

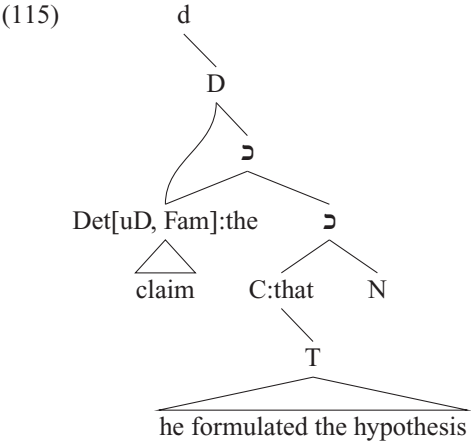
In these cases there is no movement of Det to D, so D has a 2-part “slot” that the extractee can avail itself of (the awkwardness of (113b) has, I think, to

do with the oddness of the indefinite article with *rumor* when it has a clausal complement more generally, as shown by the parallel awkwardness of (113a)).

Recall that there, however, are cases where extraction is possible even though there is a definite article:

- (114) a. This is a hypothesis which he made the claim that he had formulated himself.
- b. Who did Kerry start the rumor that Kelsey is fond of?

Why should these be well formed in this system? Again, we can adopt the proposal I made in the last section that verbs of creation of this sort select *d*, which provides an extra 2-part slot:



Overall, then, there’s nothing special about CNPCs; complex nominal phrases containing clauses work in just the same way as simple nominal phrases containing PPs. The primary constraint is syntactic definiteness: Does the *D* category have a 2-part? I’ve argued that the Russian exceptions to the CNPC involve the same *d* category, which introduces a new Discourse Referent, that we saw in the case of extraction from PPs.<sup>22</sup>

6.9 Summary

Overall, the approach I’ve defended here derives a set of effects close to, but not identical with, the effects captured by Fiengo and Higginbotham’s Specificity Condition. I’ve argued that the mereological system improves over the Specificity Condition and over Bowers’s *Barriers*-style syntactic implementation of that condition in that it can explain why extraction can still be possible

from semantically specific nominal phrases and from nominals that are morphologically definite nominal constituents; in both cases there is a free 2-part slot, so the nominal is not *syntactically* definite, though it may be semantically or morphologically definite. Further, the system does this using just the same technology as was applied to capture wh-island effects: Dimensionality, Transitivity, Angular Locality, and the idea that feature checking takes place in a 2-part relation.



## 7 Subject Islands

### 7.1 Introduction

Classical  $\bar{X}$ -theory and Bare Phrase Structure both make a certain theoretical cut in the way that constituents relate to their head: the distinction between specifiers and complements. In both theories, there is an irreducible element (a lexical item) that serves as the head of the structure, and the complement is a sister of that head, while the specifier c-commands it. These classical systems allow arguments to be either specifiers or complements, which means that it is straightforward to map specifiers and complements to what are traditionally thought of as subjects and objects. These systems build in an idea that the syntactic relation between a verb and its object is the same phrase-structural relation as that between a functional head and its complement.

The mereological system in contrast takes the fundamental syntactic relation of 1-parthood to correspond with the relationship of Extended Projection complementation (for example, the relationship between C and T, or between D and N). The relationship of 2-parthood, on the other hand, corresponds to the introduction of one set of Extended Projection relationships into another. This means that DPs are introduced into clauses via the same kind of phrase-structural relations irrespective of their status as subjects and objects: both are 2-parts, so the distinction between subjects and objects cannot be connected to the geometry of structure, unlike in the more classical approaches to phrase structure.

The structural differentiation made in the classical systems allows them to quite straightforwardly tackle what have become known over the decades as “subject-object asymmetries.” They have a means of embedding, into the theory of phrase structure, the intuition behind Huang’s (1982) Condition on Extraction Domains (CED): head-complement structures are transparent to syntactic processes in a way that other structures are not. That is not true for the mereological system. Without this capacity, the mereological system is faced

with the problem of how to analyze the subject-object asymmetries that the CED was designed to handle.

In this chapter, I'll look at CED effects as they relate to the islandhood of subjects. I'll argue that the CED intuition is, in fact, wrong (that is, appeal to complement/noncomplement asymmetries leads to an incorrect analysis of the data; see also Stepanov 2007). I'll show that the opacity of subjects is more complex than the CED predicts, and that the island effects should be understood in terms of the theory of specificity already introduced together with an analysis of the topicality of subjects. I'll argue that this consequence of the mereological system's weaker predictions is a positive one.

## 7.2 Extraction from Subjects

Consider the contrast between the (a) and (b) examples in (1) and (2) (the latter from Chomsky 1973, p. 249):

- (1) a. This is the only book that the shop invited [an author of].  
       b. \*This is the only book that [an author of] visited the shop.
- (2) a. Who did John hear stories about?  
       b. \*Who did stories about terrify John?

To deal with such examples, Chomsky (1973, p. 250) formulated the Subject Condition, simplified somewhat in (3):

- (3) Subject Condition:  
       No rule can involve X, Y in the structure  $\dots X \dots [\alpha \dots Y \dots] \dots$  where  $\alpha$  is a subject phrase properly containing Y.<sup>1</sup>

The condition in (3) basically stipulates that subjects are islands.<sup>2</sup> As the theory developed, the Subject Condition was replaced by more general constraints that appealed to the core structural asymmetry noted above: objects are the complements of verbs, while subjects are specifiers (either of the verb, a functional category that licenses them thematically, such as *v*, or a higher functional category that case licenses them and/or that they agree with, such as *T*). Various analyses of Subject Island effects appeal to the complement/noncomplement distinction, most obviously those that descend from Huang's CED, such as the Barriers system of Chomsky (1986a), but also those that have developed the idea that specifiers are opaque to syntactic processes for other reasons, such as mapping to the spellout systems (e.g., Uriagereka 1999; Johnson 2003; Privoznov 2021). Some of these theories take only certain specifiers to be opaque, rather than all specifiers; Chomsky (2008), for example, takes phase edges to be opaque.

Huang states the CED as follows (Huang 1982, p. 505):

- (4) Condition on Extraction Domain (CED): A phrase A may be extracted out of a domain B only if B is properly governed

In (4) “properly governed” holds of an item if that item is the sister of a lexical category (like a verb) or if it is appropriately local to an antecedent. Ignoring the latter, the CED will predict that complements are transparent for extraction while specifiers, which are never the sister of lexical categories, are not. The details are somewhat irrelevant, as the theory underwent much modification over the years, but the intuition, that complements are more transparent for extraction than specifiers, is the core of the CED.

The extraction data in (1)–(2) seems to be a *prima facie* problem for the system I’ve developed so far if the intuition behind the CED is correct, because, in the mereological system, there is no structural difference between subjects and objects: both are 2-parts. One might appeal to the nature of the category of which subjects vs objects are 2-parts, but the only difference is that between the categories *v* (or *T*) and *O*. Tying down a constraint to 2-parts of *v* vs *O* or *T* vs *O* of course just stipulates, rather than derives, the desired effects (why is it not the other way around?). One might argue that the stipulation might be justified, perhaps since *v* is special because it is a phasal category (Chomsky 2008), but that would be unwarranted in a system whose theory of locality is not phase theoretic.

There is an alternative to dealing with Subject Islands, however, that does not rely on the base-structural difference between subjects and objects, but on their surface position. Stepanov (2007), for example, argues that the issue is not specifier vs complement, but rather whether the constituent from which subextraction takes place is moved vs unmoved. He suggests that what creates a Subject Island is that extraction from a moved constituent is barred (cf. an idea developed by Culicover and Wexler 1977, p. 17ff; Wexler and Culicover 1981, under the rubric of their Freezing Principle). Since subjects are usually assumed to have moved from the specifier of *vP* to the specifier of *TP*, extraction from a subject will be impossible.

Stepanov adopts a version of the system in Takahashi (1994) to derive this effect. Takahashi proposed that the various elements of a movement chain have to be uniform in a certain sense, so that extraction from just one element will be impossible. More technically, Takahashi proposed that adjunction to part of a nontrivial chain is disallowed. To subextract from a subject, in Takahashi’s system, a potential extractee has to adjoin to that subject to meet a locality constraint. However, since the subject in the specifier of *TP* is part of a nontrivial (movement) chain, this is impossible. To see how this works, consider the difference between (5a) and (5b):

- (5) a.  $X \dots [_{\text{ZP}} X [_{\text{ZP}} \dots \langle X \rangle ] ]$   
 b.  $X \dots [_{\text{ZP}} X [_{\text{ZP}} \dots \langle X \rangle ] ] \dots \langle \text{ZP} \rangle$

In (5a), we extract the category  $X$  from inside  $\text{ZP}$ . To do this, Takahashi assumes that  $X$  must adjoin to  $\text{ZP}$  (essentially so that it is at the edge of  $\text{ZP}$ ) before it then moves to its final position. In (5b), we have exactly the same derivation except that  $\text{ZP}$  is first itself moved. Extraction from the moved  $\text{ZP}$  will mean that the two copies of  $\text{ZP}$  are not uniform (one has an extra “edge”), and this is ruled out by Takahashi’s condition. It follows from this that any phrase that has moved will be “frozen” for extraction from inside it.

There are examples that suggest that the Freezing approach is plausible. McNally (1992, p. 65) noted that extraction from subjects of existentials, which are lower than  $\text{TP}$ , is generally well formed in English, which would follow from Takahashi’s system, if these subjects are in situ:

- (6) a. Which book was there [an author of] on TV?  
 b. It was Elizabeth Taylor that there was [a feature on] in this week’s *People*.

Stepanov argues that Takahashi’s system also extends to cover cases that are otherwise captured by the Specificity Condition, adopting Diesing’s (1992) suggestion that presuppositional DPs must move to a position outside of the verb phrase, as I discussed in the previous chapter. If that is correct, these DPs will be islands by Freezing. For Stepanov, the movement is the crucial issue, with the presuppositionality being a side effect of movement. For Diesing, the presuppositionality is the “trigger,” with the final moved position being rendered an island via a CED-type system (Diesing adopts Chomsky’s Barriers theory, where a moved phrase is ungoverned, and hence acts as a barrier to movement).

My own proposal for Specificity Condition effects, laid out in chapter 6, is somewhere in the middle of these. I argued that scrambled nominal phrases are triggered to (internally) subjoin to some category outside of the verb phrase by a featural requirement of that category (it bears the feature  $[\text{uFam}]$ ). However, the matching  $[\text{Fam}]$  feature that can satisfy that requirement is internal to  $D$ , and so the derivation will not converge unless the category bearing the matching feature subjoins to  $D$ . This makes  $D$  opaque to extraction, and derives the Specificity Condition effects, where they hold. The crucial difference between the mereological system and the Freezing system is in how they capture these effects. In the former, only certain movements (subjuncts), those that check  $[\text{Fam}]$ , will give rise to an opaque object, so it will be possible to extract from moved non-specific DPs; in Freezing systems, by contrast, no element in a movement chain is transparent from movement from inside it.<sup>3</sup> I will return to this difference when I come to develop the mereological approach to subject islands in more detail in section 7.7.

The clearest cases for testing whether the Takahashi/Stepanov Freezing approach is more empirically adequate than a CED-type approach are languages where transitive subjects may remain in situ. From a CED perspective, such subjects would be predicted to be opaque for subextraction, since they are specifiers. From a Freezing perspective they should be transparent for extraction, since they are unmoved, hence not part of a nontrivial chain. Stepanov provides various examples from German, Hindi, and elsewhere to support his proposal, but the data in this area are notoriously murky. We will see though that it is possible to extract from a moved subject, a fact predicted by neither Freezing nor the CED.

### 7.3 CED vs Freezing

Jurka (2010) investigated *was-für* constructions in German, looking at examples where subjects and objects are moved vs (putatively) in situ. Following Webelhuth (1990), Grewendorf (1989), Diesing (1992), and Müller (2010) among many others, Jurka adopts the idea that the subject in German may remain inside the verb phrase and that if the subject is positioned following certain particles, that signals that it is low. The empirical generalization is that when the subject remains inside the verb phrase, it must be indefinite, or bear contrastive focus. Subjects outside the verb phrase may be definite, and, if indefinite, are interpreted as specific, almost exactly as we saw for scrambling in chapter 6 (this correlation between interpretation and position is, as we've seen, part of the motivation for the way that Diesing's system is set up).

One of the arguments for the general claim that subjects may remain inside the verb phrase is that verbal constituents containing indefinite, but not definite, subjects may front (Wurmbrand 2004, Grewendorf 1989):

- (7) a. ?[Ein Bauchredner einen Vortrag gehalten] hat bei der LSA noch  
       a ventriloquist a talk given has at the LSA never  
       nie.  
       before  
       'As far as ventriloquists are concerned, none has ever given a talk at  
       the LSA.'
- b. \*[Der Bauchredner einen Vortrag gehalten] hat bei der LSA noch  
       the ventriloquist a talk given has at the LSA never  
       nie.  
       before  
       for 'As far as the ventriloquist is concerned, he has never given a talk  
       at the LSA.'

This can be explained if indefinite subjects remain inside the verb phrase and then are pied-piped by fronting of a verb phrase. Wurmbrand attributes the

ungrammaticality of examples like (7b) to the general property of German that verb phrases can front but TPs cannot (Abels’s 2003 version of Antilocality—captured in the current system because TP subjunction to CP violates the combination of Angular Locality and Irreflexivity).

With this conclusion in hand, Jurka attempts to test whether in situ subjects differ from moved subjects in terms of their transparency to extraction processes via the *was-für* split construction. Consider the following examples:

- (8) a. Was für ein Schnitzel hat der Hermes verspeist?  
      what for a schnitzel has the Hermes eaten.up  
      ‘What kind of schnitzel did Hermes eat up?’
- b. Was hat der Hermes für ein Schnitzel verspeist?  
      what has the Hermes for a schnitzel eaten.up  
      ‘What kind of schnitzel did Hermes eat up?’

In (8a), the whole constituent *was für ein Schnitzel* has been extracted to the specifier of CP, while in (8b), this constituent has been “split,” with the *wh*-expression *was* extracting and stranding the remainder of the constituent.

Jurka tested cases where there was extraction with no *was-für* split from both subject and object positions, cases where there was a *was-für* split involving subjects in their putative base positions and in their moved position, and also those involving moved and unmoved objects. The raw judgment data (means with standard deviations) that he reports are as follows:<sup>4</sup>

| (9) |               | Mean | SD   |
|-----|---------------|------|------|
| 1   | Sub, –split   | 6.35 | 1.21 |
| 2   | Obj, –split   | 6.64 | 0.83 |
| 3   | Sub, +in situ | 3.55 | 1.81 |
| 4   | Sub, –in situ | 2.28 | 1.51 |
| 5   | Obj, +in situ | 6.14 | 1.29 |
| 6   | Obj, –in situ | 2.84 | 1.71 |

The first two lines of the table give the cases where there is no *was-für* split: the sentences’ average acceptability is high. The split from an in situ object (line 5 of the table) has roughly the same status. Splits from moved subjects and objects (lines 4 and 6) are much less acceptable, and one might think this backs up the Takahashi/Stepanov Freezing approach. It also provides some experimental evidence for the island status of scrambled objects discussed in the previous chapter. However, contra the expectations of a Freezing account, a split from a transitive in situ subject (line 3) is degraded compared to in situ objects.

Jurka draws the conclusion that there are two constraints at play: extraction is degraded when it is from a moved element, or from a noncomplement. The classical CED account would have trouble explaining why extraction

from in situ subjects is better than extraction from moved subjects, while, on the Stepanov/Takahashi account, it is not obvious why unmoved subjects and unmoved objects should differ. On Jurka's two-constraint proposal, it would seem that the constraints have different levels of input to grammaticality, since extraction from an in situ specifier (an in situ subject) has a different level of grammaticality from that from a moved object, the latter being worse. Exactly how the effects of the two constraints are to be combined is something Jurka leaves open.

#### 7.4 Phase Edge Accounts

Chomsky (2008) suggests an alternative to both the CED and Freezing approaches using notions from phase theory. Though his approach borrows certain aspects of both the CED and Freezing theories, it is set up differently and makes a different set of empirical predictions.

Chomsky suggests that a subject may move to the specifier of TP and, in parallel, a subpart of that subject may move to the specifier of CP. In this system, a derivation is possible where the two movements effectively take place in parallel. Recall that Bare Phrase Structure takes movement to involve a copy of the moved item. If we examine an unaccusative, for example, we have a representation that looks as in (10) (I have indexed the relevant copies for clarity, though the structure itself does not contain indices, and have enclosed lower copies in angled braces):

- (10)  $[_{CP} Wh_1 \dots [_{TP} [_{DP_2} \dots Wh] \dots [_{VP} \langle [_{DP_2} \dots \langle Wh_1 \rangle \rangle ] ] ] ] ]$

One copy of  $DP_2$  ends up as the subject, but the *wh*-expression extracted from DP is extracted from the copy of  $DP_2$  in object position. This would be ruled out by Takahashi's version of Chain Uniformity, so Chomsky does not adopt that. The rule for spelling out copies spells out the highest element of the chain, and all the lower copies of the chain are not spelled out.

Chomsky, using examples where P's are pied-piped rather than stranded, claims that subextraction from subjects of unaccusatives is better than subextraction from subjects of transitives:

- (11) a. It was the car (not the truck) of which the driver was awarded a prize/  
arrived late.  
b. \*It was the car (not the truck) of which the driver caused a scandal.

Putting aside the pied-piping, and indeed the status of the data, which is fairly delicate, it's worth looking at how this theory works in detail.<sup>5</sup> Chomsky proposes that phase theory rules out extraction from the base position of a transitive subject as in (11b). Subjects are in the specifier of *vP*, *vP* is a phase, and,

Chomsky claims, it is not possible to extract from inside the specifier of a phase (Chomsky 2008, p. 148). Gallego and Uriagereka (2007, p. 55) state this condition as:

(12) Edge Condition

Syntactic Objects in phase edges are internally frozen.

The Edge Condition in (12) will block subextraction in (11b) from the base position of the subject, while allowing extraction from the complement position in a passive or unaccusative. There is some motivation for (12) from the fact that, generally, subextraction from the specifier of CP is ruled out, and Chomsky points out that examples like (13) are not well formed:

- (13) \*[<sub>PP</sub> Of which car] did you wonder [<sub>CP</sub> [<sub>DP</sub> which driver <PP> ] ] [ <DP> caused a scandal]]

This same constraint is appealed to in order to explain why we don't find, in English, prepositions stranded in an intermediate specifier of CP:

- (14) a. Who did you believe that Anson listened to?  
b. \*Who did you believe to that Anson listened?

However, recall that the equivalent of (14b) is, in fact, well formed in Gaelic, undermining a Bare Phrase Structure type system incorporating (12):

- (15) Cò a bha thu a' crèidsinn ris an do dh'èist Anson?  
who REL be.PST you ASP believe.VN to.DEF C PRT listen Anson  
'Who did you believe that Anson listened to?'

Even if we adopt the Edge Condition, there is one more potential derivation for (11b) that needs to be ruled out: this involves subextraction of the *wh*-PP from the *surface* subject position, the specifier of TP (i.e., extraction of the bolded *Wh* in (16)). The specifier of TP is not a phase edge, so should be transparent to extraction:

- (16) [<sub>CP</sub> Wh<sub>1</sub> ... [<sub>TP</sub> [<sub>DP</sub> ... <Wh<sub>1</sub>>] ... [<sub>VP</sub> <[<sub>DP</sub> ... Wh] > ] ] ]

Here the subject has moved to the specifier of TP, and then the *wh*-expression has extracted. But if this is a possible derivation, then there should be no contrast between (11a) and (11b), since, in both, a derivation where the PP extracts from the subject (i.e., from the copy in the specifier of TP) should be possible.

Chomsky rules this derivation out by a further condition, which he calls the Inactivity Condition. This is essentially a version of Boeckx's (2012) idea that it is the feature-checking relation between a subject and T that leads to T's opacity. For Boeckx it is agreement features, and for Chomsky, it is an uninterpretable (case) feature.

## (17) Inactivity Condition (Chomsky 2008, p. 150)

An A-chain becomes invisible to further computation when its uninterpretable features are valued.

Chomsky's system makes different predictions from either the CED or the Freezing approaches. It blocks subextraction from higher copies of A-moved constituents via the Inactivity Condition, and from elements in phase edges via the Edge Condition, but allows it elsewhere, including from the base position of chains that are not in phase edges.<sup>6</sup>

Chomsky's appeal to the Inactivity Condition allows him to account for what he takes to be grammatical subextractions from raised and ECM subjects, as in (18):

- (18) a. [<sub>PP</sub> Of which car] is [the driver ⟨PP⟩] likely to cause a scandal?  
 b. [<sub>PP</sub> Of which car] did they believe [the driver ⟨PP⟩] to have caused a scandal?

Both of these examples have a transitive subject but, Chomsky argues, are grammatical because there is a derivation where extraction of the PP can take place before the subject's uninterpretable case features are checked. In (18a), there is a point in the derivation where *the driver of which car* is in the lower specifier of TP. However, since it does not have its case features checked there, it is possible to extract *of which car* to the higher specifier of CP, and then to raise the DP [*the driver* ⟨PP⟩] to its final (case checking) position. A similar derivation underlies (18b), with the only difference being that the final case-checking position for the ECM subject is a position in the matrix vP that the DP [*the driver* ⟨PP⟩] raises to for ECM case checking.

Although the empirical base for Chomsky's approach as presented in Chomsky (2008) was fairly limited, a number of experimental studies since have provided interesting evidence that is supportive of some of these predictions, although not all.

Jurka (2010), as well as testing in situ vs moved subjects and objects as discussed above, ran a different experiment along the same protocols to examine *was-für* splits in German for different kinds of in situ subjects. The raw means are:

| (19)         | no Split | Split |
|--------------|----------|-------|
| Unaccusative | 5.79     | 4.65  |
| Object       | 5.39     | 4.36  |
| Passive      | 5.84     | 4.74  |
| Transitive   | 5.52     | 3.51  |
| Unergative   | 5.85     | 4.32  |

These are slightly harder to interpret than the previous raw scores we saw, without some statistical help. The left-hand column shows the judgments of

the sentences without any *was-für* split, and are all pretty high, as expected. The right-hand column shows the judgments for splits from different kinds of subjects (as well as objects). Jurka shows, via 2 x 2 ANOVA tests, that subextraction from transitive in situ subjects is significantly more difficult than any other extraction (compare the raw mean score of 3.51 with the other scores), which is predicted by both the CED approach and by Chomsky's, but is a surprise in a theory based on Freezing. All of the other interactions are either very marginal (unergatives compared to transitive subjects or with objects) or nonexistent.

Jurka conducted a further experiment focusing on just unergative vs unaccusative subjects in *was-für* constructions, where he attempted to control for various potential processing factors (such as a processing cost connected to the processor dealing with two vs one arguments). This again showed a small effect of the type of predicate, with extraction from subjects of unergatives slightly worse than from subjects of unaccusatives, though Chomsky's theory would predict subjects of unergatives to be as different from subjects of unaccusatives as subjects of transitives are. The direction of the contrasts is what both the CED and Chomsky's system would predict, though why the size of the effect is small is not obvious. Jurka also looked at this same question in a separate experiment on English subextraction, focusing on the difference between passivized objects and subjects of transitives (p. 164ff). These are predicted by Chomsky's theory to be quite different. Jurka again found that there was a very small effect of the type of predicate, which he dismisses as being too weak to draw conclusions from.

Finally, Jurka ran an experiment looking at extraction from ECM subjects of transitives in English, as in (18b). These received a low score, comparable to, though slightly lower than, extraction from transitive subjects in general (means of 2.24 vs 2.61, respectively). This actually comports with previous informal judgments in the literature (for example, Chomsky marks as unacceptable *\*Who do you expect stories about to terrify John* in Chomsky 1973, p. 249), and is the opposite of what Chomsky's system predicts.

This final experiment undermines the Inactivity Condition: whether a transitive subject is in an ECM context or not, it seems to resist subextraction, though the Inactivity Condition predicts that there is a derivation where subextraction is possible. One might be simply inclined to remove the Inactivity Condition from Chomsky's system, but then the question arises of how to block a derivation for extraction from a transitive subject where the *wh*-expression raises from the specifier of TP. Without Inactivity, such a derivation is possible, but then the theory loses any way of capturing the core contrast; with Inactivity, the predictions made about ECM turn out to be incorrect. The *On Phases* theory as it stands doesn't appear to be supported by Jurka's experimental evidence.

Another language where grammatical subjects have been argued to be able to remain in situ is Russian. In a recent study using traditional judgment elicitation techniques, as opposed to experimental techniques, Privoznov (2021) argues that, in Russian, subextraction of a possessor works in the same way as Jurka reports for *was-für* split in German: subjects of transitives are more difficult to extract from than are subjects of unaccusatives and objects of transitives.<sup>7</sup> Privoznov also finds that subjects of unergatives behave like subjects of transitives (which was not a strong effect in Jurka's results). If the relevant subjects are indeed in situ in Russian, then this would favor a model like Chomsky's (or indeed a CED-based model) over a Freezing approach.

First Privoznov shows a clear contrast between extraction of a *wh*-possessor from preverbal subjects vs postverbal objects in transitives (I have bolded the *wh*-possessor and the DP it is extracted from, and italicized the verb):

- (20) vot tot čelovek,  
 here.is that man  
 'Here is the man ...'
- a. **č-ju** ty xote-l-a, čto-by ja *pročita-l* [**knigu**]  
 whose.ACC you want-PST-F that-SBJ I read-PST.M book.ACC  
 '...[whose book] you wanted me to read.'
- b. \***č-ej** ty xote-l-a, čto-by [**brat**] *pročita-l*  
 whose.NOM you want-PST-F that-SBJ brother.NOM read-PST.M  
 moju knigu  
 my.ACC book.ACC  
 Intended: '...[whose brother] you wanted to read my book.'

He then notes that the contrast is identical when the subject is postverbal. The examples he tests involve a preverbal object, which allows the subject to be postverbal. Crucially Privoznov assumes that postverbal transitive subjects are in situ (p. 115):

- (21) vot tot čelovek,  
 here.is that man  
 'Here is the man ...'
- a. **č-ju** ty xote-l-a, čto-by ja *pročita-l* [**knigu**]  
 whose.ACC you want-PST-F that-SBJ I read-PST.M book.ACC  
 '...[whose1 book]2 you wanted me to read 2.'
- b. \***č-ej** ty xote-l-a, čto-by moju knigu  
 whose.NOM you want-PST-F that-SBJ my.ACC book.ACC  
*pročita-l* [**brat**]  
 read-PST.M brother.NOM  
 Intended: '...[whose brother] you wanted to read my book.'

If we assume that what is at issue in (21b) is extractability from the subject (as opposed to something to do with the preposed object in that example), then this suggests that what we have here is not a Freezing effect, since the subject is assumed to be in situ, but rather an effect that blocks extraction from transitive subjects more generally.

When the object is preposed, it is an island, a result which is predicted by a Freezing account (and also by a Specificity Condition type account, since preposed objects are topical and specific). It is also predicted by a CED-type account that simply makes specifiers islands in general. On Chomsky's approach, one might say that if the object preposing is an A-movement, Inactivity would ensure it is opaque.

- (22) *eto tot samyj človek,*  
 this that.NOM same.NOM person.NOM  
 'This is the very same person...'
- a. *č-ju ty xote-l-a, čto-by [knigu] pročita-l*  
 whose.ACC you want-PST-F that-SBJ book.ACC read-PST.M  
*moj brat*  
 my.nom brother.nom  
 '...[whose book] you wanted me to read.'
- b. *\*č-ej ty xote-l-a, čto-by [brat] pročita-l*  
 whose.NOM you want-PST-F that-SBJ brother.NOM read-PST.M  
*moju knigu*  
 my.ACC book.ACC  
 Intended: '...[whose brother] you wanted to read my book.'

Finally, Privoznov shows that when we look at postverbal subjects of intransitives, it is possible to extract from subjects of unaccusative verbs but not unergatives, consistent with CED and Chomsky's phase-based account, but not straightforwardly compatible with Freezing approaches:

- (23) a. *..človek čja ty xote-l-a čto-by u nas na vitrine*  
 person.NOM whose.NOM you want-PST-F that-SBJ our on display  
*leža-l-a [kniga]*  
 lie-PST-F book.NOM  
 '...the person [whose book] you wanted to lie on our display.'
- b. *...\*človek čej ty xote-l-a čto-by u nas v kafe*  
 person.NOM whose.NOM you want-PST-F that-SBJ our in café  
*rabota-l [brat]*  
 work-PST brother.NOM  
 Intended: '...the person [whose brother] you wanted to work in our store.'

In an experimental study also focused on Russian data, Polinsky et al. (2013) examined *wh*-extraction from subjects and objects in Russian of a sort that is similar to *was-für* splits, and find results that are broadly consistent with Jurka's and Privoznov's. The core examples they used were of the sort in (24), where subextraction has taken place from the subject of an unaccusative:

- (24) **kakie**            ty    mečtaeš            [čto-by [aktjory] okazalis na  
       what-kind-of you dream.PST.2.SG that-SBJ actors    appear on  
       scene]?  
       stage  
       'What kind of actors do you hope to appear on the stage?'

Polinsky et al. varied the type of predicate, and also the position of the subject (whether it was postverbal or preverbal). For postverbal subjects they found extraction from transitives was in general significantly worse than extraction from intransitives, and, within intransitives, extraction from unergatives was worse than extraction from unaccusatives. As expected, extraction from transitive objects is significantly better than extraction from transitive subjects. These results for the *in situ* cases look very similar to both Privoznov's informal judgment tests and, to a certain extent, Jurka's results for *in situ* subjects in German.

Interestingly, in the order where the subject is preverbal, the same results hold. Extraction from transitive subjects is worse than from intransitive ones, and there is a difference, though more marginal than for the *in situ* cases, between subjects of unaccusatives and of unergatives. In cases where the object is placed to the left of the verb, extraction from that object is significantly degraded. These findings suggest that, in Russian, extraction from transitive and unergative subjects is degraded compared to unaccusative subjects irrespective of position, again consistent with Chomsky's approach, but in this case not with either Freezing or classical CED-type theories.

There are, however, some potential issues with the assumption, made by both Privoznov and Polinsky et al., that postverbal subjects in Russian are *in situ*. Although Bailyn (2004) argues that postverbal subjects are *in situ*, Slioussar (2011) shows, through a careful corpus study, that postverbal subjects of transitives and unergatives have to be *in focus* in Russian. She argues that they move to that position, while postverbal subjects of unaccusatives are *in situ*. In addition to the corpus evidence, Slioussar adduces evidence from scope and binding that postverbal subjects of unergatives and transitives are actually syntactically high in the structure.

There is further evidence that postverbal subjects are probably not *in situ* in Russian. Erechko (2003) noted that there are no definiteness effects on subjects

in OVS orders, which might be expected if they were low in the structure. She further shows that postverbal subjects must take scope over negation, suggesting that they are (interpreted) high:

- (25) *Etu zadaču ne rešili dva studenta.*  
 this problem.ACC NEG solve.PST.PL two students.GEN.SG  
 “Two students didn’t solve this problem.”  
 \*“It is not the case that two students solved this problem.”

However, if these subjects are high, then, of course, we can’t conclude that the results of Privoznov and Polinsky et al. support Chomsky’s idea that it is the Edge Condition that makes extraction from transitive subjects impossible. It may rather be the case that these subjects have actually moved, and that this is why subextraction is degraded. Further, Borik (2016) argues that bare singular nominals in Russian in preverbal position are typically interpreted as topics, and that they are also always definite. If this is the case, then there may be distinct reasons for the low level of acceptability of extraction from preverbal and postverbal transitive subjects (specificity for the former, and Freezing for the latter). It seems, then, that Russian, although it has been the subject of much investigation and therefore important to discuss, is not helpful in adjudicating between the various approaches we are considering.

Hungarian however does not suffer from the issue of whether postverbal subjects are low or not, as noted by Surányi and Turi (2018) in their experimental investigation of extraction in that language. The general consensus for Hungarian is that the verb raises outside of the verb phrase, and that any constituents following the verb are vP-internal (e.g., É. Kiss 1994, É. Kiss 2008). Preverbal arguments are assumed to be adjoined (or in specifiers) above the verb phrase and are, in the general case, interpreted as topics. There is also a preverbal contrastive focus position, where *wh*-expressions move to, but that can be diagnosed by the morphology of the verb and was controlled for in Surányi and Turi’s experimental design. The general structure of a Hungarian sentence is then roughly in (26), where Topic\* represents iterating Topic projections.

- (26) [Topic\* ... [Focus XP Verb Prt [vP ... ] ] ]

If the particle precedes the verb, then that signals that there is no Focus projection, so we simply have:

- (27) a. [Topic\* ... Prt-Verb [vP ... ] ] ]

Surányi and Turi carried out a series of experiments where an adpositional phrase is extracted from an argument, as in the following example where the PP *Melyik politikus salszeretnéd*, “with which politician” is extracted from the argument of an unaccusative:

- (28) Melyik           politikus salszeretné, hogy az újságban  
       'With.which politician want.2           that   the press.INESS  
       meg-jelenjen   [egy eltitkolt   interjú]  
       PRT-appear.SUBJ a       concealed interview  
       'With which politician do you want the newspaper to publish a concealed  
       interview.'

In this example, the DP *egy eltitkolt interjú*, 'a concealed interview' appears postverbally as the subject of an unaccusative verb and has had its complement *melyik politikus*, 'with which politician' extracted to the matrix wh-position. The topic in the embedded clause is the locative (inessive) *az újságba*, 'in the press.' Compare this with (29), where the subjects of the unaccusative and the locative change places:

- (29) Melyik           politikus salszeretné, hogy [egy eltitkolt   interjú]  
       With.which politician want.2           that a       concealed interview  
       meg-jelenjen   az újságban  
       PRT-appear.SUBJ the press.INESS  
       'With which politician do you want the newspaper to publish a concealed  
       interview.'

Surányi and Turi's analysis of the experimental results showed that that extraction from a transitive subject is far less acceptable than from an object or unaccusative subject (which behave the same with respect to extraction).<sup>8</sup> This is consistent with Chomsky's approach and the CED, since in both of these approaches (certain) specifiers are opaque for extraction; it is not predicted by Freezing, since extraction is degraded even though transitive subjects are in situ.

In addition, Surányi and Turi show that the preverbal vs postverbal position of the constituent that undergoes subextraction has no effect on the overall acceptability of the extraction. These results make the Freezing account difficult to maintain, unless somehow the preverbal constituents are not the result of movement, and the transitive subject has obligatorily moved, while the object and unaccusative subject are in situ. There is no other reason to adopt such an approach to Hungarian clause structure, making Freezing an unattractive route to explain the patterns.

Assuming that the preverbal arguments are moved to their positions, and are specifiers, a CED account would predict that they should be islands, but that is not what the findings of the experiments were. Topicalization has no effect on the opaqueness of the arguments to extraction. Crucially, an unaccusative subject (or an object) is as transparent for extraction when it is topicalized as when it is not. This is so even though the topicalized phrases are obligatorily interpreted as specific, suggesting, as Surányi and Turi note, that a blanket condition blocking

extraction from nominals that are specific or are surface topics cannot be correct. This is further evidence, parallel to the Persian phenomena I discussed in the last chapter, that Specificity Condition effects are syntactic, not semantic.

As noted by Surányi and Turi, Chomsky's phase-based approach seems to cut the empirical cake in the right way: transitive subjects are opaque, while unaccusative subjects and objects are transparent for subextraction, and movement to a higher position does not change that. Recall that in Chomsky's original approach the higher copy of a subject that has been extracted from has to be itself opaque for extraction. This was why Chomsky adopted the Inactivity Condition: when the subject moves to the specifier of TP, the copy in that specifier becomes opaque. In the absence of this stipulation, a transitive subject would always allow extraction, since extraction could take place from the higher copy, which is not at a phase edge.

Something similar will need to be said for the Hungarian data. If the specifiers of Topic projections are also opaque to extraction, this will mean that only the base position is relevant, and the patterning in the data will follow from the constraints on the base position of the arguments.

One issue is that Surányi and Turi do not investigate unergative subjects. Standardly, these are considered to be Merged in the specifier of the phasal category *v*. However, in Hungarian, the complements of nouns that are subjects to unergative verbs are usually possessors, not PPs, according to Surányi and Turi. Possessors apparently extract very easily in Hungarian, and it may be that this is not to be analyzed as movement, but rather as binding of a null pronominal. For this reason, subjects of unergatives were excluded from the study, which weakens it somewhat as support for Chomsky's theory. Recall, also, that there was only a very slight difference in extraction from unergatives vs unaccusatives in Jurka's studies, which also undermines the phase-theoretic account.

A second issue is that Surányi and Turi used specific indefinites throughout their examples. They note that a potential confound in Jurka's experiments is that the arguments of unaccusatives are often most easily read as nonspecific (that is, they are posited as new Discourse Referents), while the subjects of transitives are, they say, "typically interpreted as specific" (p. 327). To control for this, they use indefinites throughout, with a particular adjective that should enhance a specific reading (*a previous charge, a former debate, an undisclosed interview*, etc.). The idea is that this strategy should force specific readings of even postverbal nominal phrases, though it's not clear to me that it actually would. Consider (30), modeled on Hintikka's (1986) discussion of *certain*:

- (30) a. Each politician remembered a former debate—the one that had destroyed their career.

- b. Each politician remembered an undisclosed interview—the one that would certainly destroy their career if it came out.

Here the politicians each have a particular debate/interview in mind, though each has (at least potentially) a different one in mind from the one that the others do. It is not obvious that these are specific in the relevant way, as they take scope below a higher quantifier.

Even more so, unlike modifiers such as *certain*, *specific*, or *particular*, modifiers like *previous* or *former* seem to allow nonspecific readings (the following modeled on Enç 1991, ex. (8)):

- (31) a. Anson talked to a politician about this problem.  
       b. Anson talked to a certain politician about this problem.  
       c. Anson talked to a former politician about this problem.
- (32) a. Mona wrote about a girlfriend of Lara's in a poem.  
       b. Mona wrote about a certain girlfriend of Lara's in a poem.  
       c. Mona wrote about a previous girlfriend of Lara's in a poem.

In the examples with *certain*, a particular individual is identified, while in the examples with *former* or *previous*, no particular individual needs to be assumed to be present in the speaker's or addressee's mind.

For this reason, it's not clear to me that specificity was controlled for in the experiment. Given that, Surányi and Turi's comment that the subjects of transitives are "typically interpreted as specific" might well still be responsible for their findings, as opposed to the idea that such subjects are susceptible to the Edge Condition.

## 7.5 Further Empirical Problems

I've sketched out some recent evidence for the kinds of theories of subject islandhood that are on the market within minimalist syntax. However, although Chomsky's (2008) theory as outlined in *On Phases* seems to be closest to the empirical patterns that I've summarized here, it is not without its difficulties.

One is the point I made in the previous section, but which is worth repeating here. In Chomsky's system, the solution to the (putative) difference between extraction from transitive subjects and from other subjects is to allow parallel derivations, where a DP may be in one position for subextraction but another for full extraction. However, if that is possible, then we need a way to block subextraction from certain specifiers via phase theory (vP, the Edge Condition) and from others via feature theory (TP, the Inactivity Condition). Phase Theory will not rule out extraction from the specifier of TP, but if such extraction

were allowed, then a derivation with extraction from the subject of a transitive verb would converge with a grammatical outcome. However, the Inactivity Condition, at least as stated, incorrectly allows extraction from ECM subjects of transitives; Jurka's results (and informal judgments in the literature) show the opposite, so something else is needed, and it's not clear what.

In addition to this issue, there are some other, more straightforward, empirical patterns that are potentially problematic for any CED-type theory, including Chomsky's.

The first is just how acceptable PP pied-piping from transitive subjects is. It was known, even from the earliest discussions (Ross 1967, p. 133), that subjects are not islands for the extraction of PPs. Ross gave an example with a passive in (33), but examples of transitives seem fairly well formed, contrary to the predictions of the Edge Condition. For me at least, the difference between the (a) and (b) examples in (34) and (35) leans, in fact, toward the extractions from the subject being marginally better:

- (33) Of which cars were the hoods destroyed in the explosion?
- (34) a. Of which book did the shop invite [the author]?  
       b. Of which book did [the author] visit the shop?
- (35) a. This book, of which the shop invited [the author], is worth reading.  
       b. This book, of which [the author] visited the shop, is worth reading.

This observation is backed up by experimental evidence, which shows that PP extraction from subjects in English is rated fairly similarly to PP extraction from objects. In an experimental judgment task, Jurka (2010) showed that there is a striking difference in acceptability between subjects and objects of transitive predicates in preposition stranding extractions: extraction from objects is better than extraction from subjects. Jurka contrasted examples like (36), where the preposition has been stranded, with (37), where it has been pied-piped:

- (36) a. Phil wondered which topic a documentary about had swayed the voters last year.  
       b. Phil wondered which topic Scott had filmed a documentary about last year.
- (37) a. Phil wondered about which topic a documentary had swayed the voters last year.  
       b. Phil wondered about which topic Scott had filmed a documentary last year.

The raw mean judgment scores from Jurka's experiment are as follows:

|      |         |               |             |             |
|------|---------|---------------|-------------|-------------|
| (38) |         | no extraction | pied-piping | p-stranding |
|      | Subject | 6.24          | 3.29        | 2.51        |
|      | Object  | 6.54          | 3.86        | 5.08        |

As can be seen even from these raw scores, the contrast between the stranding and the pied-piping cases is stark. While stranding a preposition in a subject is degraded compared to stranding a preposition in an object, pied-piping the preposition from inside subjects or objects appears to make very little difference. Jurka carried out statistical tests (2-way repeated measures ANOVA subanalyses) that showed that the effect was very strong with stranding, and, although there was an effect with pied-piping, it was much weaker. In fact, while pied-piping decreases the acceptability of extraction from an object, it increases it when extraction is from a subject.

Jurka himself actually concludes from these data that apparent pied-piping does not involve extraction at all, but rather a PP topic attached at clausal level (following an analysis developed by Broekhuis 2005), so that the underlying form of an apparent subject extraction is as in (39):

- (39) Phil wondered [<sub>CP</sub> [<sub>TP</sub> [about some topic] [<sub>TP</sub> a documentary had swayed the voters last year.

The idea is that the PP headed by *about* is simply base generated attached to part of the clausal spine, and then is pragmatically connected to the subject noun. If this is the right analysis, there is no subextraction from the subject (see McInnerney 2023 for an extension of this idea).

In more recent work, Abeillé et al. (2020) control for this possibility in their experimental design, using relational nouns like *color* that take PP complements headed by the preposition *of*. They argue that such PPs cannot be interpreted as topics. Nevertheless, their results are similar to Jurka's: PPs extracted from objects are worse than when the P is stranded, while PPs extracted from subjects are better than when the P is stranded. It should be noted, however, that, for wh-extraction, Abeillé et al. found extraction from subjects to be markedly worse than extraction from objects in general (whereas they found that the result for relative clause extraction goes in the other direction when the PP is pied-piped). However, the materials used in the experiments involved non-agentive subjects of psychological predicates like *delight*, *please*, and *frighten*, as opposed to subjects of agentive transitives, so this may introduce another complexity.

Overall, it seems that it is possible to subextract PPs from subjects of transitives, contrary to Chomsky's approach in *On Phases*.

Turning to P-stranding in subjects in the specifier of TP, Polinsky et al. (2013) found no effect of transitivity at all. They did find a very marginal

difference between unergative and unaccusative subjects, but at such a low level of significance that they were unable to draw conclusions from the difference.

Neither result is expected on Chomsky's system: pied-piping should be impossible from transitive subjects, and it should not be the case that extractions from subjects in the specifier of TP, stranding P, should behave the same, irrespective of transitivity.

There are also a number of counterexamples in the literature of surprisingly acceptable extractions from subjects of transitives even when the preposition is stranded (the following are a subset of cases mentioned in Chaves 2021):

- (40) a. [Which car] did [some pictures of] cause a scandal (Jiménez Fernández 2009, p. 111)  
 b. [What] did [the attempt to find] end in failure? (Hofmeister and Sag 2010, p. 370)  
 c. [Which president] would [the impeachment of] cause more outrage? (Chaves 2012, p. 467)

Turning to other languages, Starke (2001) reports extractions from transitive subjects as well formed in French (though Tellier 1990, p. 307, gives extraction from subjects of transitives as unacceptable):

- (41) a. De quel film est-ce que tu as raté la première partie?  
 of which film is-it that you have missed the first part  
 'Which film did you miss the first part of?'  
 b. De quel film est-ce que tu crois que la première partie va  
 of which film is-it that you think that the first part goes  
 créer une scandale?  
 to-make a scandal  
 'Which film do you think that the first part of caused a scandal?'

Jiménez-Fernandez (2009, pp. 127–129) suggests that extraction from transitive subjects is acceptable in Spanish as long as the subject is indefinite:

- (42) a. De qué cantante te parece que algunas fotos han  
 of which singer you seem that some pictures have  
 escandalizado a la audiencia?  
 shocked to the audience  
 'Of which singer does it seem to you that some pictures shocked the audience?'  
 b. ??De qué cantante te parece que las fotos han escandalizado  
 of which singer you seem that the pictures have shocked  
 a la audiencia?  
 to the audience

for: ‘Of which singer do you think that the pictures shocked the audience?’

Finally, Spyropoulos and Stamatogiannis (2011) give the following examples from Greek of extraction of gentive marked arguments from transitive subjects:

- (43) a. pjanu aftokinitu su fanike [CP oti o odigos  
 which car-GEN you.GEN seem-PAST.3SG that the driver-NOM  
 paraviase to stop]?  
 violate-PAST.3SG the stop  
 ‘Of which car did it seem to you that the driver has disobeyed the stop sign?’
- b. pjanu aftokinitu su fanike [CP oti o odigos  
 which car-GEN you.GEN seem-PAST.3SG that the driver-NOM  
 xtipise mia griula]?  
 hit-PAST.3SG an old.woman-ACC  
 ‘Of which car did it seem to you that the driver has hit an old woman?’

It appears, then, that there is a great deal of variability in judgments, both informal and experimental, and both within languages and across languages, when it comes to subject islands, suggesting that the various factors that the major theories appeal to are incorrect. Chaves (2021) suggests that the variability in subject island judgments is a result of a number of interacting factors, some pragmatic, some processing, that all conspire in complex ways to give rise to the pattern of judgments we find. A more thorough-going syntactic approach that also attributes judgment variability to multiple interacting constraints is taken by Haegeman et al. (2014); their conclusions are that a number of different syntactic constraints are at play, and that acceptability depends on how many of these are violated. However, Greco, Marelli, and Haegeman (2017) attempted to test experimentally the set of constraints proposed by Haegeman et al. and argued that that theoretical system in fact was unable to explain, in a satisfying way, the patterns of judgments they found. Greco et al. suggest instead that the classification of verb types into transitive, unergative, and unaccusative is not helpful, and speculate that richer event representations may play a role.

I think this is correct. The representations that syntactic theory typically uses connect structural configurations to thematic classifications of verb types: unaccusative subjects are similar to objects in that both occur as complements, while transitive and unergative subjects are similar in that both are specifiers. This may well be correct, although of course more nuanced ways to connect syntactic structure and verbal argument structure have been developed (Ramchand 2008; Borer 2005b; and others). However, it appears that the rather blunt categories offered to us by such theories (complement of V vs specifier of *v*) are not

particularly successful in explaining the effects. There are, however, other ways in which semantic structure and syntactic structure are tied together that offer, I will argue, an alternative and more satisfying theory.

## 7.6 Locality at the Interface

One system that goes beyond the transitive/unaccusative difference is that of Bianchi and Chesi (2014), who argue that the most useful way of understanding the variability of acceptability judgments for subject islands is via appeal to the notion of thetic vs categorical interpretations of sentences. Their proposal builds from an observation due to Diesing (1992, p. 40), that subextraction from a subject of a predicate that is a stage-level intransitive predicate is better than subextraction from the subject of an individual-level intransitive predicate, as is shown in the contrast between (a) and (b) in (44):

- (44) a. \*Was sind für Schuhe wasserdicht?  
           what are for shoes waterproof  
           for: ‘Which kind of shoes are waterproof?’  
       b. Was sind für Karotten im Kühlschrank?  
           what are for carrots in.the refrigerator  
           ‘What carrots are in the refrigerator?’

Stage-level predicates (Carlson 1977, Chierchia 1995, Kratzer 1995) are predicates whose natural interpretation is that they hold of individuals in a way that might change across time. Individual-level predicates are naturally thought of as holding of an individual throughout its existence. In (44) *wasserdicht*, “waterproof,” is individual level, while *im Kühlschrank* “in the refrigerator,” is stage level. (44) shows that the subject of the individual-level predicate (44a) resists extraction in a *was-für* construction, while the subject of the stage-level predicate (44b) allows it.

Bianchi and Chesi give an analogous example from English, with an informal judgment (which I share).

- (45) Context: An art collector has ordered reproductions of a number of masterpieces: some big-size reproductions and a small-size one for each.  
       a. Of which masterpiece is [one reproduction] already available?  
       b. ?\*Of which masterpiece is [one reproduction] absolutely perfect?

Bianchi and Chesi argue that the stage- vs individual-level predicate distinction is what is relevant for extraction, as opposed to the notion of whether the subject is an internal or external argument. They show that the subject of a stage-level predicate like “useful” can be fully transparent for extraction even

if the predicate fails classical tests for unaccusativity. (46b) shows that *utile*, “useful” in Italian, does not allow *ne*-cliticization, one of Cinque’s (1990a) diagnostics for unaccusativity. By that test, the adjective is unergative. Bianchi and Chesi show that Cinque’s other unaccusativity tests lead to the same classification for this predicate. Nevertheless, (46a) shows that it does allow for extraction.

- (46) a. [Di quale legge] ritieni che sarebbe utile [una revisione]  
           of which law think that would.be useful a revision  
           ‘Of which law do you think that a revision would be useful’  
       b. ?\*(Di questa legge), **ne** sarebbe utile [una revisione]  
           of this law of.it would.be useful a revision  
           for: ‘As for this law, a revision of it would be useful’

*Utile*, then, is an unergative predicate that allows extraction. In addition, there are unaccusative predicates whose subjects are opaque for extraction, like *appartenare*, “belong” in (47). These are, Bianchi and Chesi argue, opaque because the predicate is individual level, irrespective of its unaccusativity.

- (47) \*[Di quale regione] sembra che vi appartengano [alcuni dialetti]  
           of which area seems that in.it belong some dialects  
           for: ‘Of which area do some dialects seem to belong in it?’

To test these ideas Bianchi and Chesi ran a series of experiments and concluded that the primary distinction relevant to extraction is the stage- vs individual-level interpretation of the predicate, as opposed to internal vs external argumenthood of the subject.

Why should it be the case that subjects of individual-level predicates are opaque for extraction? On Diesing’s account, these predicates have their subject generated outside the VP, and, given her Mapping Hypothesis, they are then presuppositional. Stage-level predicates generate their subjects VP internally, and those subjects may then move higher, in which case they are presuppositional (unless they reconstruct). Under Diesing’s *Barriers*-style account, these high subjects are all opaque for extraction.

Bianchi and Chesi do not, however, adopt Diesing’s proposals, but rather Ladusaw’s (1994) reinterpretation of them. Ladusaw argued that the crucial factor is not the nature of the predicate, but the nature of the kind of propositional structure that the predicate is compatible with. Ladusaw himself drew on Kuroda’s (1972) suggestion that there is a grammatically relevant semantic distinction between statements that are fundamentally about situations or events (thetic statements) and those that are fundamentally about individuals (categorical statements). Ladusaw, rethinking Diesing’s approach from this

perspective, suggested that “thetic judgments draw their basis from material analyzed as originating in the VP,” while categorical judgments involve a subject that is generated outside the VP. The idea that Bianchi and Chesi pursue is that it is the thetic/categorical divide that is crucial, and that judgments on subject islands are often somewhat variable because various contextual factors influence how people construe sentences as thetic vs categorical.

I think this is (close to) exactly right, though I’ll develop the same intuition in a slightly different way, which I think will allow us to derive aspects of what Bianchi and Chesi have to stipulate. Before turning to my own proposal though, it’s useful to look more closely at Bianchi and Chesi’s.

Bianchi and Chesi assume the standard syntactic view that all subjects are to be analyzed as being Merged verb phrase internally. In some languages or constructions such subjects raise to a higher position. To comport with Ladusaw’s theory, for a thetic construal of the sentence, a raised subject must be interpreted in its base position, irrespective of its surface syntactic position; for a categorical construal, it must be interpreted in its raised position.

Bianchi and Chesi implement categorical construal via a proposal made by Cardinaletti (2004) that there is a dedicated position for subjects (which Cardinaletti calls SubjP) in the higher clause structure. SubjP is associated with categorical interpretation. Cardinaletti suggests that there is another position for the subject (which she calls AgrSP), where the subject checks agreement features, but which is not associated with a categorical interpretation. Adopting this proposal for clause structure, Bianchi and Chesi combine it with Rizzi’s (2006) suggestion that, once a DP has moved to SubjP, it is frozen in place by a general constraint on how checking works (which Rizzi calls Criterial Freezing). It follows that DPs that have moved to SubjP are interpreted there, as opposed to being interpreted in their base position. This is what gives rise to a categorical statement. If, however, the subject is raised to a different position outside the verb phrase (say the specifier of Cardinaletti’s AgrSP, as opposed to the specifier of SubjP), it may be interpreted in its base position, inside the verb phrase, and the sentence then receives a thetic interpretation.

In order to capture the fact that extraction is blocked in categorical statements, Bianchi and Chesi propose the following constraint (similar to the proposal in Adger 2003, p. 397):

(48) Extraction from Subject Constraint (ESC)

Only a subject occupying a thematic position *at the interface* is transparent for extraction.

The italicized portion of the constraint highlights that, for a subject to be transparent for extraction, it needs to “totally reconstruct” inside the verb phrase.

With this theory in hand, Bianchi and Chesi ran an experiment to test its predictions. Italian is a language, like Hungarian or German, where subjects may sometimes remain in a postverbal, putatively *in situ*, position. This allowed Bianchi and Chesi to combine the pre-vs postverbal position of the subject with predicates that were individual vs stage level. For example, they presented their participants with sentences like the following (in a randomized Latin square design with distractors), where the predicate is individual level and the subject is pre-vs postverbal:

- (49) Context: A discussion between two experts on constitutional law.
- a. Di quale articolo ritieni     che [una revisione] sarebbe  
     of which section think-you that a    revision    would.be  
     incostituzionale?  
     unconstitutional  
     ‘Of which section do you think that a revision would be  
     unconstitutional?’
  - b. Di quale articolo ritieni     che sarebbe    incostituzionale [una  
     of which section think.you that would.be unconstitutional a  
     revisione]?  
     revision  
     ‘Of which section do you think that a revision would be  
     unconstitutional?’

The predictions of this system are:

- (50) a. The subject of an individual-level predicate is not transparent for extraction, irrespective of position, since it has to be interpreted as categorical, and hence is frozen in the specifier of SubjP, either on the surface, or at Logical Form (LF)
- b. The subject of a stage-level predicate is transparent for extraction only if it is part of a *thetic* structure—that is, it is non-presuppositional.

Interestingly, the results of the experiment showed no effect of predicate type on its own, though there was some effect of pre- vs postverbal subject position. They did, however, show a significant interaction between subject position and predicate type of the following sort: for the individual-level predicates, syntactic position was not significant while for stage-level predicates it was. The finding for individual-level predicates follows from the theory, since these should

always be opaque for extraction always. For stage-level predicates, if the subject moves to SubjP, it should be opaque, but if it moves to AgrSP, it should be able to reconstruct, and hence extraction should be possible. However, the results of the experiment suggested that the subject of a stage-level predicate that has moved preverbally is not transparent for extraction. It seems then that such subjects also resist reconstruction in Italian.

That conclusion seems to be independently warranted, given that there are differences between Italian and English with respect to whether subjects can reconstruct below sentential operators like Negation. Bianchi and Chesi give examples like (51), to show that while subjects can reconstruct in English, they cannot in Italian.

- (51) a. A unicorn seems [ to be in the garden]. ( seem >  $\exists$ )  
 b. Un unicorno sembra [ essere nel giardino].  
     a unicorn seems to.be in.the garden  
     ‘A unicorn seems to be in the garden.’ ( ?\*seem >  $\exists$ )

Bianchi and Chesi’s account (which they go on to implement in a top-down system) relies on a special constraint, their Extraction from Subject Condition in (48). This essentially blocks extraction unless the subject is reconstructed inside the verb phrase. They derive this, in their top-down implementation, from a more general condition on nested phases. I’ll argue in the next section that, in fact, it can be derived via the same mechanisms I have used to capture wh-islands and specificity islands: subjunction to a higher position requires previous subjunction internal to the DP, and this previous subjunction blocks extraction of anything else.

## 7.7 A Mereological Account

I think that the broad thrust of Bianchi and Chesi’s proposal is along the right lines. I agree that the variability in Subject Island judgments suggests that speakers are accessing contexts in different ways when they are judging the relevant sentences, and that it is this access to context that is influencing their judgments: if certain construals of the sentence are available, then certain extraction possibilities are available.

There is, of course, a long tradition that attempts to explain island phenomena not via mechanisms of the syntax, but rather via constraints on information structure/pragmatic interpretation (Erteschik-Shir 1973, Van Valin 1986, Erteschik-Shir and Lappin 1979, Kuno 1987, Deane 1991, Goldberg 2013 among many others). The core idea that is shared in these accounts is that certain constituents in a sentence, when uttered in discourse, are typically understood

to have certain pragmatic properties (for example, they are somehow backgrounded, given, semantically undominant, etc.). Those pragmatic properties are inconsistent with the pragmatic properties that are associated with syntactic displacement in some way. Though there are compelling critiques of the idea that syntax is irrelevant to islandhood (e.g., Boeckx 2012 for overview), clearly there is an interaction with information structure.

I'll argue here for a kind of middle way: the idea will be that the mechanisms of displacement are syntactic, and the constraints on them are syntactic, but, for certain information-structural interpretations, the syntax must be configured in certain ways. Those configurations are inimical to extraction. Different grammatical systems have different syntactic properties in terms of their distribution of features, and this is why we find variation both across languages (parametric variation) and within languages and speakers (individual variation and variation that is dependent on context). My proposal will be, then, resolutely syntactic, but the syntax will feed information structure, accounting for the correlations we see. I will not, however, appeal to a rich system of information structural features that are hierarchically arranged in particular syntactic positions, as in Cartographic approaches (Rizzi 1997 and work developed from there). Instead, I assume that information structural properties are essentially instructions given by syntactic configurations to the pragmatic systems.

There is, in fact, a fairly straightforward way of capturing the empirical patterns in the mereological system. The idea takes its guiding intuition from various sources, including Diesing's (1990) suggestion that there are two types of Infl associated with different syntactic and semantic interpretations, Cardinaletti's (2004) suggestion that there are two subject positions, and Miyagawa's (1997) notion that languages differ in whether they drive movement via discourse related or agreement related features. All of these proposals essentially suggest that, outside of the verb phrase, there are two (at least) syntactically and semantically distinct kinds of subject.

To implement this, I propose two kinds of T in a language like English, both of which have a requirement that they have a 2-part. One has a straightforward EPP-like requirement that it have a category D as a 2-part, so it is a T category that bears an uninterpretable [D] feature. There is also another T. This also requires a 2-part but places the restriction on it that it is interpreted as a sentence topic (Reinhart 1981).

For Reinhart, a sentence topic is what the sentence is about. She assumes Stalnaker's notion of Context Set (Stalnaker 1978), which is a set of propositions that the interlocutors hold to and which is updated as the conversation proceeds. Reinhart gives the Context Set a structure, whereby the entities discussed in the conversation are represented by file cards upon which are inscribed the various

properties the entities have, as expressed by the sentences in the conversation. With this theory in place, she takes a sentence topic to identify the entity in the Context Set to which information will be added by the sentence that is uttered (see also Tomioka 2007; Portner and Yabushita 1998; and others; cf. the “link” of Vallduví 1994). The Context Set is more usually, in current work, called the Common Ground (Krifka 2008), and I’ll use that term rather than Reinhart’s in what follows.<sup>9</sup> For example, imagine I utter (52):

(52) Your cat is hungry again.

We assume that Common Ground already contains an entity (a Discourse Referent in the sense of Karttunen 1976 or Kamp 1981, or a file, as in Reinhart 1981; Heim 1982; Erteschik-Shir 2007). This Discourse Referent has the information already attached to it that it is your cat, either because we’ve been talking about the cat (so it is discourse familiar) or it’s miaowing in the kitchen right now (so it is context familiar). Sentence topics are familiar in one of these ways, though that is not an innocent or uncontroversial assumption.<sup>10</sup> When I utter (52), my use of *your cat* allows identification of an existing Discourse Referent/file, and new information, that it is hungry, is attached to it. The phrase *your cat* is the sentence topic, and its nature as the sentence topic is what provides the instruction to manage the Common Ground in this way (Krifka 2008).<sup>11</sup> Sentence topics can also identify/accommodate Discourse Referents via bridging reference, as in:

(53) I have a shed. The roof is leaking.

I will assume, then, that a sentence topic identifies an entity that already exists in the Common Ground or can be accommodated to the Common Ground (Jäger 2001). It follows that a sentence with such a topic will be a categorical statement about an entity that is discourse (or context) familiar, so it can be identified. If the sentence does not have an entity as a topic, it is *thetic* (and the topic is a situation given in the context).

Since I take sentence topics to be discourse or situation familiar, I will reuse the Discourse Familiarity feature [Fam], introduced in the last chapter, which marks categories as being interpreted in that way. In fact, the proposal for subjects will be very close to the proposal I made for scrambling. I will say that, in addition to T with uninterpretable [uD], there is also a T that bears uninterpretable [uFam]. A speaker of English, then, may place a categorical construal on a sentence or not by interpreting (or by using) T\*[uFam] or T\*[uD]. Obviously, sometimes the syntactic configuration is incompatible with T\*[uFam], in existentials for example, which is why, in such circumstances, a categorical interpretation is impossible.

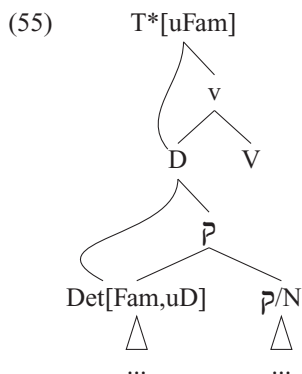
We can give an interpretation schema for the structures the two Ts will find themselves in:

(54) T-Interpretation

- a.  $[T_{[uFam]} D [x \dots]] \rightarrow$  Identify the (individual) Discourse Referent in the Common Ground that is anchored by D and add the information in X to it.
- b.  $[T_{[uD]} D [x \dots]] \rightarrow$  Identify the situation Discourse Referent in the Common Ground that is anchored by T and add the information in X to it.

I use the locution “anchored by” here to mean that the content of D or T is sufficient to pick out the relevant Discourse Referent (or file) in the Common Ground to be updated.

If X is a vP subjoined to  $T^*[uFam]$ , then in order to check the  $[uFam]$  feature, T needs to have a category specified with  $[Fam]$  as a 2-part. Since D does not bear interpretable  $[Fam]$  in English, this means that some category bearing  $[uD, Fam]$  within the DP has to subjoin to become a 2-part of D. The category could be Det, or a cardinal quantifier, as we’ve seen before, or Dem or a possessor (that is, another D). D will then subjoin to  $T^*[uFam]$ . Since  $T^*[uFam]$  is checked by  $[Fam]$  via Transitivity going through D to the relevant category, the sentence topic of T is the individual anchored by D, via (54), and we have what Bianchi and Chesi would take to be a categoral construal.



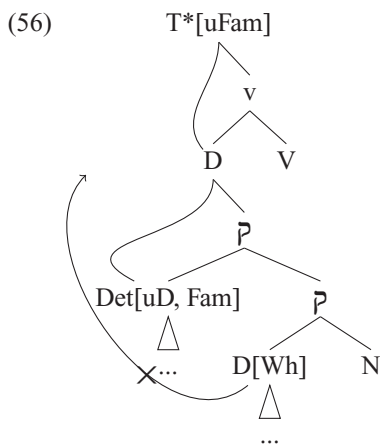
The way T-Interpretation works is not via a  $[topic]$  feature, but rather via the interpretation of the configuration: D is a 2-part of  $T^*[uFam]$  and the systems that manage where information goes in the Common Ground take advantage of that structure. This is very close to the configuration I proposed for scrambling except of course that scrambling involves  $Scr^*[uFam]$  not  $T^*[uFam]$ . The

difference between the two is T-Interpretation which makes subjects suitable sentence topics, as opposed to objects.

This system predicts that it will be impossible to subextract from D when  $T^*[uFam]$  is involved, since, to check T, D's 2-part slot will be filled with whatever category bears  $[uD, Fam]$ , capturing Bianchi and Chesi's (and ultimately Diesing's) generalization that it is not possible to subextract from the subject of an individual-level predicate/categorical statement. This is true also for topics that are subjoined to  $T^*[uFam]$  but pronounced in a lower position, if the language allows this: since there is just one object, in order to subjoin it to  $T^*[uFam]$ , D must either bear  $[Fam]$  (on which, see below) or have a 2-part that does. In the latter situation, extraction will be blocked.

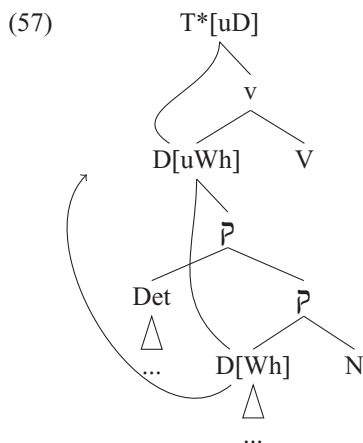
In general, this system captures the opacity of Topics that is often appealed to in the more pragmatically oriented accounts via feature checking, making it more flexible than the purely pragmatic accounts. I'll argue below that this is an advantage of this system.

Schematically, then, we have the following structure for extraction from a sentence topic/subject of a categorical statement:



Let's now turn to the structure with the other kind of T,  $T^*[uD]$ . In the statement of T-Interpretation, X is again a vP subjoined to  $T^*[uD]$  but the Discourse Referent identified is not an individual Discourse Referent anchored by D, but rather a situation Discourse Referent (temporally) anchored by T. This leads to athetic construal of the sentence, incompatible with individual-level predicates.

In terms of extraction, in order to check its uninterpretable D feature, T simply requires a D as a 2-part. If D is indefinite, or is a weak definite, it has a free 2-part, and extraction from the subject is possible.



In (57), the *wh*-expression subjoins to *D*, and so is a 2-part of *v*; by Angular Locality, it can then subjoin higher to *C*.

Of course, it might be that *Dem* or (optionally) *Det* bears [*uD*, *Fam*], in which case it will subjoin to *D*, resulting in a strong definite or specific indefinite. In such cases extraction is blocked. However, that extraction is blocked by the syntactic definiteness of *D* (i.e., the fact *D* has a filled 2-part), as opposed to a requirement imposed at the clausal level. It follows, correctly, that extraction will be blocked from strong definite subjects, irrespective of their topicality.

The general predictions of this system, for English and similar languages, are that non-discourse-familiar subjects will be transparent to extraction, and that discourse familiar subjects will block extraction. If a subject is interpreted as a topic, it will be discourse familiar (at least in English, where *T* bears [*uFam*]), irrespective of its determiner, and opaque for extraction. However, the system doesn't force subjects to be topics in English. *T*\*[*uD*] may trigger subjunction to *T*, but in such cases extraction from inside *D*, depending on the determiner in *D*, can be well formed.

With this theory in hand, let me briefly reconsider some of the experimental evidence reviewed above. Recall that Jurka showed that, in German *was-für* split constructions, subjects that had moved tended to be judged as unsplittable. This is predicted if, in German, *T*\*[*uFam*] is the only *T*\* available (which would make sense of the fact that indefinite high subjects in German are obligatorily specific). Subjects that remain inside the verb phrase are correctly predicted to be easier to split, since they are not forced to be discourse-familiar. In such cases *T* does not bear [*uFam*] or \*. Of these internal subjects, Jurka found only a mild difference between unergative and unaccusative subjects, and that may be related to unergatives tending to be construed categorically (that is,

their subjects are more easily interpreted as topics, given they are typically Agents). That may lead to a preference to parse them as having subjoined to T[uFam] even though they are pronounced low, making subextraction impossible. Equally, it may be that participants did not interpret unergative subjects topically, and so allowed the subextraction parse.

Jurka also found an effect that transitive subjects were more difficult to split regardless of their position (though not as difficult to split as moved subjects). I return to this below.

In Hungarian (and perhaps Russian) it's not clear that we have a T position, since Topics may iterate in that language, but whatever category hosts Topics also bears [uFam]: É. Kiss (1998) shows that topics in Hungarian are interpreted as discourse familiar.<sup>12</sup> It may be that the relevant category outside vP in Hungarian is more like Scr, which iterates in the language, and that T-interpretation applies to Scr in Hungarian. Whatever the details of the categories for the analysis, it seems that specificity is not a block to extraction in Hungarian. Recall that Surányi and Turi found that extraction from preverbal topics was possible. This suggests that, like Persian, D bears an interpretable feature [Fam] in Hungarian, unlike in German or English. This means that, when D subjoins to Scr[uFam], its 2-part is still free, allowing extraction. In general, the mereological system does not enforce discourse-familiar DPs to be islands, though it allows them to be.

What though are we to make of the differences between subjects of transitive predicates and of unaccusative ones that Surányi and Turi found in their experiments? In fact, in much of the experimental evidence, there was an asymmetry between transitive subjects and intransitive subjects (the evidence for an unergative vs unaccusative distinction was much weaker across the board) with the evidence pointing to in situ subjects of transitives being more transparent to extraction than moved ones, but still less so than objects or subjects of unaccusatives. Why is this?

It's interesting to note that, in typological surveys of languages,thetic sentences generally do not occur with transitive verbs with full nominal objects; rather the object has to be pronominal, cliticized, or incorporated (see Lambrecht 2000; Sasse 1987). The generalization seems roughly to be the following:

- (58) a. [ DP V DP ] : Categorical  
       b. [ DP V-pro], [DP V-N] : Thetic

Whatever is responsible for this, if it is correct, may impact on the design of the experimental materials. The various sentence types used to test transitives, as far as I can tell, involved full DP objects. Given that, their subjects were likely to be interpreted as topics in a categorical construal, which means in turn that

they are likely to be interpreted as discourse or context familiar. That entails that subextraction would have been incompatible with that parse. In order to test this, of course, the experiments would need to be rerun with pronominal objects, in which case the prediction would be that the extraction from transitive subjects would look more like extraction from unergative subjects.

I think this may well be what explains the effects found by Surányi and Turi, where transitive subjects were more difficult to extract from, and it may also underlie Chomsky's judgments.

I don't have a full explanation of the pied-piping cases. Again, this seems to be something that is variable across languages and speakers. Recall that the experimental evidence shows that English speakers allow pied-piping from both subjects and objects, something that is reported for Greek too (where pied-piping is the only option). French seems to be variable from the informal judgments, and Spanish seems to be like Greek, but only with indefinites. Again, to determine what is going on, a cross-linguistic experimental investigation that controls for whether the subjects are sentence topics or not is needed.

The various counterexamples to Subject Islands discussed by Chaves are, however, I think, expected on the system I've laid out here. He provides the following examples culled from the literature (see Chaves 2021, p. 683, for full citations):

- (59) a. [What] were [pictures of] seen around the globe?  
 b. It's [the kind of policy statement] that [jokes about] are a dime a dozen.  
 c. There are [certain topics] that [jokes about] are completely unacceptable.  
 d. [Which car] did [some pictures of] cause a scandal?  
 e. [What] did [the attempt to find] end in failure?  
 f. [Which president] would [the impeachment of] cause more out-rage?  
 g. I have a question that [the probability of you knowing the answer to] is zero.

As far as I can tell, most of these examples involve indefinites and weak definites (definites in nominalizations or uniqueness, as opposed to discourse-familiar, definites). On the theory discussed in the last chapter, they should not be islands, if subjects in general are not islands. It is easy to construe these subjects as nontopics, and none are of a sort that would require the 2-part of D to be filled on the proposal developed so far.

Finally, the system I've sketched out here of course is consistent with Bianchi and Chesi's results, although no special Extraction from Subjects Constraint needs to be stated. While Bianchi and Chesi restrict extraction to reconstructed vP internal positions, my system bars it from Topic positions, using just the

same mechanisms as are used to block movement from *wh*-islands, or from scrambled constituents: the 2-part of the relevant category has been filled because of a feature checking relation that category has to enter into. Bianchi and Chesi's system predicts that subextraction from subjects of individual-level predicates is impossible irrespective of where the subject is, for a language like Italian. In my system, this corresponds to the claim that such subjects subjoin to T[uFam] but may be pronounced either high or low. Their system also predicts that a subject of a stage-level predicate is transparent for subextraction irrespective of height, which turns out not to be correct. This is the reason that they add the requirement that, in Italian, movement to a preverbal position does not reconstruct. In the mereological system, a subject of a stage-level predicate will be a topic when it has subjoined to T[uFam], in which case subextraction will correctly be predicted to be impossible.

## 7.8 Summary

One of the consequences of the mereological approach developed here is that the CED, and indeed Freezing, are unstatable. The former because there is no geometrical difference between subjects and objects, and the latter because there is only one SO, which is subjoined in multiple places, so there is no way of distinguishing "chain" links. This means that these subtheories are unavailable as explanations for subject islands, and I've argued that this is actually a positive result. Subject islands, it turns out are not consistent islands, and, as was suggested by Haegeman et al. (2014), the various situations where subjects are opaque for extraction should be tied down to other factors (see also McInnerney and Sugimoto 2022). I've suggested that the fundamental relevant factor is, as we saw in the previous chapter, having a D whose 2-part is filled. This comes about in various ways for subjects, but the central point is that subjects that are difficult to extract from are sentence topics, and the syntax of topicalization is such that, in many cases, an item bearing a [Fam] feature must subjoin to D, making it opaque for further extraction.

## 8 Conclusion

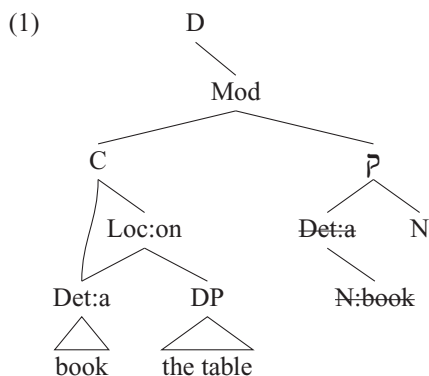
In this book, I've argued that we should think seriously about replacing Merge with Subjoin, hence making the SOs that the system builds mereological, rather than set theoretic. I've shown that this has an immediate theoretical advantage: it makes the system simpler, as there is no need for a theory of labeling or for a theory of the difference between copies and repetitions. However, the mereological system is in other respects very similar to the Merge-based system, in that a single operation builds and modifies structures.

I proposed replacing Phase Theory with a geometrical constraint on what makes a well-formed SO, Angular Locality. I argued that Angular Locality was a very natural constraint in this system, in that it was simply an extension of the properties of the most local configurations to nonlocal ones. I showed how it does an enormous amount of work in restricting the kinds of possible SO that can be built, blocking downward dependencies, dependencies that are too short, and dependencies that are, in a particular sense, too long. It also derives as a theorem the successive cyclicity of long-distance dependencies, and answers a number of quite fundamental theoretical questions that classical Phase Theory doesn't have answers to.

I argued that Angular Locality also allows us to have a fairly unified theory of islands, a goal that has been rather sidelined over the past few decades of minimalist syntax. In particular, I argued that *wh*-islands, specificity islands, and subject islands, together with their various exceptions can be brought within a single theory: they all emerge because of an interaction between Angular Locality and the Dimensionality and Transitivity properties of parthood.

There are, however, many areas that I have not touched on, but that are, at least arguably, relevant to the general narrative of the book. One such area is adjunct islands, including relative clauses and clausal adjuncts. These, like subject islands, have been argued to result from the CED, and since the mereological system eschews even the intuition behind the CED, I owe at least a sketch of how these islands might be tackled.

For relative clauses, I mentioned in the discussion of extraction from locative PPs inside nominal phrases that these might be analyzed as reduced relatives, where the PP's "subject" is subjoined to a C projection that takes the PP as a 1-part, and then that CP is subjoined to a modificational category in the nominal Extended Projection. This will mean that *a book on the table* has the following structure:



This is a matching relative clause structure, so the Det constituent in the main projection is deleted under identity with that of the relative clause. This structure will not allow subjunction of the DP that is a 1-part of the locative PP (or the PP itself) out of the nominal phrase, as that would violate Angular Locality. Matching relative clauses, then, are predicted to be islands in the system.

However, there are well-known exceptions to relative clause islandhood, especially in situations where the nominal that the relative modifies is in a presentational context:

- (2) This is the child that there is [nobody who is willing to accept ]. (Kuno 1976, p. 423)

Sichel (2018), in an investigation of the vitiation of relative clause islands in Hebrew, argues that their island status depends on two things: whether the relative clause is interpreted *in situ*, and whether it has the characteristics of a raising or matching analysis. She shows that extraction from a relative is possible just in case it is analyzed via a raising strategy, and the DP it modifies remains *in situ* in the vP and does not scramble out of it. The second claim comports with the discussion in chapter 6, and I will put it aside here, as it is already predicted by the theory.

As evidence that extraction is possible where the relative clause can have a raising analysis, Sichel gives examples like (3):

- (3) [al lexem šaxor], ani makira rak [gvina levana axat] še-efšar  
 on bread black I know only cheese white one that-possible  
 limroax ⟨gvina levana axat⟩ ⟨al lexem šaxor⟩  
 to.spread  
 ‘On black bread, I know only one white cheese that can be spread.’  
 (Sichel 2018, p. 336)

In this example, an argument PP has been extracted from an infinitival clause one of whose arguments has been relativized:

- (4) PP ... [ NP [<sub>CP</sub> ... [ PRO V-inf ⟨NP⟩ ⟨PP⟩ ] ] ]

Sichel uses PPs to ensure that a *pro* analysis, à la Cinque (1990b), is not available, and she ensures that this PP is neither a general sentence topic nor is selected by the higher predicate, by carefully choosing the matrix verbs so that they are incompatible with such analyses. She further argues that the PP must have moved, because putting in a strong island degrades the sentence more than putting in a non-island structure does. In (5a) the PP trace is inside the relative clause, which is itself inside a complement clause, while in (5b) it is inside an adjunct clause:

- (5) a. ?[al lexem šaxor], yeš [rak gvina axat] še-karati še-keday  
 on bread black BE only cheese one that-read.I that-worth  
 limroax ⟨rak gvina axat⟩ ⟨al lexem šaxor⟩.  
 to.spread  
 ‘On black bread, there’s only one cheese that I read was worth spreading.’  
 b. \*[al lexem šaxor], yeš [rak gvina axat] še-ta’amti ⟨rak gvina axat⟩  
 on bread black BE only cheese one that-tasted.I  
 [bli limroax ota ⟨al lexem šaxor⟩].  
 without to.spread it  
 for: ‘On black bread, there is only one cheese that I tasted without spreading.’

Sichel also shows that in such cases you can have reconstruction for condition A, again suggesting that the PP has moved:

- (6) [al acmam<sub>1</sub>], yeš nora me’at talmidim<sub>1</sub> še-muxanim lixtov  
 about themselves BE very few students that-willing to.write  
 ⟨al acmam⟩.  
 ‘About themselves, there are very few students who are willing to write.’

Having established that the extractee from the relative is indeed moved, rather than base generated, Sichel shows that, in certain circumstances, relative clauses disallow extraction. She draws a correlation between the possibility of extraction, and whether the nominal “head” of the relative clause is forced to be interpreted high, in its surface position, or low, in the trace position. Consider the following examples:

- (7) a. Those are the bar mitzvah photos of John<sub>1</sub> that he<sub>1</sub> stole from my office.  
 b. He<sub>1</sub> tried to deny the rumors about himself<sub>1</sub> that Mary spread at school.

In both cases the head of the relative (*the bar mitzvah photos of John*, or *the rumors about himself*), must be interpreted high. If it is interpreted low, then there will be a Condition C violation in (7a) or a Condition A violation in (7b). This means that these structures can receive only a matching analysis, since only in that analysis is it possible to interpret the relative head high.

Sichel shows that in such circumstances, where the relative head is interpreted high, extraction from the relative clause is unacceptable in Hebrew:

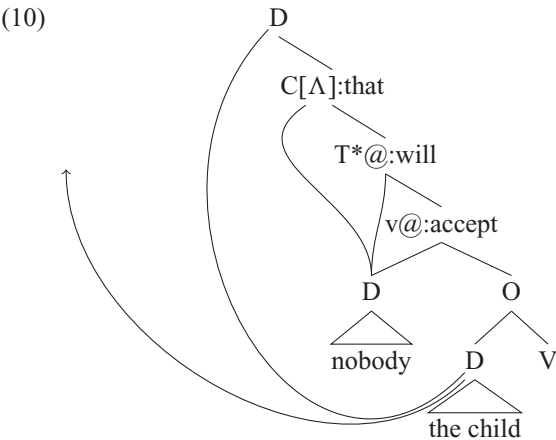
- (8) a. \*[me-ha-doda ha-zot], yeš [kama tmunot bar micva šel dani]<sub>1</sub>  
 from-the-aunt the-this be few photos bar mitzvah of Dani  
 še-hu<sub>1</sub> ša'al ⟨kama tmunot bar micva šel dani<sub>1</sub>⟩  
 that-he borrowed  
 ⟨me-ha-doda ha-zot⟩.  
 for: ‘From this aunt, there are a few bar mitzvah photos of Dani’s that he borrowed.’  
 b. [me-ha-doda ha-zot], yeš [kama tmunot bar micva šelo]<sub>2</sub> še-hu<sub>1</sub>  
 from-the-aunt the-this be few photos bar mitzvah of.his that-he  
 ša'al ⟨kama tmunot bar micva šelo<sub>1</sub>⟩ ⟨me-ha-doda ha-zot⟩.  
 borrowed  
 ‘From this aunt, there are a few bar mitzvah photos of his that he borrowed.’

In (8a), the PP has been extracted across the relative clause, but crucially the relative clause requires high interpretation of the head, since the head *kama tmunot bar micva šel dani*, “few photos of the bar mitzvah of Dani” contains the R-expression *Dani* which would be bound by the pronoun *hu*, “he,” giving rise to a Condition C effect. To the extent that Condition C effects are found under reconstruction in Hebrew, this forces a high interpretation of the head that correlates with a matching structure for the relative clause, and we see extraction is then blocked.

To the extent that Sichel’s conclusions are correct, they can be implemented in the current system in a straightforward fashion. Sichel’s own approach to locality involves a combination of Antilocality of the sort that requires movement to cross a full maximal projection, and the phasehood of NP, following the proposals outlined in Bošković (2017). However, there is a way of translating Sichel’s core insight into the system developed here that does not require such stipulations. As is standard, we can treat raising relatives as involving a D with a CP complement. Take a version of Kuno’s example in (2):

(9) This is the child that there is nobody that will accept.

The embedded relative clause here is *nobody that will accept (the child)*. (10) shows how the subject *nobody* is relativized via subjunction to C. Because C is a 1-part of D, it is possible to subjoin *the child* outside of that relative clause, and from there, if required, to successively higher positions.



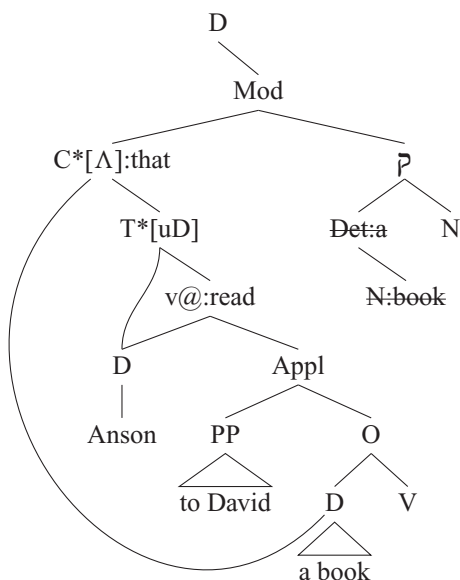
I use the [Λ] feature from (Adger and Ramchand 2005) on C to mark C as a relative clause with the interpretation:  $[\lambda x. \text{ nobody will accept } x]$ . This relative clause will eventually combine with the meaning of the extracted D, to derive the outcome:  $[\lambda x. x \text{ is a child and nobody will accept } x]$ .

This structure captures Sichel’s claim that it is relatives with raising analyses that allow extraction.

On the matching analysis, we take the relative clause to be a 2-part of a modificational head, as we did for the locative PPs mentioned above. Since the relative clause is a 2-part, and its C has a filled 2-part, any further subextraction is ruled out. An example like (11), where the indirect object is a 2-part of an applicative category, will have the representation in (12):

(11) A book that Anson read to David

(12)



Since  $C^*[\Lambda]$  has a 2-part, extraction from the PP *to David* to some higher position will violate Angular Locality:

(13) \*Which linguist did you buy a book that Anson read to?

Overall, the system I've already outlined can, I think, be readily extended to capture the island status of relative clauses, and the reasons for their exceptions, following Sichel. There is of course a great deal of empirical work still to do to determine if the various consequences of such a move are empirically borne out, especially in figuring out exactly when the different analyses of relatives as raising or matching are forced, possible, or impossible in different languages and different constructions.

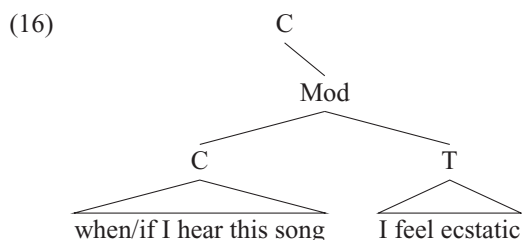
The second kind of CED island that I haven't yet dealt with is clausal adjuncts. Consider the examples in (14):

- (14) a. When I hear this song, I feel ecstatic.  
 b. If I hear this song, I feel ecstatic.  
 c. I feel ecstatic when I hear this song.  
 d. I feel ecstatic if I hear this song.

It is impossible to extract the object of the clausal adjunct in either case:

- (15) a. \*Which song did you say that when you hear, you feel ecstatic?  
 b. \*Which song did you say that if you hear, you feel ecstatic?  
 c. \*Which song did you say that you feel ecstatic when you hear?  
 d. \*Which song did you say that you feel ecstatic if you hear?

Let's assume that these clausal adjuncts are also introduced as the 2-parts of a modificational category in the clause (which again, I'll just call Mod, leaving aside the possibility, indeed likelihood, that there are different types of Mod at different "heights" in the Extended Projection). We might think of the different orderings of the clauses as either involving subjunction to some higher category or as involving different linearizations in situ. Sentences like those in (14) will then have the structure in (17):



The question to be answered is why extraction from inside the 2-part of Mod leads to unacceptability.

There is a long tradition in generative syntax, beginning with Geis (1970), of analyzing temporal and conditional adjuncts as involving movement of some operator to an initial position in that clause. Haegeman (2012) surveys some of the evidence for this perspective and gives examples involving a temporal relative clause as evidence that the temporal *wh*-expression *when* moves in such clauses:

- (17) Abdul Barr said Beah was at school in 1993, the year [when he said he was a child soldier]
- a. high construal: 'Beah was at school when he said this.'  
the year [<sub>CP</sub> when [<sub>TP</sub> he said [<sub>CP</sub> [<sub>TP</sub> he was a child soldier]] <when>]]
  - b. low construal: 'Beah was at school at the time of allegedly being a soldier.'  
the year [<sub>CP</sub> when [<sub>TP</sub> he said [<sub>CP</sub> <when> [<sub>TP</sub> he was a child soldier <when>]]]]]

(17) has two readings for the construal of the time of the temporal relative: in the high construal, the event time of the nontemporal embedded clause (i.e., the time of Beah being at school) is equated with the time of Beah saying he was a child soldier, while in the low construal it is equated with the time he was a child soldier. This can be understood as a case of structural ambiguity that depends on the position of *when*. Haegeman shows, again following Geis, that the same ambiguity arises with temporal adjuncts, drawing on Larson (1987).

Consider the following, modeled on Larson's examples. I have altered the examples a little to make the relevant contrast sharper, to my ears at least. The context is that Mary is a hacker who is engaged in some nefarious activities:

- (18) I saw Mary in New York when [<sub>TP</sub> Peter said [<sub>CP</sub> that [<sub>TP</sub> she was defrauding Lucy.]]]
- a. high construal: 'I saw her at the time Peter talked to me.'
  - b. low construal: 'I saw her at the time of her crime.'

Beyond the ambiguity, there is further evidence that these temporal adjuncts also involve movement of *when*. If a *wh*-island is inserted between the lowest clause and the surface position of *when*, only the high construal is available. For the following, imagine a context where Mary is a hacker working on a bunch of anonymous accounts, so she doesn't know that she's actually defrauding her own wife, Lucy:

- (19) I saw Mary in New York when [<sub>TP</sub> she found out [<sub>CP</sub> who [<sub>TP</sub> she was defrauding.]]]
- a. high construal: 'I saw her at the time of the revelation'
  - b. \*low construal: 'I saw her at the time of the crime.'

Haegeman also draws on Geis's original observations to show that this same kind of ambiguity, and same blocking effect of islands, is found with other temporal connectives that are not so obviously *wh*-expressions, such as *until*, *before*, *after*, and *since*.

If this approach is along the right lines, then we can assimilate many clausal adjuncts to relative clauses, and appeal to the same structural factors that rule out extraction from matching relatives to rule out extraction from clausal adjuncts.<sup>1</sup> It may also be the case that we can appeal to the matching vs raising ambiguity of relatives to explain why, in some languages, it is possible to extract from clausal adjuncts (see, for example, Bondevik and Lohndal 2023 on Norwegian and Schurr et al. 2024 on Shupamem). I leave exploration of these issues for future work.

These two sets of phenomena (relative clause islands and clausal adjunct islands) have potential analyses in the system developed here, though there is much to be explored. However, there are other locality effects that seem quite different. One particularly interesting case is Improper Movement effects (and the more generalized version of this sometimes known as the Williams Cycle, after Williams 2003). Why is it not possible to subjoin a *wh*-expression from a subordinate clause to a higher T, to give examples like (20):

- (20) \*Who seems that we saw?

There have been many different approaches to these effects over the years, involving binding (Chomsky 1981, p. 195; p. 201ff), feature accessibility (Adger 2003, p. 388; Keine 2020b), the timing of the building of Extended Projections (Williams 2003; Meadows 2024), among others. The mereological system as it stands has nothing in particular to say in terms of phrase structure about such examples, though one intriguing possibility arises given the theory of \*-Linearization I laid out in chapter 3.

Recall that Linearization of a sequence of 1-parts triggered by @ operates on categories organized in the Extended Projection order. One way of thinking about Improper Movement might be that Linearization triggered by \* (that is of 2-parts, not 1-parts) also needs to have the relevant categories organized in the Extended Projection order. For example, if some wh-expression is a 2-part of O and C in an embedded clause, and it is a 2-part of T in the higher clause, as in (20), then there will be a sequence of categories that looks as follows:

(21)  $T^* \frown C \frown O$

(21) is of course not legitimate sequence in terms of the Extended Projection order. We might say, then, that the wh-expression cannot be linearized at  $T^*$  since  $T^*$  is not the highest category in the sequence qua Extended Projection (C is) but that it also cannot be linearized lower, since it must be linearized at the highest category bearing \*. If we say further that every subsequence in such a structure must have a well-formed linearization, that will block the Improper Movement example in (21) and will extend, at least in principle, to other Williams Cycle-style effects.

If we were to define \*-Linearization in this way, it would be more complex than the current theory, but it would straightforwardly extend a constraint that holds perforce on 1-part linearization to 2-part linearization. In that sense, it is a fairly natural extension of the theory of Linearization in general. However, I'll leave this idea as speculation here, unfortunately adding linearization to binding, feature accessibility, and timing as a possible, but probably unsatisfactory, explanation for the Improper Movement effect.

Finally, the mereological approach has little to say about whether there is a Relativized Minimality-type constraint that limits the span of dependencies in a way that goes beyond what Angular Locality requires. One reason to assume that there is such a constraint is that subjects and not objects subjoin to T. If there were a constraint requiring the dependency that T enters into with D to be “short” in some sense, this would follow. On the other hand, it may be that this is simply a matter of parameterization. If ergative languages involve subjunction of the object to T, as proposed in some theories of ergativity, the option of raising the subject vs the object might be just determined by different kinds of T

in accusative vs ergative languages (see, e.g., Bittner and Hale 1996; Aldridge 2008; however, see Legate 2012; Polinsky 2016 for alternative theories of ergativity). Again, I leave this as an unresolved question, just noting that, for the phenomena that have most concerned me here, Relativized Minimality or similar theories do not seem to be required, though they are not incompatible with the system.

As must be evident from this short concluding chapter, the mereological approach opens up at least as many questions as it (hopefully) provides answers to. I'm sure there is much to be improved in the system, but I hope it constitutes a small step in a new and interesting direction for the theory of phrase structure.

## Notes

### 2 Sets

1. Semanticists reading this may be perplexed by the idea that mereological objects, which are typically used in semantics to model undifferentiated structures, could be useful in syntax, given that SOs are highly differentiated. This is true for Classical Extensional mereological theories (see, e.g., Cotnoir and Varzi 2021). I'll be adopting a view of mereological objects that traces its roots back to Aristotle and takes mereologically organized objects to involve both substance (what they are built out of) and form (how they are built). See chapter 3 for discussion.

2. There are of course transitive relations, such as dominance, that have been central to defining SOs as phrase structure trees (see especially Chametzky 1996; Chametzky 2000), but I accept the minimalist argument from Chomsky (1994) that Phrase Structure-theoretic approaches contain too many stipulations to serve as the basis for a minimalistically inclined syntactic theory.

3. There are actually some contentful differences between the kinds of information encoded in trees vs sets of strings, but these can be put aside. See Lasnik and Kupin (1977) and Chametzky (1996) for some discussion.

4. As an example of a theory adopting a set-theoretic perspective, Chomsky mentions Haskell Curry's (1961) proposals. However, Curry does not, in fact, suggest that SOs are sets; rather, he defines a formal system of objects that are built from atoms via operations on the atoms and already constructed objects. Curry's suggestion was that SOs should be understood as being built from a binary operation and are to be thought of "not as strings of atoms, but as structures like a genealogical tree" (Curry 1961, p. 65). Such objects are certainly linearly unordered, but they are not set theoretic. Curry further suggested that their relationship to linear strings can be given by a function that associates parts of the tree with parts of the string, so that a grammar has two components: one determines grammatical structures (what Curry called tectogrammatics), and the other determines their relationship to linear order (phenogrammatics). Curry remarks, "No doubt someone will propose better terms later." One might think that Merge and Externalization (e.g., Chomsky 2007), which are indeed better terms, can be thought of in a way not dissimilar to Curry's suggestion (see also Dowty 1982 and more recently Needle 2022).

5. There are some technical issues with the definition of maximal in Speas's system, and it is not obvious that the relation between a word and the category containing it is one of domination, but these issues are fixable with some tweaking of the system. For example, Speas proposes that each distinct word in a tree comes with a lexical index, ensuring that two instances of what might look like the same category (e.g., a DP within a DP) are actually distinct, so that each can be maximal. X in Project Alpha is then a variable over the properties of the lexical item (category features, theta-related features, case features, the lexical index, etc.).

6. There are many technical questions that this account leaves open, but my purpose here is not so much to critique the account as to outline its theoretical motivation and sketch its core mechanisms.

7. Again, this may not be the correct analysis of this structure, and, of course, there is controversy about whether nominal phrases are headed by the determiner or by the noun. The example here is intended just to show the mechanisms, rather than to be an analysis.

8. Aissen notes that it may be that it is first raised to become an applied argument of the clause before extraction, though this is immaterial to the point here.

9. As noted by Chomsky (2005), an overarching constraint on the system, usually referred to as No Tampering, will disallow any change to the SO from which *b* is extracted. No Tampering simply says that an operation cannot destroy information that has already been constructed, and is seen as a constraint on the general monotonicity of the system. Given No Tampering, the outcome in (45b) is guaranteed.

### 3 Parts

1. There are versions of set theory, in fact the standard Zermelo-Fraenkel axiomatizations, where there is no distinction between sets and objects: everything is a set. However, my reading of the syntactic literature is that syntacticians in general assume that there is a distinction between the basic units of syntax (categories, features, perhaps lexical items, which are not set theoretic, and constituents, which are). Many thanks to Giuseppe Varaschin for pointing out the issue to me.

2. There are versions of the Merge approach that take lexical items to be “turned into” sets before Merge operates on them via a special operation, such as Hornstein (2024, pp. 71–74), but this is not the usual approach.

3. Thanks go to the participants of a party hosted by Kristina Bentzen and Peter Svenonius where this name got comprehensively workshopped!

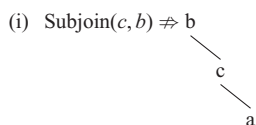
4. This system of axioms is somewhat redundant, since Irreflexivity follows from Asymmetry, and Asymmetry follows from Irreflexivity and Transitivity combined, but it will be useful to have the three concepts distinguished when it comes to discussing the implications of the system.

5. I put “tree” here in scare quotes as these diagrams are not trees in the standard sense of directed acyclic graphs (Partee et al. 1990), as we will see directly.

6. One might also ask why not have an operation  $\text{Subjoin}(c, a)$ , constructing this object. I address this in section 3.2.3.

7. Binariness here is not the same as classical binary branching, as in Kayne (1984); as we will see, it is more akin to a claim that there are maximally two fundamental syntactic dependencies from a single head.

8. Note that Dimensionality will rule out a kind of “Tucking-in” derivation, where subjunction of *c* to *b* “displaces” *a*. Take the complex object *b* in (6). We can’t derive (i) as follows:



In (i), the subjunction targets *b* and subjoins *c* to it. But, by Dimensionality, this would be the second subjunction to *b*, so it should be in the second dimension, whereas in (i), *c* is a 1-part of *b*.

9. There is an important set of issues that I will put aside here to do with the relation between the V category and certain complements, such as infinitival complements of raising verbs, small clause complements, complements of restructuring predicates, and so on (see Wurmbrand and Lohninger 2023 for recent discussion). These seem to involve semantic parthood, and, I think, are probably best understood as 1-parts of V (that is, subjoined to V). However, they are all clearly higher in any Extended Projection relation than V, so, as noted above, it is not the case that 1-parts are always lower in the Extended Projection. In some sense, the Extended Projection here is “reset” (Svenonius 2024a). I’ll leave the analysis of nonfinite clausal complementation for another time, and, for the

purposes of this book, I will mainly talk about cases where, if  $x$  is a 1-part of  $y$ ,  $x$  is lower in the relevant Extended Projection than  $y$ .

10. One might think that, since the objects that Merge creates are all terms, terms are given by the system. However, in order for Merge to apply to a term  $x$  within another term  $y$ , some extra operation needs to “search” inside  $y$ . That operation must go beyond Merge, and effectively be transitive in nature, or be able to otherwise access previously constructed units (e.g., in the way that Bobaljik 1995 suggests, or via a notion of accessibility as in Chomsky 2022).

11. The Strict Cycle Condition in fact has two separate aspects to it, when viewed through a more modern lens. It blocks derivations that extend embedded structures (the interior of lower cyclic nodes), and it also disallows access to elements inside those embedded structures. The former is handled in modern theories by Extension, and the latter by Phase Theory. Sometimes Extension-type cyclicity is referred to as Strict Cyclicity, but I’ll avoid that terminology, since the Strict Cycle Condition in fact entailed both Extension-type cyclicity and Phase-type cyclicity.

12. These kinds of constraints actually make Merge asymmetric, as they ensure that one of the arguments is undominated (a “root node”). In a case where both arguments are undominated, Merge looks symmetric, but that is simply a side effect of what is underlyingly an asymmetric constraint (the constraint requires *one* argument of Merge to be undominated). Of course, if Phrase Structure requires asymmetry, that’s another reason not to model it via binary set formation.

13. As Collins notes, these various requirements on Merge that bar replacement of categories don’t immediately rule out a countercyclic operation that Merges some object in parallel to an embedded object. He suggests that this is ruled out because it is unlinearizable. I return, below, to the question of Parallel Merge operations (c.f. Citko 2005, and many others).

14. I leave the other logical possibilities aside for the moment: Parallel Merge/Sideways Movement, where some element  $B$  is drawn from an object  $A$  but is Merged with an external object  $C$  and a derivation where objects  $B$  and  $C$  inside objects  $A$  and  $D$ , respectively, are Merged to create a new object  $E$ .

15. It may well be possible to defend a mereological system built along such lines, taking the part relation to be similar to the dependency relation of Dependency Grammar, but that is not the approach I will develop here.

16. There is an open question about whether there are “roots” in Phrase Structure. I followed Borer (2013) in Adger (2013) and took roots to be the “base” of Extended Projections; in the current system this is not necessary. I’ll develop the system here without assuming roots, adopting the idea that phonology works on the exponents of sequences of SOs, mainly to keep the ontology simple. It would, I think, be possible to incorporate roots into this system, but the issues I focus on here, which are fundamentally connected to  $\bar{A}$ -locality, are orthogonal to the questions that raises.

17. To allow checking between category features, we can think of each of the categories in an Extended Projection as a set containing a single interpretable category feature, plus various other features (so  $T$  would be  $\{T, \text{Past}, uD\}$ , with the  $T$  category being singled out as having a particular position in the relevant Extended Projection). I’ll continue, however, for familiarity’s sake, to notate SOs with just the category, putting the other features in square brackets.

18. Bye and Svenonius (2012) distinguish the @ diacritic, which specifies the linearization point from a separate diacritic,  $w$ , which accesses the lexicon and instructions to run a phonological cycle. I abstract away from this complexity here.

19. Evidence for the exponent of  $T[\text{Past}]$  being  $/\gamma/$  is that in vowel initial verbs (or verbs beginning with  $f$  followed by a vowel),  $/\gamma/$  surfaces phonetically. For the verb  $\partial l$ , ‘drink’  $/\partial:l/$ , the past is  $dh\ \partial l$ , ‘drank’  $/\gamma\partial:l/$ . For an  $f$  initial verb like  $f\bar{a}g$ , ‘leave,’  $/f\bar{a}:k/$ ,  $T$  causes lenition, and  $f$  lenites to zero, making the verb vowel initial so that  $/\gamma/$  surfaces: the past is  $dh\ f\bar{h}\bar{a}g$ , ‘left’  $/\gamma\bar{a}:k/$ . See Adger (2010) for discussion.

## 4 Angles

1. The theory is more sophisticated than the sketch given in (4), and there are a number of different ways of implementing it, but I abstract away from that here.
2. There are other versions of the PIC, including one that makes the complement of a phase head only inaccessible to operations that are triggered by the next phase head up (and higher), as in Chomsky (2001, p. 13). Other proposals advocate for more flexible notions of what a phase is, further complicating the theory (e.g., den Dikken 2007; Gallego 2010).
3. We could restrict Angular Locality to allow only internal subjunctions that have exactly one angle, by rewording it to the effect that  $\gamma$  can subjoin to  $\beta$  only if there is an  $\alpha$  s.t.  $\gamma$  is a 2-part of  $\alpha$  and  $\alpha$  is a 1-part of  $\beta$ . This would limit subjunction so that it doesn't change the Dimensionality of an object's part relation with other objects (a 1-part of an object cannot become a 2-part of another object, though a 2-part may). This may be the right outcome if, as I argued in Adger (2013), roll-up analyses should be ruled out in general. I leave this set of issues aside here.
4. There are other, stronger, Anti-Locality constraints, such as those that require a movement operation to cross a category (e.g., Bošković 2013). The system does not rule out such structures as it stands, so either would need to be strengthened to capture this, or some other explanation for the phenomena that motivate such stronger Anti-Locality constraints is needed.
5. Since we have derived the result that all internal subjunction is upward, I'll generally leave off arrows in the diagrams from now on, unless the discussion warrants their use.
6. Unlike in the pied-piping example, the preposition in this third construction appears in a special definite form, not a default form. The preposition *ri*, 'to/with' is syncretic for these two forms, but we can see the difference in other prepositions. For example, the preposition *do*, 'for/to' has the default form *dha*, but the definite form *dhan*, so we have the following contrast:

- (i) a. Cò dha a thug thu an leabhar?  
 who to.3SM REL give.PAST.REL you the book  
 'Who did you give the book to?'  
 b. Cò dhan an tug thu an leabhar?  
 who to.DEF C give.PAST.DEP thu an leabhar  
 'Who did you give the book to?'

I'll abstract away from this in what follows. I think the form of the P is an allomorph that is dependent on the following C, but this is not the place for defending that proposal.

7. This is, of course, very similar to the scope-marking strategies found in German dialects, Romani, Hindi, Bangla, some Arabic dialects, and elsewhere (Dayal 1993). I leave the complexities of this construction for future investigation.
8. There is much more complexity in these structures that I will leave for another time. The true pied-piping construction is only available with the *wh*-expression *cò*, and in such cases *cò* loses its usual obligatory humanness interpretation (so can mean 'who' or 'what'). That is not the case for the PP fronting examples. The nasalization found with dependent forms in some dialects is realized as voicing of unvoiced stops, and is traditionally known as "eclipsis."
9. Thanks to Luke Adamson for the confirmation that this is indeed possible and to Stan Zoppi for the suggestion.

## 5 Wh-Islands

1. One potential puzzle for this kind of system is the class of languages that allow multiple *wh*-elements to move overtly to the C domain, suggesting multiple specifiers of C. A subclass of these languages still show the *wh*-island effect (see Rudin 1988 on Polish, Serbo-Croatian, and Czech). However, there is some evidence that this particular subclass of languages does not actually move all *wh*-elements to the specifier of the same C; rather some raise to a lower position. If that is correct, such languages are brought within the same system (Bošković 2000 et seq.).

2. I leave aside here wh-islands where the embedded clause is headed by a wh-complementizer, such as *whether* or *if*. I take it that these cases, which are typically much better in terms of the levels of acceptability assigned to them, are the result of other factors, perhaps semantic (e.g., Abrusán 2014).

3. At time of publication, twenty-three speakers, from both the UK and North America, linguists and nonlinguists. Speakers who do not allow wh-island violations also, unsurprisingly, do not allow these cases.

4. There is one case I know of where such double extraction of wh-expressions is reported grammatical though unacceptable, which is a Swedish example given in Engdahl (1986). Most of Engdahl's cases of double extraction from a clause involve one of the extractions being local, to the edge of that clause, complying with the generalization expressed by WIRE. However, she gives the following case of topicalization, where there are three extractions.

- (i) Var det Södersjukhuset som hälsovårdsmyndigheterna lät undersöka [CP  
 was it South-Hospital-DEF that Public-Health-Department-DEF was investigating  
**vilka färskvaror** det var oklart [CP (vilka färskvaror) [CP *vilken grossist* som  
 which produce it was unclear which caterer that  
 (*vilken grossist*) levererat (**vilka färskvaror**) till (Södersjukhuset)  
 deliver.PRFX to

'Was it the South Hospital that the Department of Public Health was investigating which produce it was unclear which caterer had delivered that produce to that place.'

Engdahl remarks that (i) would "sound extremely odd" but suggests that the theory should not mark it as ungrammatical because that would involve "counting." Instead she puts its unacceptability down to processing factors. The current theory rules it out as a straightforward WIRE effect: the subject of the most embedded clause *vilken grossist*, "which caterer" stays local to that clause, but two other expressions in the clause are extracted: the object (in bold in the example) and the goal (underlined).

5. Thanks to Stan Zompi for pointing this out to me, and for discussion of the complexities.

6. Stan Zompi notes that the Italian examples, for him, improve to almost being good when they are nonfinite.

7. Marcel Den Dikken pointed out to me at GLOW 2023 that the system developed in Lasnik and Saito (1984), with one minor tweak, can also capture the observation discussed here, though that paper doesn't itself, as far as I can tell, discuss the relevant data. Den Dikken pointed out that multiple movement out of a single clause in Lasnik and Saito's system will leave two traces in the specifier of that clause, and that one of these will necessarily not be antecedent governed by the wh-expression in the next higher complementizer phrase. The tweak is that in Lasnik and Saito's paper, one of these traces could be deleted at Logical Form (LF), so the effect wouldn't emerge, but if the system were to allow the trace to remain, it could capture the effect. I think this is actually Den Dikken's own observation, as opposed to Lasnik and Saito's, but in any event the current system predicts the effect rather more straightforwardly.

8. I have assumed here that the Goal argument of the verb is some kind of Dir(ectional) category and that it appears lower than the position where the Agent is introduced.

9. The alternative offered by Richards (1997) is that there are multiple specifiers of CP in these languages. However, that kind of analysis is not possible in the mercological system because of Dimensionality.

10. The system predicts that the WIRE effect should appear in [−MFS] languages, but these languages do not allow multiple extraction from a clause anyway, so cannot generate the relevant structures.

11. There is an ongoing discussion about this kind of data in footnotes to which I now add: Dayal (1996, p. 99 fn 16) claims that (5.52a) does not have the pair list reading for the first and last wh-expressions, contra a claim by Mahajan (1990, p. 180 fn 6), who attributes to David Pesetsky the judgment that examples like (5.52a), but with bare wh-expressions, do actually have the relevant

pair list reading. Dayal points out that it is necessary to use singular *wh*-expressions, to avoid an interfering plural reading, and I tend to agree with her that when one does this, the pair-list reading is unavailable. English native speaker linguists I have asked also tend to agree, though more general informal testing hasn't led to judgments that are consistent enough for me to be sure.

## 6 Nominal Islands

1. There are some languages, like Japanese or Mandarin, that allow initial APs, as noted by Whitman (1981), but these plausibly involve displacement, perhaps subjunction to D, as in the Mandarin phenomenon discussed in the next section. As far as I know, initial APs usually have a semantic effect on the interpretation of the whole phrase in these languages and so are irrelevant to the PP Peripherality generalization, which is about the base positions of categories (though see Belk 2017 for an alternative view of the data and a different proposal for capturing it).

2. There is an alternative, which is that movement takes place for case or other licensing reasons, as in Kayne 2004; Kayne's proposal derives a very similar surface structure, but I argued that the *p* analysis had certain advantages for understanding certain binding and reconstruction facts.

3. Peter Svenonius pointed out to me examples like "a man with bushy eyebrows" that are equally impossible to extract from but are not immediately susceptible to a simple reduced relative analysis. However, if the preposition *with* encodes a kind of dative, it may be that "with bushy eyebrows" is underlyingly "to-whom are bushy eyebrows," and that this too is a reduced relative.

4. There is an interesting question as to the lack of evidence for Ds that have been extracted from being marked morphologically in a particular way. Note, however, in the approach to nominal structure I've been developing, the article is not a spellout of D, and D is often, in fact, unpronounced.

5. Kuno (1987) notes some exceptions to this effect with possessors:

- (i) a. This is the event that I liked CBS's reporting of best of all.
- b. This is the story I haven't been able to get Mary's version of.

Kuno notes that the relevant factor seems to be whether the possessive is contrastive or not. One possible avenue for analysis would be that DPs allow a further category in the Extended Projection of D encoding contrast, and this means that the D category itself has a free 2-part slot, though I find myself skeptical of this analysis, since there isn't any clear syntactic evidence for it. One might take this as a good argument that these extraction effects are not to be handled syntactically (which is Kuno's point, see also Goldberg 2006; Chaves and Putnam 2021), but there are other reasons to take the effect to be syntactic, as I argue below. I leave Kuno's examples as a puzzle pending further examination.

6. In the system developed here the analysis of (6.43b) will involve subjunction of the constituent *Dem*, which contains the classifier and the *N*, to *D*, and then linearization of the constituent *Dem* to the immediate right of the relative clause.

7. Bowers suggests that strong quantifiers have the same effect. I don't share the judgments he gives and find the following fine, though the literature is variable on these judgments:

- (i) a. Who did you buy every statue of?
- b. Who did you buy few statues of?

The theory will make extraction in these cases depend on subjunction to D or not, so the issue will be exactly what drives that subjunction.

8. There is an interesting question for all these accounts about how to account for the fairly systematic variance in judgments here, where extraction from a DP with a definite article is less bad than from a DP with a prenominal possessor. I return to this below.

9. In later work, Bošković (2013) argues that NP is in fact a phase, and that APs etc. are base-generated at the phase edge, rather than moving there.

10. It is actually something of an open question as to whether examples like (52) involve extraction of the AP from the NP. Bondarenko and Davis (2023) argue that they involve the whole nominal phrase raising, but only parts of each spelling out (a distributed deletion analysis), while Abels (2012a), among others, argues that it involves movement of (a constituent containing) the N to a position outside the nominal phrase, followed by remnant fronting of (the constituent containing) the adjective.

11. To be strictly accurate, Bošković (2013) adds the rider “like in (6)” to this generalization, where his example (6) is the Serbo-Croatian version of my (52) above. The reason for this is to exclude possessors since, if possessors are generated below D in a language with articles, but not adjoined to N (say in the specifier of a special functional projection), then movement to the specifier of D will cross a category and not violate Anti-Locality, allowing Possessor Raising even in a language with articles.

12. An alternative would be to have a specialized Det in the Extended Projection of A take A as a 1-part, and then subjoin to Det in the Extended Projection of N, and finally to D, then to a higher clausal category. This involves less work for the morphology at the expense of an extra projection in the syntax, and there are of course many languages that have something akin to a definiteness marking on the A that agrees with that on the N.

13. There is some discussion of the solidity of this data in Otero (1996) and in Uriagereka’s (1996a) response, the upshot of which, as far as I can ascertain, is that the judgments are nuanced but consistent.

14. There is a huge literature that provides evidence for ascribing the specificity and definiteness properties of a nominal phrase to phi-features or clitics associated with that phrases’ clausal licensing positions, e.g., Mahajan (1991), Adger (1994), Anagnostopoulou (2006), López (2012) among many others.

15. I have given the translations here as supplied by Setayesh Dashti, which are more precise than those supplied by Karimi.

16. Karimi and Lobeck (1998) note that overt scrambling from Ezafe constructions is ungrammatical in general.

17. Recall that the checking system I adopt here just requires that a feature that needs to be checked, and the feature that checks it, are in a 2-part relation. It does not require that, for example, the checker is a 2-part of the checkee, or vice versa.

18. On a side note, if we assume that the Persian scrambling clausal head bears [uFam], just like the German one does, this will predict that any D spelled out as *râ* can scramble or not, but that true indefinites cannot scramble, which appears to be the case (Browning and Karimi 1994).

19. This style of explanation is one that I’ve proposed elsewhere to tackle sociolinguistic variation in grammars, the so-called Combinatorial Variation model (see Adger 2006 and especially Adger 2007).

20. The notion of weakness here is distinct from what is sometimes called weak definiteness, which involves definites that have a kind of generic interpretation as in the following examples:

- (i) a. My grandmother often listens to (vs. fixes) [the radio].
- b. Someone needs to open (vs. clean) [the window].

Schwarz (2014) notes that weak definiteness is signaled by weak definite articles, but the two are conceptually distinct.

21. Hankamer and Mikkelsen (2021) suggest something similar for NPs with clausal complements in Danish. See footnote 22.

22. Hankamer and Mikkelsen (2021) in their analysis of DPs that have CPs inside them in Danish show that in that language finite CP complements to N appear with the complementizer *at* but may also have a prepositional element *om*:

- (i) a. den ide at ingefær gavner fordøjels-en  
     the idea that ginger aids digestion-DEF  
     ‘the idea that ginger aids digestion’

- b. ide-en **om** at ingefær gavner fordøjels-en  
 idea-DEF about that ginger aids digestion-DEF  
 ‘the idea that ginger aids digestion’

The presence and absence of the preposition correlates with a number of morphosyntactic and semantico-pragmatic facts. Morphosyntactically, (ia) only occurs with the free-standing article *den*, while (ib) occurs with suffixed definites, indefinites, possessors, and demonstratives:

- (ii) en / hans / denne ide \*(om) at ingefær gavner fordøjels-en  
 a / his / this idea (about) that ginger aids digestion-DEF  
 ‘a / his / this idea that ginger aids digestion.’

Semantico-pragmatically, only (ib) can be used with a linguistic antecedent. If examples like (ia) involve little *d*, while examples like (ib) do not, that might be an analytical hook for the morphosyntactic and pragmatic distinctions that Hankamer and Mikkelsen discuss.

## 7 Subject Islands

1. Chomsky’s actual formulation required that the subject phrase properly contain the minimal major category containing *Y* and also that *Y* be subjacent to *X*. These additional requirements were imposed to accommodate the behavior of certain reciprocals, which were assumed to involve a movement rule at the time. I abstract away from this here.

2. Ross (1967, p. 133) in his discussion of islands does not impose such a condition, giving the example *Of which cars were the hoods damaged by the explosion?* I return to such cases below.

3. There are variants of the Freezing approach where only certain positions in movement chains are opaque for extraction (Boeckx 2003). Boeckx argues that once a subject has moved to a position where it agrees (for example, the specifier of TP), it becomes opaque (technically, via early spell out of the TP). Chomsky (2008) also assumes that different copies in a movement chain may have different conditions on them, as discussed below. However, neither of these systems connect DP internal movement with DP movement and opacity to subextraction in the way the mereological system does.

4. Jurka’s subjects were thirty-one native nonlinguist speakers of German who used a seven-point Likert scale to rate the sentences. The six conditions below were split across three lexicalizations, leading to 108 items that were distributed via a Latin Square design and (pseudo)-randomized with sixty other items. Jurka shows that individuals in general follow the pattern of the group, suggesting that the results are not a side effect of averaging across idiolects.

5. As discussed below, Abeillé et al. show that pied-piping from transitive subjects is actually fairly well accepted. However, it should be noted that their experiment did not test unaccusative subjects, so we don’t have a relative comparison. We will see, interestingly, that Chomsky’s suggested distinction shows up more clearly in experimental work on a range of other languages, though I will argue that those results are ultimately not to be explained by the position of the low subject.

6. Zyman (2021) retheorizes Chomsky’s system, proposing to derive an extended set of results, though maintaining some of the core predictions of Chomsky’s system. His approach involves a constraint that bars a phrase subextracting from a phase edge to a position inside a higher projection that immediately selects the phase (his Phasal Antilocality constraint) together with some particular assumptions about which categories count as phases, about clause structure, and about feature checking. He assumes that *vP* is a phase, including in unaccusatives and passives, but that *ProgP* in existentials is not (contra Harwood 2013 who provides evidence for *ProgP* having phasal characteristics); that *TP* as well as *CP* is a phase and that embedded *CP*s in English include a low Force head that allows Anti-Locality to be satisfied when something is subextracted from the specifier of *T*. He also assumes that certain attracting features on heads can be ordered. With this much in place, he argues that the system can make more nuanced predictions than Chomsky’s system does, extending to some novel cases, such as when an adverb intervenes:

- (i) (?)Of which car, according to your recollection, did the driver cause a scandal?

Zyman claims such examples are an improvement over identical sentences without the adverb. I haven't been able to replicate this result via informal judgment tasks with native speakers, and don't get the judgment myself, so I leave Zyman's proposal aside in what follows and focus on Chomsky's.

7. Privoznov focuses his attention on literary Russian, arguing that colloquial Russian allows an extra mechanism for displacement, which muddies the waters somewhat, but which can be controlled for given careful design of the test sentences used.

8. These experiments, like Jurka's, involved judgments by native nonlinguist speakers who used a seven-point Likert scale. The sentences were varied using five different lexicalizations, seventy-four fillers were added, and the sentences were pseudo-randomized. The results of forty-five speakers underwent statistical analysis.

9. Certain entities can also be assumed to always be available in the Common Ground (such as the speaker, hearer, the sun, earth, galaxy, etc., and important entities varying from community to community, or even speaker to speaker (Erteschik-Shir 2007, p. 18).

10. Reinhart gives some examples of sentence topics that appear to introduce new Discourse Referents/files. For example, in (i), the phrase *Pat McGee* introduces a new file, but is clearly the sentence topic:

- (i) Pat McGee, I don't know if you know him, he - he lives in Palisades—he used to go to the school I did...

Reinhart concludes from this that sentence topics don't need to be discourse or context familiar. I'm not sure the argument goes through. It may be that *Pat McGee* is actually in focus in a Left Dislocation construction, introducing a new Discourse Referent, and then the pronoun *he* is the sentence topic. Similarly, Krifka and Musan (2012) give the following example that is, they suggest, a new information sentence topic:

- (ii) A good friend of mine married Britney Spears last year.

Even here, though, we have a specific indefinite anchored to the speaker by *mine*. Compare with (iii), which doesn't seem to allow the subject to be construed as the sentence topic (unless *a werewolf* is discourse familiar):

- (iii) A werewolf married Britney Spears last year.

Tomioka (2016) notes that, in Japanese, the subject in (ii) cannot be marked with the *-wa* topic marker.

11. In a Discourse Representation Theory–type system (e.g., Kamp and Reyle 1993), uttering a DP introduces a new Discourse Referent, but because it is a sentence topic, it is specified as discourse familiar to a previously introduced Discourse Referent that has the relevant information about cathood and ownership attached.

12. They may also be interpreted as generic. I leave generic interpretations for future work.

## 8 Conclusion

1. Haegeman proposes a raising analysis of these clausal adjuncts, but I don't think that this is integral to the basic claim, nor is there evidence that these are not matching relatives.



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